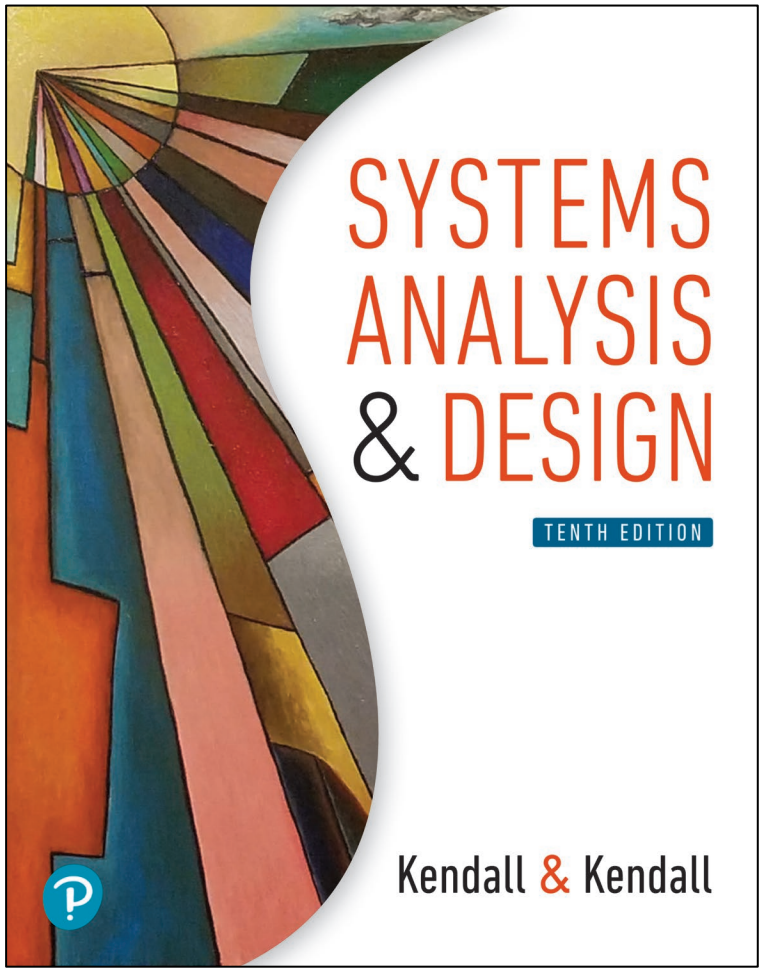
Systems Analysis & Design

Tenth Edition



Chapter 1

Systems, Roles, and Development Methodologies

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Learning Objectives

1.1 Understand the need for systems analysis and design in organizations

1.2 Realize what the many roles of the systems analyst are

1.3 Comprehend the fundamentals of three development methodologies:

- **1.3.1** The systems development life cycle (SDLC)
- **1.3.2** The agile approach including Scrum
- **1.3.3** Object-oriented systems analysis and design

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Information – A Key Resource

- Fuels business and can be the critical factor in determining the success or failure of a business
- Needs to be managed correctly
- Managing computer-generated information differs from handling manually produced data

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Major Topics

- Fundamentals of different kinds of information systems
- Roles of systems analysts
- Phases in the systems development life cycle as they relate to Human-Computer Interaction (HCI) factors
- Agile development
- Object-oriented Systems Analysis and Design

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Need for Systems Analysis and Design

- Installing a system without proper planning leads to great user dissatisfaction and frequently causes the system to fall into disuse
- Systems analysis and design lends structure to the analysis and design of information systems
- User involvement throughout a systems project is critical to the successful development of computerized systems

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Roles of the Systems Analyst

- The analyst must be able to work with people of all descriptions and be experienced in working with computers
- Three primary roles:
  - Consultant
  - Supporting Expert
  - Agent of change

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Qualities of the Systems Analyst

- Problem solver
- Communicator
- Strong personal and professional ethics
- Self-disciplined and self-motivated

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Systems Development Life Cycle (SDLC)

- The systems development life cycle is a phased approach to solving business problems
- Developed through the use of a specific cycle of analyst and user activities
- Each phase has unique user activities

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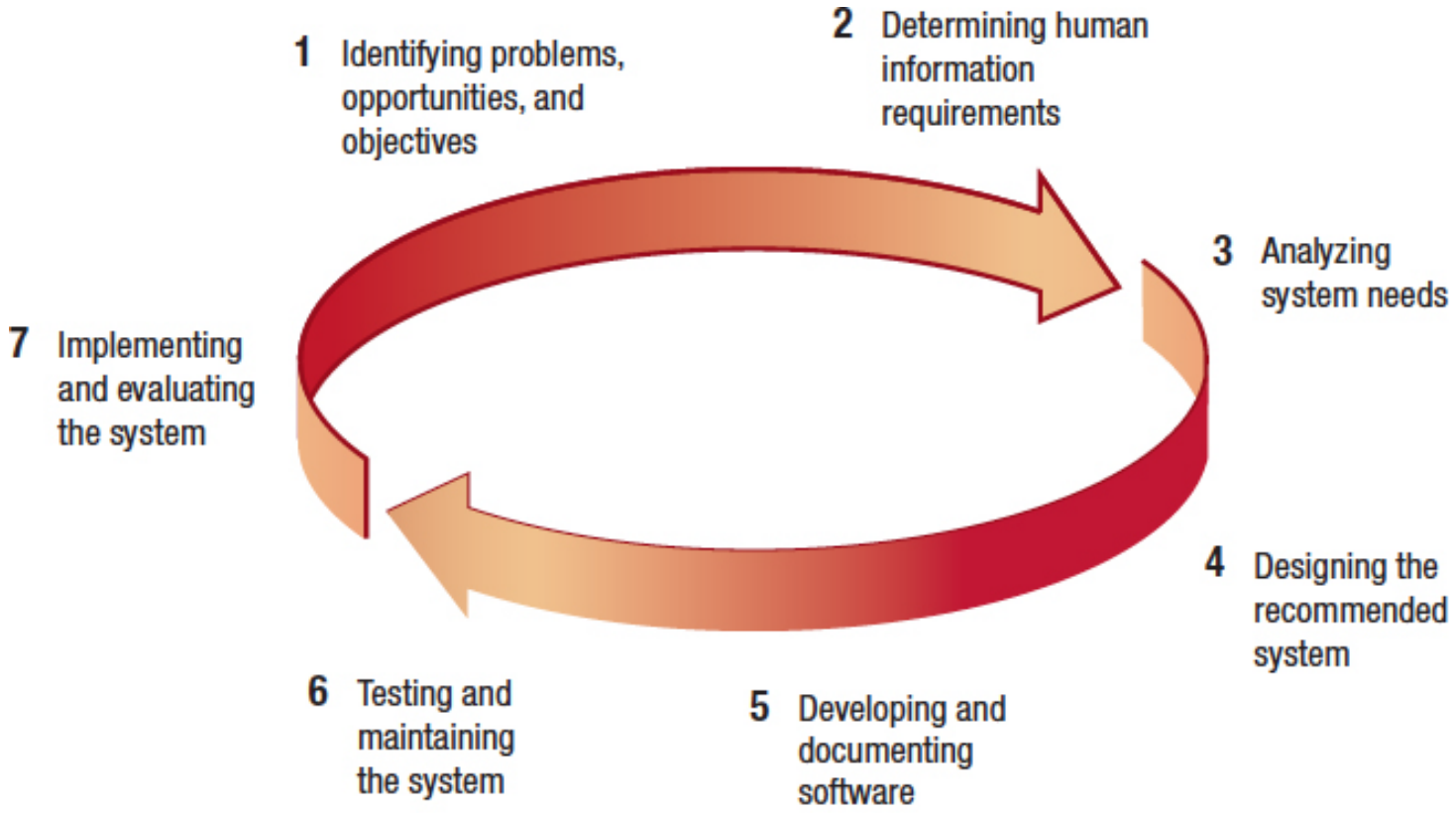
Figure 1.1 The Seven Phases of the Systems Development Life Cycle



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1. Identifying Problems, Opportunities, and Objectives

- Activity:
  - Interviewing user management
  - Summarizing the knowledge obtained
  - Estimating the scope of the project
  - Documenting the results
- Output:
  - Feasibility report containing problem definition and objective summaries from which management can make a decision on whether to proceed with the proposed project

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2. Determining Human Information Requirements (1 of 2)

- Activity:
  - Interviewing
  - Sampling and investing hard data
  - Questionnaires
  - Observe the decision maker's behavior and environment
  - Prototyping
  - Learn the who, what, where, when, how, and why of the current system

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Determining Human Information Requirements (2 of 2)

- Output:
  - Analyst understands how users accomplish their work when interacting with a computer
  - Begin to know how to make the new system more useful and usable
  - The analyst should also know the business functions
  - Have complete information on the people, goals, data and procedure involved

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3. Analyzing System Needs

- Activity:
  - Create data flow diagrams
  - Complete the data dictionary
  - Analyze the structured decisions made
  - Prepare and present the system proposal
- Output:
  - Recommendation on what, if anything, should be done

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4. Designing the Recommended System

- Activity:
  - Design procedures to accurately enter data
  - Design the human–computer interface (HCI)
  - Design files and/or database
  - Design backup procedures
- Output
  - Model of the actual system

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5. Developing and Documenting Software

- Activity:
  - System analyst works with programmers to develop any original software
  - Works with users to develop effective documentation
  - Coders design, code, and remove syntactical errors from computer programs
  - Document software with procedure manuals, online help, Frequently Asked Questions (FAQ) and Read Me files
- Output:
  - Computer programs
  - System documentation

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6. Testing and Maintaining the System

- Activity:
  - Test the information system
  - System maintenance
  - Maintenance documentation
- Output:
  - Problems, if any
  - Updated programs
  - Documentation

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7. Implementing and Evaluating the System

- Activity:
  - Train users
  - Plans the conversion from old system to new system
  - Review and evaluate system
- Output:
  - Trained personnel
  - Installed system

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THE IMPACT OF MAINTENANCE

- Maintenance is performed for two reasons
  - Remove software errors, and
  - Enhance existing software
- Enhance software for three reasons
  - Include additional features
  - Address business changes over time
  - Address hardware and software changes

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Maintenance Impact

- Over time the cost of continued maintenance will be greater than that of creating an entirely new system
- At that point it becomes more feasible to perform a new systems study

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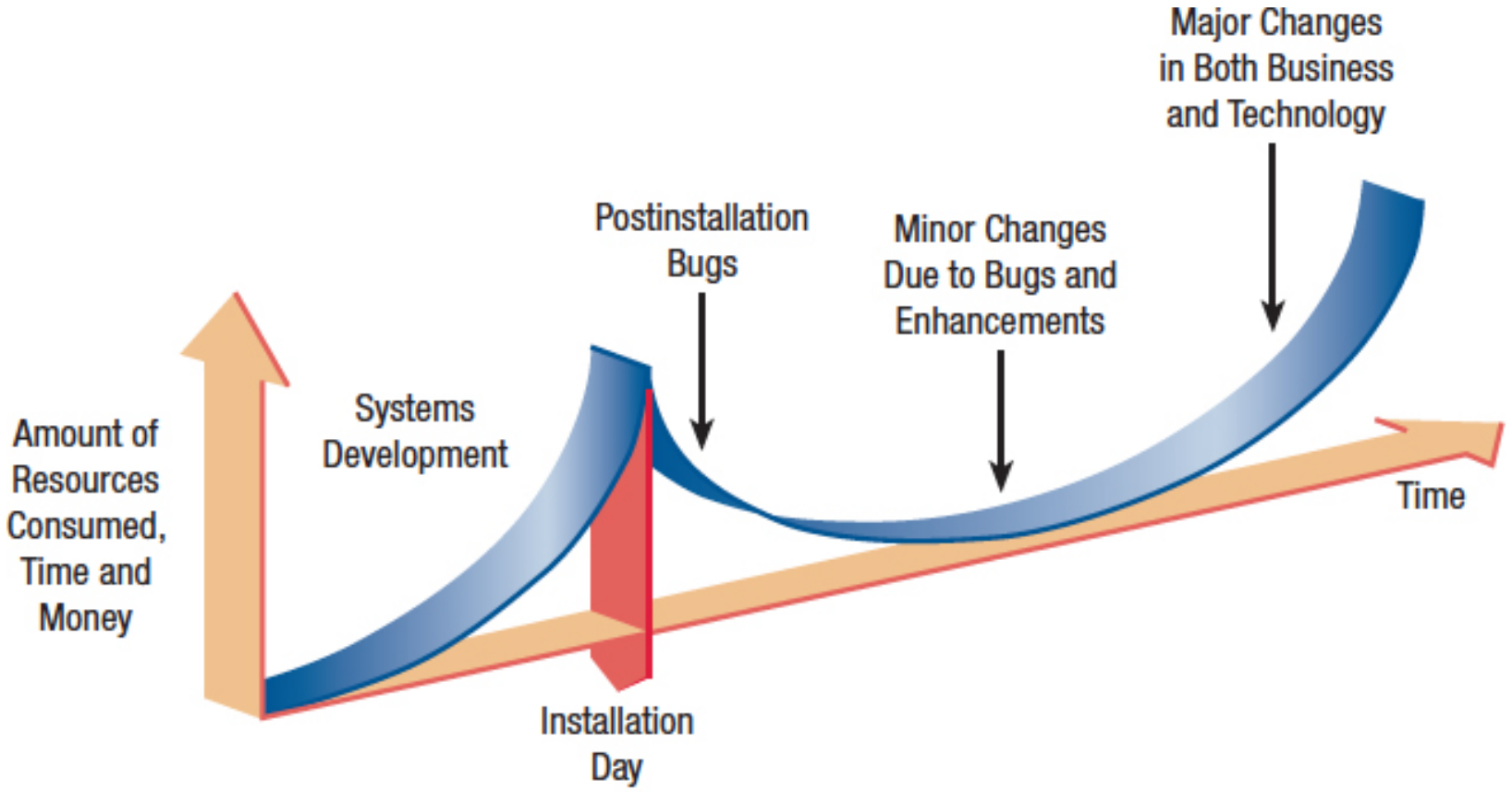
Figure 1.2 Resource Consumption Over the System Life



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CASE TOOLS

- CASE (computer aided software engineering) tools are productivity tools for systems analysts that have been created explicitly to improve their routine work through the use of automated support
- Reasons for using CASE tools
  - Improving Analyst–User Communication
  - Help support modeling functional requirements
  - Assist in drawing project boundaries

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THE AGILE APPROACH

- The agile approach is a software development approach based on
  - Values
  - Principles
  - Core practices

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Agile Values

- The four values are
  - Communication
  - Simplicity
  - Feedback
  - Courage

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Agile Approach

- Agile approach is
  - Interactive
  - Incremental
- Frequent iterations are essential for successful system development

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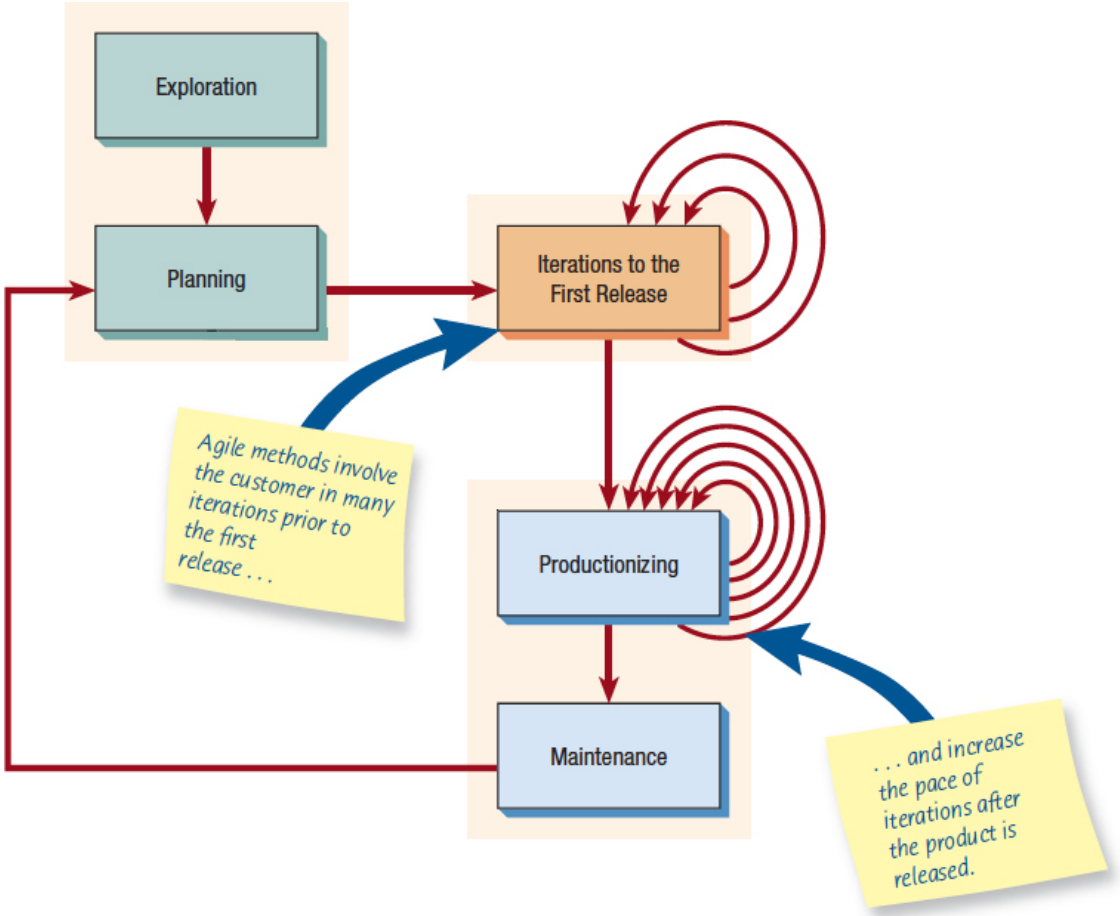
Figure 1.4 The Five Stages of the Agile Modeling Development Process



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1. Exploration

- Assemble team
- Assess skills
- Examine potential technologies
- Experiment with writing user stories
- Adopt a playful and curious attitude toward the work environment, its problems, technologies, and people

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2. Planning

- Rules that can help formulate the agile development team's relationship with their business customers
- Maximize the value of the system produced by the agile team
- Main players are the development team and the business customer

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3. Iterations to the First Release

- Iterations are cycles of
  - Testing
  - Feedback
  - Change
- One goal is to run customer-written function tests at the end of each iteration

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4. Productionizing

- The product is released in this phase
- May be improved by adding other features

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5. Maintenance

- New features may be added
- Riskier customer suggestions may be considered
- Team members may be rotated on or off the team

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OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN

- Alternate approach to the structured approach of the SDLC that is intended to facilitate the development of systems that must change rapidly in response to dynamic business environments
- Use unified modeling language (UML) to model object-oriented systems
- Each object is a computer representation of some actual thing or event

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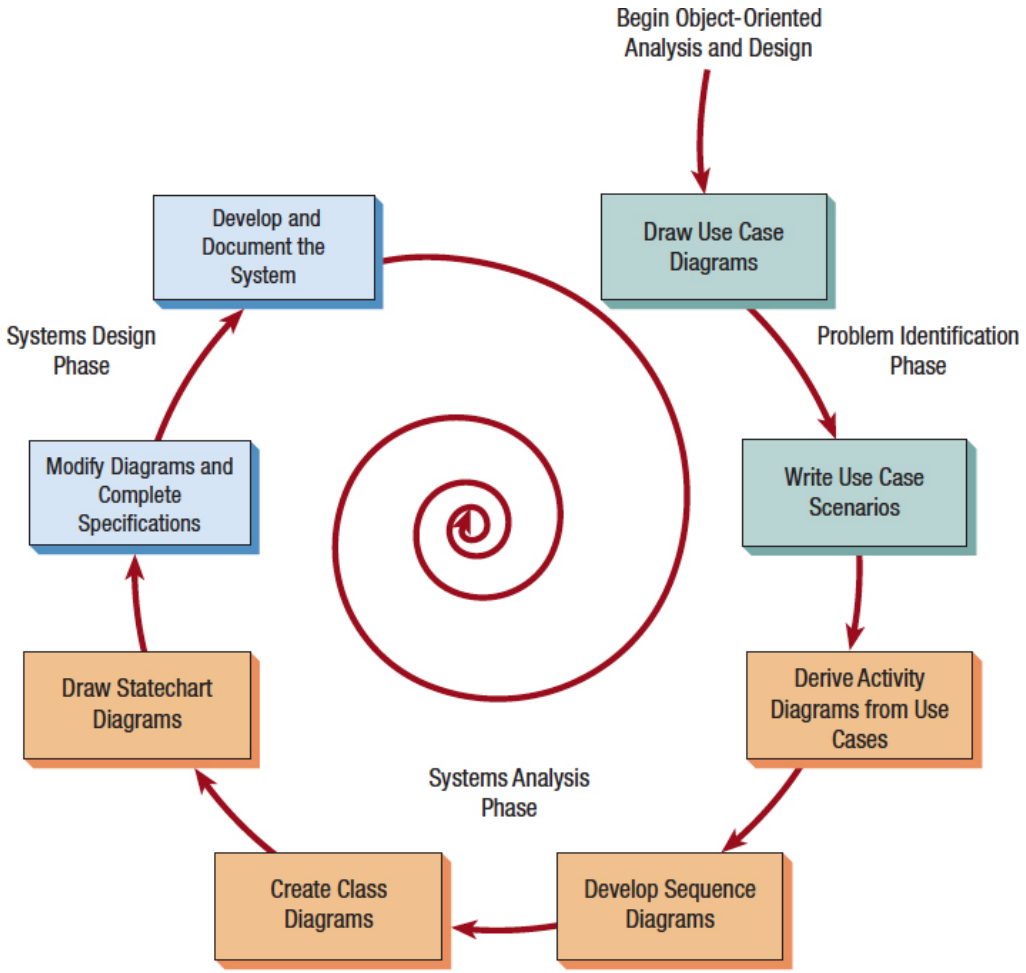
Figure 1.5 The Steps in the UML Development Process



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Define the Use Case Model

- Identify the actors and the major events initiated by the actors
- Draw a use case diagram
- A diagram with stick figures representing the actors and arrows showing how the actors relate

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Begin Drawing UML Diagrams

- Draw activity diagrams, which illustrate all the major activities in the use case
- Create one or more sequence diagrams for each use case that show the sequence of activities and their timing
- Review the use cases, rethink them, and modify them if necessary

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Analysis Phase

- Develop class diagrams
- Draw statechart diagrams

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System Design

- Modifying the existing system
- Modifying the diagrams drawn in the previous phase
- Write class specifications for each class

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Develop and Document the System

- The more complete the information you provide to the development team through documentation and UML diagrams, the faster the development and the more solid the final production system

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Figure 1.6 How to Decide Which Development Method to Use

Choose	When
The Systems Development Life Cycle (SDLC) Approach	systems have been developed and documented using SDLC it is important to document each step of the way upper-level management feels more comfortable or safe using SDLC there are adequate resources and time to complete the full SDLC communication of how new systems work is important
Agile Methodologies	there is a project champion of agile methods in the organization applications need to be developed quickly in response to a dynamic environment a rescue takes place (the system failed and there is no time to figure out what went wrong) the customer is satisfied with incremental improvements executives and analysts agree with the principles of agile methodologies
Object-Oriented Methodologies	the problems modeled lend themselves to classes an organization supports the UML learning systems can be added gradually, one subsystem at a time reuse of previously written software is a possibility it is acceptable to tackle the difficult problems first

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Open Source Software (1 of 6)

- An alternative to traditional software development
- Many users and coders can study, share, and modify the code
- Characterized as a philosophy rather than simply the process of creating new software
- Examples: Linux Operating System, Apache Web Server, Mozilla Firefox

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Open Source Software (2 of 6)

- Categorized open source communities into four community types
  - Ad hoc
  - Standardized
  - Organized
  - Commercial

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Open Source Software (3 of 6)

- Six different dimensions
  - General structure
  - Environment
  - Goals
  - Methods
  - User community
  - Licensing

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Open Source Software (4 of 6)

- Reasons for Participating in Open Source Communities
  - Rapidity with which new software can be developed and tested
  - Faster to have a committed group of experts develop, test, and debug code
  - This fosters creativity
  - Have many good minds work with innovative applications

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Open Source Software (4 of 6)

- Reasons for Participating in Open Source Communities
  - Potential to reduce development costs
  - Bolster their self-image
  - Contribute something worthwhile to the software development community

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Open Source Software (5 of 6)

- Reasons for Analyst Participation in the Open Source Community
  - Curiosity about software benefits
  - Achieve collective design
    - Incorporate open source software design into:
      - Proprietary products
      - Processes
      - Knowledge
      - IT artifacts

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Open Source Software (6 of 6)

- Collective Design
  - Through a process of collective design the IT artifact is imbued with
    - Community and organizational structures
    - Knowledge
    - Practices

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Summary

- Systematic approach to identifying problems
- Systems analysts are required to take on many roles
- Analysts possess a wide range of skills
- The systems Development Life Cycle
- CASE tools
- Agile approach
- Object-oriented analysis and design
- Open source systems

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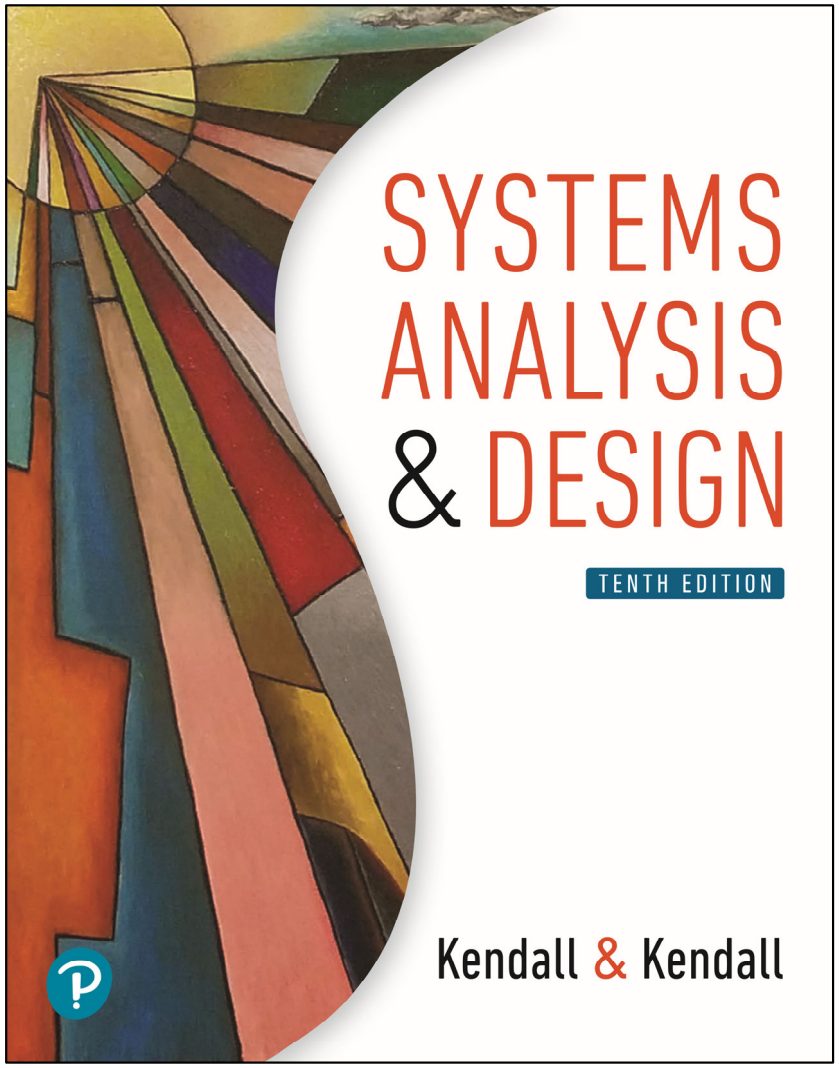

Chapter 2

Understanding and Modeling Organizational Systems

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Learning Objectives

2.1 Understand that organizations and their members are systems and that analysts need to take a systems perspective.

2.2 Depict systems graphically using context-level data flow diagrams, and entity-relationship models, use cases, and use case scenarios.

2.3 Recognize that different levels of management require different systems.

2.4 Comprehend that organizational culture impacts the design of information systems.

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Three Main Forces Interacting to Shape Organizations

- Levels of management
- Design of organizations
- Organizational cultures

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Organizations are Composed of Interrelated Subsystems

- Influenced by **levels** of **management decision makers** that cut horizontally across the organizational system
  - **Operations**
  - Middle management
  - **Strategic** management
- Influenced by organizational **cultures** and **subcultures**

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Major Topics

- Organizations as systems
- Depicting systems graphically
  - **Data flow** diagram
  - **Entity-relationship** model
  - **Use case** modeling
- Levels of management
- Organizational culture

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Organizations as Systems

- Conceptualized as systems designed to accomplish predetermined **goals** and **objectives**
- Composed of **smaller**, **interrelated** systems serving specialized functions
- Specialized functions are reintegrated to form an **effective** organizational **whole**

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Organizations as Systems

- *https://youtu.be/WxqDqJBPZA4?si=Eeled7UjO5cah3he*

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Interrelatedness and Independence of Systems

- All systems and subsystems are interrelated(相关) and interdependent (相互依存)
- All systems process inputs from their **environments**
- All systems are contained by boundaries **separating** them **from** their **environments**
- System feedback for planning and control
- An ideal system **self-corrects** or **self-regulates** itself

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Figure 2.1 System Outputs Serve as Feedback That Compares Performance with Goals

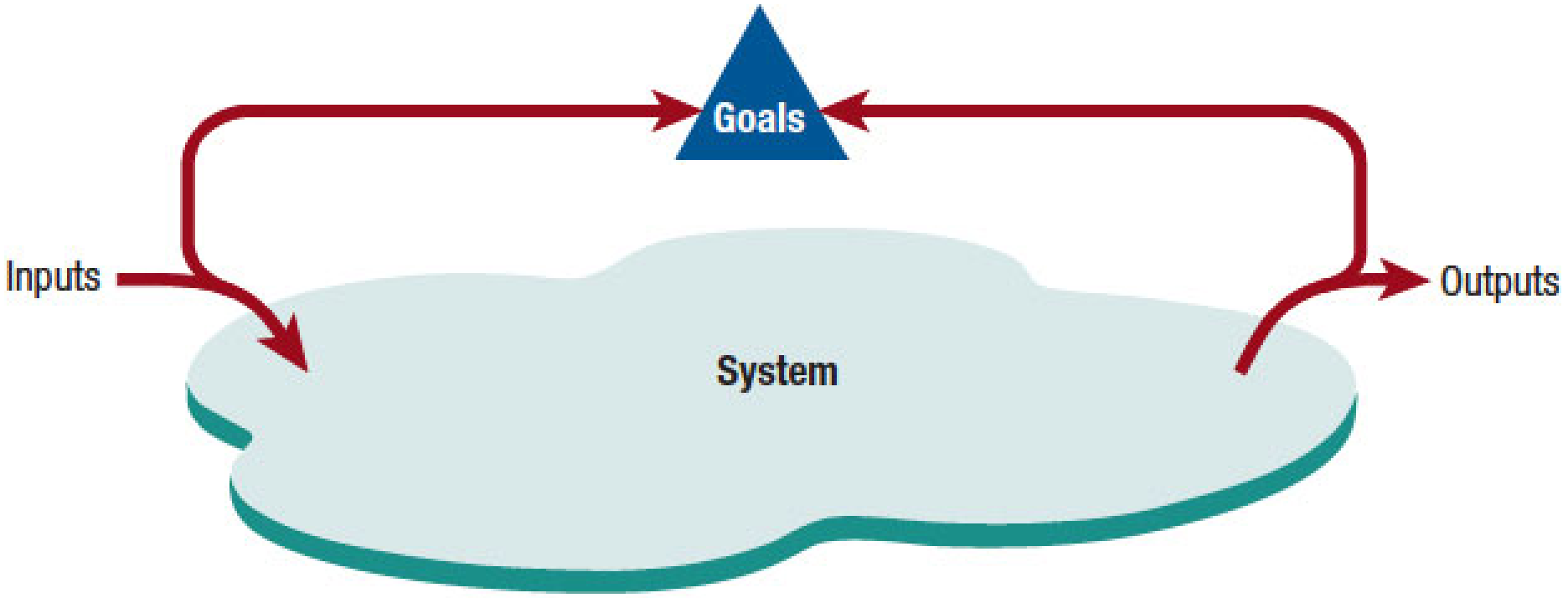

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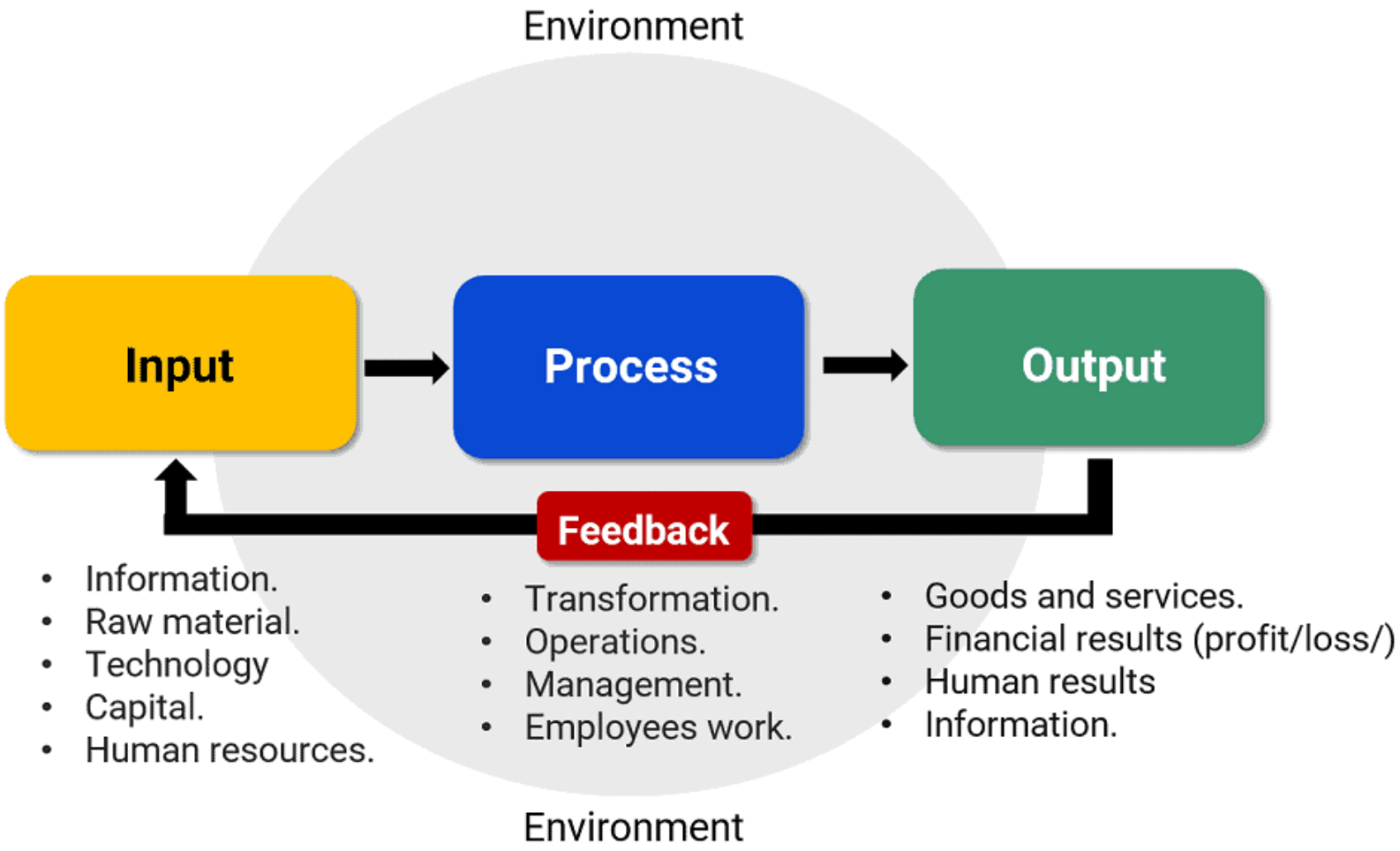
Figure 2.1a Model of System Theory For Organization Management



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Organizational Environments

- **Community**
  - **Location:** Affects access to talent, customers, and logistics.
  - **Demographics:** Local education & income levels influence workforce quality and consumer behavior.
- **Economic**
  - **Market Factors:** Customer demand, trends, and pricing influence strategy.
  - **Competition:** Drives innovation, pricing, and positioning.
- **Political**
  - **State & Local Government:** Policies on taxes, infrastructure, and incentives shape business climate.
- **Legal**
  - **Laws & Guidelines:** Federal to local regulations impact operations, HR, and compliance.

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Virtual Organizations and Virtual Teams

- A **virtual organization** has parts of the organization in different physical locations (like **branches**)
- Computer networks and communications technology are used to bring **virtual teams together** to **work** on **projects**

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Benefits of Virtual Organizations and Teams

- Possibility of **reducing costs** of physical facilities
- More **rapid response** to customer needs
- Helping virtual employees to **fulfill** their **familial obligations** to children or aging parents

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Looking into a Systems Perspective

- Allows system analyst to **understand businesses they will come into contact with**
- It is important that members of subsystems realize that they are **interrelated** with **other subsystems**
- Problems occur when **each manager** thinks that his/her **department** is the most important
- Bigger **problems** may **occur** when that **manager rises through the ranks**

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Figure 2.2 Looking into a Systems Perspective

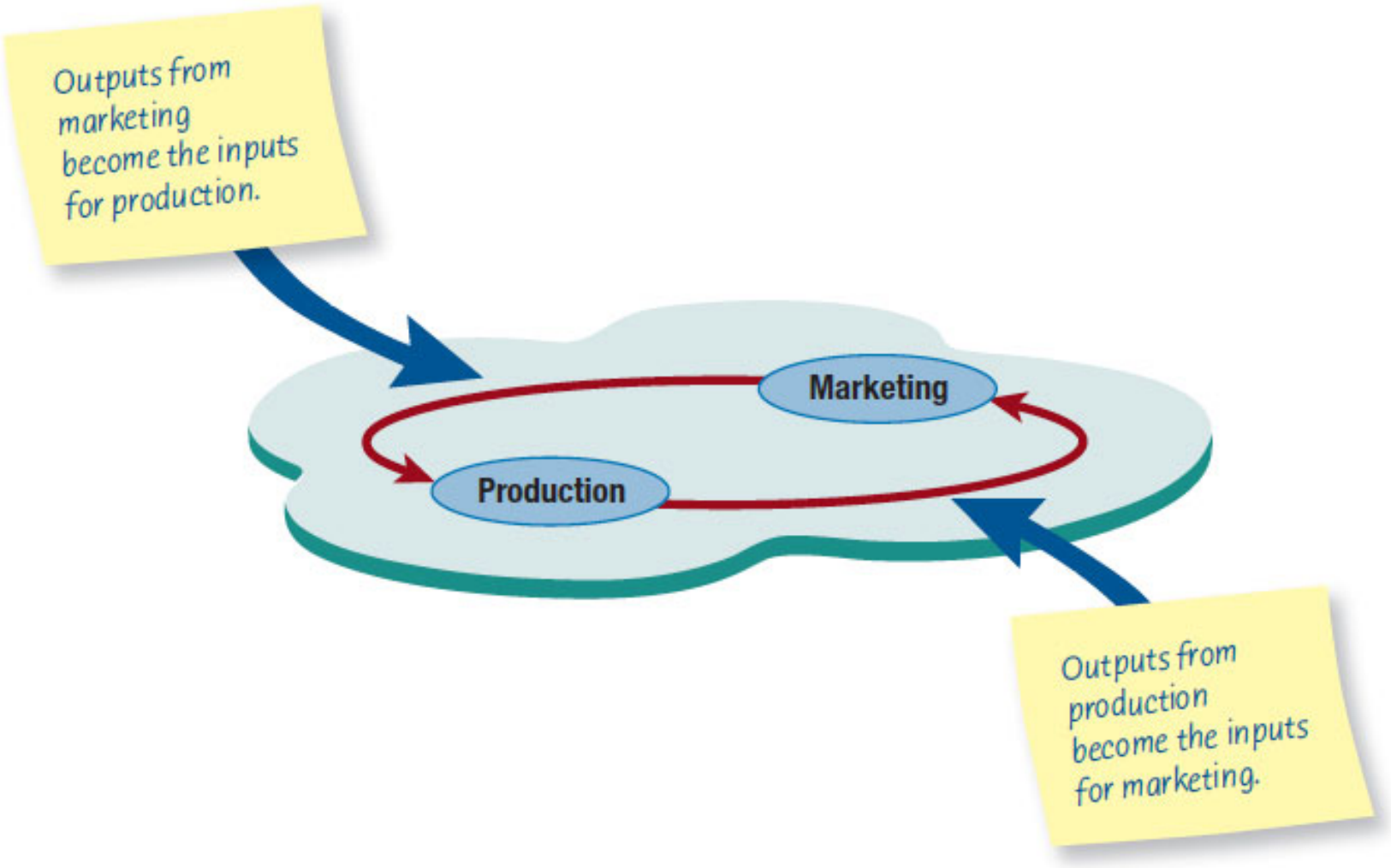

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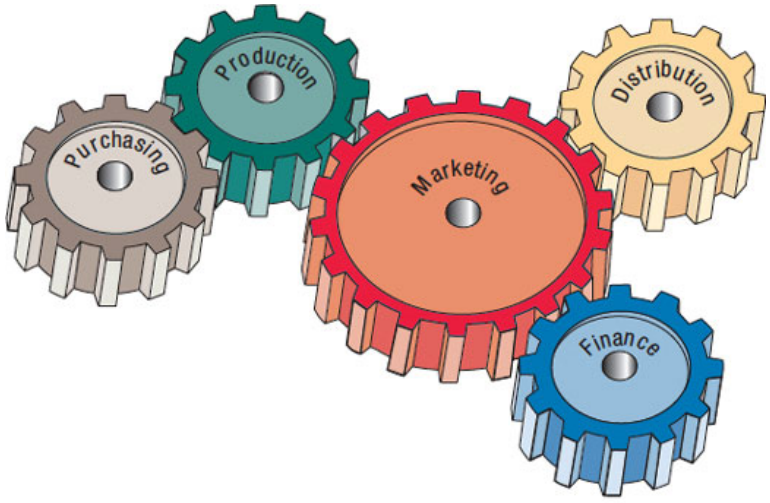
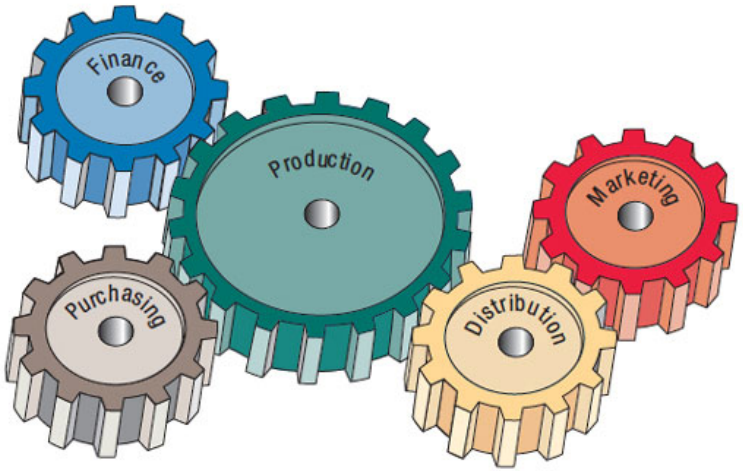
Figure 2.3 Perspective of Functional Managers



How a Marketing Manager May View the Organization



How a Production Manager May See the Organization

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Figure 2.3a: So How to Solve Different Perspectives?

Figure 2.3b: To Solve using ERP system

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Enterprise Resource Planning

- **Enterprise Resource Planning** (ERP) is a platform companies use to manage and integrate the essential parts of their businesses.
- ERP is also an integrated organizational **information system**
- An ERP software system can also integrate planning, purchasing inventory, sales, marketing, finance, human resources, and more.
- Software that helps the **flow of information** between the **functional areas within the organization**
- Recently ERP systems are **moving** to **cloud computing**

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ERP and the Organization

- ERP can affect every aspect of the organization, including:
  - **Design** of **employees' work**
  - **Skills required** for **job competency**
  - **Strategic positioning** of the company

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The Benefits of ERP to Organization

1 Improves Accuracy and Productivity

ERP connects all business processes, removing duplicate work and errors. Teams can share data easily, work faster, and produce better results.

2 Improves Reporting

With real-time data in one system, reports are more accurate and up to date. This helps companies plan, budget, and make smarter decisions.

3 Increases Efficiency

Employees can quickly find information about customers, suppliers. and operations. This saves time, reduces mistakes, and lowers costs.

4 Provides Better Visibility

Managers can see what's happening in every part, of the business in real time. This helps in making quick and confident

5 Increases Collaboration

Departments work together more effectively and understand each other's roles. Routine tasks are automated, allowing employees to focus on more meaningful work

Issues to Be Overcome for ERP Success

- Many **issues must be overcome** for the **ERP installation** is to be declared a success:
  - **User acceptance**
  - **Integration** with legacy systems and the supply chain
  - Upgrading **functionality** (and complexity) of ERP modules
  - Reorganizing **work life** of **users** and **decision makers**
  - **Expanded** reach across **several organizations**
  - **Strategic repositioning** of the company

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Tools to plan ERP Systems Graphically

- Context-level **Data Flow Diagrams (DFD)**
- **Entity-relationship model (ER)**
- **Use case modeling**

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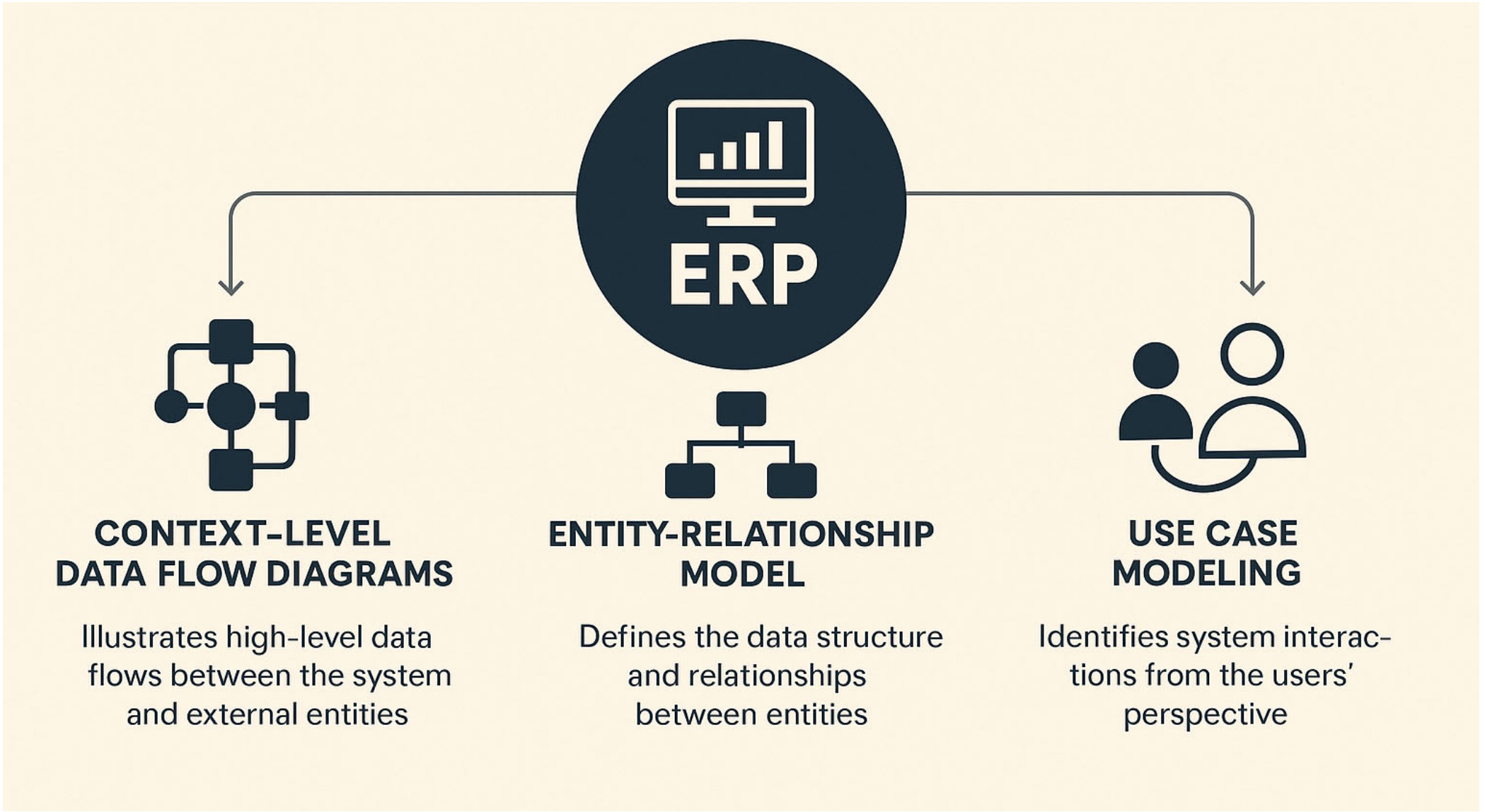
ERP Planning & Modeling Graphically



Context-Level Data Flow Diagrams

- **Focus** is on the **data flowing into** and **out** of the system and the **processing** of the data
- **Shows** the **scope** of the system:
  - **What** is to be **included in** the **system**
  - The **external entities** are **outside** the **scope** of the system

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Figure 2.4 The Basic Symbols of a Data Flow Diagram

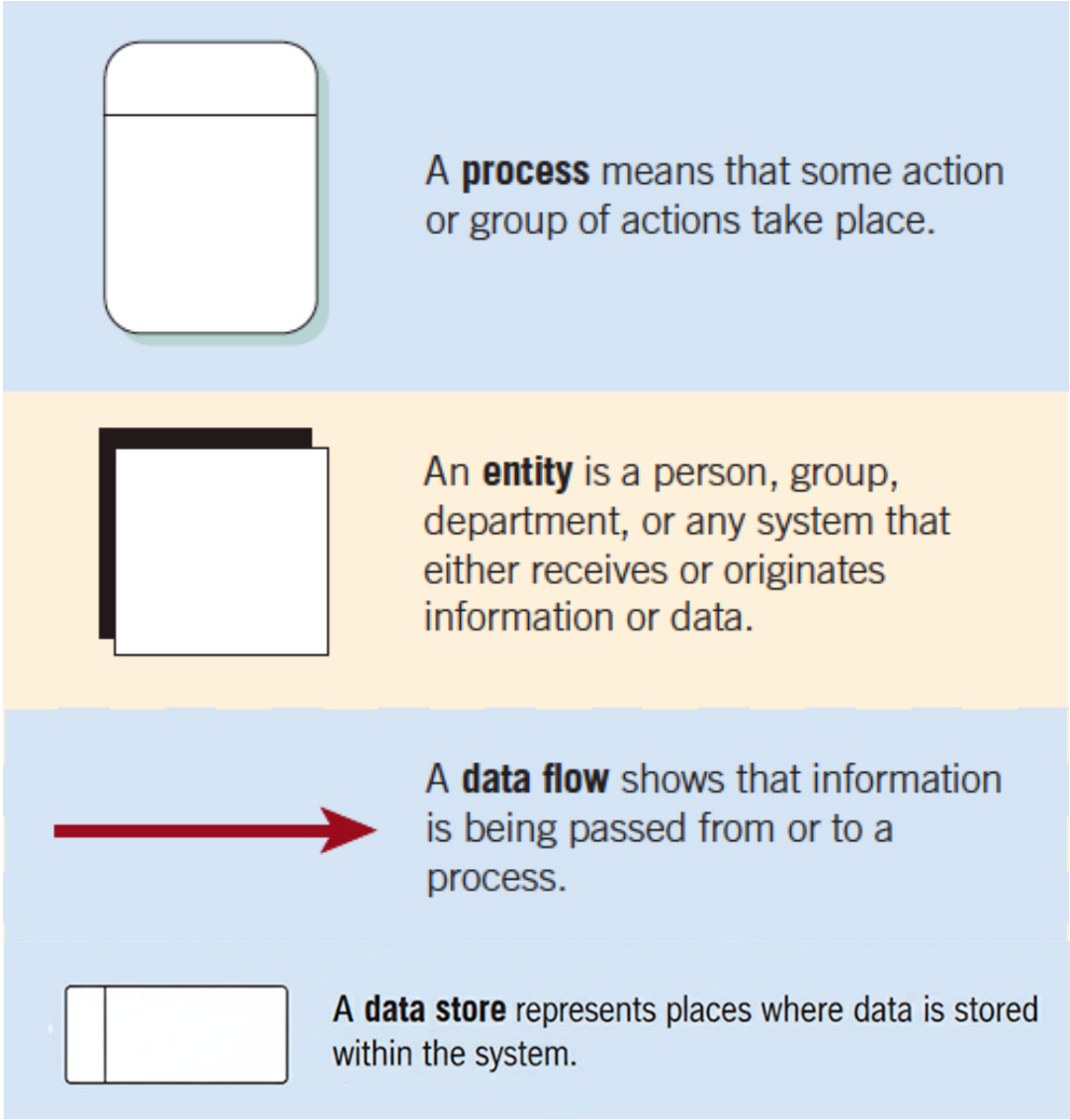

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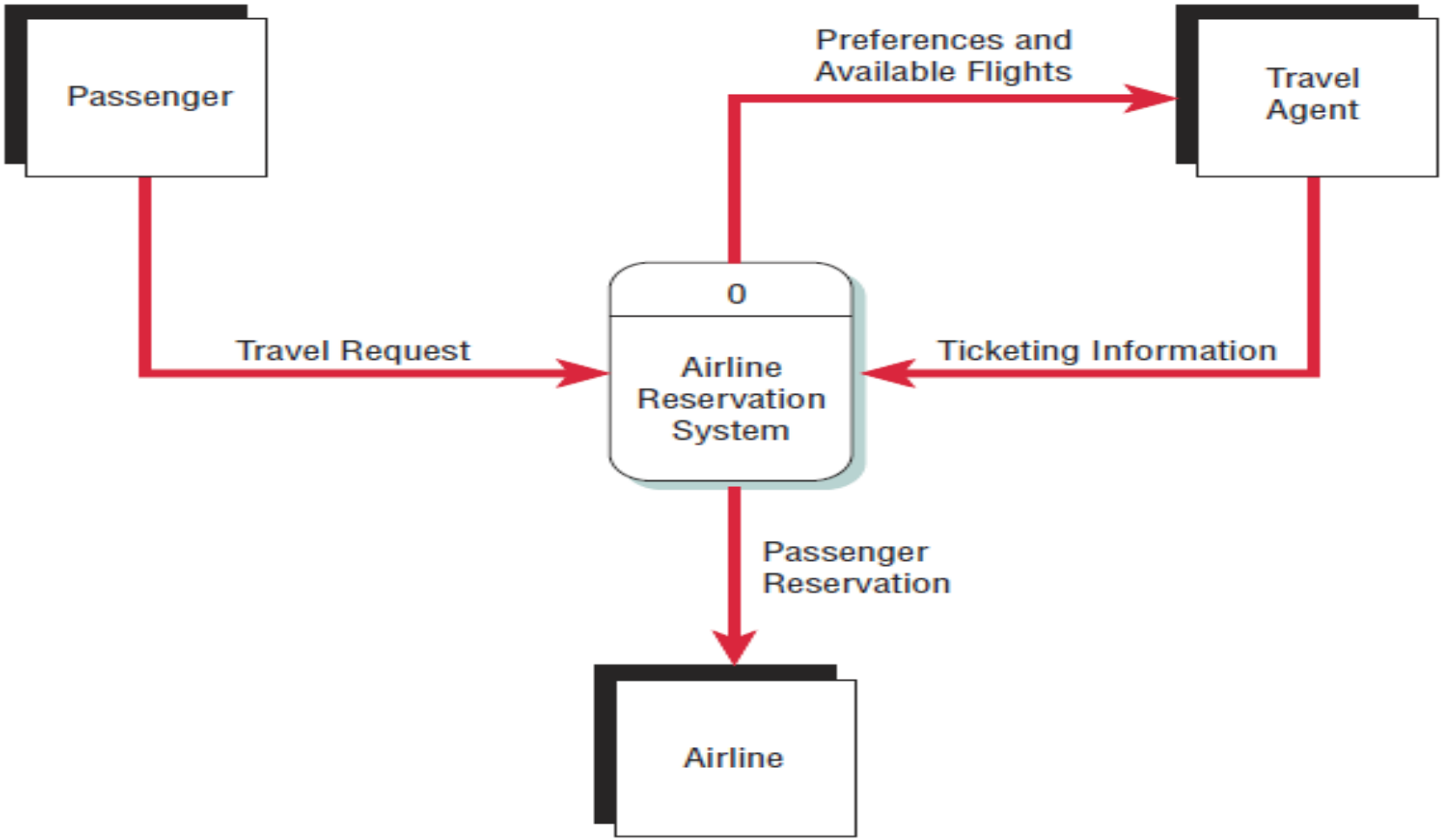
Figure 2.5 Airline Reservation System

A **context-level data flow diagram** for an **airline reservation system**



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Entity-Relationship Model

- **Focus** is on the **entities** and their **relationships** within the **organizational** system
- **Another way** to **show** the **scope** of a system
- **Entity** can represent a real-world object or concept (e.g., Customer, Order) that data is stored about.
- Each entity has **Attributes** (details), and at least one **primary key** to identify it uniquely.
- Also helps define the **structure of data** in the ERP system - which is crucial when planning the database side of ERP.

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Relationships

- Relationships show **how** the **entities** are **connected**
- Three Main types of relationships:
  - **One-to-one**
  - **One-to-many**
  - **Many-to-many**

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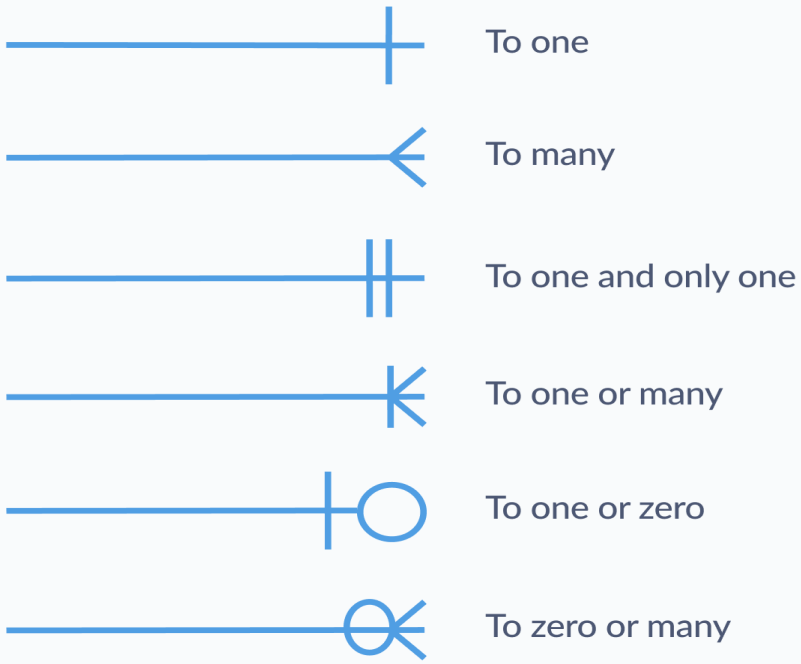
Crow's Foot Notation



One card holder can have: 0 or 1 — Credit card

One card holder can have: Only 1 — Credit card

One card holder can have: Atleast 0 — Credit card

One card holder can have: Atleast 1 — Credit card

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Figure 2.7 Entity-Relationship Example

An entity-relationship diagram showing a many-to-one relationship

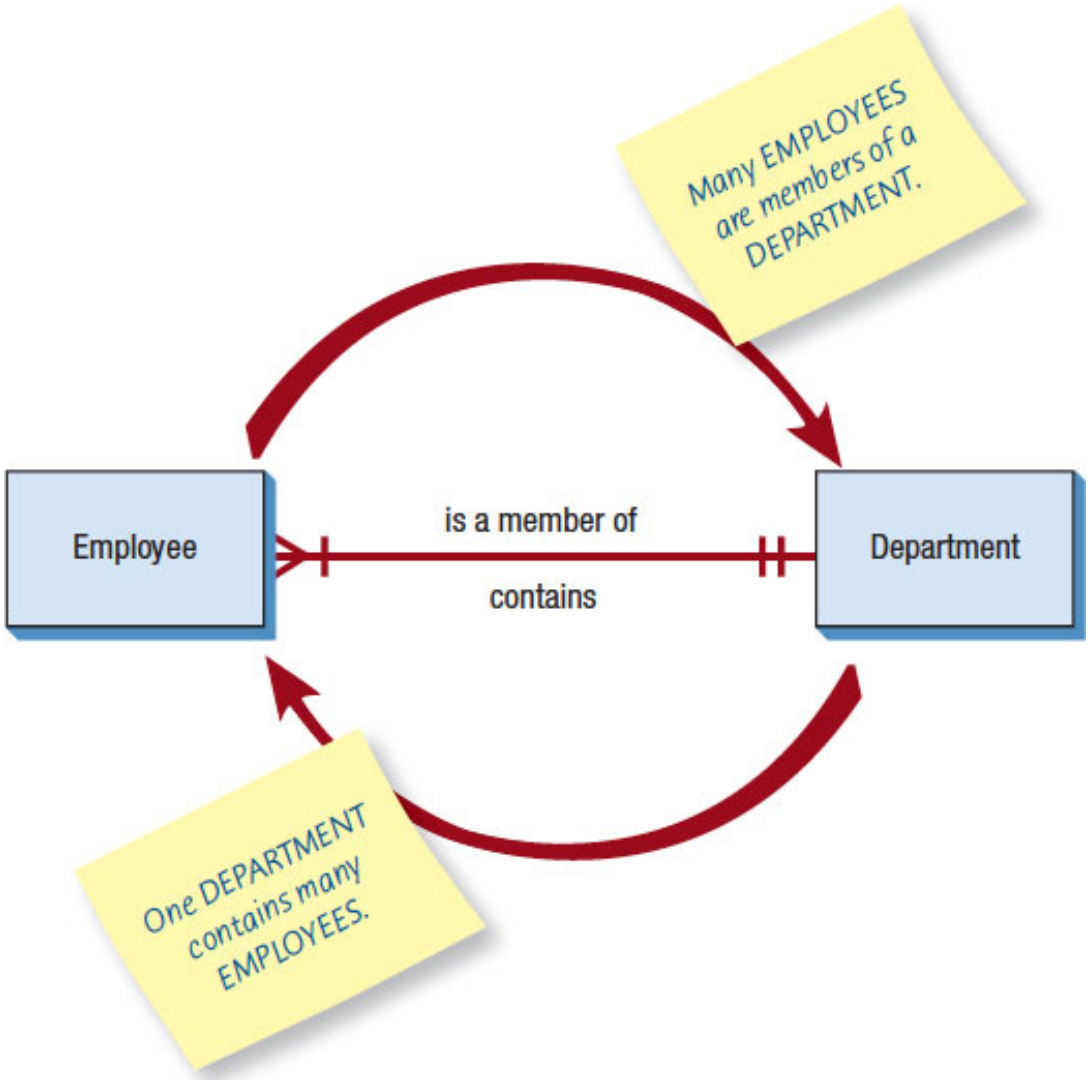

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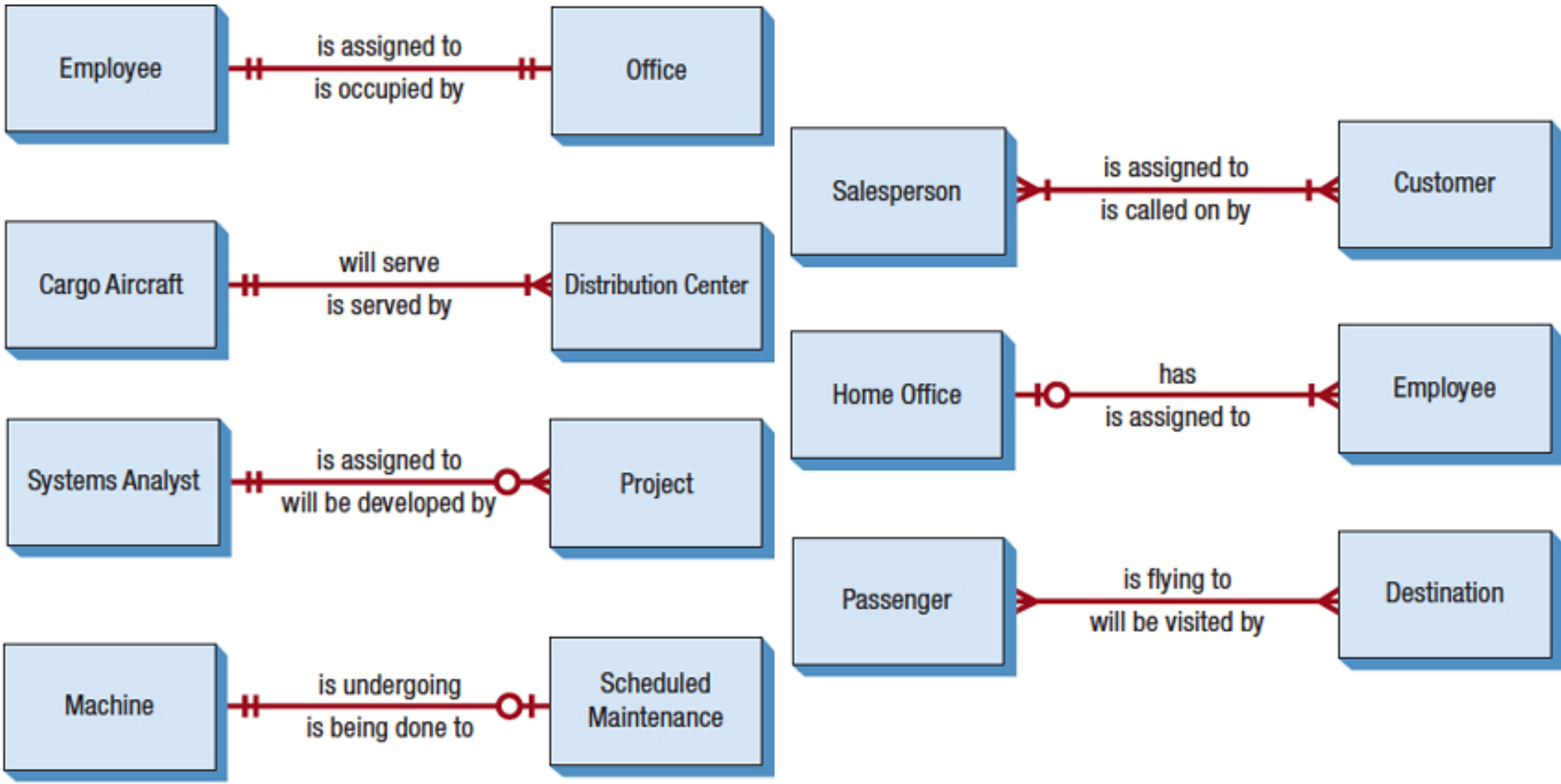
Figure 2.8 Examples of Different Types of Relationships in E-R Diagrams



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Entities

- **Fundamental** entity (**Strong** Entity)
  - Exists independently of any other entity
  - Has a primary key (unique identifier)
  - Represents real-world objects or concepts
- **Associative** entity (**Bridge/Junction)** Entity.
  - Links two entities
  - Often has its own attributes
  - Usually has a composite key made from foreign keys of the linked entities
- **Attributive** entity (**Weak Attribute Group** Entity)
  - Used to model multi: valued or complex attributes
  - Not standalone: always tied to a main (parent) entity
  - Not always considered a "true" entity in classic ER modeling

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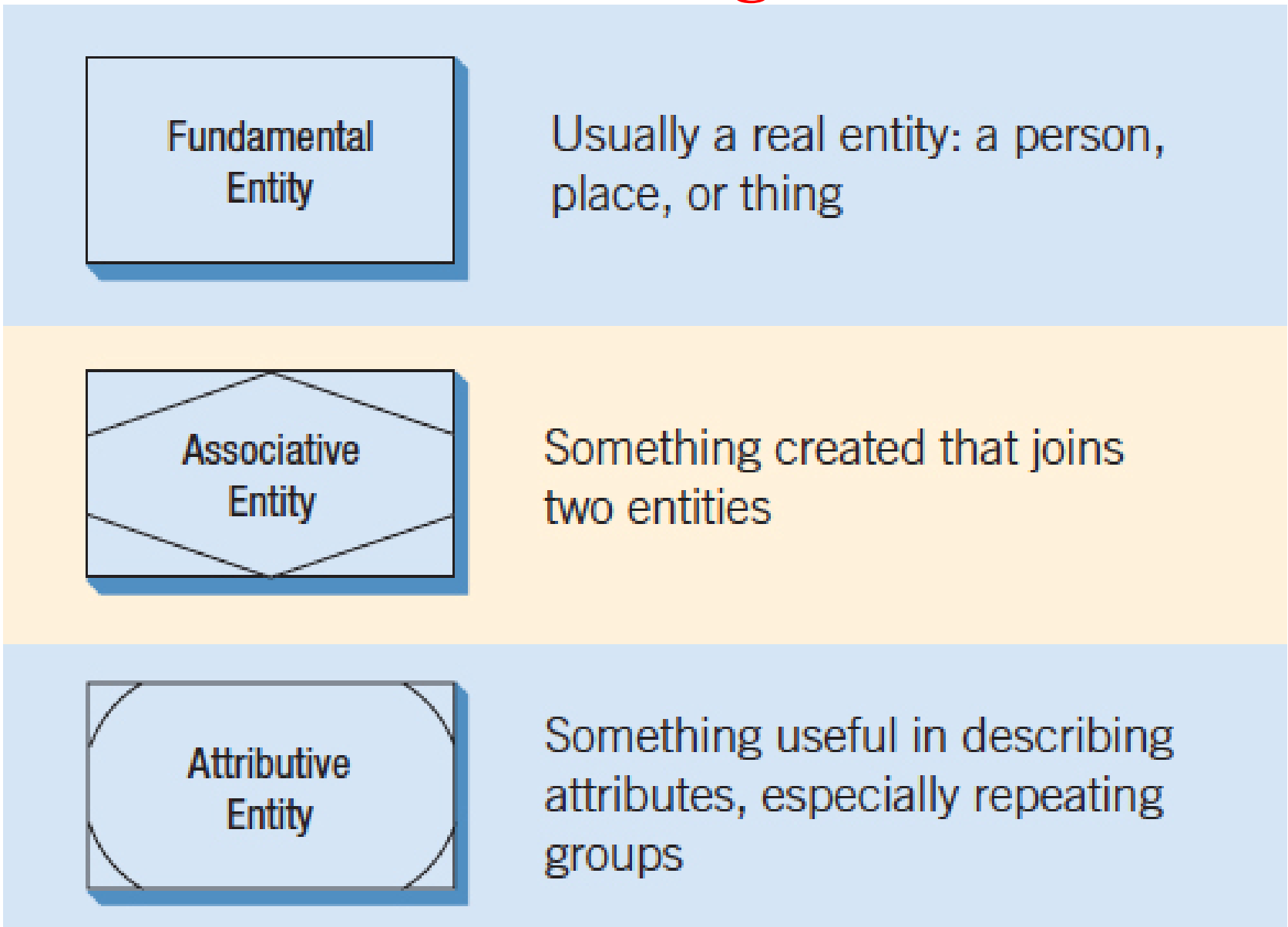
Figure 2.9 Three Different Types of Entities Used in E-R Diagrams



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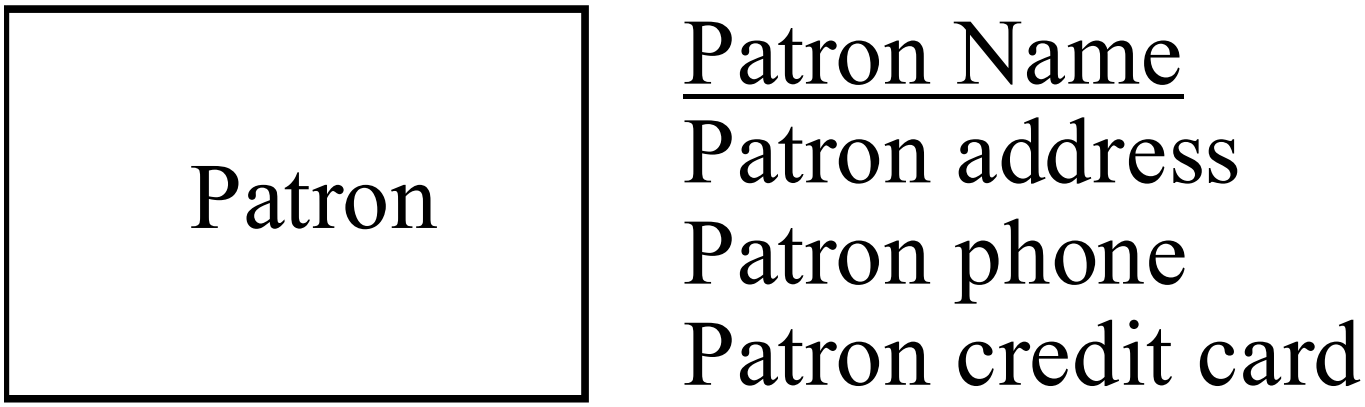
Attributes

- **Data attributes** may be **added** to the **diagram**.



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Creating Entity-Relationship Diagrams

1. **List** the **entities** in the **organization**
2. **Choose** **key entities** to **narrow** the **scope** of the **problem**
3. **Identify** what the **primary entity** should be
4. **Confirm** the **results** of the **above** through **data gathering**

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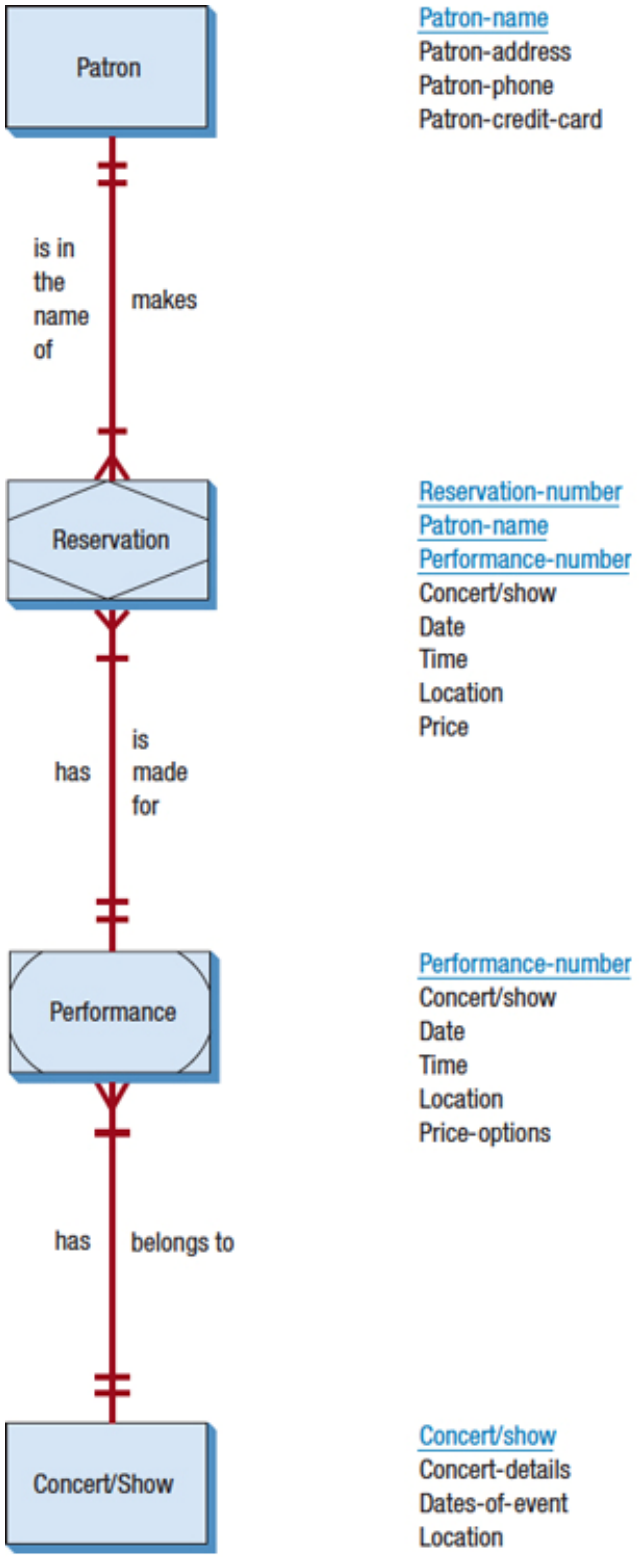
Figure 2.12 A More Complete E-R Diagram Showing Data Attributes of the Entities



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Use Case Modeling

- **Part** of the **unified modeling language (UML)**
- Focuses on **user behavior** and **system functionality**
- Capture how users interact with the system.
- Identify who uses the system (**actors**) and what they do **(use cases).**
- Describes what a system does without describing how the system works
- A **view** of the **system requirements**
- Analyst works with business experts to **develop requirements**

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Use Case Diagram

- **Actor**
  - Refers to a **particular role** of a user of the system
  - **Similar** to **external** **entities;** they **exist** **outside** of the **system boundary**
- **Use case** symbols
  - An **oval** **indicating** the **task**, **function** or **action** performed by the system
- **Connecting lines**
  - **Arrows** and **lines** used to diagram **behavioral relationships**
  - **Connects** actors and use cases

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Actor

- Divided into **two groups**
  - **Primary** actors:
    - **Supply data** or **receive information** from the system
    - **Provide details** on what the use case should do
    - Initiates interaction to achieve a goal.
  - **Supporting** actors:
    - Help to **keep** the **system running** or **provide help**
    - Supports or provides data for the interaction.
    - The **people** who run the **help desk**, the **analysts**, **programmers**, and so on

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A Use Case Always Provides Three Things

- An **actor** that **initiates** an **event**
- The **event** that **triggers** a **use case**
- The **use case** that **performs** the **actions triggered** **by the event**
- Describes a **specific action or goal** performed by or for an actor.
- Represents **functional requirements** — what the system must do. **Examples:**
  - "Process Payroll"
  - "Create Sales Order"
  - "Generate Report"

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Use Case Relations (1 of 2)

- **Behavioral relationships**
  - **Communicates**
    - Used to **connect** an **actor** to a **use case**
  - **Includes**
    - **Describes** the **situation** in which a **use case contains behavior** that is **common** to **more than one use case**

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Use Case Relations (2 of 2)

- Behavioral relationships [continued]
  - **Extends**
    - Describes the **situation** in which one use case **possesses** the **behavior** that allows the new case to **handle** a **variation** or **exception** from the basic use case
  - **Generalizes**
    - **Implies** that **one thing** is **more typical** **than the other thing**

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Figure 2.13 Four Types of Behavioral Relationships and the Lines Used to Diagram Each

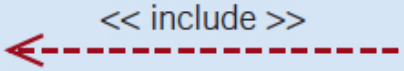
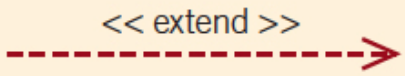
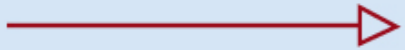
Relationship	Symbol	Meaning
Communicates		An actor is connected to a use case using a line with no arrowheads.
Includes		A use case contains a behavior that is common to more than one other use case. The arrow points to the common use case.
Extends		A different use case handles exceptions from the basic use case. The arrow points from the extended to the basic use case.
Generalizes		One UML "thing" is more general than another "thing." The arrow points to the general "thing."

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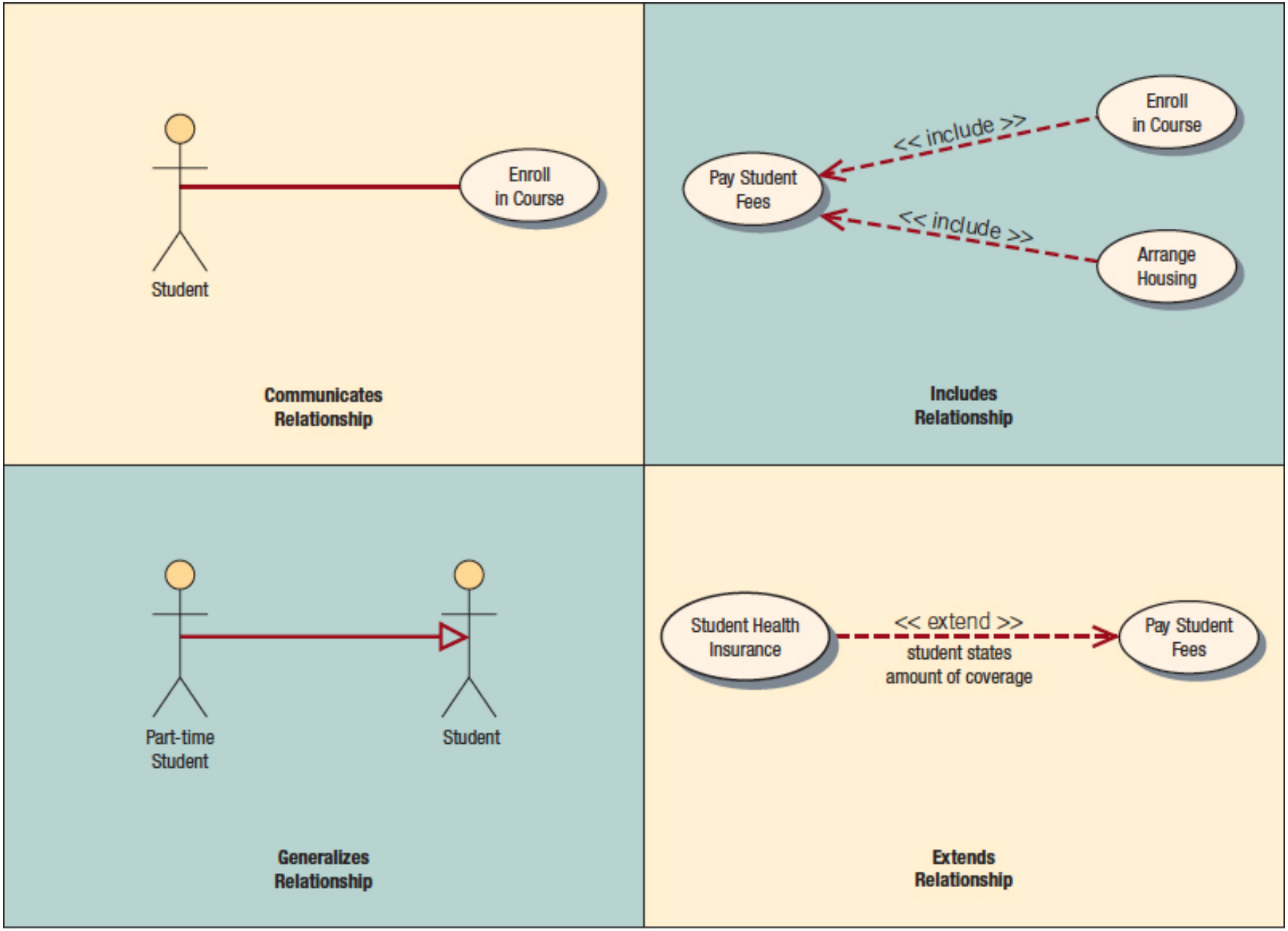
Figure 2.14 Actors, Use Cases, and Relationships for a Student Enrollment Example



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Why Use-Case Relationship matters in ERP systems?

Avoid redundancy
– by reuse steps (via include)

Add flexibility
– by managing optional flows (via extend)

Support modular design
– by organizing parent-child behaviors

Clarify user interactions
– making the system's scope and logic clear

Scope

- System **scope** defines its **boundaries**:
  - What is **in** or **outside** the system
  - Project has a **budget** that **helps** to **define scope**
  - Project has a **start** and an **end** time
- **Actors** are always **outside** of **scope**
- **Communication lines** are the **boundaries** and **define** the **scope**

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Why Scope in ERP systems?

Prevents Overlap

– Keeps each module (HR, Finance, etc), clear in function

Ensures Integration Efficiency

– Clarifirs how data flows, between departments

Supports Accurate Budgeting

– Reduces surprises during implementation

Improves Stakeloter Communication

– Everyone understands what system will and won't do

Developing Use Case Diagrams

1. **Review** the **business specifications** and **identify** the **actors involved**
2. **Identify** the **high-level events** and **develop** the **primary use cases** that **describe** those **events** and **how** the **actors initiate them**
3. **Review** each **primary use case** to **determine** the **possible variations** of **flow** through the use case
4. The context-level **data flow diagram** could **act** as a **starting point** for **creating a use case**

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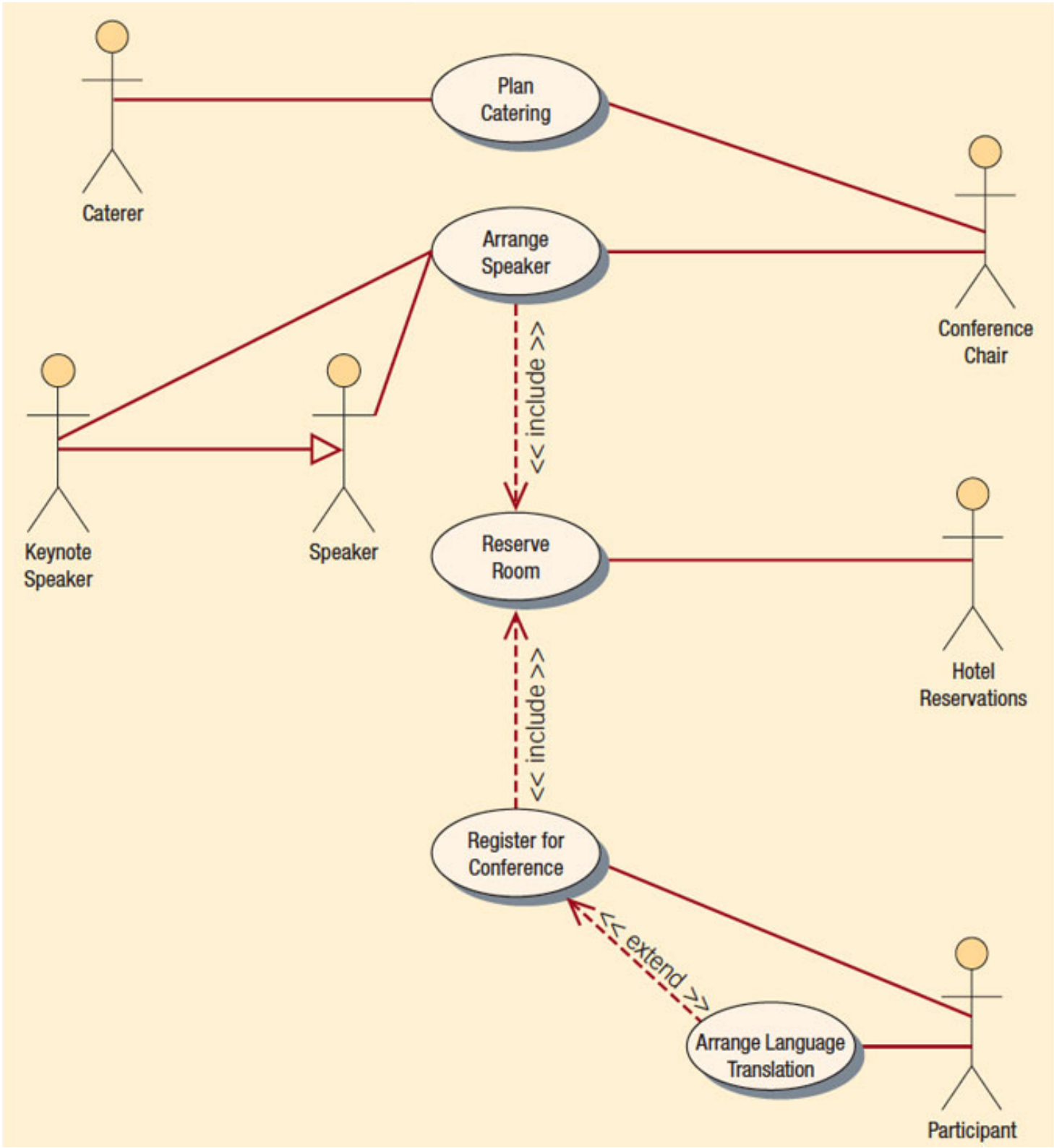
Figure 2.15 A Use Case Diagram Representing a System Used to Plan a Conference



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Developing the Use Case Scenarios

- The **description** of the use case
- Three main areas:
  - Use case **identifiers** and **initiators**
  - **Steps** performed
  - **Conditions, assumptions**, and **questions**

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Figure 2.16 A Use Case Scenario is Divided into Three Sections

Use case name:	Register for Conference	**UniqueID: Conf RG 003**
Area:	Conference Planning	
Actor(s):	Participant	
Stakeholder:	Conference Sponsor, Conference Speakers	
Level:	Blue	
Description:	Allow conference participant to register online for the conference using a secure website.	
Triggering Event:	Participant uses Conference Registration website, enters userID and password, and clicks the logon button.	
Trigger type:	External Temporal	
Steps Performed (Main Path)		**Information for Steps**
1. Participant logs in using the secure web server.		userID, Password
2. Participant record is read and password is verified.		Participant Record, userID, Password
3. Participant and session information is displayed on the Registration web page.		Participant Record, Session Record
4. Participant enters information on the Registration Web form and clicks **Submit** button.		Registration Web Form
5. Registration information is validated on the Web server.		Registration Web Form
6. Registration Confirmation page is displayed to confirm registration information.		Confirmation Web Page
7. Credit card is charged for registration fees.		Secure Credit Card Web Page
8. Add Registration Journal record is written.		Confirmation Web Page
9. Registration record is updated on the Registration Master.		Confirmation Web Page, Registration Record
10. Session record is updated for each selected session on the Session Master.		Confirmation Web Page, Session Record
11. Participant record is updated for the participant on the Participant Master.		Confirmation Web Page, Participant Record
12. Successful Registration Confirmation web page is sent to the participant.		Registration Record Confirmation Number
Preconditions:	Participant has already registered and has created a user account.	
Postconditions:	Participant has successfully registered for the conference.	
Assumptions:	Participant has a browser and a valid userID and password.	
Success Guarantee:	Participant has registered for the conference and is enrolled in all selected sessions.	
Minimum Guarantee:	Participant was able to logon.	
Requirements Met:	Allow conference participants to be able to register for the conference using a secure website.	
Outstanding Issues:	How should a rejected credit card be handled?	
Priority:	High	
Risk:	Medium	

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Use Case Header Area

- Has a **name** and a **unique ID**
- Include **application** area
- List **actors**
- Include **stakeholders**
- Include the **level**
- Has a **brief description** of the **use case**

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Use Case Levels

- Use case levels **describe** **how** **global** or **detailed** the use case description is:
  - **White** (like **clouds**): enterprise level
  - **Kite**: business unit or department level
  - **Blue** (**sea level**): user goals
  - **Indigo** (or **fish**): functional or subfunctional
  - **Black** (or **clam**): most detailed

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Alternative Scenarios

- **Extensions** or **exceptions** to the main use case
- **Number** with an **integer**, **decimal point**, **integer**
- **Steps** that **may** or **may not always be used**

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Use Case Footer Area

- **Preconditions**—need to be met **before** **use case** can be **performed**
- **Postconditions** or the **state** of the system **after** the **use case** has **finished**
- **Assumptions**
- **Minimal** **guarantee**
- **Success** **guarantee**
- **Outstanding** **issues**
- Optional **priority** and **risk**

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Four Steps Used to Create Use Cases

1. Use **agile stories**, **problem definition objectives**, user **requirements**, or a **features list**
2. **Ask** about the **tasks** that **must be done**
3. **Determine** if there are any **iterative** or **looping actions**
4. The **use case ends** when the **customer goal** is **complete**

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Why Use Case Diagrams Are Helpful

- **Identify** all the actors in the **problem domain**
- **Actions** that **need** to be **completed** are also clearly shown on the use case diagram
- The use case scenario is also **worthwhile**
- **Simplicity** and **lack** of **technical detail**

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Figure 2.18 The Main Reasons for Writing Use Cases are Their Effectiveness in Communicating with Users and Their Capturing of User Stories

- Use cases **effectively communicate** **systems requirements** because the **diagrams** are kept **simple**.
- Use cases **allow** people to **tell stories**.
- Use case **stories** **make sense** to **nontechnical people**.
- Use cases **do not depend** on a **special language**.
- Use cases can **describe** most **functional** requirements (such as **interactions** between **actors** and **applications**).
- Use cases can **describe** **nonfunctional** requirements (such as **performance** and **maintainability**) through the **use** of **stereotypes**.
- Use cases **help** analysts **define** **boundaries**.
- Use cases can be **traceable**, **allowing** analysts to **identify** **links between use cases** and other **design** and **documentation tools**.

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Figure 2.19 Management in Organizations Exists on Three Horizontal Levels

Operational control, **managerial planning** and **control**, and **strategic mar**

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Operations Control

- **Make decisions** using **predetermined rules** that have **predictable outcomes**
- **Oversee** the **operating details** of the **organization**

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Managerial Planning and Control

- Make **short-term planning** and **control decisions** about **resources** and **organizational objectives**
- **Decisions** may be **partly operational** and **partly strategic**

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Strategic Management

- **Look outward** from the **organization** to the **future**
- **Make decisions** that will **guide middle** and **operations managers**
- **Work** in **highly uncertain decision-making environment**
- **Faced** with **semistructured problems**
- **Define** the **organization** as a **whole**

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Managerial Levels

- Different organization **structure**
- Leadership **style**
- **Technological** considerations
- Organization **culture**
- Human **interaction**
- **All carry implications** for the **analysis** and **design** of information systems

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Collaborative Design

- **External** and **internal** stakeholders follow **processes** to share in designing a system to meet their goals
- **Giving power** to those who **possess** a **technical** or **strategic** **expertise**

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Organizational Culture

- Organizations have **cultures** and **subcultures**
- Learn from **verbal** and **nonverbal** symbolism

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Technology Impact on Culture

- **Technology** is **changing** the **culture** of organizations and teams
- **Slack** is an employer-sanctioned social media platform, or workplace-messaging app
- **Public** and **private** channels
- **Direct** or **group** messages

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Summary (1 of 2)

- Organizational fundamentals
  - Organizations as systems
  - Levels of management
  - Organizational culture
- Graphical representation of systems
  - DFD
  - ERD
  - Use case diagrams and scenarios

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Summary (2 of 2)

- Levels of managerial control
  - Operational
  - Middle management
  - Strategic
- Organizational culture

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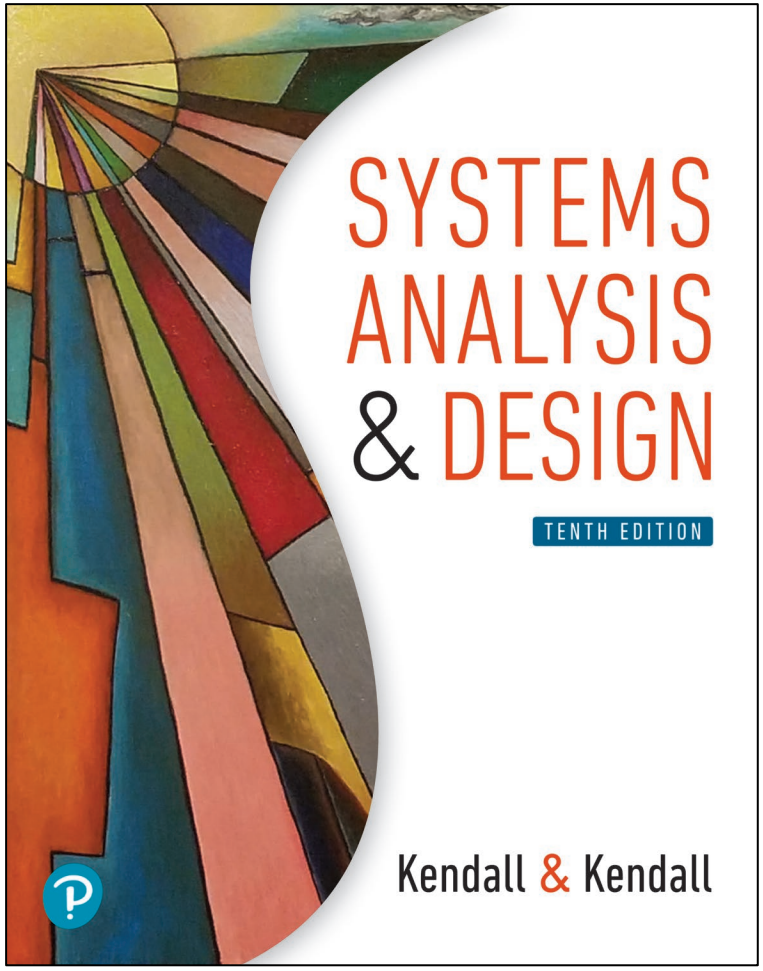
Systems Analysis & Design

Tenth Edition



Chapter 3

Project Management

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Learning Objectives (1 of 2)

3.1 Understand how projects are initiated and selected, **define** a business **problem**, and **determine** the **feasibility** of a proposed project

3.2 Evaluate hardware and software and the way it supports **human interactions** with **technology**

3.3 Forecast and analyze **tangible** and **intangible** **costs** and **benefits**

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Learning Objectives (2 of 2)

3.4 Manage a project by **preparing** a **budget**, creating a **work breakdown structure**, **scheduling activities,** and **controlling** the **schedule** and **costs**

3.5 Build and manage a project team

3.6 Professionally **write** and **present** an **effective** **systems** proposal, concentrating on both **content** and **design**

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Project Management Fundamentals

- Project **initiation**
- Determining project **feasibility**
- Activity **planning** and **control**
- Project **scheduling**
- **Managing** systems analysis team **members**

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Major Topics

- Project **initiation**
- Determining **feasibility**
- Determining **resources**
- Activity **planning** and **control**
  - **Gantt charts**
  - **PERT diagrams**
- Managing **analysis** and **design** activities
- The **agile** approach

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Project Initiation

- Problems in the organization
  - **Problems** that **lend** themselves to **systems solutions**
- Opportunities for improvement
  - Caused through **upgrading**, **altering**, or **installing** new systems

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Figure 3.1 Checking Output

Observing employee **behavior and listening** to **feedback** are all ways to help the analyst pinpoint systems problems and opportunities.

To Identify Problems	Look for These Specific Signs:
Check output against performance criteria.	Too many errors Work completed slowly Work done incorrectly Work done incompletely Work not done at all
Observe behavior of employees.	High absenteeism High job dissatisfaction High job turnover
Listen to external feedback from: Vendors and service providers. Customers. Suppliers.	Complaints Suggestions for improvement Loss of sales Lower sales

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Problem Definition

- **Problem statement**
  - Paragraph or two **stating** the **problem** or **opportunity**
- **Issues**
  - **Independent pieces pertaining** to the **problem** or **opportunity**
- **Objectives**
  - **Goals** that **match** the **issues point-by-point**
- **Requirements**
  - The things that must be accomplished along with the possible **solutions**, and **constraints**, that **limit** the **development** of the system
- Use the problem definition to **create** a **preliminary test plan**

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Problem Definition: Identifying Issues

- Find a **number of points** that may be included in **one issue**
- State the objective
- Determine the **relative importance** of the issues or objectives
- Identify which **objectives** are **most critical**

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Selection of Projects

- Backing from management
- Appropriate timing of project commitment
- Possibility of improving attainment of organizational goals
- Practical in terms of resources for the system analyst and organization
- Worthwhile project compared with other ways the organization could invest resources

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Selection of Projects: Improving Attainment of Organizational Goals

- Improving corporate profits
- Supporting the competitive strategy of the organization
- Improving cooperation with vendors and partners
- Improving internal operations support
- Improving internal decision support so that decisions are more effective
- Improving customer service
- Increasing employee morale

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Defining Objectives

Many possible objectives exist including:

- Speeding up a process
- Streamlining a process
- Combining processes
- Reducing errors in input
- Reducing redundant storage
- Reducing redundant output
- Improving system and subsystem integration

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Determining Feasibility

- Assessed in three principal ways
  - Operationally
  - Technically
  - Economically

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Figure 3.3 The Three Key Elements of Feasibility Include Technical, Economic, and Operational Feasibility

The Three Key Elements of Feasibility

- Technical Feasibility
  - Add on to present system
  - Technology available to meet users' needs
- Economic Feasibility
  - Systems analysts' time
  - Cost of systems study
  - Cost of employees' time for study
  - Estimated cost of hardware
  - Cost of packaged software or software development
- Operational Feasibility
  - Whether the system will operate when put in service
  - Whether the system will be used

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Technical Feasibility

- Can current technical resources be upgraded or added to in a manner that fulfills the request under consideration?
- If not, is there technology in existence that meets the specifications?

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Economic Feasibility

- Economic feasibility determines whether value of the investment exceeds the time and cost
- Includes:
  - Analyst and analyst team time
  - Business employee time
  - Hardware
  - Software
  - Software development

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Operational Feasibility

- Operational feasibility determines if the human resources are available to operate the system once it has been installed
- Users that do not want a new system may prevent it from becoming operationally feasible

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Estimating Workloads

- Systems analysts formulate numbers that represent both current and projected workloads for the system so that any hardware obtained will possess the capability to handle current and future workloads

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Figure 3.4 Comparisons of Workloads between Existing and Proposed Systems

	Existing System	Proposed System
Task	Compare performance of distribution warehouses by running the summary program.	Compare performance of distribution warehouses on the Web-based dashboard.
Method	Computer programs are run when needed; processing is done from the workstation.	Updates occur immediately; processing is done online.
Personnel	Distribution manager	Distribution manager
When and how	Daily: Enter shipments on Excel spreadsheet; verify accuracy of spreadsheet manually; and then write files to backup media. Monthly: Run program that summarizes daily records and prints report; get report and make evaluations.	Daily: Enter shipments on the Web-based system using automatically backed up drop-down boxes. Data are to remote location. Monthly: Compare warehouses online using the performance dashboard; print only if needed.
Human time requirements	Daily: 20 minutes Monthly: 30 minutes	Daily: 10 minutes Monthly: 10 minutes
Computer time requirements	Daily: 20 minutes Monthly: 30 minutes	Daily: 10 minutes Monthly: 10 minutes

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Ascertaining Hardware and Software Needs

- Steps used to determine hardware and software needs:
  - Inventory computer hardware currently available
  - Estimate current and future system workloads
  - Evaluate available hardware and software
  - Choose the vendor
  - Acquire the computer equipment

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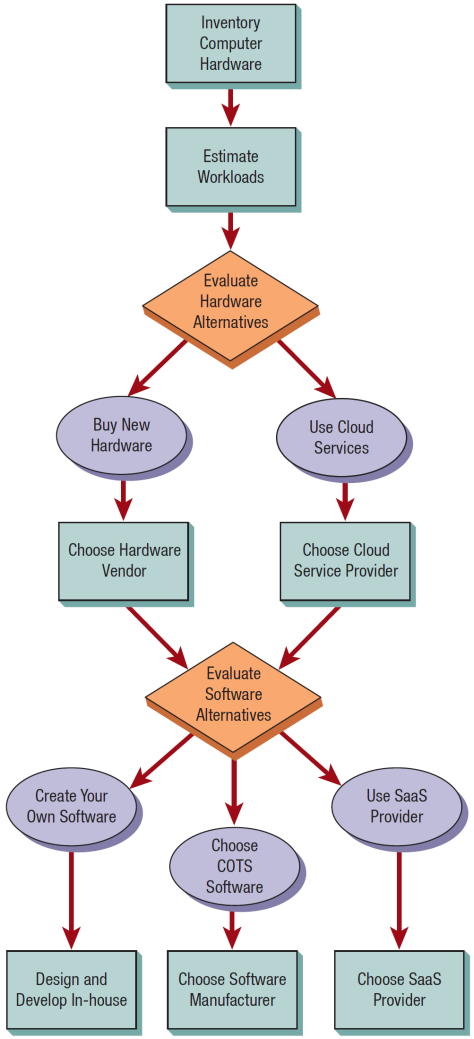
Figure 3.5 Steps in Choosing Hardware and Software



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Inventorying Computer Hardware

- Type of equipment
- Operation status of the equipment
- Estimated age of equipment
- Projected life of equipment
- Physical location of equipment
- Department or person responsible for equipment
- Financial arrangement for equipment

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Evaluating Hardware

- Time required for average transactions
- Total volume capacity of the system
- Idle time of the CPU or network
- Size of memory provided

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People That Evaluate Hardware

- Management
- Users
- Systems analysts

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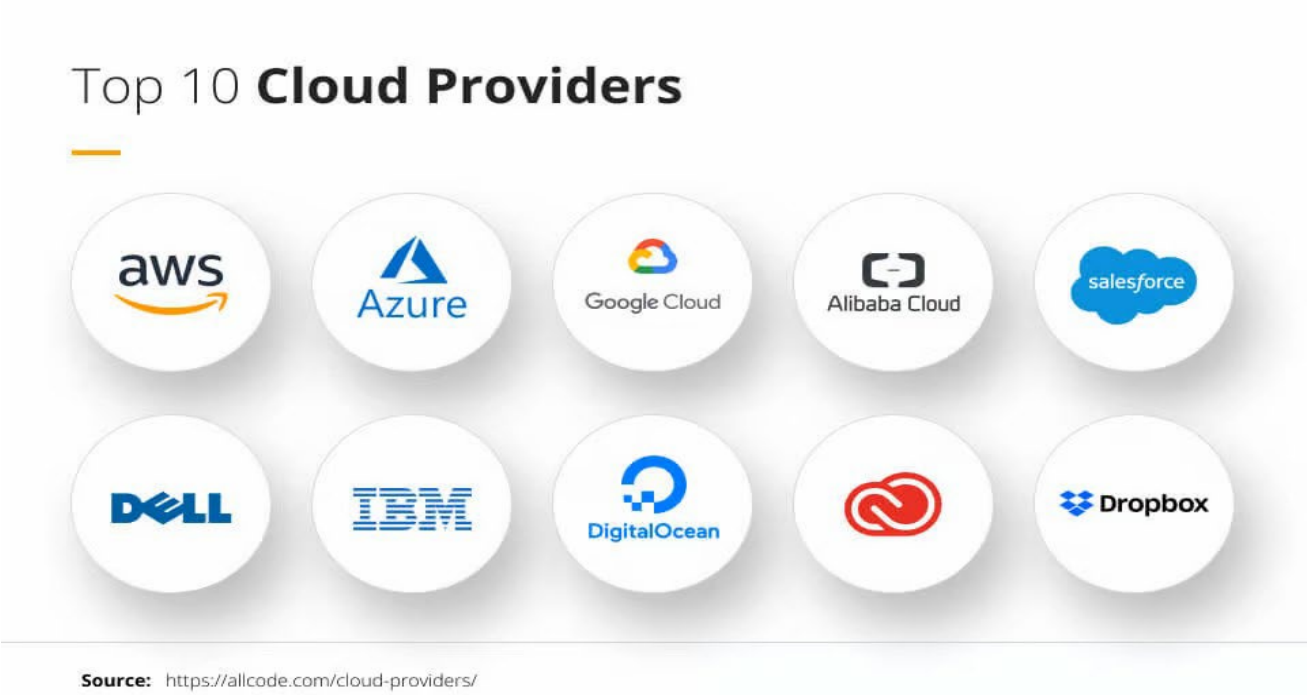
Acquisition of Computer Equipment

- Purchasing
- Using Cloud Services
  - on-demand IT resources, such as storage, software, and processing power, delivered over the internet by a third-party provider



Available Cloud Services

- Available cloud services may include:
  - Web hosting
  - Email hosting
  - Application hosting
  - Backup
  - Storage and processing of databases
  - Archiving
  - E-commerce

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Three Main Categories of Cloud Computing

- Software as a Service (SaaS)
  - a cloud-based delivery model where applications are hosted by a provider and accessed by users over the internet on a subscription basis, eliminating the need for local installation and maintenance
- Infrastructure as a Service (IaaS)
  - a cloud computing model where a vendor provides on-demand access to foundational computing resources over the internet, including servers, storage, and networking
- Platform as a Service (PaaS)
  - a cloud computing model that provides a complete, cloud-based environment for developers to build, test, deploy, and manage applications without the need to manage the underlying infrastructure

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Three Main Categories of Cloud Computing

Aspect	**SaaS** (Software as a Service)	**PaaS** (Platform as a Service)	**IaaS** (Infrastructure as a Service)
Definition	Ready-to-use software delivered over the internet	Cloud-based platform for developers to build, test, and deploy apps	Virtualized computing resources (servers, storage, network) provided online
Target Users	End users / organizations	Developers / programmers	IT admins / system engineers
User Focus	Using software	Building software	Hosting and managing infrastructure
You Manage	Just your data & usage	Application code & data	OS, applications, data
Provider Manages	Everything (app, runtime, OS, hardware)	Infrastructure, runtime, middleware	Hardware, virtualization, networking
Examples	Gmail, Microsoft 365, Salesforce, Zoom, Canva	Google App Engine, Heroku, AWS Elastic Beanstalk, Azure App Service	AWS EC2, Google Compute Engine, Azure Virtual Machines
Purpose	Use complete applications without installing locally	Build and deploy apps faster with built-in tools	Provision computing infrastructure without owning hardware
Advantages	○ Easy access via browser ○ No installation or maintenance ○ Automatic updates ○ Cost-effective	○ Fast development & deployment ○ Scalable environment ○ Developers focus on coding ○ Supports multiple languages	○ Full control of OS & apps ○ Scalable & flexible ○ Pay-as-you-use ○ Ideal for hosting legacy systems
Disadvantages	▪ Limited customization ▪ Internet dependency ▪ Vendor lock-in	▪ Less control over OS ▪ Framework restrictions ▪ Vendor lock-in	▪ Requires IT expertise ▪ Security management is user's responsibility ▪ Unpredictable cost spikes
Common Use Cases	Email, CRM, online office suites, video conferencing	App development, API services, automation tools	Virtual servers, storage, backup, testing environments

Strategic Cloud Computing Decisions

- Decisions on cloud computing can first be addressed on a strategic level
- Business should focus on:
  - Define a high-level business case that focuses on high-level benefits
  - Define core requirements
  - Define core technologies for the enterprise

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Benefits of Cloud Computing

- Less time spent maintaining legacy systems
- May be simpler to acquire IT services
- Scalable applications that can grow by adding more cloud resources
- Consistency across multiple platforms
- Capital is not tied up

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Drawbacks of Cloud Computing

- Loss of control of data stored in the cloud
- Potential security threats to data that is not stored on premises
- Reliability of the Internet as a platform

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Figure 3.6 Purchasing Hardware or Using Cloud Services Advantages and Disadvantages

	Advantages	Disadvantages
Buying computer Hardware	Full control over hardware and software Often cheaper in long run Provides tax advantages through Depreciation	Initial cost is high Risk of obsolescence Risk of being stuck if choice was wrong Full responsibility for operation and maintenance
Using cloud Services	Maintenance and upgrades performed by provider Ability to change software and hardware rapidly Scalable—can grow quickly Consistent over multiple platforms No capital is tied up	Company doesn't control its own data Data security is at risk Reliability risks of the Internet platform Proprietary APIs and software may make switching providers difficult

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Evaluating Vendor Support

- Hardware support
- Software support
- Installation and training support
- Maintenance support

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Figure 3.7 Guidelines for Vendor Selection

Vendor Services	Specifics Vendors Typically Offer
Hardware Support	Full line of hardware Quality products Warranty
Software Support	Complete software needs Custom programming Warranty
Installation and Training	Commitment to schedule In-house training Technical assistance
Maintenance	Routine maintenance procedures Specified response time in emergencies Equipment loan while repair is being done
Cloud Services	Web hosting Email hosting Data storage
Software as a Service	Automatic software upgrades Support services Security and antivirus protection

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BYOD and BYOT

- BYOD: Bring your own device
- BYOT: Bring your own technology (hardware + software + apps)
- Employee uses their own device access corporate networks, data, and services remotely

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Benefits of BYOD and BYOT

- Building employee morale
- Potential for lowering the initial cost hardware purchase
- Facilitating remote, around-the-clock access to corporate computer networks
- Building on a familiar user interface to access corporate computing services, applications, databases, and storage

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Drawbacks of BYOD and BYOT

- Security risks posed by untrained users
- Loss of the device
- Theft of the device and its data
- Unauthorized access to corporate networks using personal mobile devices

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When do we need BYOD and BYOT?

✅ Ideal Situations:

- Hybrid or remote work models: Employees work partly from home or offsite.
- Field or sales teams: Staff on the move need constant access to company apps.
- Education & training: Students or trainees use personal laptops/tablets for e-learning.
- Startups & SMEs: Reduces capital cost of hardware procurement.
- Temporary or contract workforce: Easier to onboard quickly with personal devices.

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When do we need BYOD and BYOT?

❌ Not Ideal When:

- Handling highly confidential or classified data.
- In industries requiring strict compliance (e.g., finance, healthcare, defense).
- Where employees lack technical literacy for secure device management.

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Software Alternatives

- Created custom software
  - In-house Development
  - Tailored to organization's exact needs.
- Purchased as COTS (commercial off-the-shelf) software
  - Ready-made and widely used.
  - Lower cost and faster deployment but limited customization.
- Provided by a software as a service (SaaS)
  - Subscription-based, hosted on the cloud.
  - Minimal maintenance and automatic updates.

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Figure 3.8 Software Alternatives

	Advantages	Disadvantages
Creating Custom Software	Specific response to specialized business needs Innovation may give firm a competitive advantage In-house staff available to maintain software Pride of ownership	May be significantly higher initial cost compared to COTS software or ASP Necessity of hiring or working with a development team Ongoing maintenance
Purchasing COTS Packages	Refined in the commercial world Increased reliability Increased functionality Often lower initial cost Already in use by other firms Help and training comes with software	Programming focused; not business focused Must live with the existing features Limited customization Uncertain financial future of vendor Less ownership and commitment
Using SaaS	Organizations that do not specialize in information systems can focus on what they do best (their strategic mission) There is no need to hire, train, or retain a large IT staff There is no expenditure of employee time on nonessential IT tasks	Loss of control of data, systems, IT employees, and schedules Concern over the financial viability and long-run stability of the SaaS provider Security, confidentiality, and privacy concerns Loss of potential strategic corporate advantage regarding innovativeness of applications

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Software Evaluation

- Performance effectiveness
- Performance efficiency
- Ease of use
- Flexibility
- Quality of documentation
- Manufacturer support

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Figure 3.9 Guidelines for Evaluating Software

Software Requirements	Specific Software Features
Performance Effectiveness	Able to perform all required tasks Able to perform all tasks desired Well-designed display screens Adequate capacity
Performance Efficiency	Fast response time Efficient input Efficient output Efficient storage of data Efficient backup
Ease of Use	Satisfactory user interface Help menus available “Read Me” files for last-minute changes Flexible interface Adequate feedback Good error recovery
Flexibility	Options for input Options for output Usable with other software
Quality of Documentation	Good organization Adequate online tutorial Website with FAQ
Manufacturer Support	Technical support hotline Newsletter/email Website with downloadable product updates

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Recommended Approaches for Evaluating Software

- Always involve end-users in evaluation to ensure acceptance.
- Include IT and management in scoring to balance technical and strategic perspectives.
- Request Proof of Concept (POC) or pilot trial before commitment.
- Document all findings for audit and traceability.
- Use a weighted scoring matrix or decision table.

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Identifying and Forecasting Costs and Benefits

- Before developing or acquiring a system, analysts must estimate all expected costs (what you'll spend) and benefits (what you'll gain) to determine whether the project is economically feasible.
- This process helps answer:
  - "Will the system pay for itself — and when**?**"

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Identifying and Forecasting Costs and Benefits (1 of 2)

- Judgment methods
  - Estimates from the sales force
  - Surveys to estimate customer demand
  - Delphi studies (gather expert opinions)
  - Creating scenarios
  - Drawing historical analogies

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Identifying and Forecasting Costs and Benefits (2 of 2)

- If historical data are available
  - Conditional:
    - There is an association among variables in the model
  - Unconditional:
    - Do not need to find or identify any relationships

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Estimation of Trends

- Is a quantitative forecasting method that predicts future costs or benefits by analyzing historical data patterns over time.
- Based on the principle that:
  - "Past behavior or data patterns can help predict future performance — unless major changes occur."
- Estimation Technique:
  - Graphical judgment
  - Moving averages
  - Time-Series Analysis

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Identifying Benefits and Costs

- Tangible benefits are advantages measurable in dollars through the use of the information system
- Intangible benefits are difficult to measure
- Tangible costs are accurately projected by the systems analyst and accounting personnel
- Intangible costs are difficult to estimate and may not be known (involve human, organizational, or reputational impacts).

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Tangible Benefits

- Advantages measurable in dollars that accrue to the organization through the use of the information system
- Examples:
  - Increase in the speed of processing
  - Access to otherwise inaccessible information
  - Access to information on a more timely basis
  - The advantage of the computer's superior calculating power
  - Decreases in the amount of employee time needed to complete specific tasks

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Intangible Benefits

- Intangible benefits are benefits from use of the information system that are difficult to measure
- Examples:
  - Improving the decision-making process
  - Enhancing accuracy
  - Becoming more competitive in customer service
  - Maintaining a good business image
  - Increasing job satisfaction

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Tangible Costs

- Those that can be accurately projected by systems analysts and the business' accounting personnel
- Examples:
  - Cost of equipment
  - Cost of resources
  - Cost of systems analysts' time
  - Cost of programmers' time
  - Employees' salaries

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Intangible Costs

- Those that are difficult to estimate and may not be known
- Examples:
  - Losing a competitive edge
  - Losing the reputation of being first
  - Declining company image
  - Ineffective decision making

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Comparing Costs and Benefits

- After estimating, analysts use financial analysis techniques to determine feasibility.

Technique	Purpose	Example
Break-Even Analysis	Finds when cumulative benefits = total costs.	"System pays back after 3.5 years."
Payback Period	How long it takes to recover initial investment.	Shorter payback = lower financial risk.
Return on Investment (ROI)	Measures profitability as a percentage.	ROI = (Net Benefits ÷ Total Costs) × 100
Net Present Value (NPV)	Converts future cash flows to today's value (discounted).	Used for long-term system investments.
Cost-Benefit Ratio (CBR)	Compares total benefits to total costs.	>1.0 means benefits outweigh costs.

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Figure 3.10 Break-Even Analysis

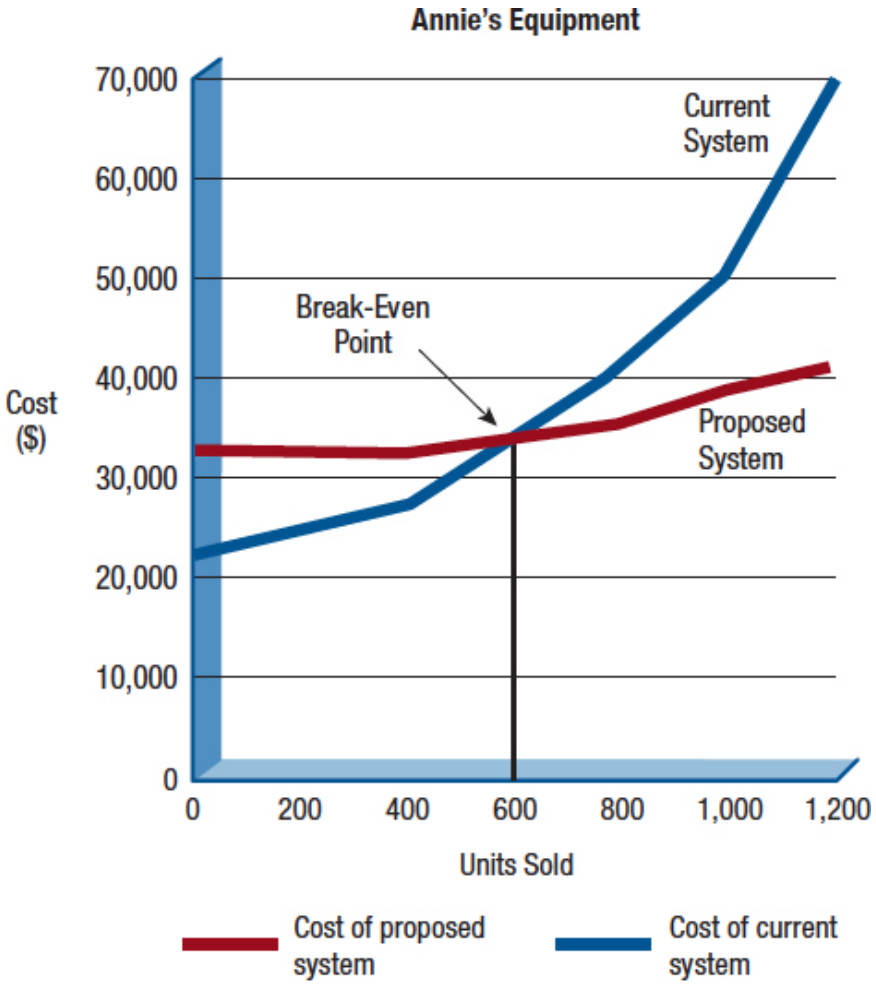

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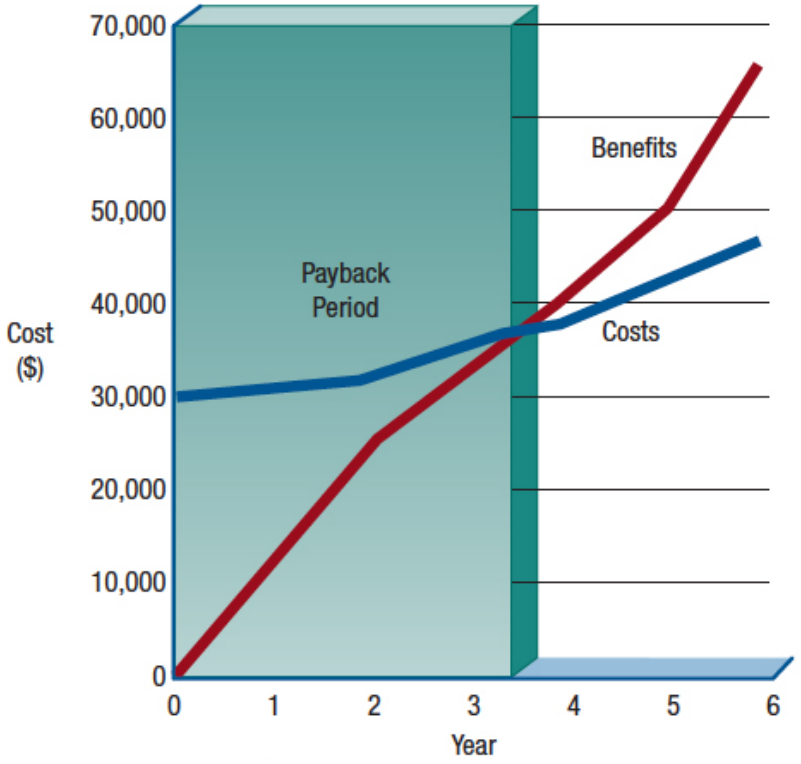
Figure 3.11 Payback Analysis Showing a Payback Period of Three and a Half Years



Year	Cost ($)	Cumulative Costs ($)	Benefits ($)	Cumulative Benefits ($)
0	30,000	30,000	0	0
1	1,000	31,000	12,000	12,000
2	2,000	33,000	12,000	24,000
3	2,000	35,000	8,000	32,000
4	3,000	38,000	8,000	40,000
5	4,000	42,000	10,000	50,000
6	4,000	46,000	15,000	65,000

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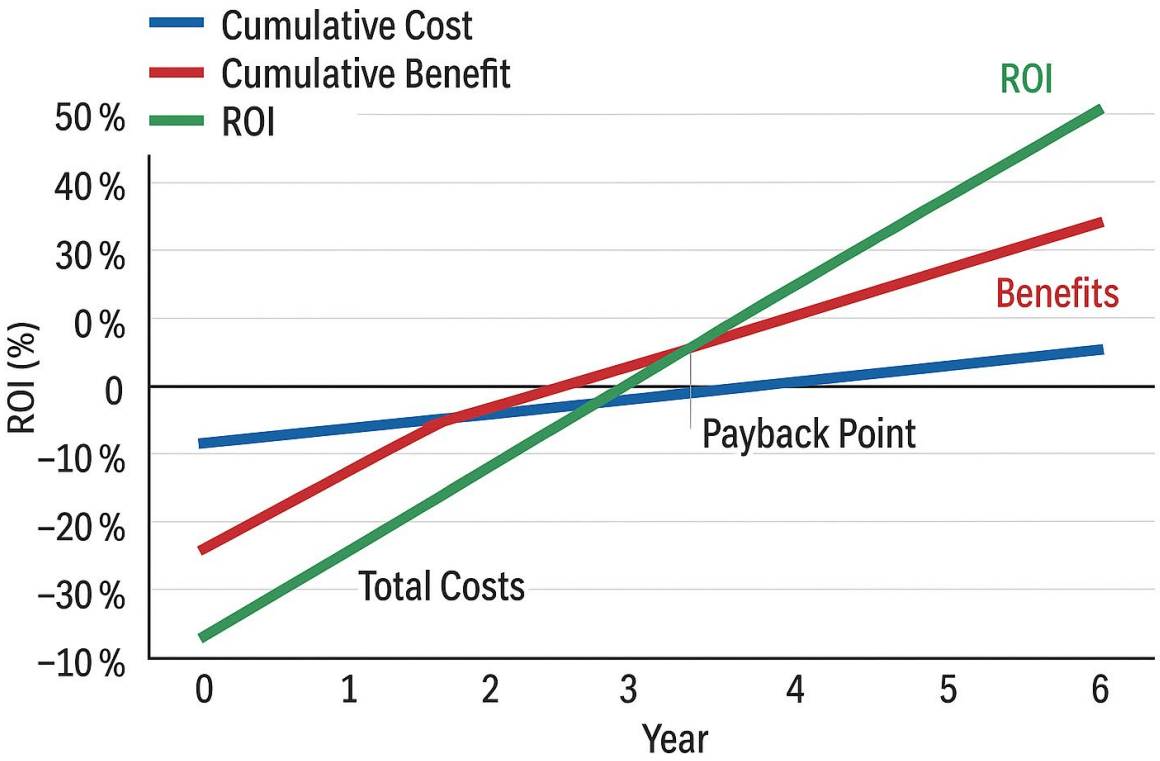
Figure 3.11a ROI Graph

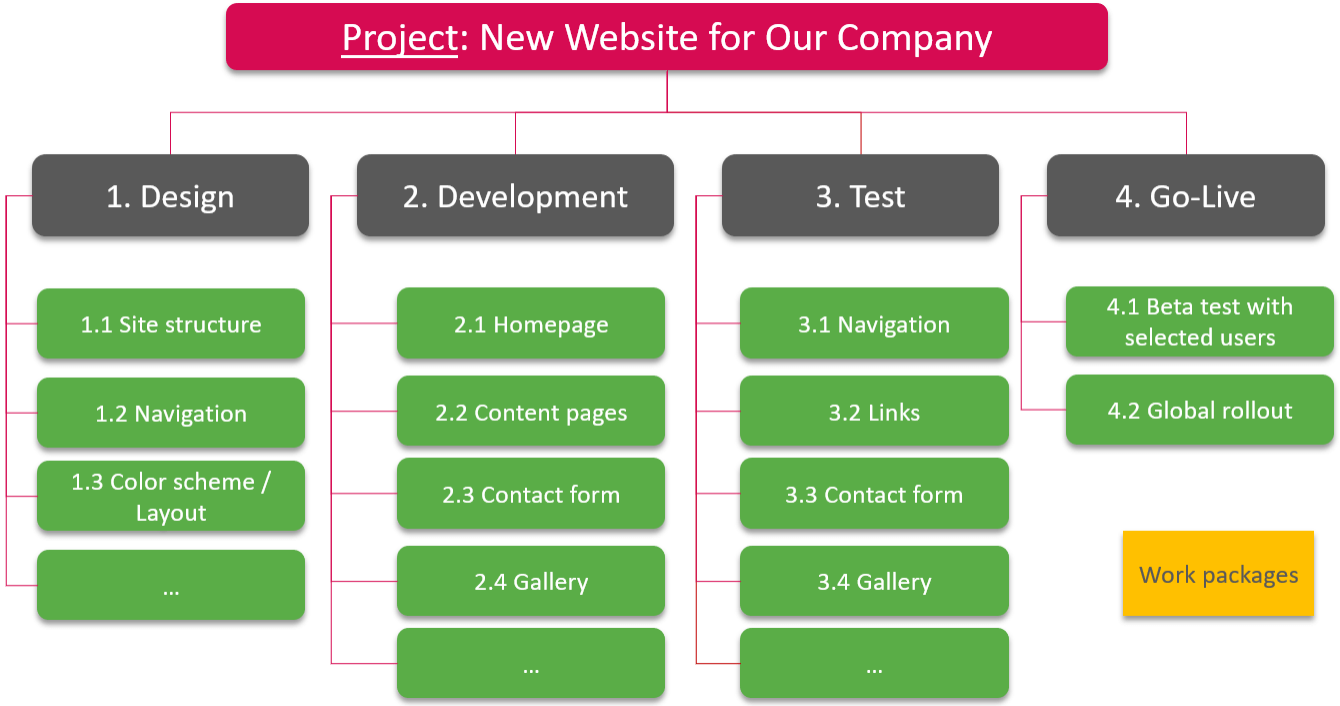


Element	Description
X-axis	Time (usually in years or months).
Y-axis	ROI (%) or cumulative net benefit ($).
Curve/Line	ROI growth or cumulative profit trend.
Payback Point	Where ROI crosses 0% (benefits = costs).
Annotations	Labels for total costs, benefits, and key ROI milestones.

Year	Cumulative Cost ($)	Cumulative Benefit ($)	Net Benefit ($)	ROI (%)
0	30,000	0	-30,000	-100%
1	31,000	12,000	-19,000	-61%
2	33,000	24,000	-9,000	-27%
3	35,000	32,000	-3,000	-9%
4	38,000	40,000	+2,000	+5%
5	42,000	50,000	+8,000	+19%
6	46,000	65,000	+19,000	+41%

Work Breakdown Structure

- Often a project needs to be broken down into smaller tasks or activities
- These tasks together make up a work breakdown structure (WBS)



Work Breakdown Structure Properties

- Key Principles (The 100% Rule)
  - The WBS must include 100% of the work required to achieve the project objective — nothing more, nothing less.
- This ensures:
  - No scope omissions (avoids “scope creep”).
  - Every task has clear accountability.

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Work Breakdown Structure Properties

- Each task or activity contains one deliverable, or tangible outcome, from the activity
- Each task can be assigned to a single individual or a single group
- Each task has a responsible person monitoring and controlling performance

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Developing a Work Breakdown Structure (WBS)

- Decomposition, starting with large ideas, then breaking them down into manageable activities
- Product oriented, building a website can be broken down into many parts
- Process-oriented, emphasizes the importance of each phase

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Figure 3.12 A Sample Work Breakdown Structure (WBS)

1.0	**Project Initiation**
1.1	Assemble and welcome project team
1.2	Conduct historical research about the business
1.3	Discuss objectives with client
2.0	**Early Planning Phase**
2.1	Investigate feasibility
2.2	Consider a make vs. buy decision
2.3	Develop a work breakdown structure
2.4	Provide time estimates
2.5	Develop a project schedule
2.6	Calculate cost estimates
2.7	Prepare project proposal to client
2.8	Present proposal to client
3.0	**Develop supporting plans**
3.1	Develop a quality management plan
3.2	Identify risks and build a risk management plan
3.3	Describe a communications plan
3.4	Develop a procurement plan
4.0	**Analysis**
4.1	Conduct interviews of key personnel
4.2	Administer questionnaires
4.3	Read company reports
4.4	Analyze data flow
5.0	**Design**
5.1	Build prototype website
5.2	Obtain reactions from client
5.3	Modify prototype website
5.4	Seek final recommendations from client
5.5	Complete website
6.0	**Launch**
6.1	Create training manual
6.2	Document website features and logic
6.3	Present final website to client

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Time Estimation Techniques

- Relying on experience
- Using analogies
  - Uses historical data from similar past projects as a reference.
- Using three-point estimation
  - Optimistic (O) – best-case scenario
  - Pessimistic (P) – worst-case scenario
  - Most likely (M) – most realistic duration
- Identifying function points
- Using time estimation software

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Function Point Analysis

- Helps estimate the effort, time, and cost of software development and maintenance by quantifying the functional requirements
- Takes the five main components of a computer system and rates them in terms of complexity:
  - External inputs : User inputs that change system data (e.g., entering a customer)
  - External outputs : Outputs sent to users (e.g., reports, invoices)
  - External queries : Requests for data with no update (e.g., search, view customer info)
  - Internal logical files : Data maintained by the system (e.g., employee records)
  - External interface files : Data used but maintained by other systems (e.g., shared product catalog)

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Figure 3.13 Beginning to Plan a Project by Breaking It into Three Major Activities

Phase	Activity
Analysis	Data gathering Data flow and decision analysis Proposal preparation
Design	Data entry design Input design Output design Data organization
Implementation	Implementation Evaluation

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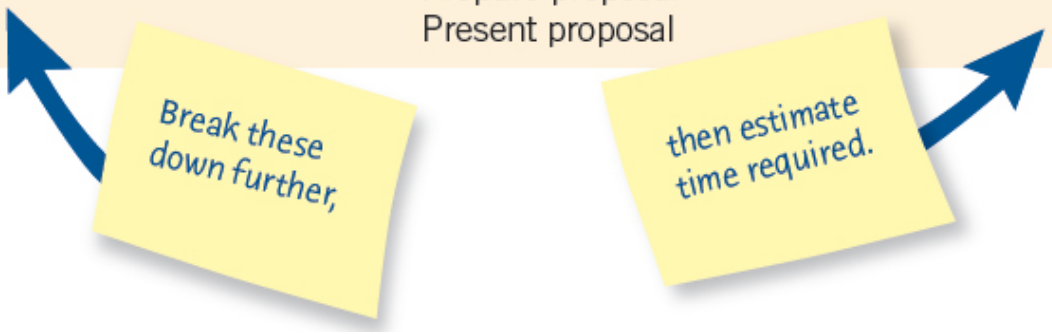
Figure 3.14 Refining the Planning and Scheduling of Analysis Activities by Adding Detailed Tasks and Establishing the Time Required to Complete the Tasks

Activity	Detailed Activity	Weeks Required
Data gathering	Conduct interviews	3
	Administer questionnaires	4
	Read company reports	4
	Introduce prototype	5
	Observe reactions to prototype	3
Data flow and decision analysis	Analyze data flow	8
Proposal preparation	Perform cost-benefit analysis	3
	Prepare proposal	2
	Present proposal	2



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What is Six Sigma?

- A data-driven methodology for process improvement and defect reduction.
- Developed by Motorola (1980s), popularized by GE.
- Aims for near-perfect quality (3.4 defects per million opportunities).

- Core Methodologies:
  - DMAIC – Define, Measure, Analyze, Improve, Control (improving existing processes)
  - DMADV – Define, Measure, Analyze, Design, Verify (designing new processes)

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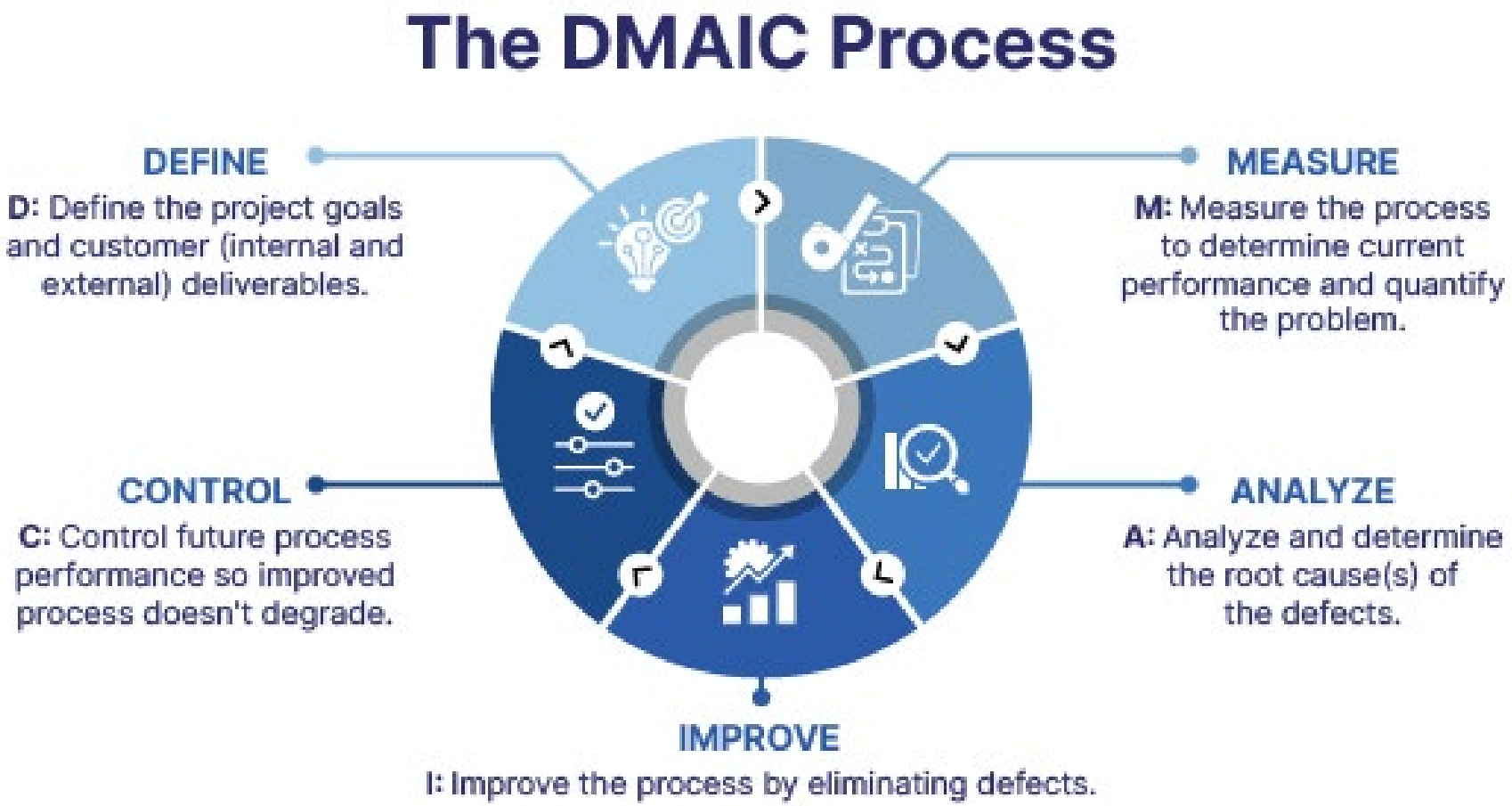
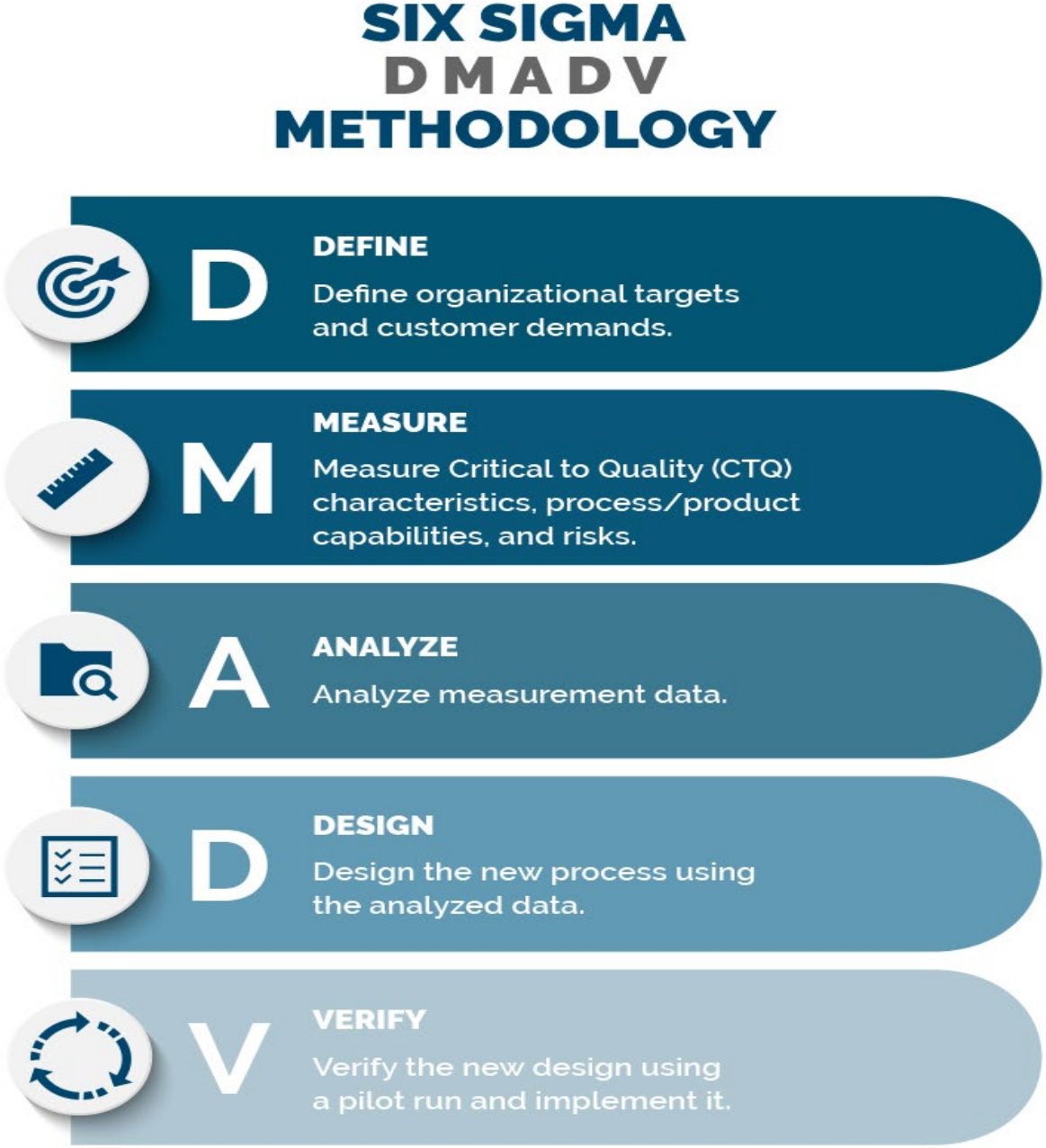
Six Sigma (DMAIC)



Six Sigma (DMADV)



Project Scheduling

- Gantt Charts
  - Simple
  - Lends itself to end user communication
  - Drawn to scale
- PERT diagrams
  - Useful when activities can be done in parallel
- Gantt chart shows when tasks occur while, PERT diagram shows how tasks are connected and dependent on each other.

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Gantt Charts

- Definition
  - A horizontal bar chart that visually represents a project schedule that showing tasks, durations, dependencies, and progress over time.
  - Introduced by Henry L. Gantt in the early 1900s and remains an essential tool in modern project management software
- Purpose of a Gantt Chart
  - To plan what tasks need to be done and when.
  - To visualize task sequences and overlaps.
  - To show dependencies (which tasks must finish before others can start).
  - To track progress and ensure project stays on schedule.
  - To communicate timelines clearly to team members and stakeholders.

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PERT Diagram

- Definition
  - A PERT Diagram (Program Evaluation and Review Technique) is a network diagram that visually represents the sequence, dependencies, and timing of project tasks.
  - It is used to analyze and schedule tasks — particularly when the duration of **activities is uncertain**.
- Purpose of a PERT Diagram
  - To Identify the minimum time needed to complete a project.
  - Determine the critical path (the longest sequence of dependent tasks).
  - Highlight task dependencies and bottlenecks.
  - Estimate probable completion times when durations are uncertain.

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Figure 3.15 Using a Two-Dimensional Gantt Chart for Planning Activities That Can Be Accomplished in Parallel

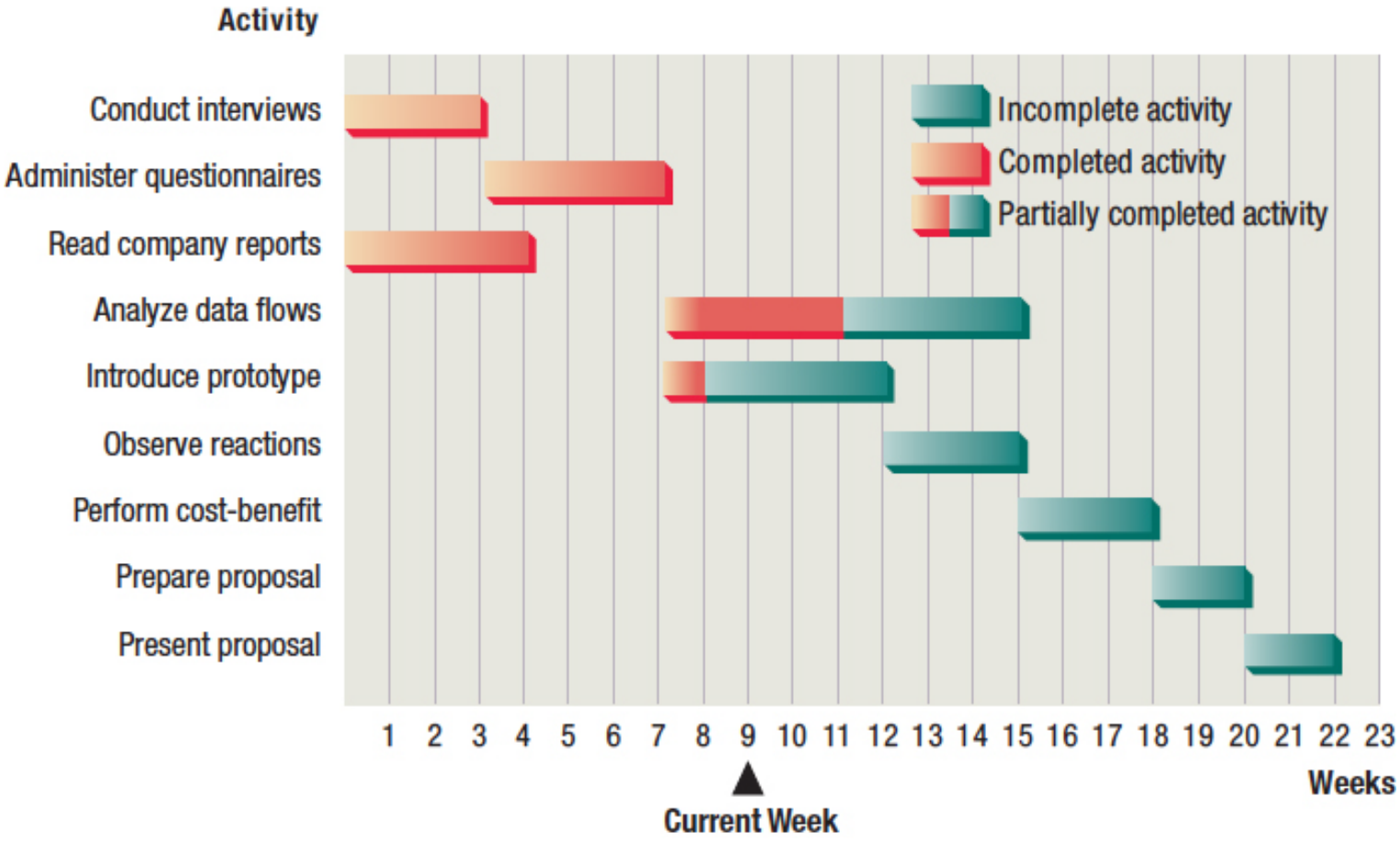

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Figure 3.16 A Gantt Chart Compared with a PERT Diagram (Network Diagram) for Scheduling Activities

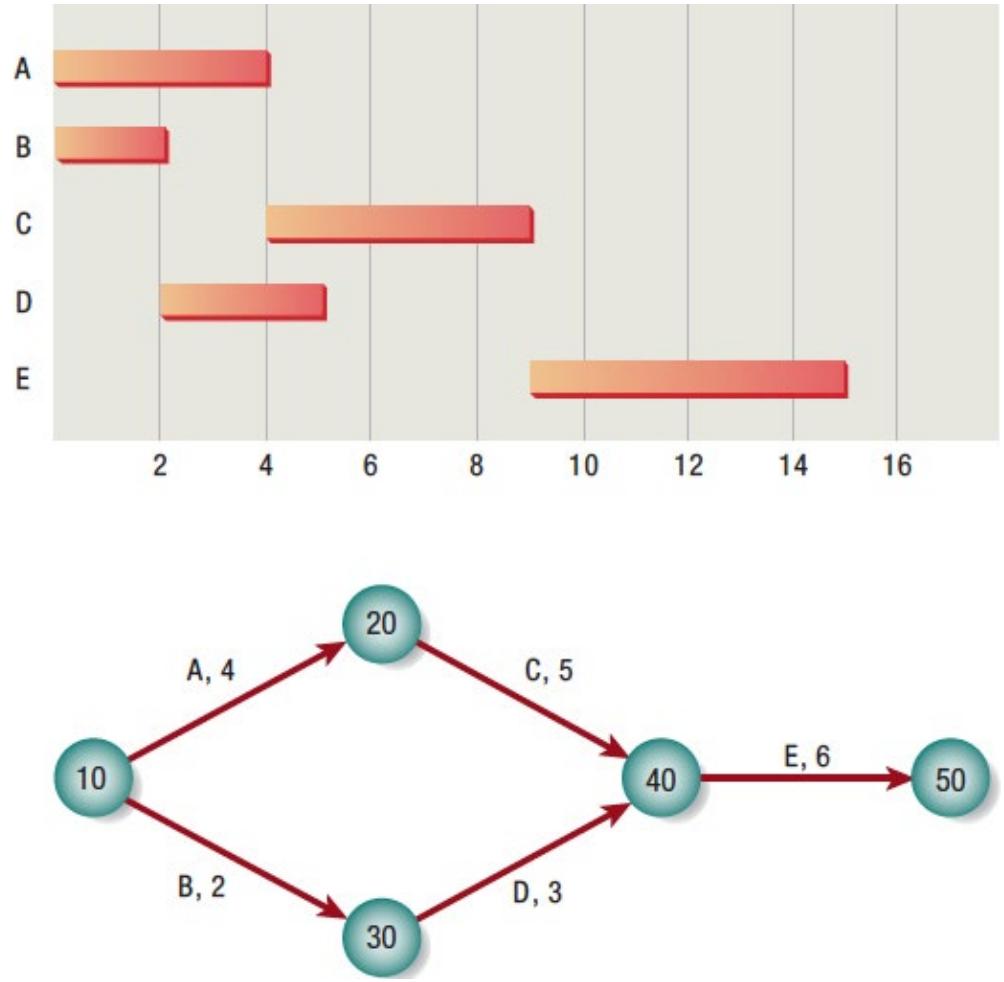

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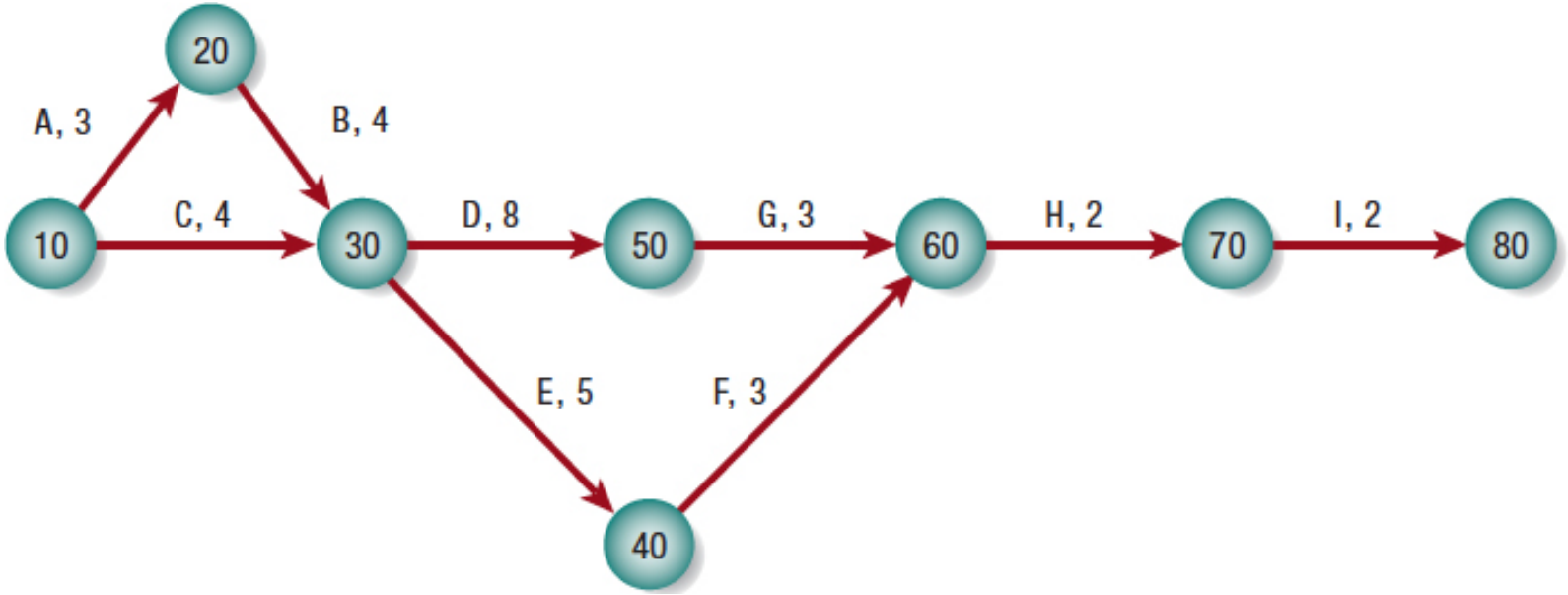
Figure 3.19 A Completed PERT Diagram for the Analysis Phase of a Systems Project



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PERT Diagram Advantages

- Easy identification of the order of precedence
- Easy identification of the critical path and thus critical activities
- Easy determination of slack time

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Estimating Costs

- Using a work breakdown structure
  - Estimate costs for each activity in the work breakdown structure
  - Prepare a budget for the project and have it approved by the organization or client
  - Manage and control the costs throughout the project

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Approaches to Cost Estimation

- Basing estimates on similar projects, also called the top-down approach
- Building bottom-up estimates
- Using parametric modeling

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Project Risk

- Project failures may be prevented by:
  - Training
  - Experience
  - Learning why other projects have failed
- Fishbone diagram (or cause-and-effect diagram, or an Ishikawa diagram) systematically lists all of the possible problems that can occur

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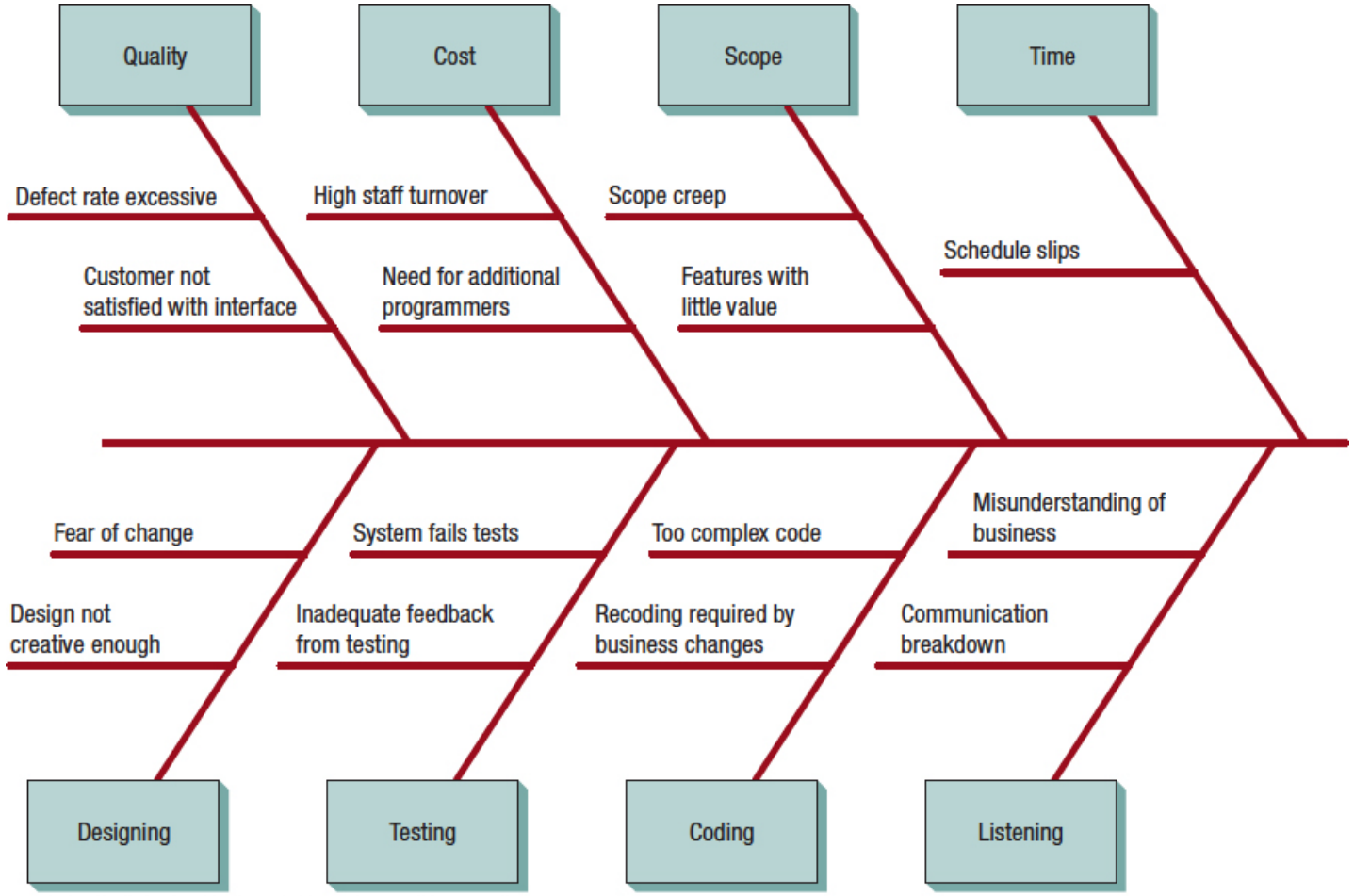
Figure 3.21 Fishbone Diagram



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Expediting

- Expediting is speeding up a process
- Expediting can help reduce the time it takes to complete an entire project
- The expedited activities have to be on the critical path

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Earned Value Management (EVM)

- A technique used to help determine progress (or setbacks) on a project
- Involves:
  - Project cost
  - Project schedule
  - Performance of the project team

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Four Key Measures in EVM

- Budget at completion (BAC) is the total budget for the project
- Planned value (PV) is the value of the work that is to be completed on the project
- Actual Cost (AC) is the total cost incurred in completing the work
- Earned value (EV) is an estimate of the value of the work performed thus far

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Managing the Team Project

- Team management
  - Assembling a team
  - Team communication strategies
  - Project productivity goals
  - Team member motivation

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Assembling a Team

- Shared value of team work
- Good work ethic
- Honesty
- Competency
- Readiness to take on leadership based on expertise
- Motivation
- Enthusiasm for the project
- Trust of teammates

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Communication Strategies

- Teams often have two leaders:
  - Task leader: leads members to accomplish tasks
  - Socioemotional leader: concerned with social relationships
- The systems analyst must manage:
  - Team members
  - Their activities
  - Their time and resources

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Project Productivity Goals and Motivation

- Successful projects require that reasonable productivity goals for tangible outputs and process activities be set.
- Goal-setting helps to motivate team members.

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Ecommerce Project Management

- Ecommerce and traditional software project management differences:
  - The data used by ecommerce systems is scattered across the organization
  - Ecommerce systems need a staff with a wide variety of skills
  - Partnerships must be built externally and internally well ahead of implementation
  - Security is of utmost importance

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Project Charter

- Describes in a written document what the expected results of the systems project are and the time frame for delivery

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Project Charter Clarifies These Questions (1 of 2)

- What does the user expect of the project?
- What is the scope of the project?
- What analysis methods will the analyst use to interact with users?
- Who are the key participants?
- What are the project deliverables?

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Project Charter Clarifies These Questions (2 of 2)

- Who will evaluate the system and how will they evaluate it?
- What is the estimated project timeline?
- Who will train the users?
- Who will maintain the system?

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The Systems Proposal

- Cover letter
- Title page of project
- Table of contents
- Executive summary
- Outline of systems study with appropriate documentation
- Detailed results of the systems study
- Systems alternatives
- Systems analysts recommendations
- Summary
- Appendices

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Using Figures for Effective Communication

- Effective use of tables
- Effective use of graphs

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Effective Use of Tables

- Integrate into the body of the proposal
- Try to fit the entire table vertically on a single page.
- Number and title the table at the top of the page.
- Label each row and column.
- Use a boxed table if room permits.
- Use footnotes if necessary to explain detailed information contained in the table.

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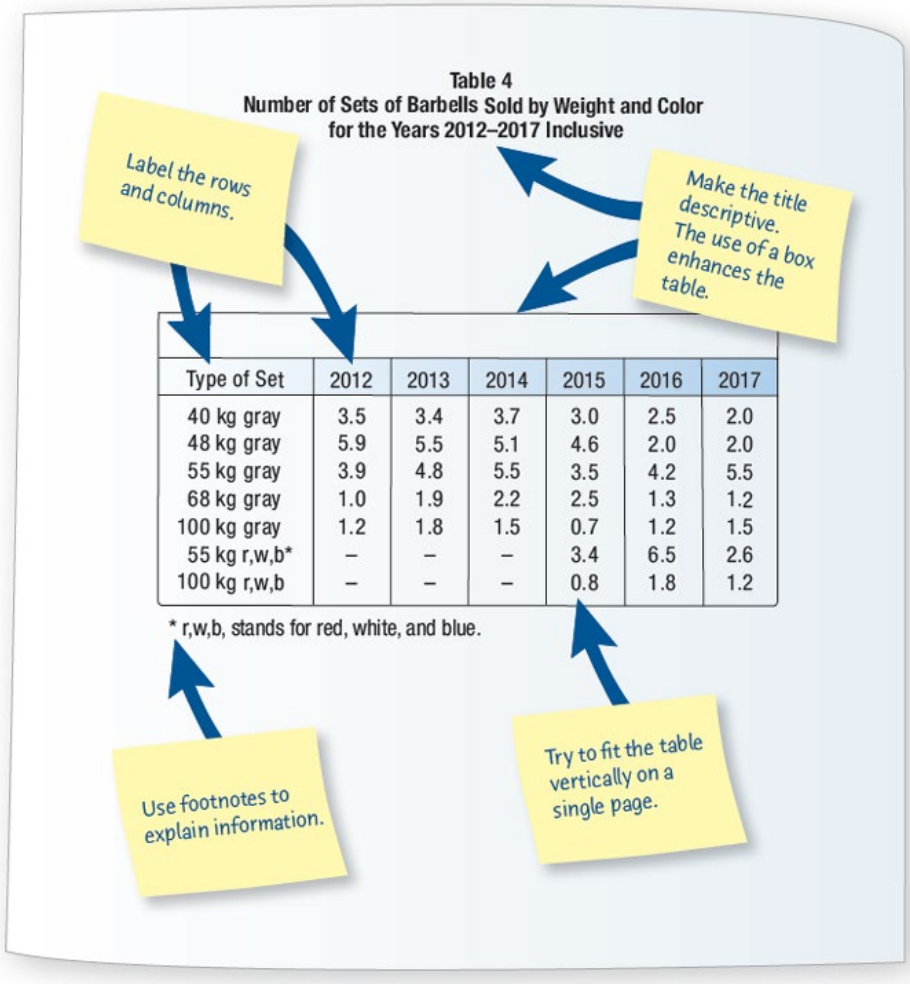
Figure 3.26 Guidelines for Creating Effective Tables



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Effective Use of Graphs

- Choose a style of graph that communicates your intended meaning well
- Integrate the graph into the body of the proposal
- Give the graph a sequential figure number and a meaningful title
- Label each axis, and any lines, columns, bars, or pieces of the pie on the graph
- Include a key to indicate differently colored lines, shaded bars, or crosshatched areas

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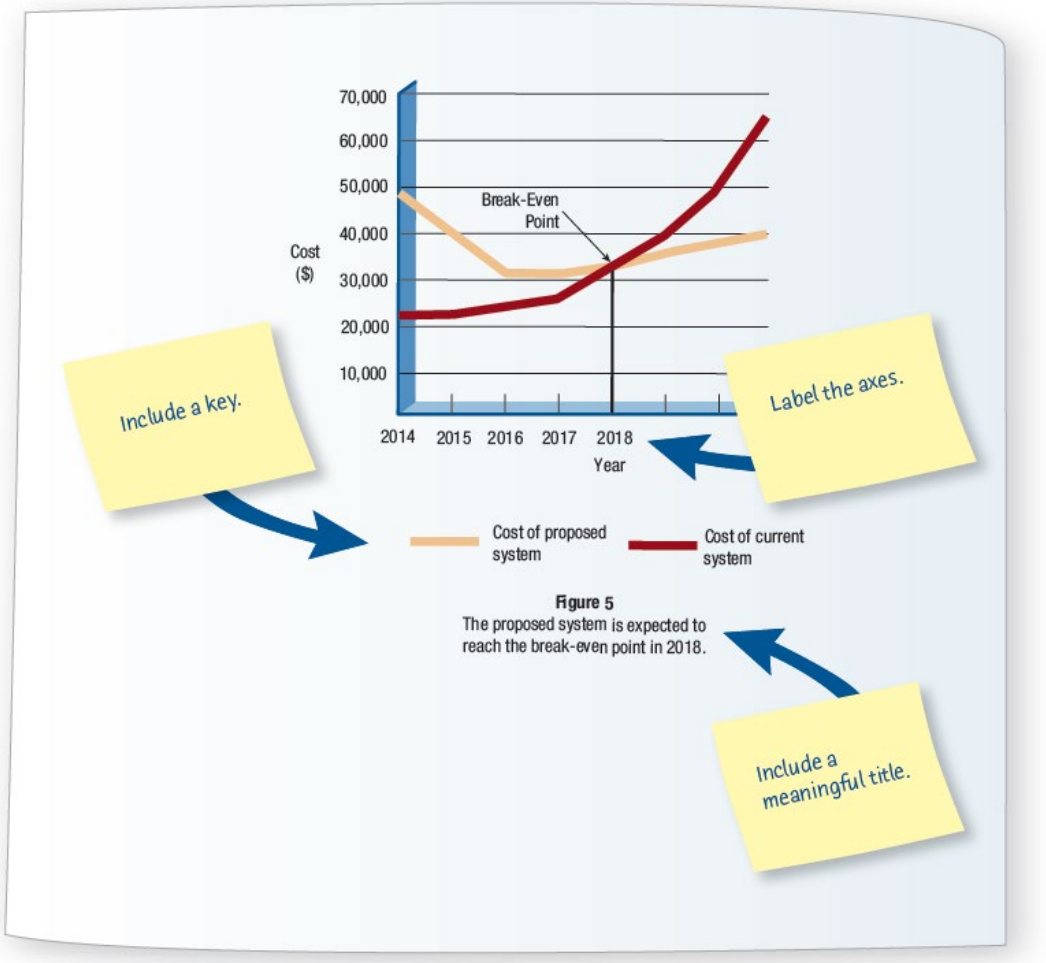
Figure 3.27 Effective Use of Graphs



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Summary (1 of 3)

- Project management fundamentals
  - Project initiation
  - Determining project feasibility
  - Activity planning and control
  - Project scheduling
  - Managing systems analysis team members
- Problem definition
  - Issues of the present system
  - The objective for each issue
  - The requirements that must be included in all proposed systems

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Summary (2 of 3)

- Project selection
  - Backed by management
  - Commitment of resources
  - Attains goals
  - Practical
  - Important
- Feasibility
  - Operational
  - Technical
  - Economic

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Summary (3 of 3)

- Acquiring hardware and software
- Work breakdown structure
- Project planning
  - Gantt charts
  - PERT
  - Function point analysis
- Team management
- Ecommerce projects
- Preparing a system proposal

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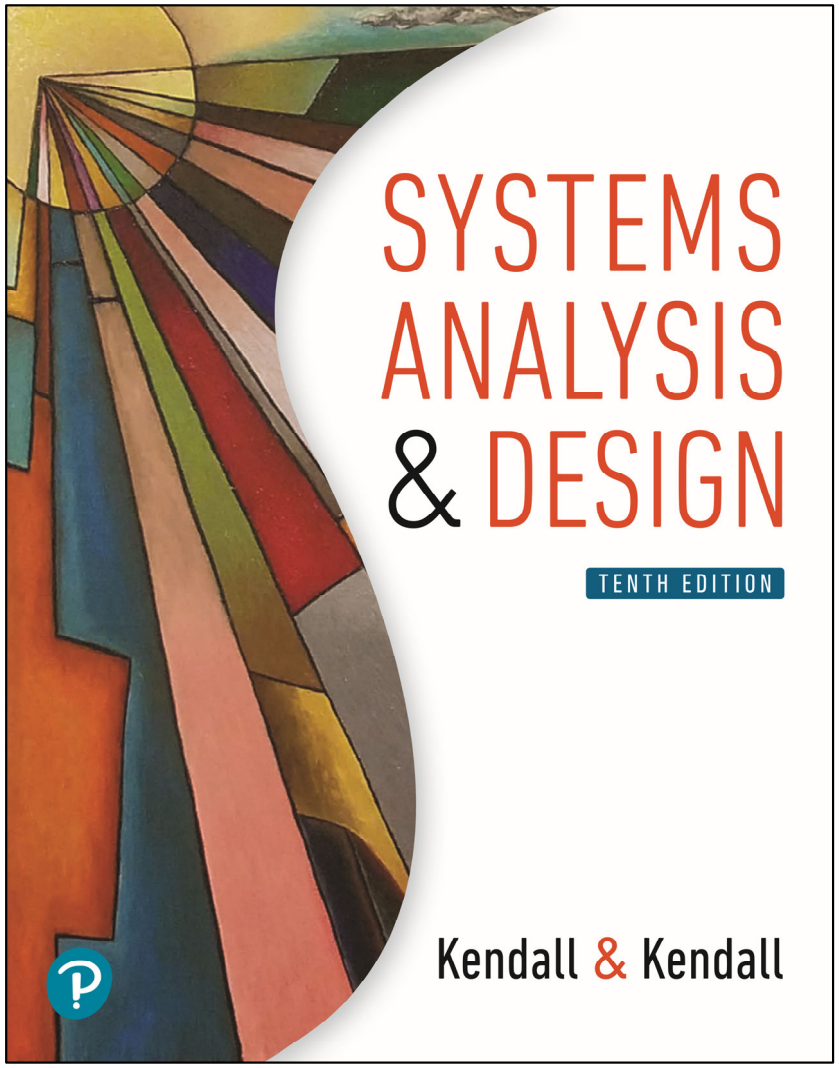
Systems Analysis & Design

Tenth Edition



Chapter 4

Information Gathering:
Interactive Methods

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Learning Objectives

4.1 Recognize the value of interactive methods for information gathering

4.2 Construct interview questions to elicit human information requirements and structure them in a way that is meaningful to users

4.3 Understand the purpose of stories and why they are useful in systems analysis

4.4 Understand the concept of JAD and when to use it

4.5 Write effective questions to survey users about their work

4.6 Design and administer effective questionnaires

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Interactive Methods to Elicit Human Information Requirements

- Interviewing
- Joint Application Design (JAD)
- Questionnaires

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Major Topics (1 of 2)

- Interviewing
  - Interview preparation
  - Question types
  - Arranging questions
  - The interview report
- User Stories

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Major Topics (2 of 2)

- Joint Application Design (JAD)
  - Involvement
  - Location
- Questionnaires
  - Writing questions
  - Using scales
  - Design
  - Administering

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Interviewing

- Interviewing is an important method for collecting data on human and system information requirements
- Interviews reveal information about:
  - Interviewee opinions
  - Interviewee feelings
  - Goals
  - Key HCI concerns

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Interview Preparation

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure

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Question Types (1 of 2)

- Open-ended
- Closed

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Open-Ended Questions

- Open-ended interview questions allow interviewees to respond how they wish without length or structure limitations
- Open-ended interview questions are appropriate when the analyst is interested in breadth and depth of reply

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Figure 4.2 Open-Ended Interview Questions

- What's your opinion of the current state of business-to-business ecommerce in your firm?
- What are the critical objectives of your department?
- Once the data are submitted via the website, how are they processed?
- Describe the monitoring process that is available online.
- What are some of the common data entry errors made in this department?
- What are the biggest frustrations you've experienced during the transition to ecommerce?

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Advantages of Open-Ended Questions (1 of 2)

- Puts the interviewee at ease
- Allows the interviewer to pick up on the interviewee's vocabulary
- Provides richness of detail
- Reveals avenues of further questioning that may have gone untapped

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Advantages of Open-Ended Questions (2 of 2)

- Provides more interest for the interviewee
- Allows more spontaneity
- Makes phrasing easier for the interviewer
- Useful if the interviewer is unprepared

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Disadvantages of Open-Ended Questions

- May result in too much irrelevant detail
- Possibly losing control of the interview
- May take too much time for the amount of useful information gained
- Potentially seeming that the interviewer is unprepared
- Possibly giving the impression that the interviewer is on a “fishing expedition”

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Closed Interview Questions

- Closed interview questions limit the number of possible responses
- Closed interview questions are appropriate for generating precise, reliable data that is easy to analyze
- The methodology is efficient, and it requires little skill for interviewers to administer

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Figure 4.3 Closed Interview Questions

- How many times a week is the project repository updated?
- On average, how many calls does the call center receive monthly?
- Which of the following sources of information is most valuable to you?
  - Completed customer complaint forms
  - Email complaints from consumers who visit the website
  - Face-to-face interaction with customers
  - Returned merchandise
- List your top two priorities for improving the technology infrastructure.
- Who receives this input?

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Benefits of Closed Interview Questions

- Save interview time
- Easily compare interviews
- Quickly get to the point
- Maintain control of the interview
- Cover a large area quickly
- Obtain relevant data

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Disadvantages of Closed Interview Questions

- May be boring for the interviewee
- May fail to obtain rich details
- May miss some main ideas
- May fail to build rapport between interviewer and interviewee

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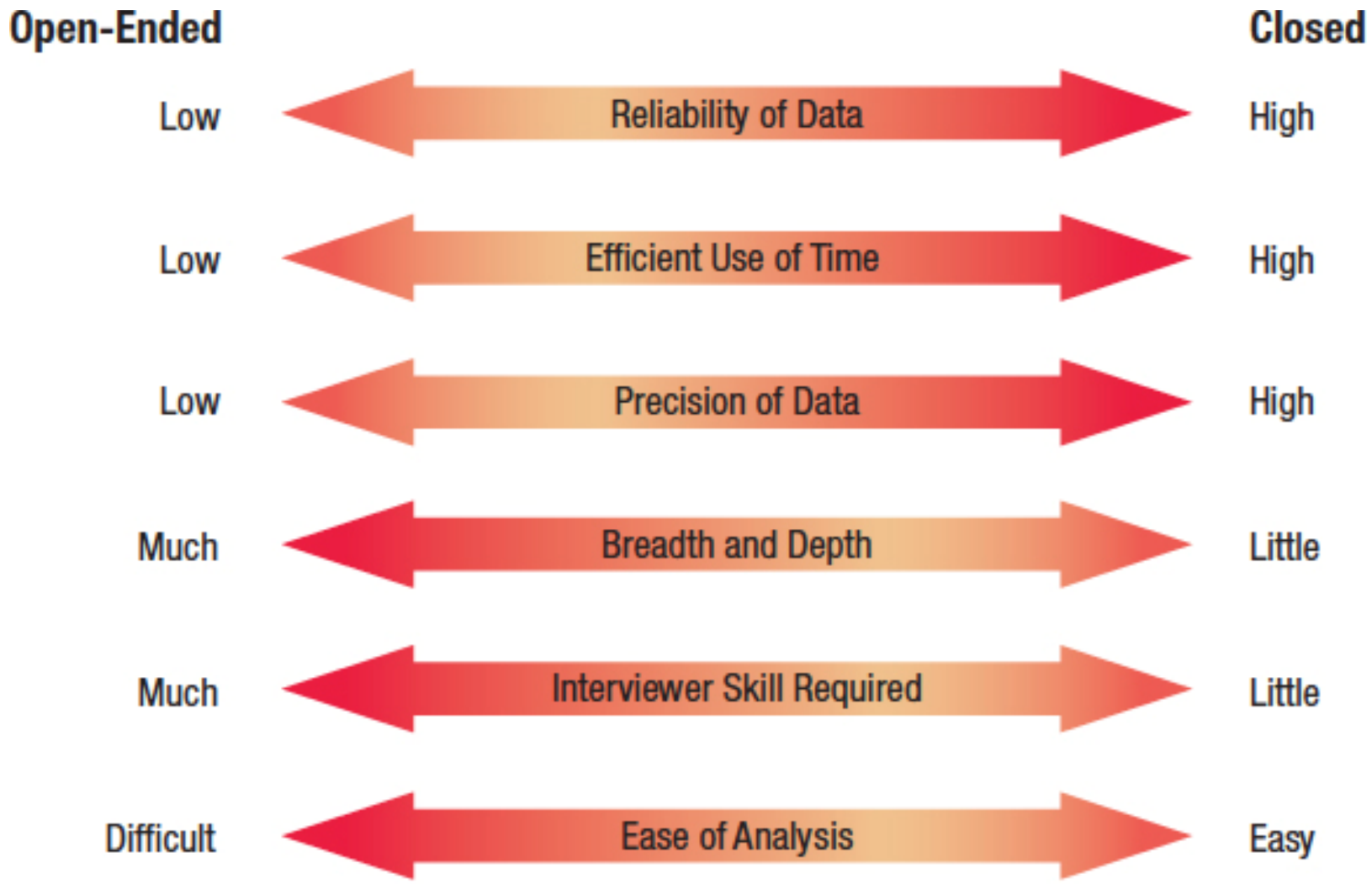
Figure 4.5 Attributes of Open-Ended and Closed Questions



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Bipolar Questions

- Bipolar questions are those that may be answered with a “yes” or “no” or “agree” or “disagree”
- Bipolar questions should be used sparingly
- A special kind of closed question

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Probes

- Probing questions elicit more detail about previous questions
- The purpose of probing questions is:
  - To get more meaning
  - To clarify
  - To draw out and expand on the interviewee's point
- May be either open-ended or closed

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Arranging Questions

- Pyramid
  - Starting with closed questions and working toward open-ended questions
- Funnel
  - Starting with open-ended questions and working toward closed questions
- Diamond
  - Starting with closed, moving toward open-ended, and ending with closed questions

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Pyramid Structure

- Begins with very detailed, often closed questions
- Expands by allowing open-ended questions and more generalized responses
- Is useful if interviewees need to be warmed up to the topic or seem reluctant to address the topic

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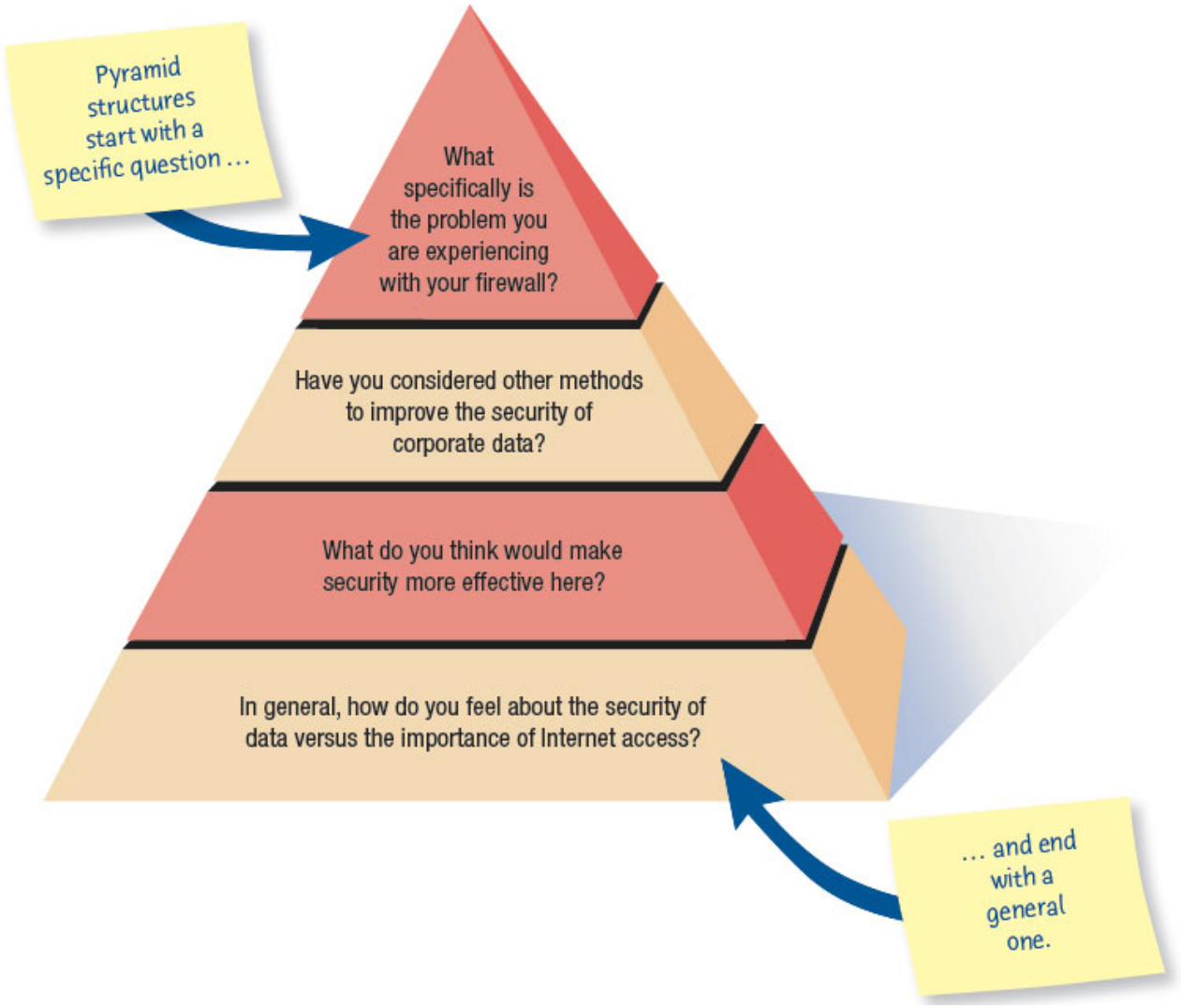
Figure 4.7 Pyramid Structure for Interviewing Goes from Specific to General Questions



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Funnel Structure

- Begins with generalized, open-ended questions
- Concludes by narrowing the possible responses using closed questions
- Provides an easy, nonthreatening way to begin an interview
- Is useful when the interviewee feels emotionally about the topic

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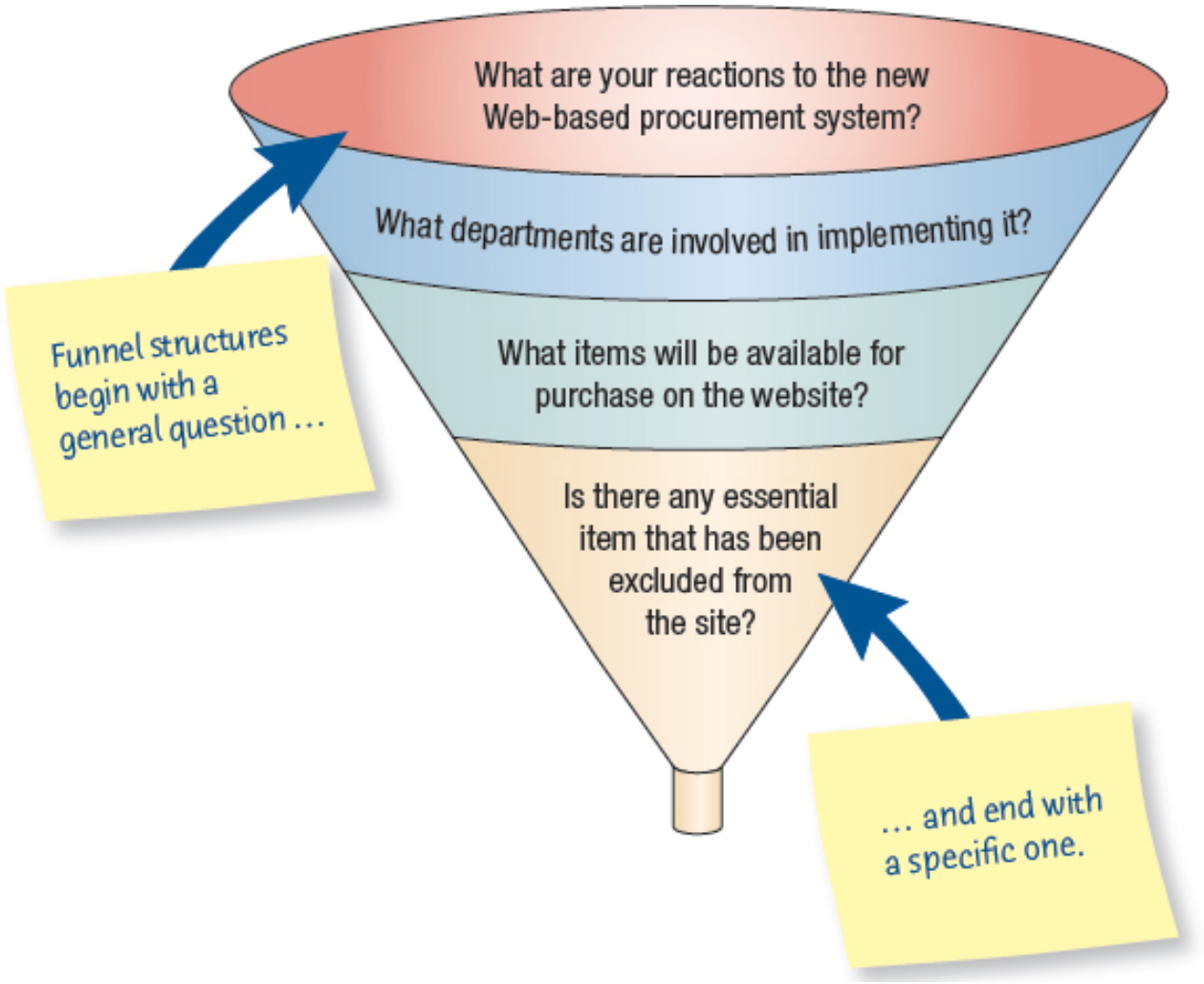
Figure 4.8 Funnel Structure for Interviewing Begins with Broad Questions Then Funnels to Specific Questions



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Diamond Structure

- A diamond-shaped structure begins in a very specific way
- Then more general issues are examined
- Concludes with specific questions
- Combines the strength of both the pyramid and funnel structures
- Takes longer than the other structures

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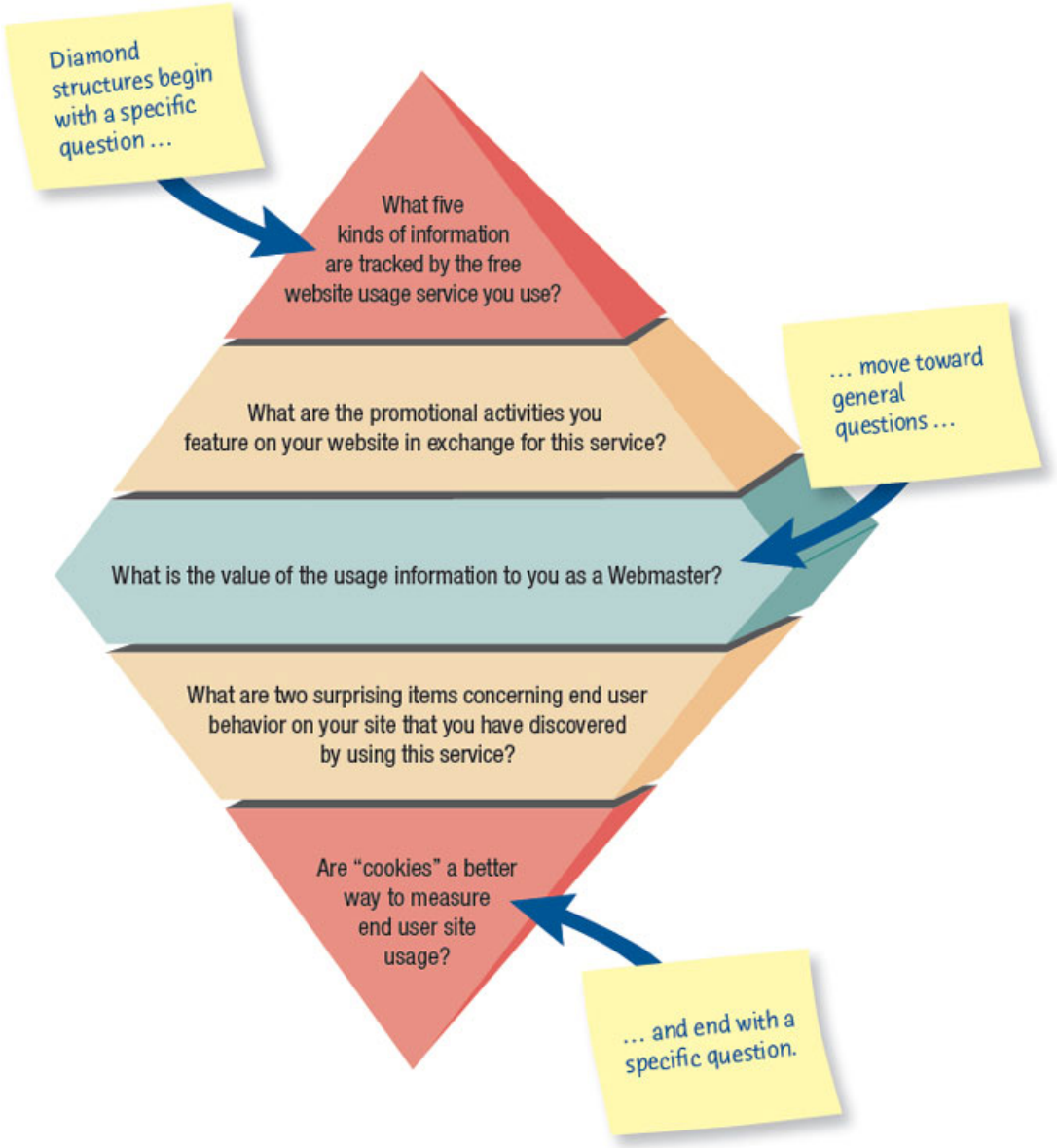
Figure 4.9 Diamond-Shaped Structure for Interviewing Combines the Pyramid and Funnel Structures



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Closing the Interview

- Always ask “Is there anything else that you would like to add?”
- Summarize and provide feedback on your impressions
- Ask whom you should talk with next
- Set up any future appointments
- Thank them for their time and shake hands.

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Interview Report

- Write as soon as possible after the interview
- Provide an initial summary, then more detail
- Review the report with the respondent

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AI Tools for Modern Interview techniques

Otter.ai	Real-time transcription, automated meeting summaries, speaker identification. Directly mentioned as a relevant tool.
Fireflies.ai	Real-time transcription, AI-powered meeting notes, search function across transcripts, and a focus on extracting key information (e.g., action items). Directly mentioned as a relevant tool.
ChatGPT/Large Language Models (LLMs)	Used for processing raw transcripts from other sources (Zoom/Teams) to generate summaries, extract themes, and refine notes. Mentioned specifically as relevant for meeting notes.
Sonix	Highly accurate transcription (supports many languages), includes AI analysis tools for summaries, thematic analysis, and topic detection.
Looppanel / Insight7	Tools specifically designed for qualitative research/UX interviews, offering transcription, automated highlight reels, and thematic analysis to generate insights quickly.
TurboScribe.ai	A platform that utilizes artificial intelligence to transcribe audio and video content into text efficiently. With a 99.8% accuracy rate, TurboScribe ensures that your transcripts are reliable and precise

Stories

- Stories originate in the workplace
- Organizational stories are used to relay some kind of information
- When a story is told and retold over time it takes on a mythic quality
- Isolated stories are good when you are looking for facts
- Enduring stories capture all aspects of the organization and are the ones a systems analyst should look be seeking

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Listening to Stories

- Listening to stories is not efficient
- It takes considerably more time than asking interview questions
- Listening to stories may be more rewarding
- Stories are more easily remembered than interview responses

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Stories are Made up of Elements

- All stories have these elements:
  - The call to adventure
  - The quest
  - The struggle
  - The transformation
  - The resolution
  - The moral
  - The epilogue

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Reasons for Telling Stories

- Rich information from listening carefully to the stories is in itself valuable
- The information gleaned from the stories will make more sense and be more valuable if seen in context

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Business Stories

- Business stories can be broken down into four important main types:
  - Experiential stories
  - Explanatory stories
  - Validating stories
  - Prescriptive stories

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Stories and the Organization

- Engage organization participants by reacting to stories
- Match one story to another by recounting it to other participants, and collaborating with the stories
- It is a way to deeply understand some of the problems associated with information systems

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Purposes for Telling a Story

- There are four purposes for telling a story:
  - Experiential stories describe what the business or industry is like
  - Explanatory stories tell why the organization acted a certain way
  - Validating stories are used to convince people that the organization made the correct decision
  - Prescriptive stories tell the listener how to act
- Systems analysts can use storytelling as a complement to other information gathering methods

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Joint Application Design (JAD)

- Joint Application Design (JAD) can replace a series of interviews with the user community
- JAD is a technique that allows the analyst to accomplish requirements analysis and design the user interface with the users in a group setting.
- Bring together key stakeholders including end-users, management, and IT specialists in a series of structured, intensive workshops (JAD sessions) to collaboratively define, refine, and agree upon project requirements and system specifications.

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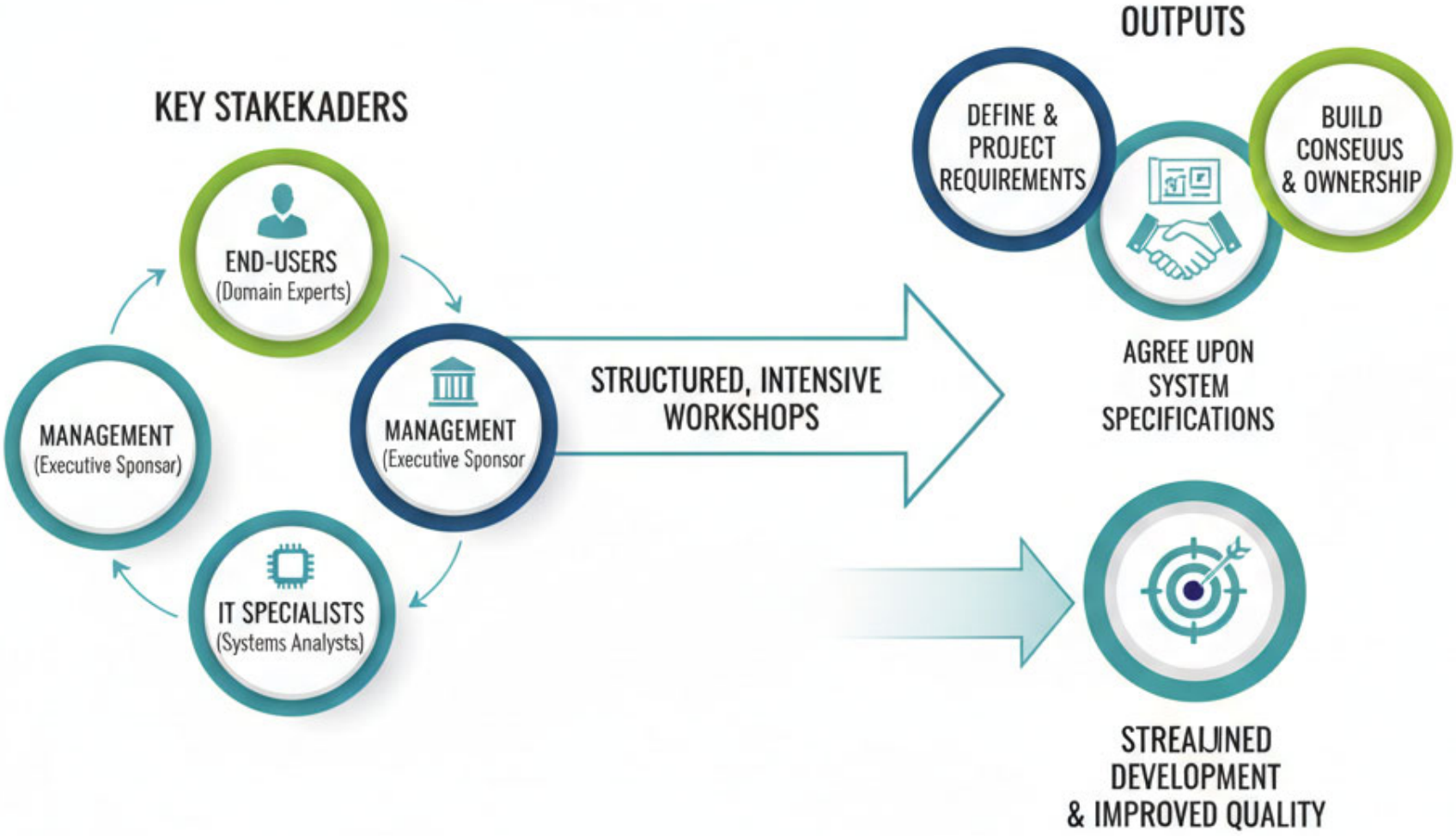
Joint Application Design (JAD)



Bringing together all key stakelders to collaborativale define & refine requirments, ensuring alighment with user needs & business objectives.

Conditions That Support the Use of JAD

- Users are restless and want something new
- The organizational culture supports joint problem-solving behaviors
- Analysts forecast an increase in the number of ideas using JAD
- Personnel may be absent from their jobs for the length of time required

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Who is Involved

- The people involved are:
  - Executive sponsor
  - IS analyst
  - Users
  - Session leader
  - Observers
  - Scribe

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Where to Hold JAD Meetings

- Offsite
  - Comfortable surroundings
  - Minimize distractions
- Attendance
  - Schedule when participants can attend
  - Agenda
  - Orientation meeting

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Benefits of JAD

- Time is saved, compared with traditional interviewing
- Rapid development of systems
- Improved user ownership of the system
- Creative idea production is improved
- Reduced Risks
- Consensus & Commitment

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Drawbacks of Using JAD

- JAD requires a large block of time to be available for all session participants
- If preparation or the follow-up report is incomplete, the session may not be successful
- The organizational skills and culture may not be conducive to a JAD session

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Questionnaires

Questionnaires are useful in gathering information from key organization members about:

- Attitudes
- Beliefs
- Behaviors
- Characteristics

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Planning for the Use of Questionnaires

- Organization members are widely dispersed
- Many members are involved with the project
- Exploratory work is needed
- Problem solving prior to interviews is necessary

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Question Types (2 of 2)

- Questions are designed as either:
  - Open-ended
    - Try to anticipate the response you will get
    - Well suited for getting opinions
  - Closed
    - Use when all the options may be listed
    - When the options are mutually exclusive

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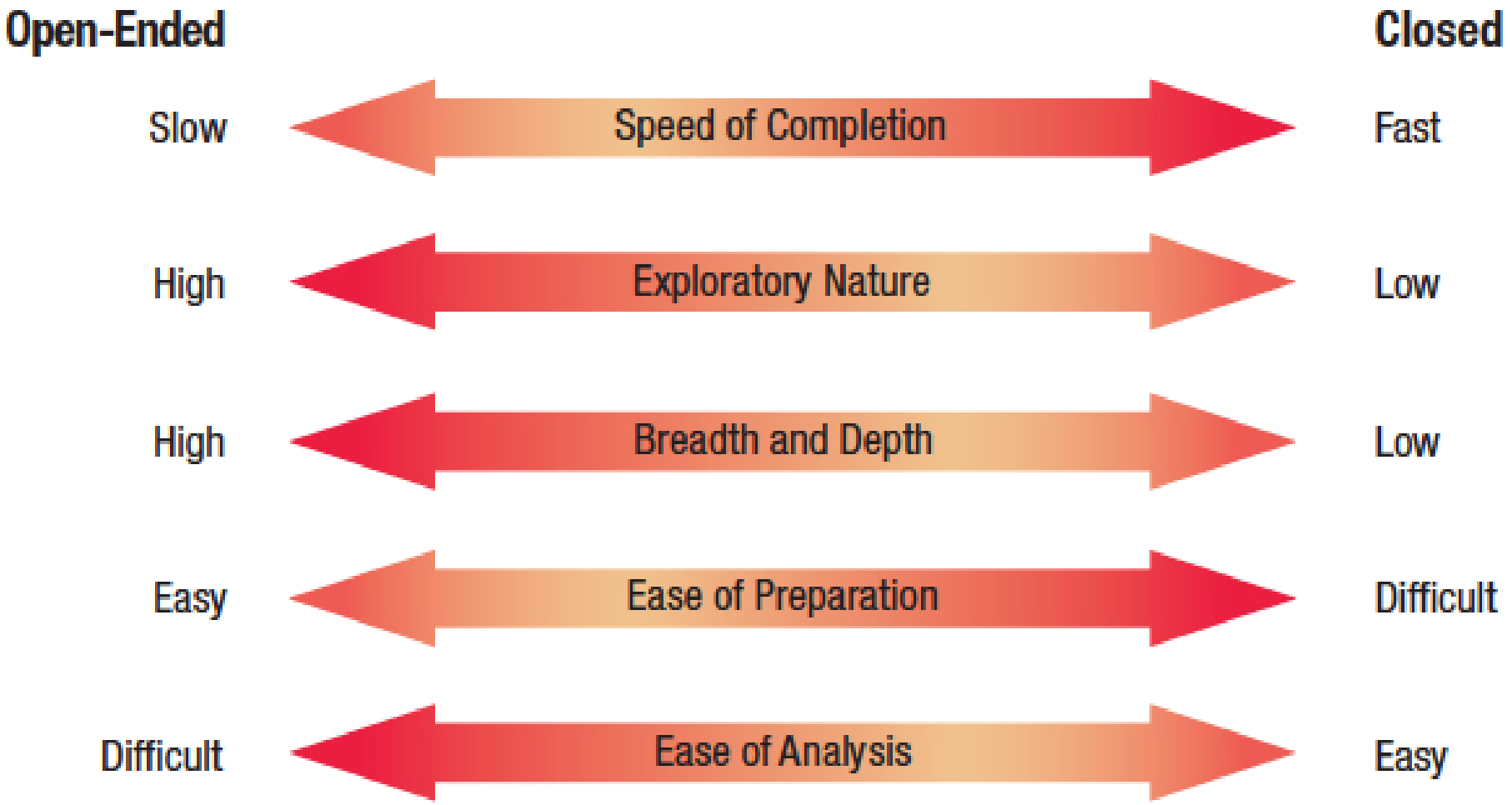
Figure 4.13 Trade-Offs between the Use of Open-Ended and Closed Questions on Questionnaires



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Questionnaire Language

- Simple
- Specific
- Short
- Not patronizing
- Free of bias
- Addressed to those who are knowledgeable
- Technically accurate
- Appropriate for the reading level of the respondent

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Measurement Scales

- The two different forms of measurement scales are:
  - Nominal
  - Interval

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Nominal Scales

- Nominal scales are used to classify things
- It is the weakest form of measurement
- Data may be totaled

What type of software do you use the most?

1 = Word Processor

2 = Spreadsheet

3 = Database

4 = An Email Program

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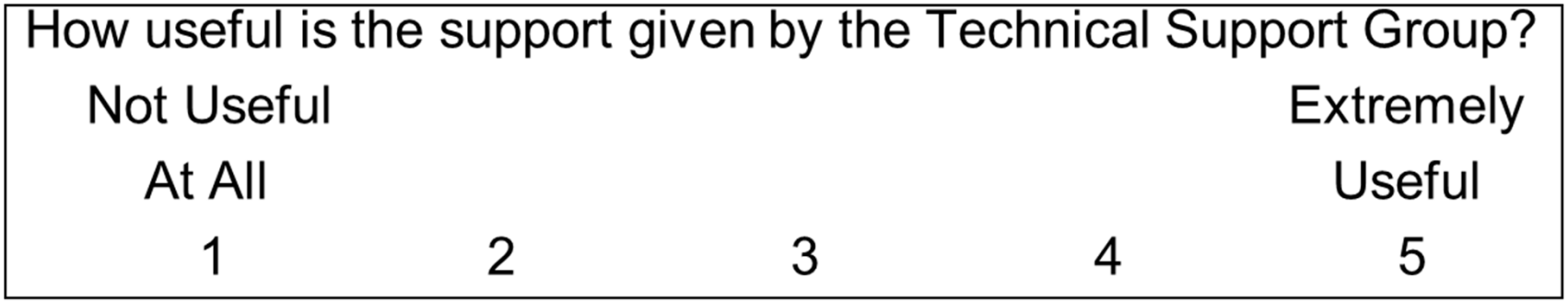
Interval Scales

- An interval scale is used when the intervals are equal
- There is no absolute zero
- Examples of interval scales include the Fahrenheit or Centigrade scale



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Validity and Reliability

- Reliability of scales refers to consistency in response—getting the same results if the same questionnaire was administered again under the same conditions
- Validity is the degree to which the question measures what the analyst intends to measure

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Problems with Scales

- Leniency
- Central tendency
- Halo effect

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Leniency

- Caused by easy raters
  - Solution is to move the “average” category to the left or right of center

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Central Tendency

- Central tendency occurs when respondents rate everything as average
  - Improve by making the differences smaller at the two ends
  - Adjust the strength of the descriptors
  - Create a scale with more points

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Halo Effect

- When the impression formed in one question carries into the next question
- the tendency for an impression created in one area to influence opinion in another area.
- Solution is to place one trait and several items on each page

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Designing the Questionnaire

- Allow ample white space
- Allow ample space to write or type in responses
- Make it easy for respondents to clearly mark their answers
- Be consistent in style

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Order of Questions

- Place most important questions first
- Cluster items of similar content together
- Introduce less controversial questions first

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Administering Questionnaires

- Administering questionnaires has two main questions:
  - Who in the organization should receive the questionnaire
  - How should the questionnaire be administered

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Figure 4.14 Ways to Capture Responses When Designing a Web Survey

Name	Appearance	Purpose
One-line text box		Used to obtain a small amount of text and limit the answer to a few words
Scrolling text box		Used to obtain one or more paragraphs of text
Check box		Used to obtain a yes-no answer (e.g., Do you wish to be included on the mailing list?)
Radio button		Used to obtain a yes-no or true-false answer
Drop-down menu		Used to obtain more consistent results (Respondent is able to choose the appropriate answer from a predetermined list [e.g., a list of state abbreviations])
Push button		Most often used for an action (e.g., a respondent pushes a button marked “Submit” or “Clear”)

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Electronically Submitting Questionnaires

- Reduced costs
- Collecting and storing the results electronically

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Summary (1 of 2)

- Interviewing
  - Interview preparation
  - Question types
  - Arranging questions
  - The interview report
- Stories
  - Story elements

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Summary (2 of 2)

- Joint Application Design (JAD)
  - Involvement and location
- Questionnaires
  - Writing questions
  - Using scales and overcoming problems
  - Design and order
  - Administering and submitting

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System Analysis and Design

CHAP 5: Using Entity Relationship Diagram

Learning Outcomes

At the end of this lesson, you will be able to:

- Analyze and model relationship based on a set of information requirements.
- Develop an ERD model based on required type of model (Chen or Crow's Foot).
- Create an ERD model using a specific data modelling tool

What is ERD?

- ERD is a data modeling technique used in software engineering to produce a conceptual data model of an information system.
- Help to visualize two important information:
  I. The major entities within the system scope,
  II. The inter-relationships among these entities.
- Way to document the entities in a database, along with the attributes.

ERD *(cont…)*

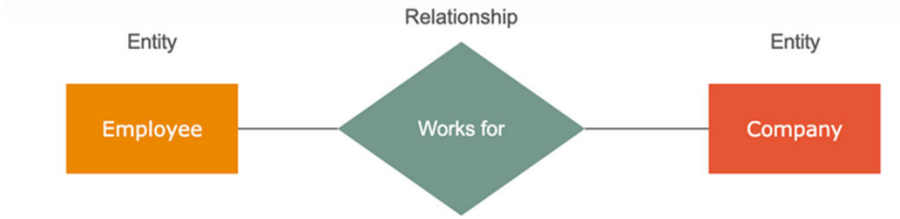
Three major methods/styles of ER

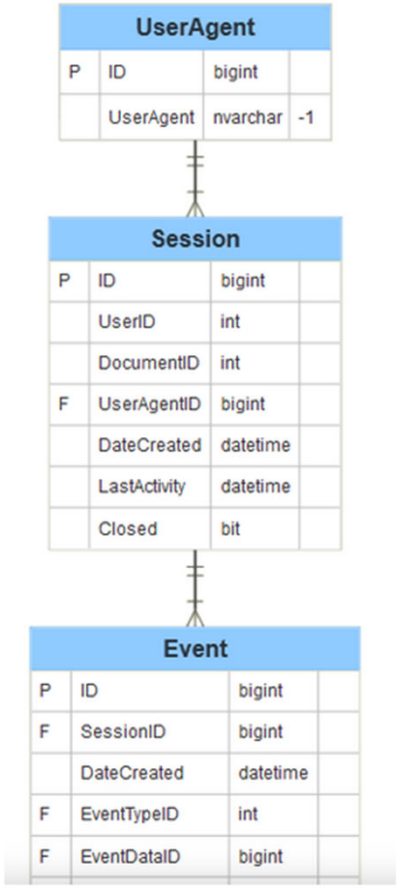
- Chen model (Dr. Peter Chen)
- Crow's Foot Model
- Unified Modelling Language (UML).

ERD *(cont…)*

Notation uses three main constructs

- Entity → table
- Attribute → column
- Relationship → line





How to find entities?

Entity:

- "...anything (people, places, objects, events, etc.) about which we store information (e.g. supplier, machine tool, employee, utility pole, airline seat, etc.)."
- Tangible: customer, product
- Intangible: order, accounting receivable
- Look for singular nouns (beginner)

Entities *(cont…)*

Entity Instance: a single occurrence of an entity.

Entity: student

6 instances

Student ID	Last Name	First Name
2144	Arnold	Betty
3122	Taylor	John
3843	Simmons	Lisa
9844	Macy	Bill
2837	Leath	Heather
2293	Wrench	Tim

How to find attributes?

Attribute:

- Attributes are data objects that either identify or describe entities (property of an entity).
- In other words, it is a descriptor whose values are associated with individual entities of a specific entity type
  - The process for identifying attributes is similar except now you want to look for and extract those names that appear to be descriptive noun phrases.

Attributes *(cont…)*

"Describe detail information about an entity "

- Entity: Student
- Attributes:
  - Student_ID
  - LastName
  - FirstName

Attributes *(cont…)*

3 attributes / columns / fields

Entity: student

Student ID	Last Name	First Name
2144	Arnold	Betty
3122	Taylor	John
3843	Simmons	Lisa
9844	Macy	Bill
2837	Leath	Heather
2293	Wrench	Tim

6 instances

Attributes *(cont…)*

Classes/Types of attributes:

- Simple attribute
- Composite attribute
- Derived attributes
- Single-valued attribute
- Multi-valued attribute

Attributes *(cont…)*

- A simple attribute cannot be subdivided.
  - Examples: Age, Gender, and Marital status
- A composite attribute can be further subdivided to yield additional attributes.
  - Examples:
    - ADDRESS --→ Street, City, State, Zip
    - PHONE NUMBER --→ Area code, Exchange number

Attributes *(cont…)*

Derived Attribute

- is not physically stored within the database
- instead, it is derived by using an algorithm.
  - Example 1: Late Charge of 2%
    - MS Access: InvoiceAmt * 0.02
  - Example 2: AGE can be derived from the date of birth and the current date.
    - MS Access: int(Date() – Emp_Dob)/365)

Attributes *(cont…)*

Single-valued Attribute

- can have only a single value.
  - Examples:
    - A person can have only one social security number.
    - A manufactured part can have only one serial number.

Attributes *(cont…)*

Multi-value Attribute

- can have many values.
  - Examples:
    - A person may have several college degrees.
    - A household may have several phones with different numbers
    - A car color

Attributes *(cont…)*

- Entity:
  - Student
- Attributes:
  - ID#: “123-45-6789” (single-valued)
  - Cell Phone: “(063)917-456-7227, (063)915-567-8255” (multi-valued)
  - Name: “Peter dela Paz” (composite)
  - Address: “14 JP Rizal St., Masipit, Calapan City” (composite)
  - Gender: “Female” (simple)
  - Age: 24 (derived)

Attributes *(cont…)*

Null Values

- Some attribute values could be optional or maybe they are not crucial to have. For example, if you have an attribute Hobbies. It is OK for the value of this attribute to be missing. In this case, we call it “NULL” value. Key Attributes cannot be NULL because they uniquely identifies an entity, so they have to have a value.

Attributes *(cont…)*

Key Attribute

- Keys should be Minimal.
- A key is minimal if it cannot be broken into smaller parts that work as a key. For example: – SectionID, courseID, Semester, Year – Is it Minimal? – Yes, because none of its smaller parts can work by itself as a key. “studentID + studentAge”, is it minimal key of Student? – No, because “studentID” by itself works as a key.

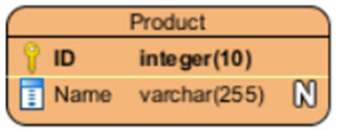
Attributes *(cont…)*

Primary Key (PK)

- Is an attribute (or field) that uniquely identifies every record in a certain table.

Primary key Rules

1. Unique
2. Never Changing
3. Never Null

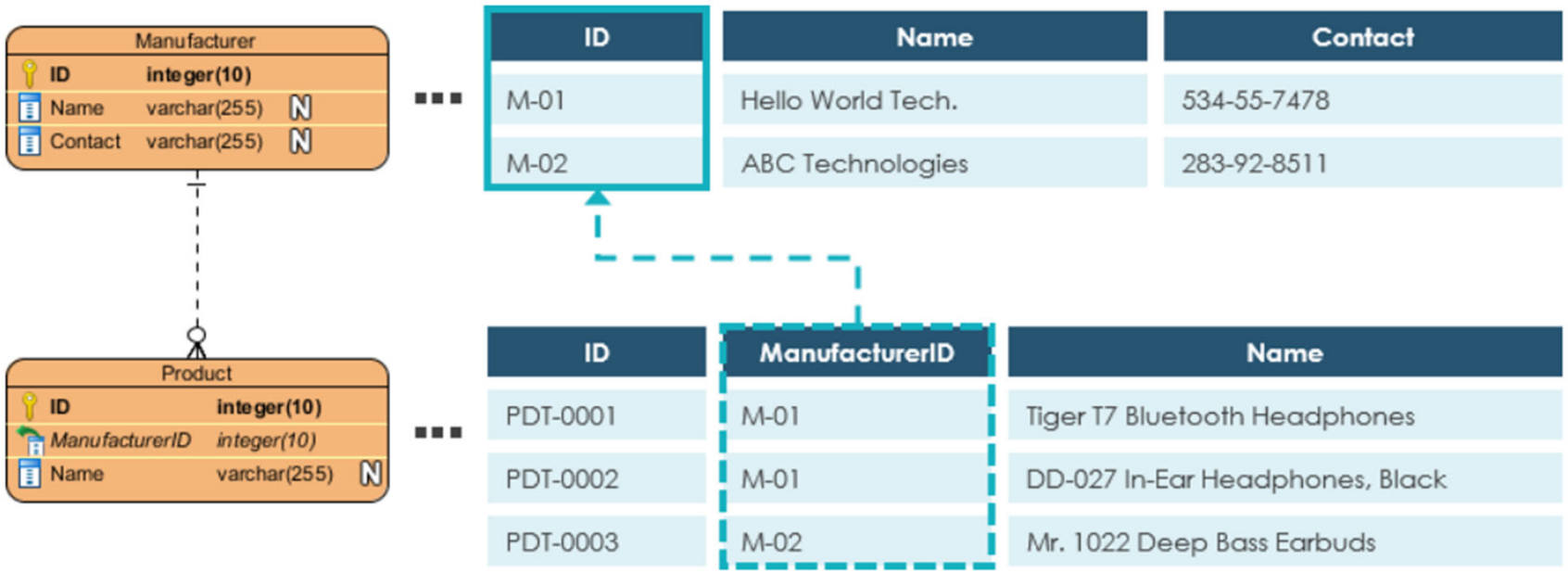


ID	Name
PDT-0001	Tiger T7 Bluetooth Headphones
PDT-0002	DD-027 In-Ear Headphones, Black
PDT-0002	SDB-21 Hi-Fi Stereo Over-ear Earphones
PDT-0003	Mr. 1022 Deep Bass Earbuds

Attributes *(cont…)*

Foreign Key (FK)

- Is the same as a primary key but just located in a foreign place (another table). But FKs don't have to be unique.



How to find relationships?

Relationship:

- Relationships are associations between entities.
- Typically, a relationship is indicated by a verb connecting two or more entities.
- **Employee** is assigned to **Project**
- Relationships should be classified in terms of cardinality.
  - One-to-one, one-to-many, etc.

Relationships *(cont…)*

Degree of Relationship:

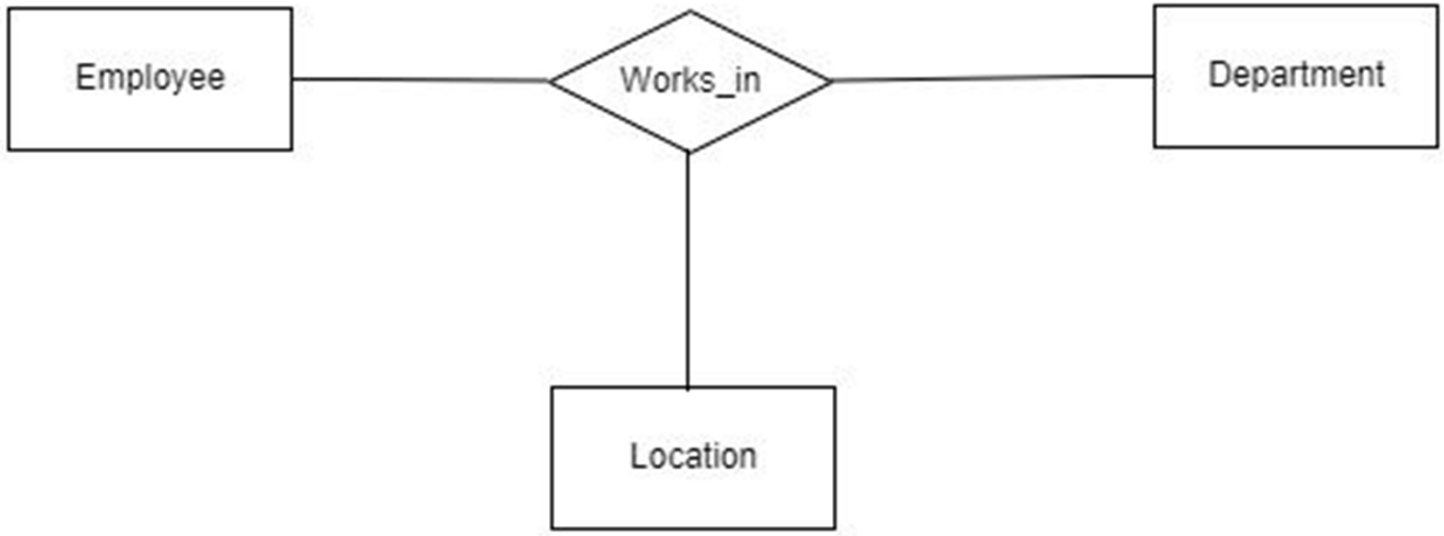
- The number of participating entities in a relationship defines the degree of the relationship.
  - Binary = degree 2
  - Ternary = degree 3
  - n-ary = degree n

Relationships *(cont…)*

Binary Relationship includes two entity types. Example: Student "takes" Class. Its notation in ER Diagram is as follows.

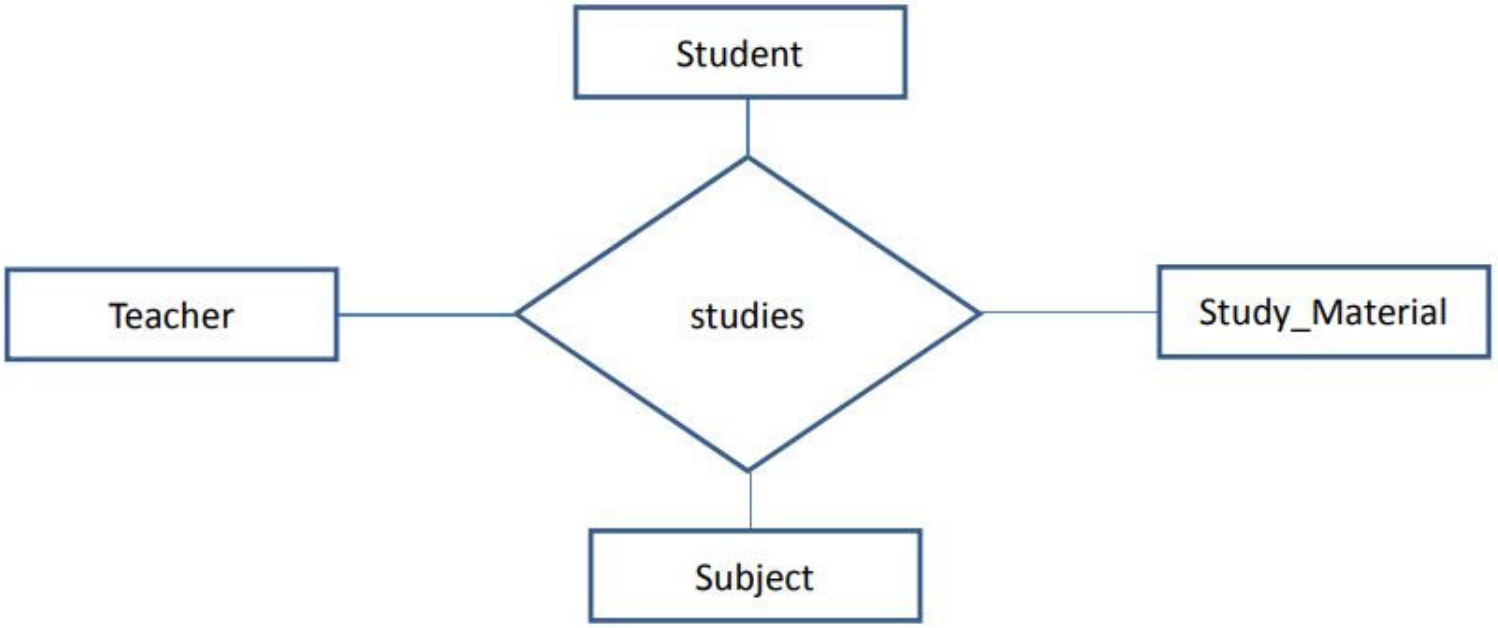
Relationships *(cont…)*

Ternary relationship includes 3 entity types. Example: Works_in: is a relationship that shows, each employee (Employee), Who worked in which department(Department) and where the department is located (Location)

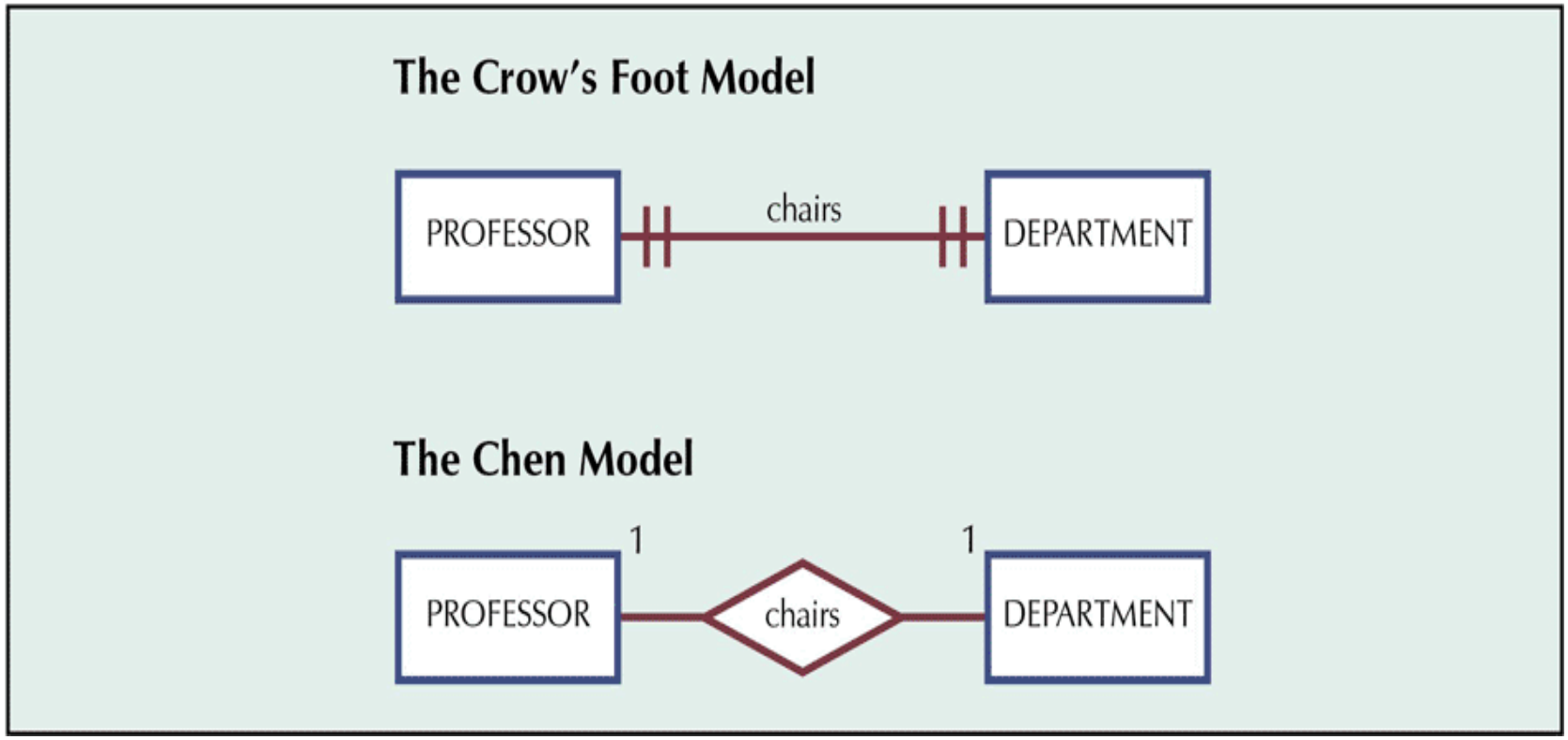


Relationships *(cont…)*

N-ary Relationship includes N entity types. Example: studies: is a 4-ary relationship that shows that a “student” studies a “subject” with a “teacher” and the help of “study_material”

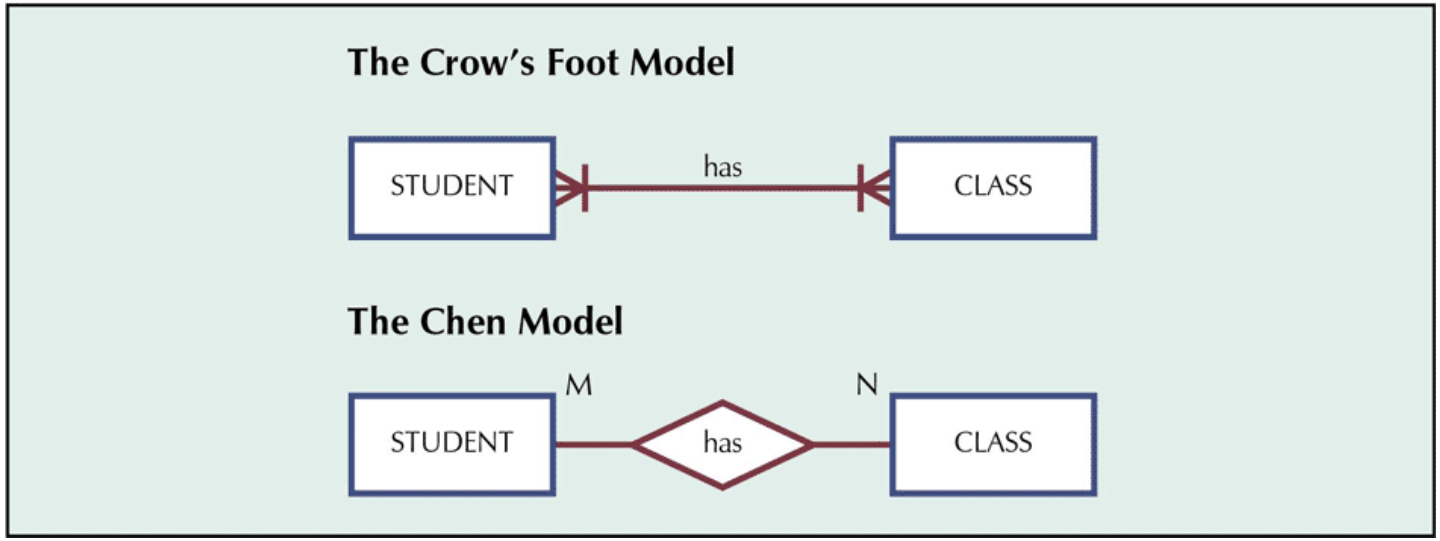


Relationships *(cont…)*

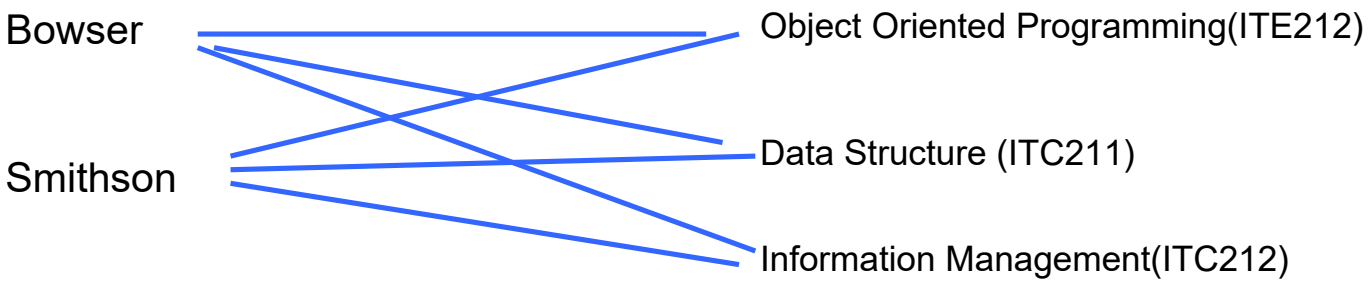


The 1:1 Relationship Between PROFESSOR and DEPARTMENT

Relationships *(cont…)*



The M:N Relationship Between STUDENT and CLASS



Relationships *(cont…)*

The tables have **many redundancies!!**

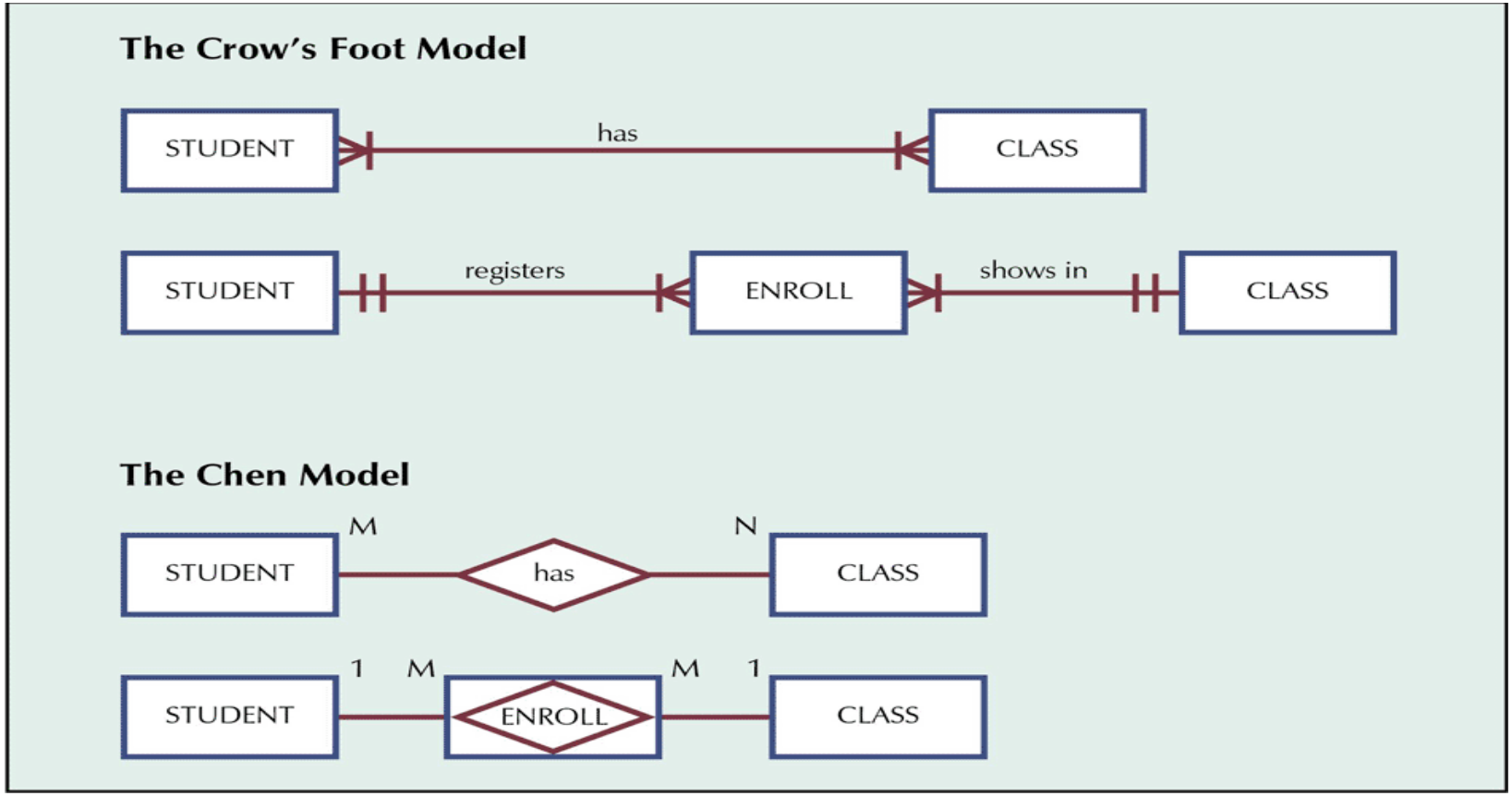
Table name: STUDENT
Primary key: STU_NUM + CLASS_CODE **Database name: Ch03_CollegeTry**
Foreign key: CLASS_CODE

	STU_NUM	STU_LNAME	CLASS_CODE
▶	321452	Bowser	10014
	321452	Bowser	10018
	321452	Bowser	10021
	324257	Smithson	10014
	324257	Smithson	10018
	324257	Smithson	10021

Table name: CLASS
Primary key: CLASS_CODE + STU_NUM
Foreign key: STU_NUM

	CLASS_CODE	STU_NUM	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
▶	10014	321452	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
	10014	324257	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
	10018	321452	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
	10018	324257	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
	10021	321452	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
	10021	324257	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114

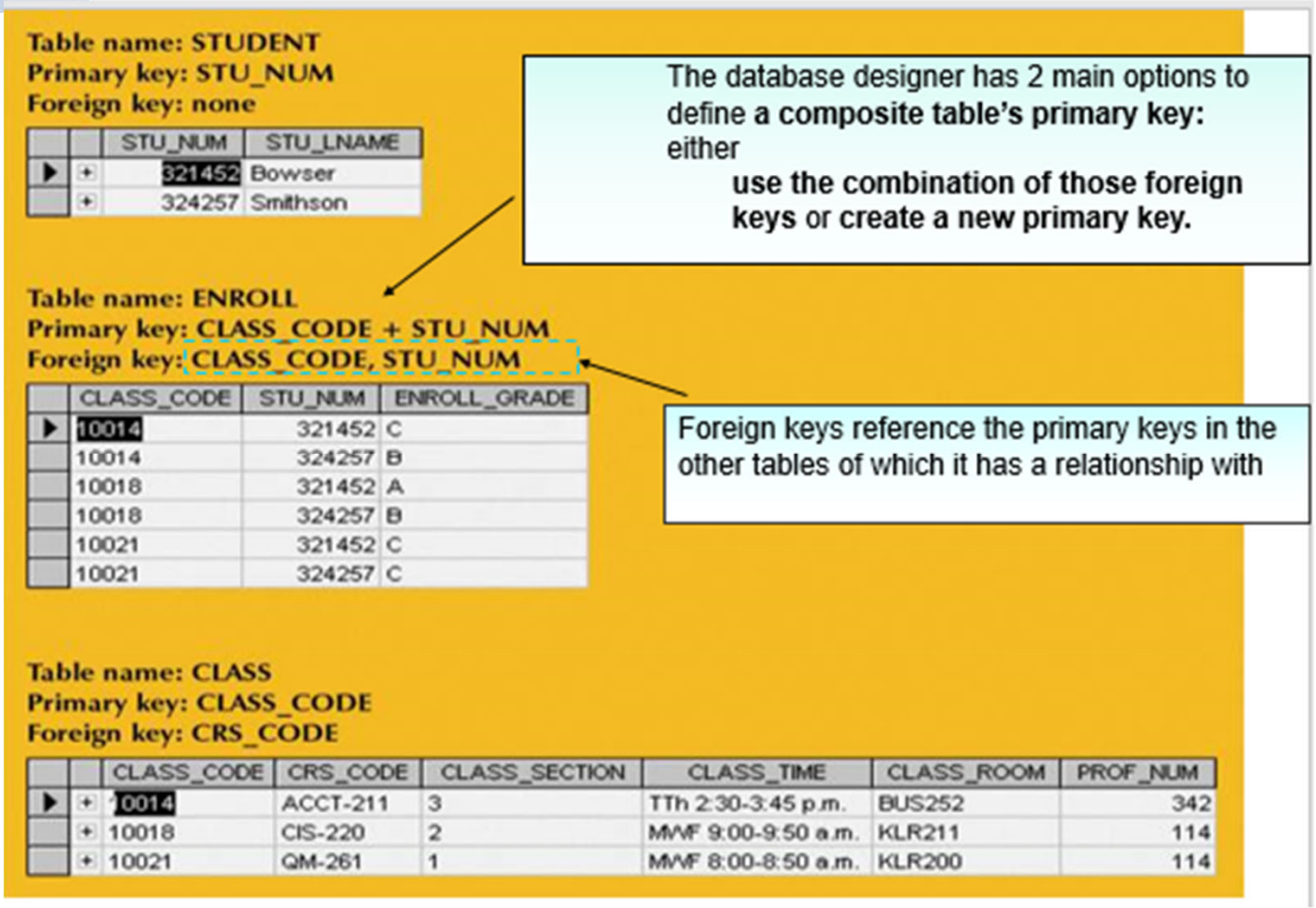
Relationships *(cont…)*



Changing the M:N relationship to TWO 1:M relationships

Relationships *(cont…)*

Converting the M:N relationship into TWO 1:M relationships



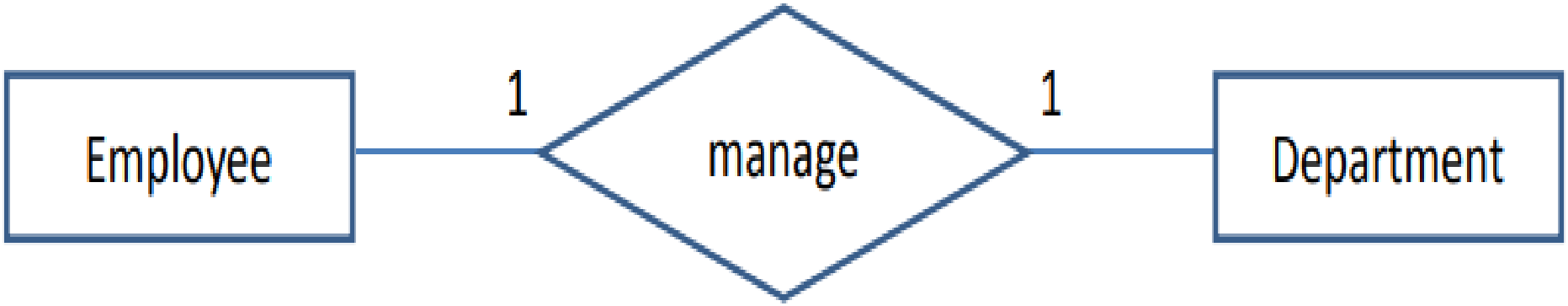
How to find cardinalities?

Cardinality:

- The cardinality is the number of occurrences in one entity which are associated to the number of occurrences in another.
- There are three basic cardinalities (degrees of relationship).
- one-to-one (1:1), one-to-many (1:M), and many-to-many (M:N)

Cardinalities *(cont...)*

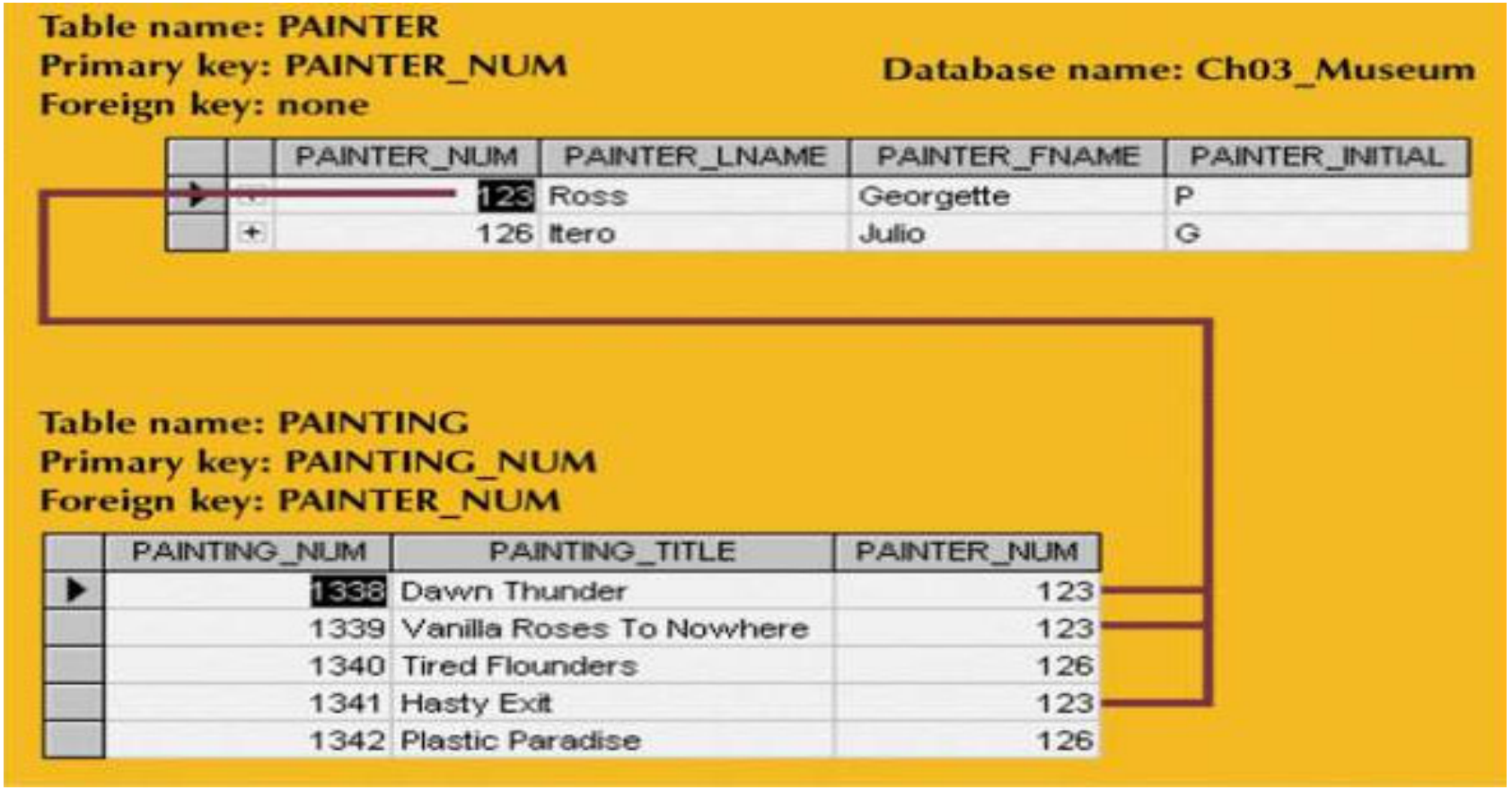
- One-to-one (1:1). One entity from entity set X can be associated with at most one entity of entity set Y and vice versa. For example, One department is managed by only One employee. One employee can manage only One department.



Cardinalities *(cont...)*

- One-to-many (1:N). One entity from entity set X can be associated with more than one entities of entity set Y but from entity set Y one entity can be associated with at most one entity. For example, One teacher can teach Many classes. One class is only taught by only One teacher.

Cardinalities *(cont...)*



The Implemented 1:M relationship between PAINTER and PAINTING

Cardinalities *(cont...)*

- **Many-to-many.** One entity from X can be associated with more than one entity from Y and vice versa. For example, One teacher can teach Many classes. One class can be taught by Many teachers

Chen's Model & Crow Foot Model

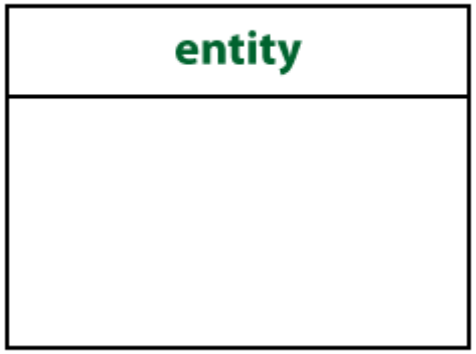
Cardinality:

- The cardinality is the number of occurrences in one entity which are associated to the number of occurrences in another.
- There are three basic cardinalities (degrees of relationship).
- one-to-one (1:1), one-to-many (1:M), and many-to-many (M:N)

Crow's Foot Notation *(cont..)*

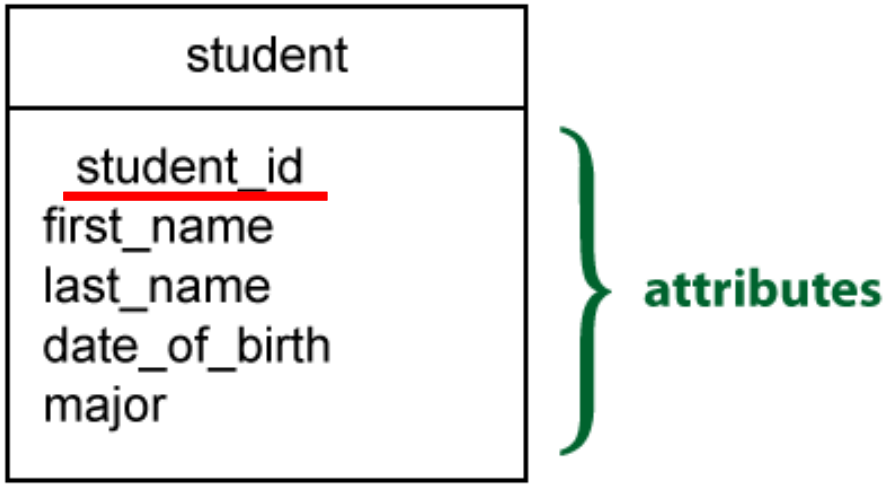
Crow's Foot Notation

- Known as IE notation (most popular)
- Entity:
  - Represented by a rectangle, with its name on the top. The name is singular (entity) rather than plural (entities).



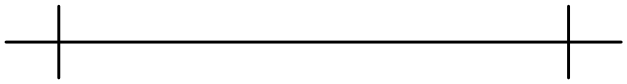
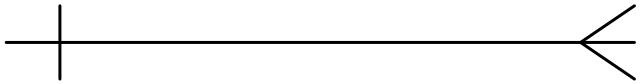
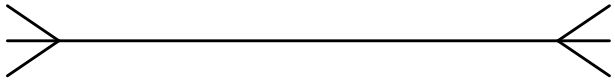
Crow's Foot Notation *(cont..)*

- Identifiers are represented by underlying the name of the attribute(s)

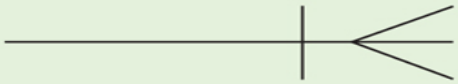
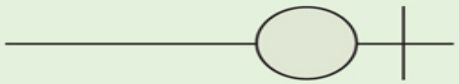
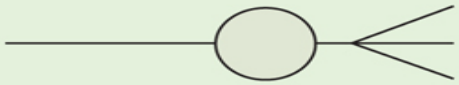


Crow's Foot Notation *(cont..)*

Basic Cardinality Type

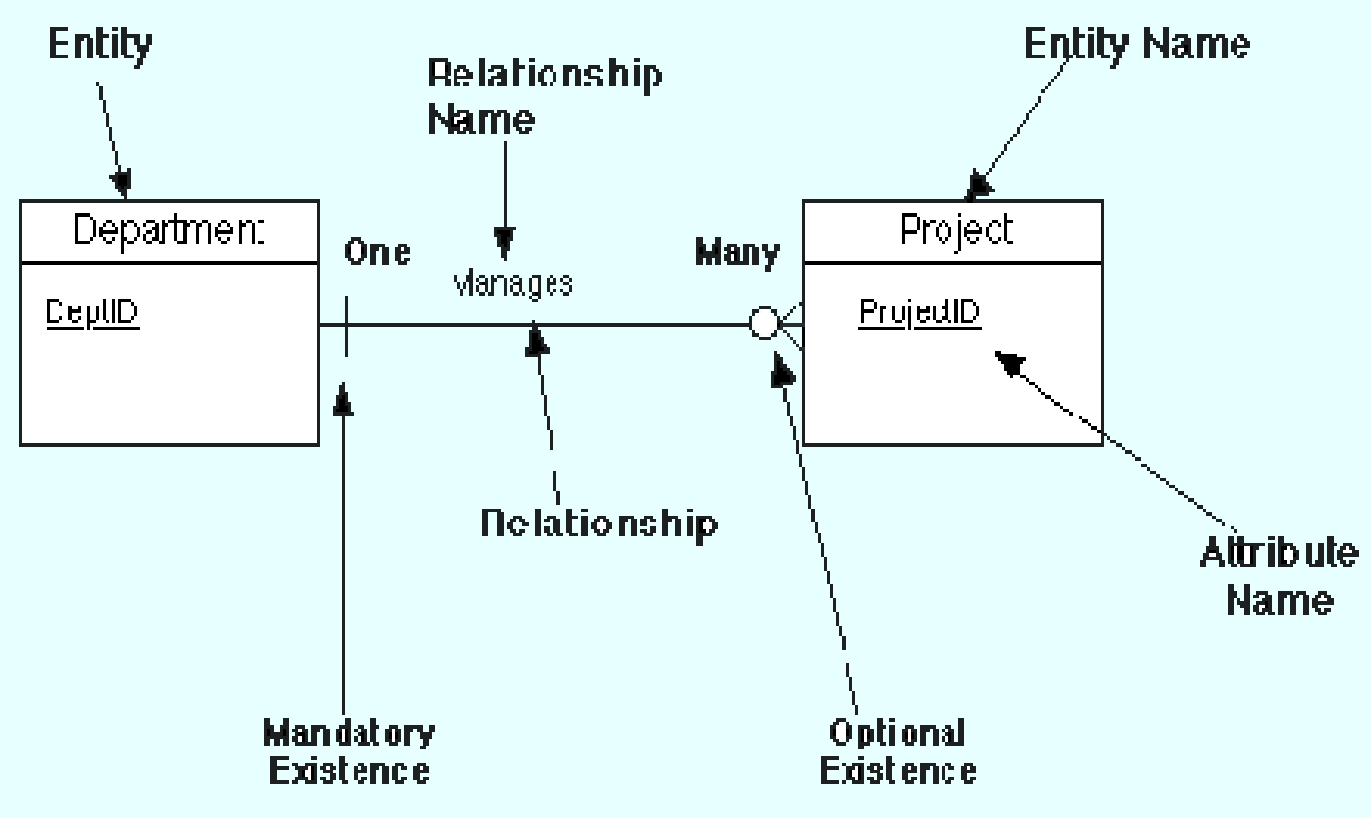
- 1-to-1 relationship 
- 1-to-M relationship 
- M-to-N relationship 

Crow's Foot Notation *(cont..)*

Symbol	Meaning
	Mandatory—One
	Mandatory—Many
	Optional—One
	Optional—Many

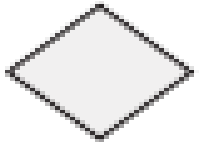
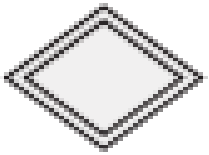
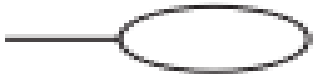
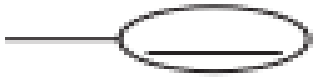
Crow's Foot Notation *(cont..)*

Example Model



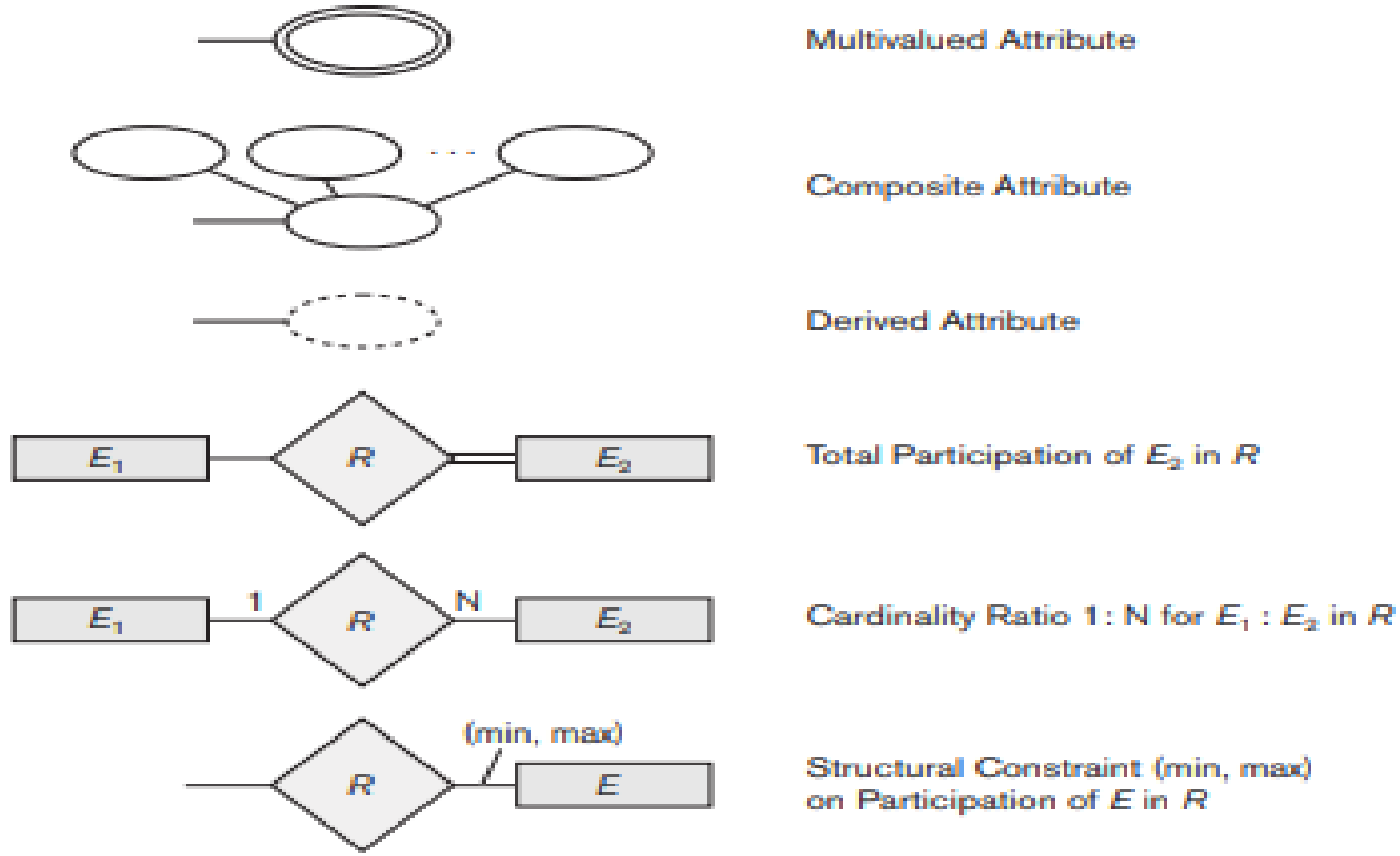
Chen Model *(cont…)*

Chen Model

Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Indentifying Relationship
	Attribute
	Key Attribute

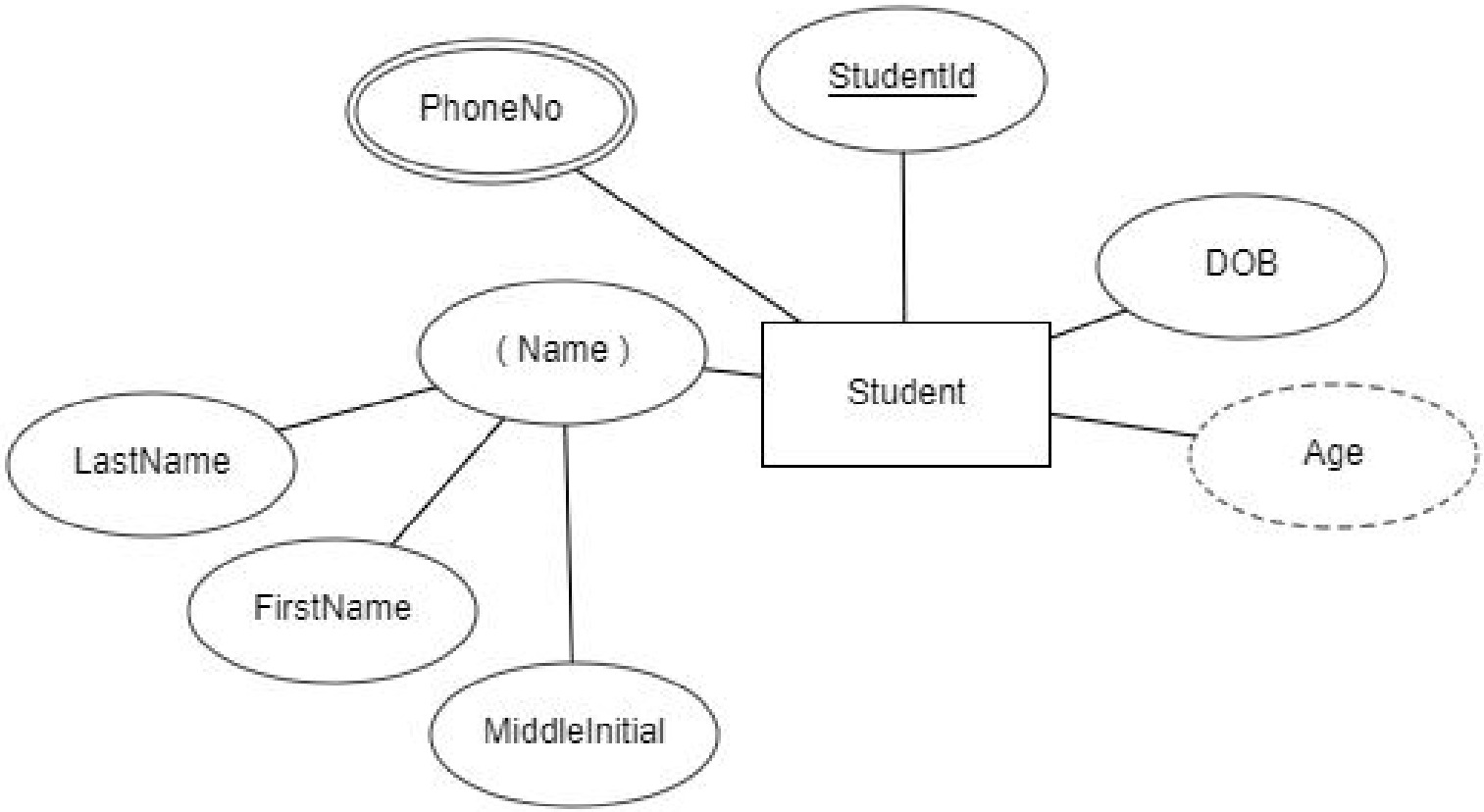
Chen Model *(cont…)*

Chen Model



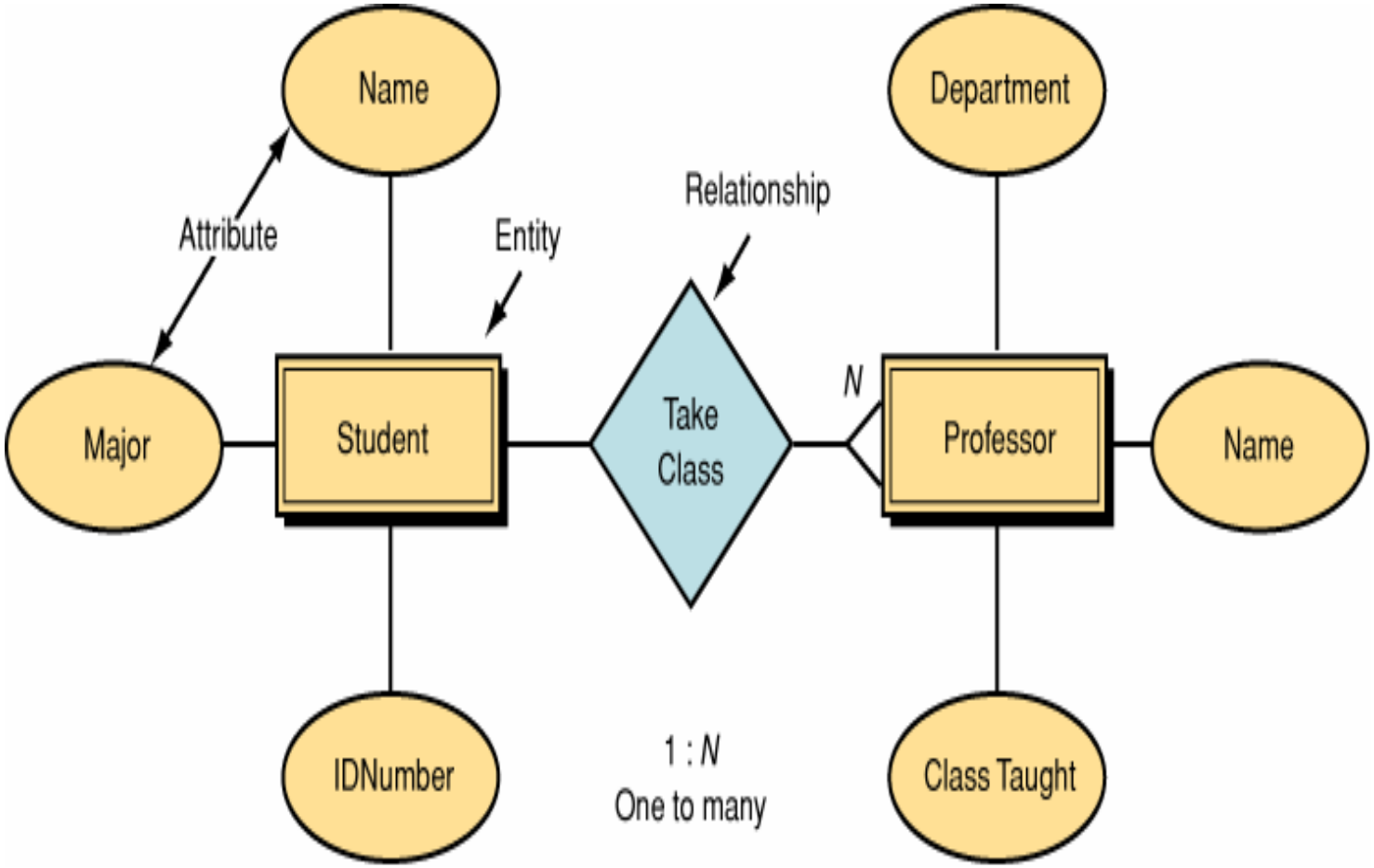
Chen Model *(cont…)*

Chen Model



Chen Model *(cont…)*

Chen Model



Levels of Models (ERD)

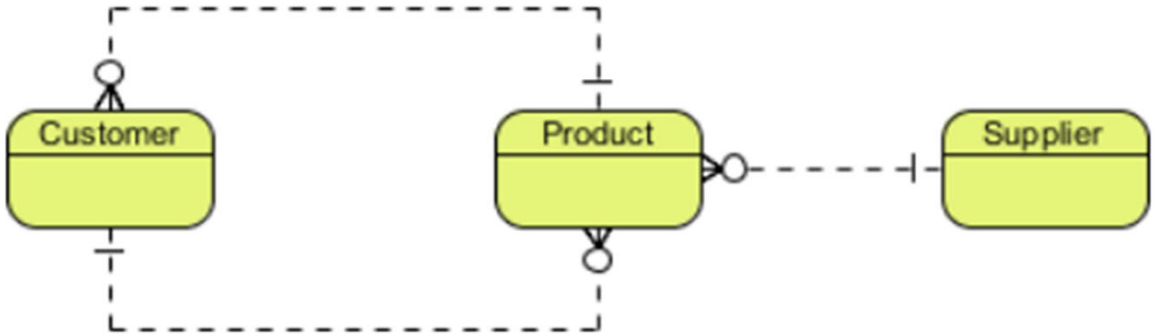
An ER model is typically drawn at up to three levels of abstraction:

- Conceptual ERD / Conceptual data model
- Logical ERD / Logical data model
- Physical ERD / Physical data model

Conceptual ERD

A conceptual ERD model outlines the key business objects (entities) in a system and their relationships, providing a high-level view without specifying database tables

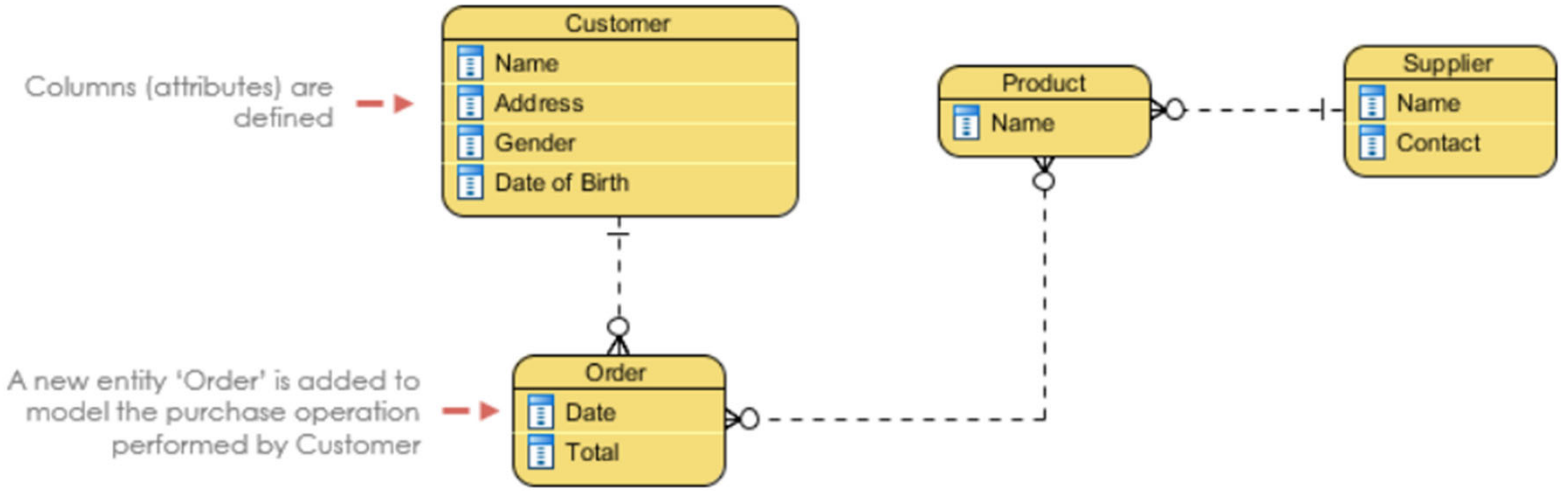
- Focuses on what entities exist, not their implementation.
- For example, many-to-many relationships are depicted simply as relationships without cardinality details, unlike logical or physical data models.



Logical Data ERD

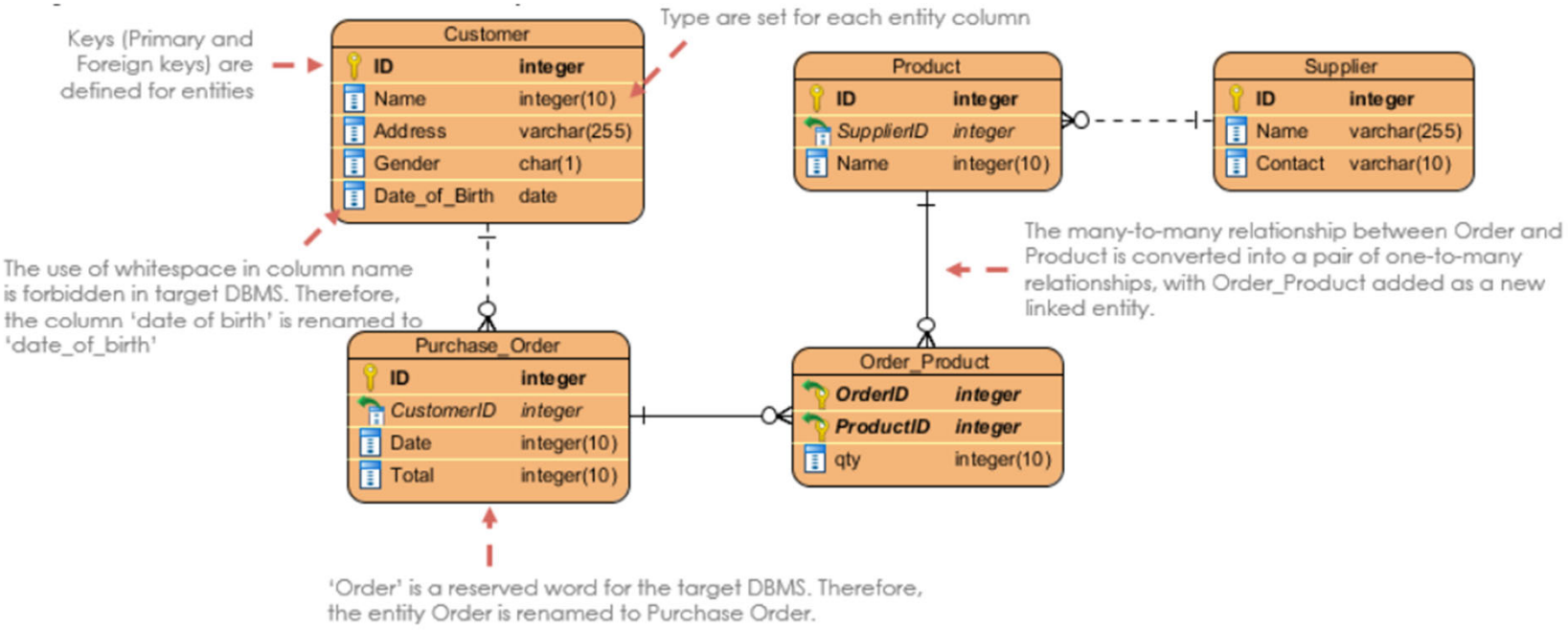
A logical ERD is a more detailed version of a conceptual ERD.

- Builds on the conceptual model by clearly defining the specific attributes (like columns) for each entity & adding operational or transactional entities.
- While still independent of the specific database system, it can consider database-related factors that might influence the design



Physical Data ERD

A physical ERD is the detailed blueprint for a relational database, building on the logical ERD by specifying column details like type, length, & whether they can be null.



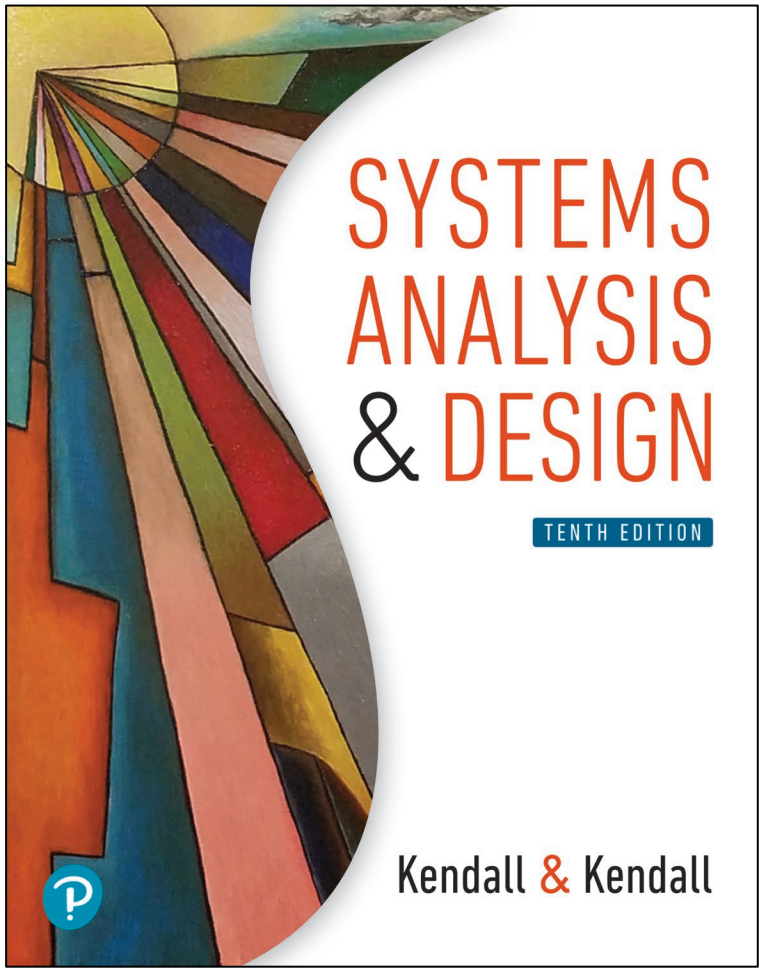
Comparisons Levels of ERD

Feature	Conceptual	Logical	Physical
Entities	✅	✅	✅
Relationships	✅	✅	✅
Attributes (Columns)	❌	✅	✅
Attribute Data Types	❌	Optional	✅
PK & FK	❌	Partial/Optional	✅ Needed

Guidelines Drawing of ERD

Systems Analysis & Design

Tenth Edition



Chapter 5 (part b)

Using Data Flow Diagrams

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Learning Objectives (1 of 2)

7.1 Comprehend the importance of using logical and physical data flow diagrams (DFDs) to **graphically depict movement** for humans and systems in an organization

7.2 Create, use, and explode logical DFDs to **capture** and **analyze** the current system through **parent** and **child levels**

7.3 Develop and explode **logical** DFDs that **illustrate** the **proposed** system

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Learning Objectives (2 of 2)

7.4 **Produce** physical DFDs based on logical DFDs you have **developed**

7.5 Understand and apply the **concept** of **partitioning** of **physical** DFDs

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Data Flow Diagrams

- **Graphically characterize** data **processes** and **flows** in a **business system**
- Depict:
  - System **inputs**
  - **Processes**
  - **Outputs**

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Major Topics

- Data flow diagram **symbols**
- Data flow diagram **levels**
- **Creating** data flow diagrams
- **Physical** and **logical** data flow diagrams
- **Partitioning**
- **Communicating** using data flow diagrams

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Advantages of the Data Flow Approach

- Freedom from **committing** to the **technical** implementation **too early**
- Understanding of the **interrelatedness** of **systems** and **subsystems**
- **Communicating** current **system** knowledge to **users**
- **Analysis** of the **proposed** system

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Basic Symbols

- A **double square** for an external **entity**
- An **arrow** for **movement** of **data** from one point to another
- A **rectangle** with **rounded corners** for the occurrence of a transforming **process**
- An **open-ended rectangle** for a **data store**

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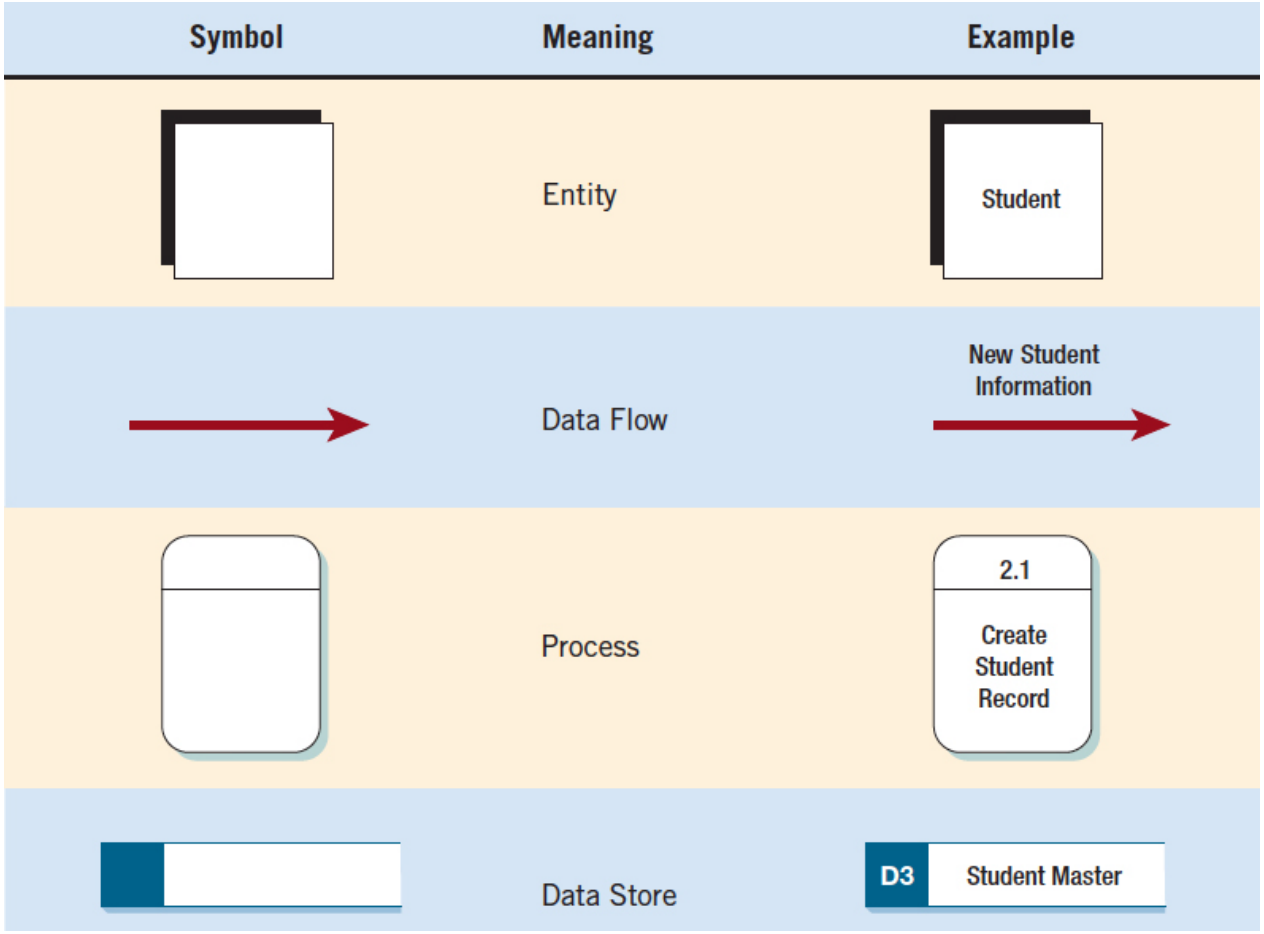
Figure 7.1 The Four Basic Symbols Used in Data Flow Diagrams, Their Meanings, and Examples



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External Entities = Actor (in Use Case)

- Represent another **department**, a **business**, a **person**, or a **machine**
- A **source** or **destination** of **data**, **outside** the **boundaries** of the **system**
- Should be **named** with a **noun**

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Data Flow

- Shows **movement** of **data** from **one point to another**
- **Described** with a **noun**
- **Arrowhead** indicates the **flow direction**
- Represents data about a **person, place, or thing**

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Process

- **Denotes** a **change** in or **transformation** of data
- **Represents** **work** being **performed** in the **system**
- Naming convention:
  - Assign the name of the whole system when naming a high-level process
  - To name a **major** subsystem **attach** the **word** subsystem to the name
  - Use the form ***verb-adjective-noun*** for detailed processes

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Data Store

- A depository for data that allows **examination**, **addition**, and **retrieval** of data
- Named with a noun, **describing the data**
- Data stores are usually given a **unique reference number**, such as D1, D2, D3
- Represents a:
  - Database
  - Computerized file
  - Filing cabinet

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Figure 7.2 Steps in Developing Data Flow Diagrams (1 of 2)

Developing Data Flow Diagrams Using a Top-Down Approach

1. Make a **list** of **business activities** and use it to **determine** various
   - External entities
   - Data flows
   - Processes
   - Data stores
2. Create a context diagram that shows **external entities** and **data flows to** and **from** the system. **Do not show** any **detailed processes** or **data stores**.
3. Draw Diagram 0, the next level. **Show processes**, but **keep** them **general**. **Show data stores** at this **level**.

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Figure 7.2 Steps in Developing Data Flow Diagrams (2 of 2)

4. Create a child diagram for **each** of the **processes** in **Diagram 0**.
5. **Check** for **errors** and make sure the **labels** you **assign** to each **process** and **data flow** are **meaningful**.
6. Develop a **physical** data flow diagram **from** the **logical** data flow diagram. Distinguish between **manual** and **automated processes**, describe **actual files** and **reports** by name, and **add controls** to **indicate** when **processes** are **complete** or **errors occur**.
7. **Partition** the **physical** data flow diagram by **separating** or **grouping** parts of the diagram in order to **facilitate programming** and **implementation**.

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Creating the Context Diagram

- The **highest level** in a data flow diagram
- Contains only **one process**, **representing** the **entire system**
- The **process** is given the **number 0**
- All **external entities**, as well as **major data flows** are shown

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Basic Rules

- The data flow diagram **must have one process**
- Must **not** be **any freestanding** objects
- A **process** must have both an **input** and **output** data flow
- A **data store** must be **connected** to **at least one process**
- **External entities** should **not** be **connected** to **one another**

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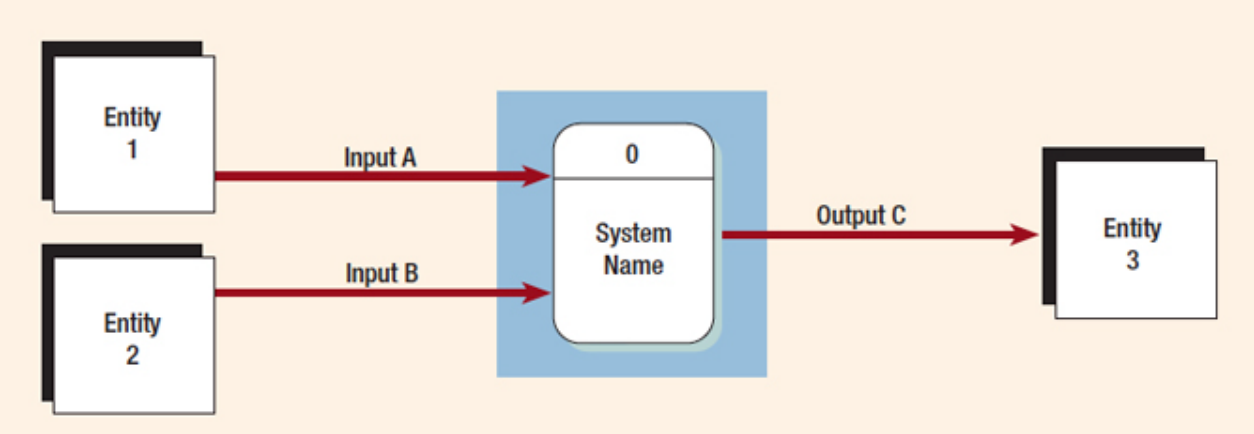
Figure 7.3 Context Diagram



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Drawing Diagram 0 (1 of 2)

- The **explosion** of the **context diagram**
- May include up to **nine** **processes**
- Each process is **numbered**
- Major **data stores** and all **external** **entities** are **included**

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Drawing Diagram 0 (2 of 2)

- Start with the **data flow** from an **entity** on the **input** side
- Work **backward** from an **output** data flow
- Examine the data flow to or from a **data store**
- Analyze a **well-defined** process
- Take note of any **fuzzy** areas

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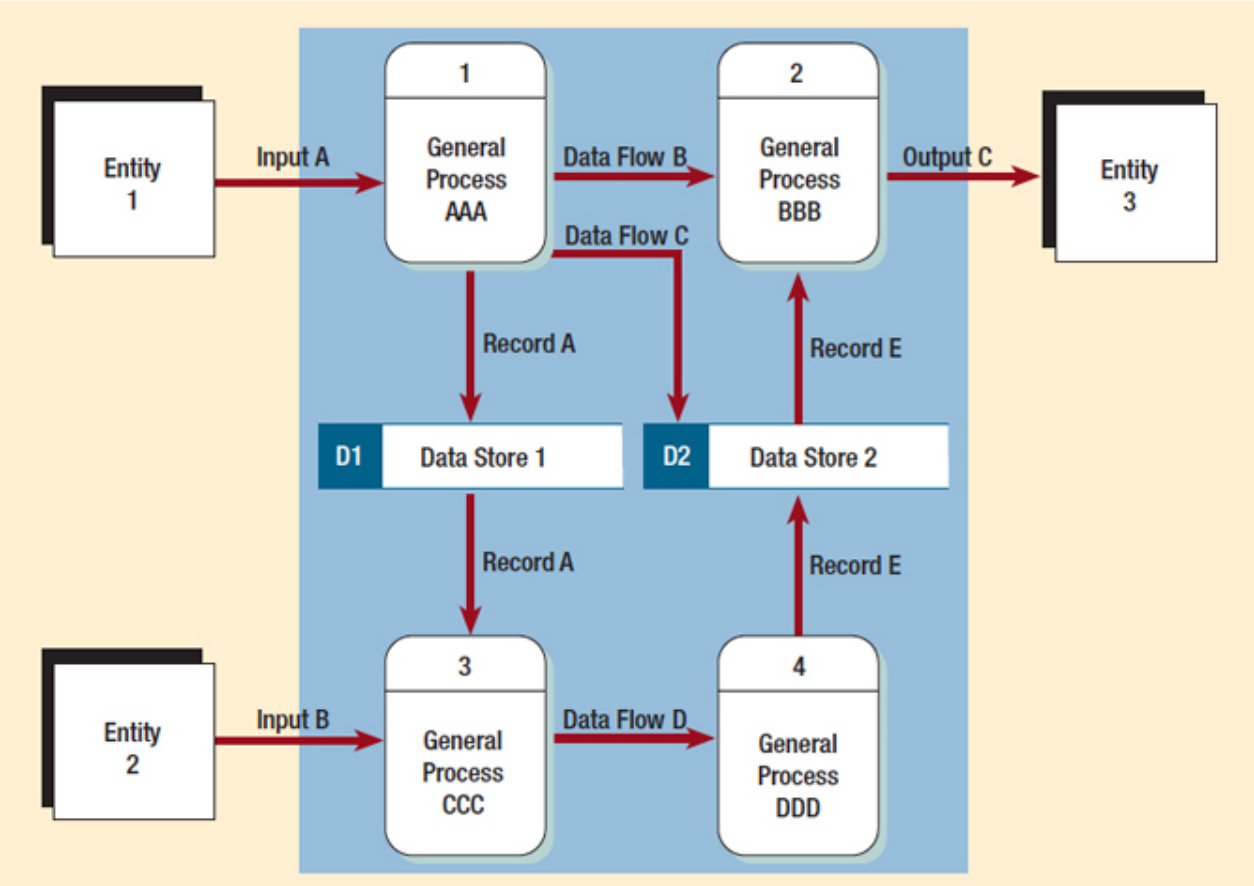
Figure 7.3 Note Greater Detail in Diagram 0



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Data Flow Diagram Levels

- Data flow diagrams are **built in layers**
- The **top level** is the **context** level
- Each **process** may **explode** to a **lower** level
- The **lower-level** diagram number is the **same** as the **parent process number**
- Processes that do not create a **child** diagram are called **primitive**

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Creating Child Diagrams (1 of 2)

- Each process on **diagram 0** may be exploded to create a **child** diagram
- A child diagram cannot produce output or **receive** input that the parent process does not also produce or receive
- The child process is given the same number as the parent process
  - **Process 3** would **explode** to **Diagram 3**

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Creating Child Diagrams (2 of 2)

- **Entities** are usually **not shown** on the child diagrams below **Diagram 0**
- If the **parent** **process** has **data flow** connecting to a **data store**, the **child** diagram may include the data store as well
- When a process is **not exploded**, it is called a **primitive** process

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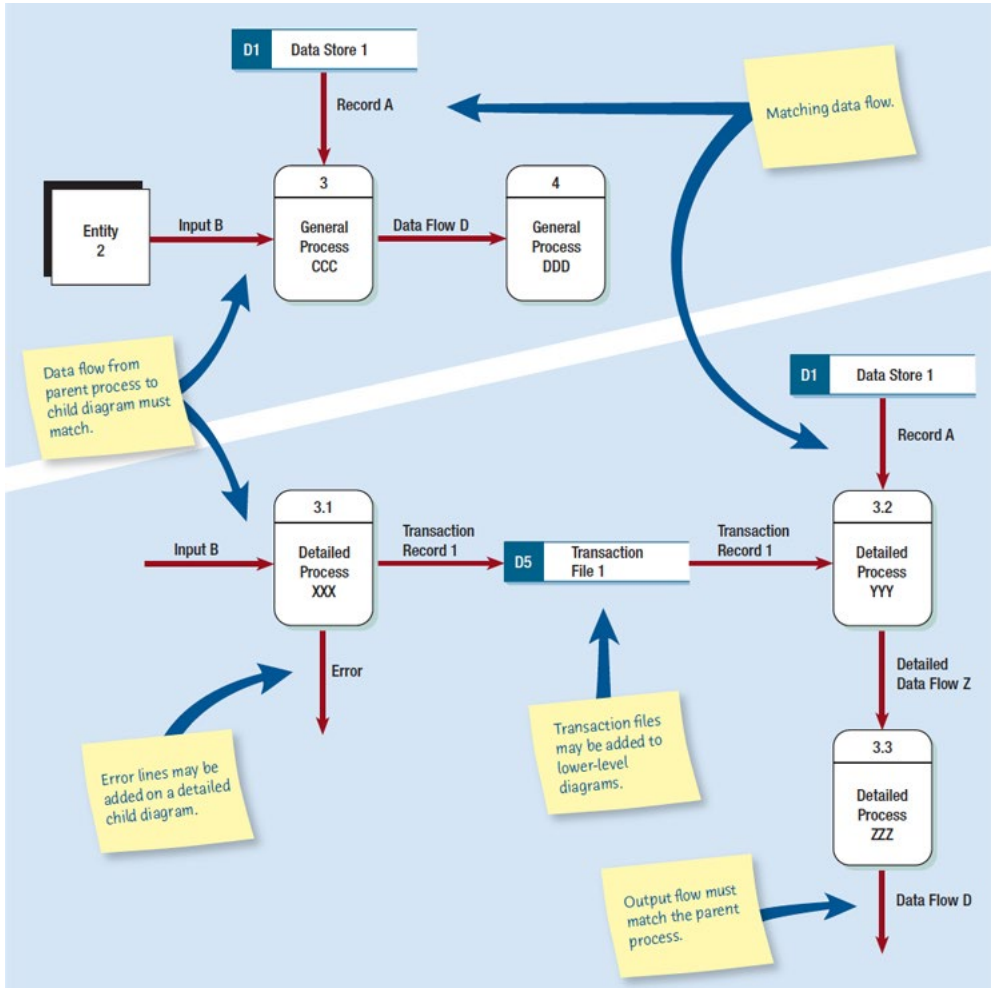
Figure 7.4 Differences between the Parent Diagram (above) and the Child Diagram (Below)



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Data Flow Diagrams Error Summary (1 of 2)

- **Forgetting** to include a **data flow** or **pointing** an arrow in the **wrong direction**
- Connecting **data stores** and **external** entities **directly** to each other
- **Incorrectly labeling processes** or **data flow**

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Data Flow Diagrams Error Summary (2 of 2)

- Including **more than nine** **processes** on a data flow diagram
- **Omitting** data flow
- Creating **unbalanced** **decomposition** (or **explosion**) in **child** diagrams

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Figure 7.5 Checking the Diagrams for Errors (1 of 2)

- Forgetting to include a **data flow** or pointing an arrow in the wrong direction

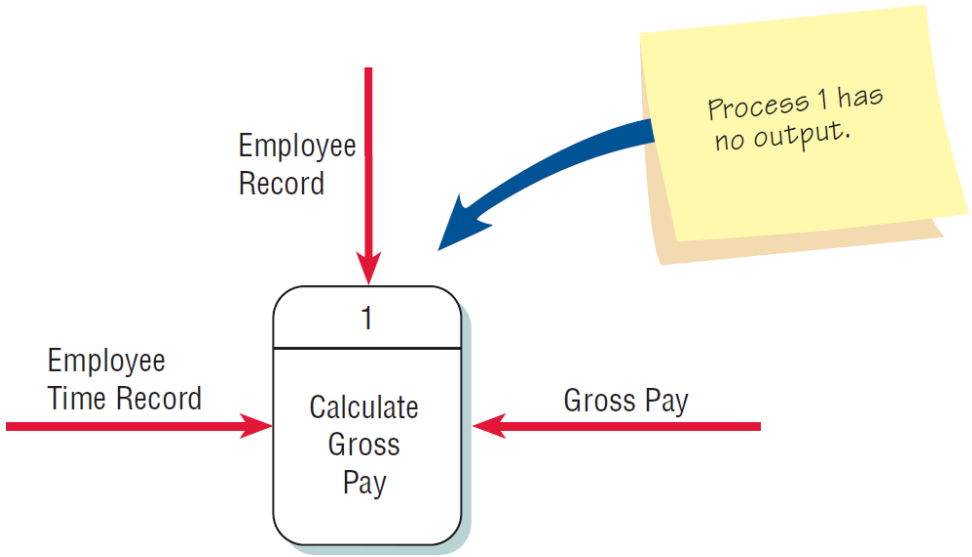

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Figure 7.5 Checking the Diagrams for Errors (2 of 2)

- Connecting data stores and external entities directly to each other

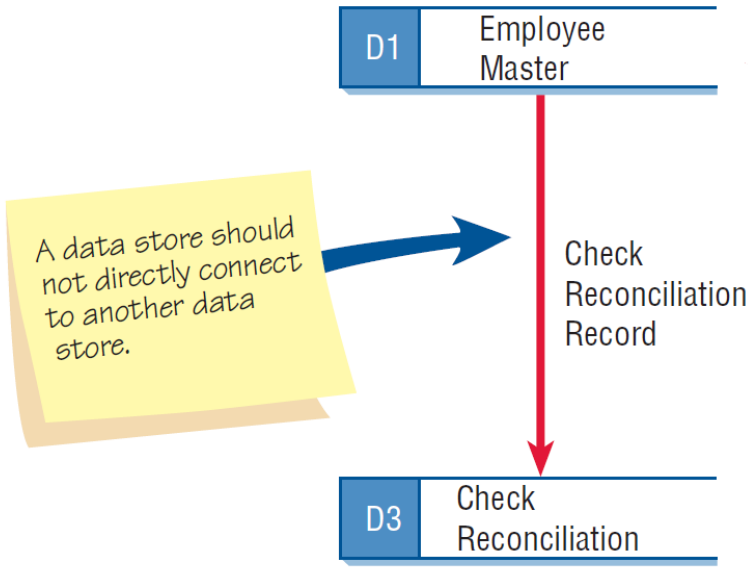

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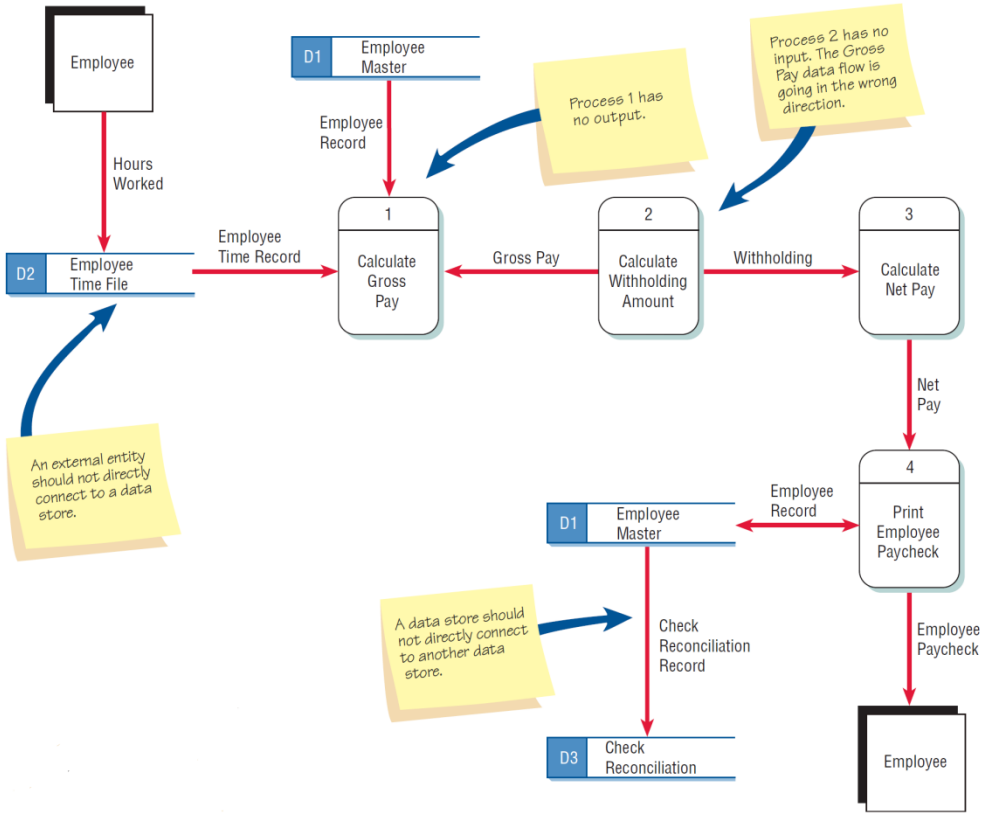
Figure 7.5 Typical Errors That Can Occur in a Data Flow Diagram



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Logical and Physical Data Flow Diagrams (1 of 2)

- **Logical**
  - Focuses on the business and **how** the business **operates**
  - **Not concerned** with how the system will be **constructed**
  - Describes the business events that take place and the data required and produced by each event

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Logical and Physical Data Flow Diagrams (2 of 2)

- **Physical**
  - Shows **how** the system will be **implemented**
  - **Depicts** the system

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Figure 7.7 Features Common of Logical and Physical Data Flow Diagrams

Design Feature	Logical	Physical
What the model depicts.	How the business **operates**.	How the system will be **implemented** (or how the current system operates).
What the processes represent.	Business **activities**.	**Programs**, program **modules**, and manual **procedures**.
What the data stores represent.	**Collections** of data regardless of how the **data** are **stored**.	Physical **files** and **databases**, **manual** files.
Type of data stores.	Show data stores representing **permanent** data **collections**.	**Master** files, **transition** files. Any processes that operate at two different times must be connected by a data store.
System controls.	Show **business** controls.	Show controls for validating input data, for obtaining a record (record found status), for ensuring successful completion of a process, and for system security (example: journal records).

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Figure 7.8 The Progression of Models from Logical to Physical

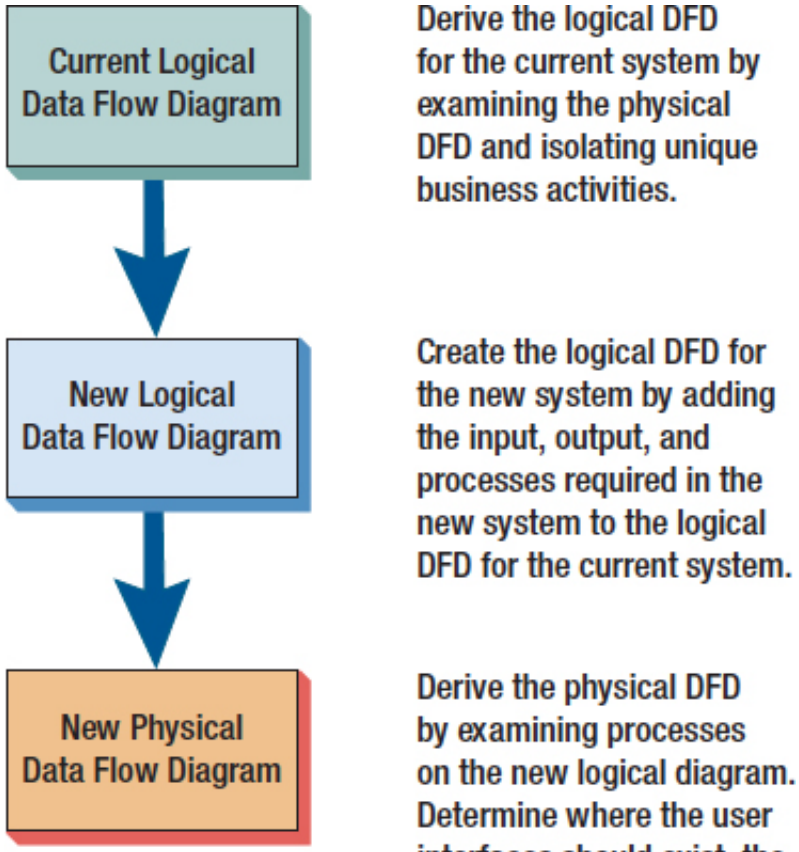

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Figure 7.9 Logical Data Flow Diagram Example

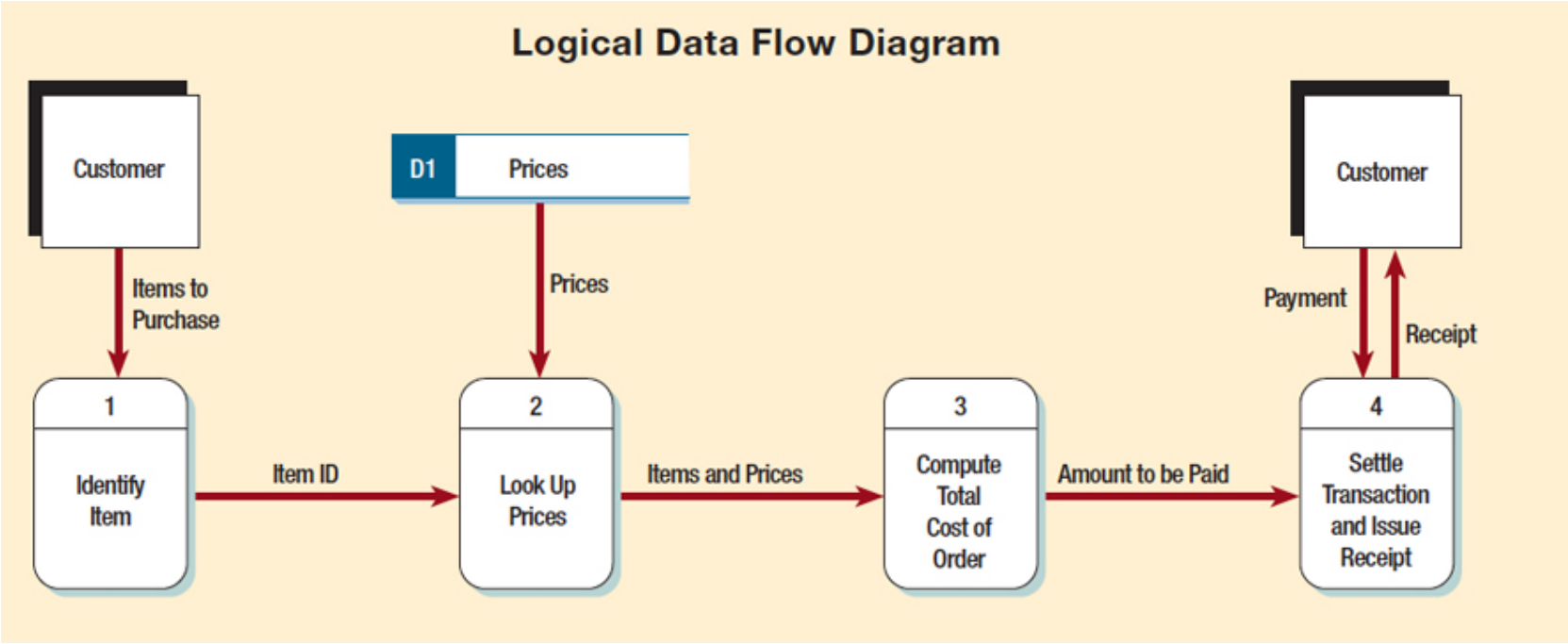

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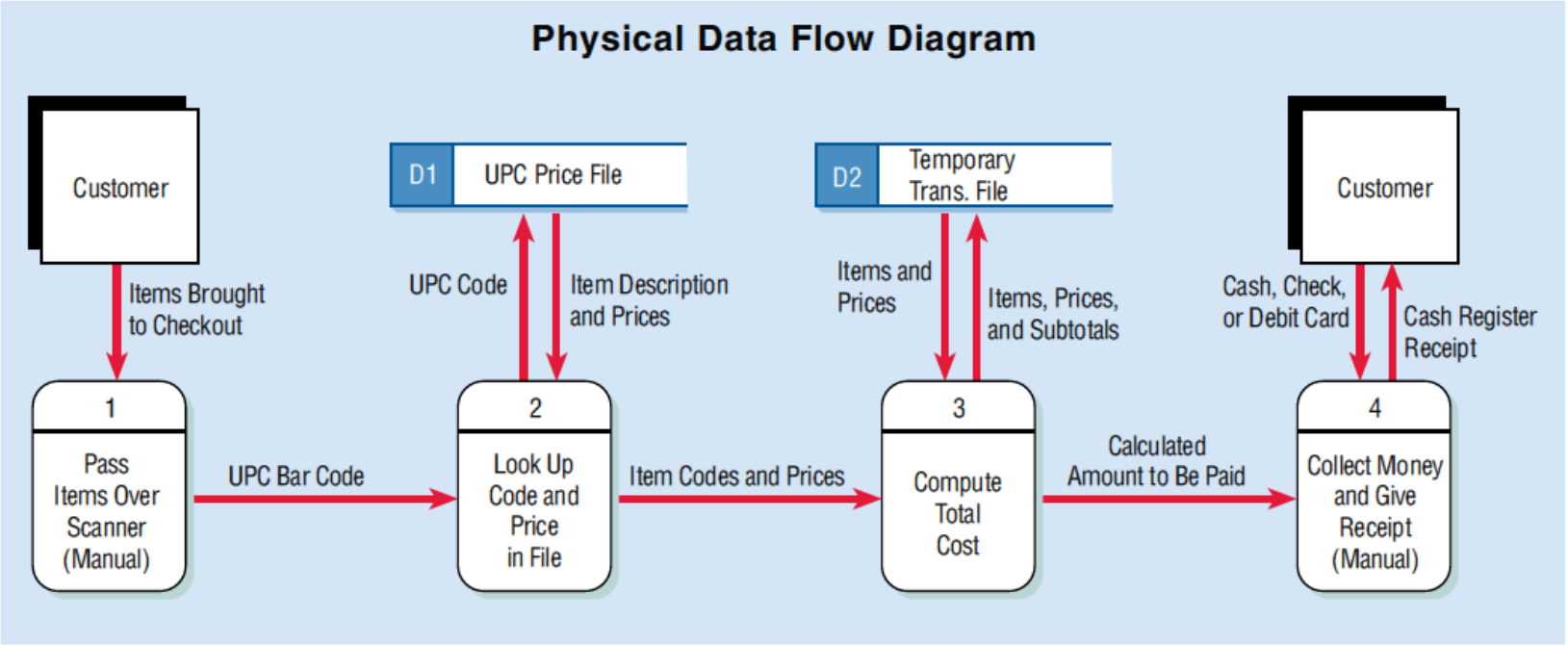
Figure 7.9 Physical Data Flow Diagram Example



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Developing Logical Data Flow Diagrams

- Better **communication** with users
- More **stable** systems
- Better understanding of the **business** by **analysts**
- **Flexibility** and **maintenance**
- Elimination of **redundancy** and **easier creation** of the physical model

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Developing Physical Data Flow Diagrams

- Clarifying which **processes** are performed by humans and which are **automated**
- Describing processes in more **detail**
- **Sequencing** processes that have to be done in a **particular order**
- Identifying **temporary** data stores
- Specifying **actual** names of **files** and **printouts**
- Adding controls to ensure the processes are done properly

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Figure 7.10 Physical Data Flow Diagrams Contain Many Items Not Found in Logical Data Flow Diagrams

Contents of Physical Data Flow Diagrams

- **Manual** processes
- Processes for **adding, deleting, changing**, and **updating** records
- Data **entry** and **verifying** processes
- **Validation** processes for ensuring **accurate** data input
- **Sequencing** processes to **rearrange** the **order** of records
- **Processes** to produce every **unique** system output
- **Intermediate** data stores
- Actual file names used to store data
- Controls to signify completion of tasks or error conditions

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CRUD Matrix

- The acronym CRUD is often used for
  - Create
  - Read
  - Update
  - Delete
- These are the activities that must be present in a system for each master file
- A CRUD matrix is a tool to represent where each of these processes occurs in a system

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Figure 7.11 CRUD Matrix

Activity	Customer	Item	Order	Order Detail
Customer Logon	R			
Item Inquiry		R		
Item Selection		R	C	C
Order Checkout	U	U	U	R
Add Account	C			
Add Item		C		
Close Customer Account	D			
Remove Obsolete Item		D		
Change Customer Demographics	RU			
Change Customer Order	RU	RU	RU	CRUD
Order Inquiry	R	R	R	R

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Event Modeling and Data Flow Diagrams

- An input flow from an external entity is sometimes called a trigger because it starts the activities of a process
- Events cause the system to do something and act as a trigger to the system
- An approach to creating physical data flow diagrams is to create a data flow diagram fragment for each unique system event

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Event Response Tables

- An event table is used to create a data flow diagram by analyzing each event and the data used and produced by the event
- Every row in an event table represents a data flow diagram fragment and is used to create a single process on a data flow diagram

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Figure 7.12 An Event Response Table for an Internet Storefront

Event	Source	Trigger	Activity	Response	Destination
Customer logs on	Customer	Customer number and password	Find customer record and verify password. Send Welcome web page	Welcome web page	Customer
Customer browses items at Web storefront	Customer	Item information	Find item price and quantity available. Send Item Response web page.	Item Response web page	Customer
Customer places item into shopping basket at Web storefront	Customer	Item purchase (item number and quantity)	Store data on Order Detail Record. Calculate shipping cost using shipping tables. Update customer total. Update item quantity on hand.	Items Purchased web page	Customer
Customer checks out	Customer	Clicks “Check Out” button on web page	Display Customer Order web page.	Verification web page	
Obtain customer Payment	Customer	Credit card information	Verify credit card amount with credit card company. Send.	Credit card data Customer feedback	Credit card company Customer
Send customer email		Temporal, hourly	Send customer an email confirming shipment.		Customer

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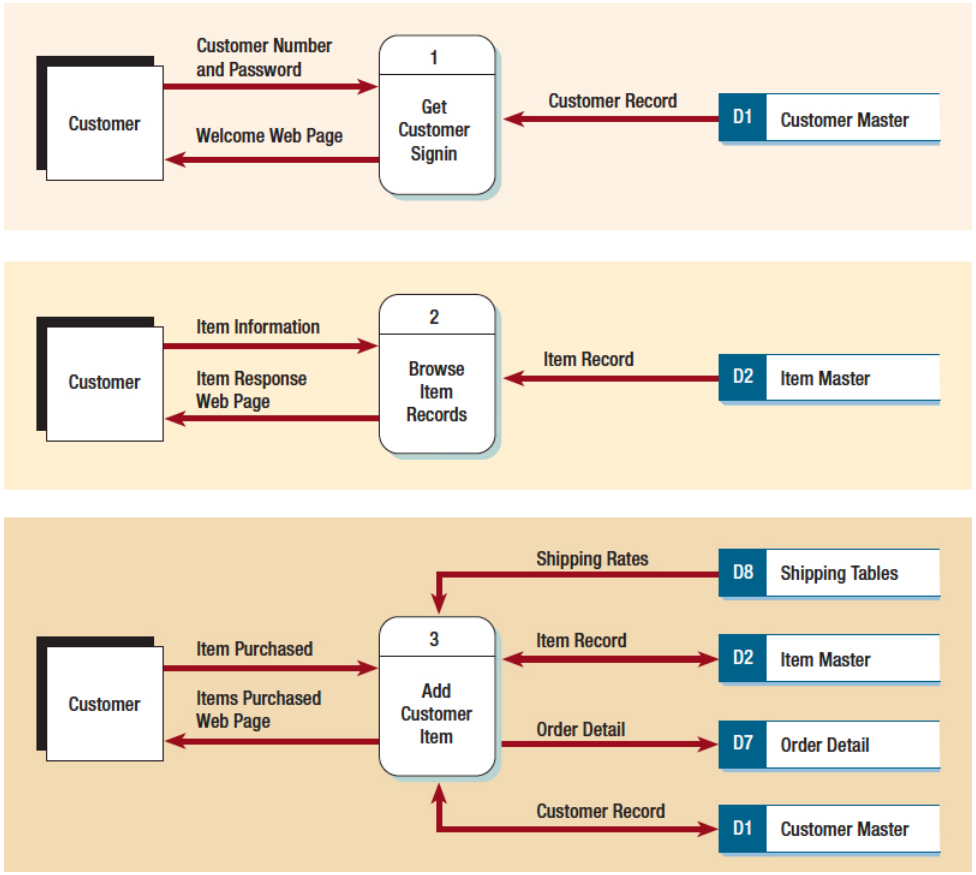
Figure 7.13 Data Flow Diagrams for the First Three Rows of the Internet Storefront Event Response Table



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Use Cases and Data Flow Diagrams

- Each use case defines one **activity** and its **trigger**, **input**, and **output**
- Allows the analyst to work with users to understand the nature of the processes and activities and then create a single data flow diagram fragment

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Partitioning Data Flow Diagrams

- Partitioning is the process of examining a data flow diagram and determining how it should be divided into collections of manual procedures and computer programs
- A dashed line is drawn around a process or group of processes that should be placed in a single computer program

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Reasons for Partitioning

- Different user groups
- Timing
- Similar tasks
- Efficiency
- Consistency of data
- Security

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Partitioning Websites

- Improves the way humans use the site
- Improves speed of processing
- Ease of maintaining the site
- Keep the transaction secure

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Communicating Using Data Flow Diagrams

- Use **unexploded** data flow diagrams early when ascertaining information requirements
- **Meaningful** labels for all data components

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Summary (1 of 3)

- Data flow diagrams
  - Structured analysis and design tools that allow the analyst to comprehend the system and subsystems visually as a set of interrelated data flows
- DFD symbols
  - Rounded rectangle
  - Double square
  - An arrow
  - Open-ended rectangle

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Summary (2 of 3)

- Creating the logical DFD
  - Context-level data flow diagram
  - Level 0 logical data flow diagram
  - Child diagrams
- Creating the physical DFD
  - Create from the logical data flow diagram
  - Partitioned to facilitate programming

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Summary (3 of 3)

- Partitioning data flow diagrams
  - Whether processes are performed by different user groups
  - Processes execute at the same time
  - Processes perform similar tasks
  - Batch processes can be combined for efficiency of data
  - Processes may be partitioned into different programs for security reasons

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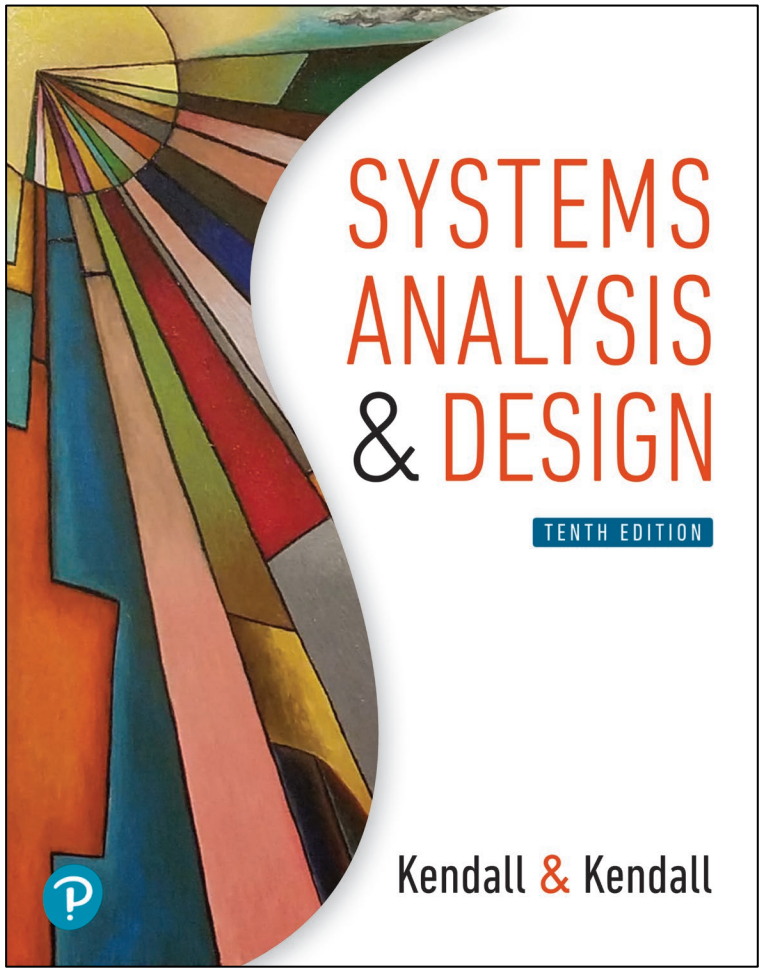
Systems Analysis & Design

Tenth Edition



Chapter 6

Analyzing Systems Using
Data Dictionaries

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Learning Objectives

6.1 Understand how analysts use data dictionaries for analyzing data-oriented systems.

6.2 Understand the concept of a repository for analysts' project information and the role of CASE tools in creating them.

6.3 Create data dictionary entries for data processes, stores, flows, structures, and logical and physical elements of the systems being studied, based on DFDs.

6.4 Recognize the functions of data dictionaries in helping users update and maintain information systems.

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Cataloging

- Data flow diagrams can be used to **catalog**:
  - **Data processes**
  - **Flows**
  - **Stores**
  - **Structures**
  - **Elements**
- Cataloging takes place with the data dictionary

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The Data Dictionary

- A **reference** **work** of **data** about data (**metadata**)
- **Collects** and **coordinates** **data terms**, and **confirms** what **each term** means to **different people** in the **organization**

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Major Topics

- The data dictionary
- The data repository
- Defining data flow
- Defining data structures
- Defining data elements
- Defining data stores
- Using the data dictionary
- XML
- JSON

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Need for Understanding the Data Dictionary

- Provide **documentation**
- **Eliminate redundancy**
- **Validate** the data flow **diagram**
- Provide a **starting point** for developing **screens** and **reports**
- Determine the **contents** of **data stored** in files
- To **develop** the **logic** for **DFD** processes
- Create **XML**
- Create JSON

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The Data Repository

- A **data repository** is a **large collection** of **project information**
- It includes:
  - **Information** about the data **maintained** by the system
  - **Procedural logic** and **use cases**
  - **Screen** and **report design**
  - **Data relationships**
  - Project **requirements** and final system **deliverables**
  - Project **management information**

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The Data Repository

- A data repository can be a **centralized, organized storage system designed to collect, manage, store, preserve**, and **provide access to datasets** or digital objects over the long term.
- Malaysia's key data repositories are:-
  - **OpenDOSM**: Platform by the Department of Statistics Malaysia providing open-sourced data for public use.
  - **data.gov.my:** One-stop data catalogue portal for browsing and accessing Malaysian government data, APIs, and related resources.
  - **NIH-DaRS:** National Institutes of Health Data Repository System for Malaysia's Ministry of Health; centralizes and preserves public health research data to support decision-making.
  - **RADARS:** MOSTI Raw Data for Research and Sciences platform, managed by MASTIC (Malaysian Science and Technology Information Centre)

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Data Dictionary Categories

- Data **flows**
  - How data moves through the system.
- Data **structures**
  - How data is organized and formatted.
- **Elements**
  - The individual pieces of data and their meaning.
- Data **stores**
  - Where data is physically or logically persisted.

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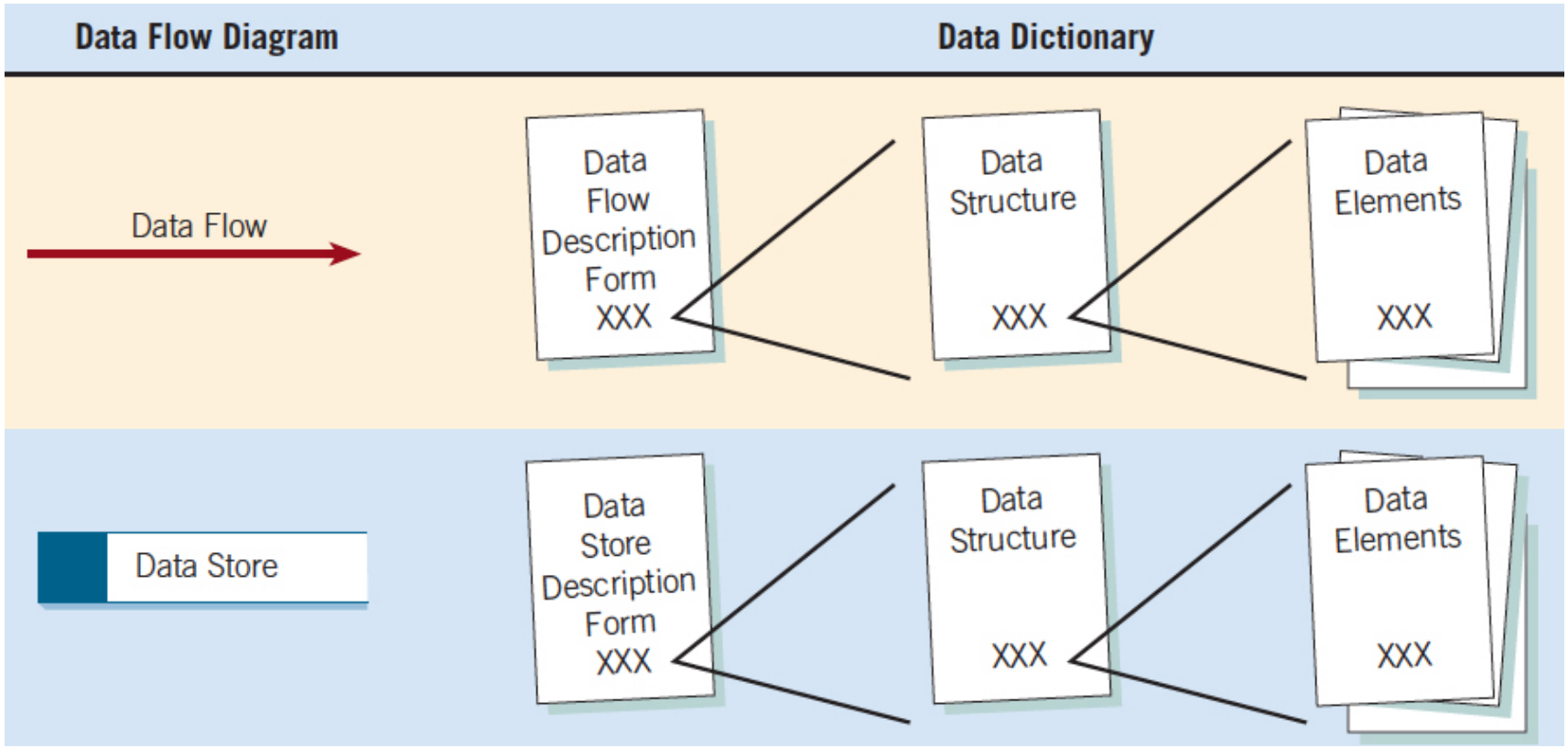
Figure 8.1 How Data Dictionaries Relate to Data Flow Diagrams



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Defining the Data Flow

- ID—identification number
- Unique descriptive name
- A general description of the data flow
- The source of the data flow
- The destination of the data flow
- Type of data flow
- The name of the data structure describing the elements
- The volume per unit time
- An area for further comments and notations

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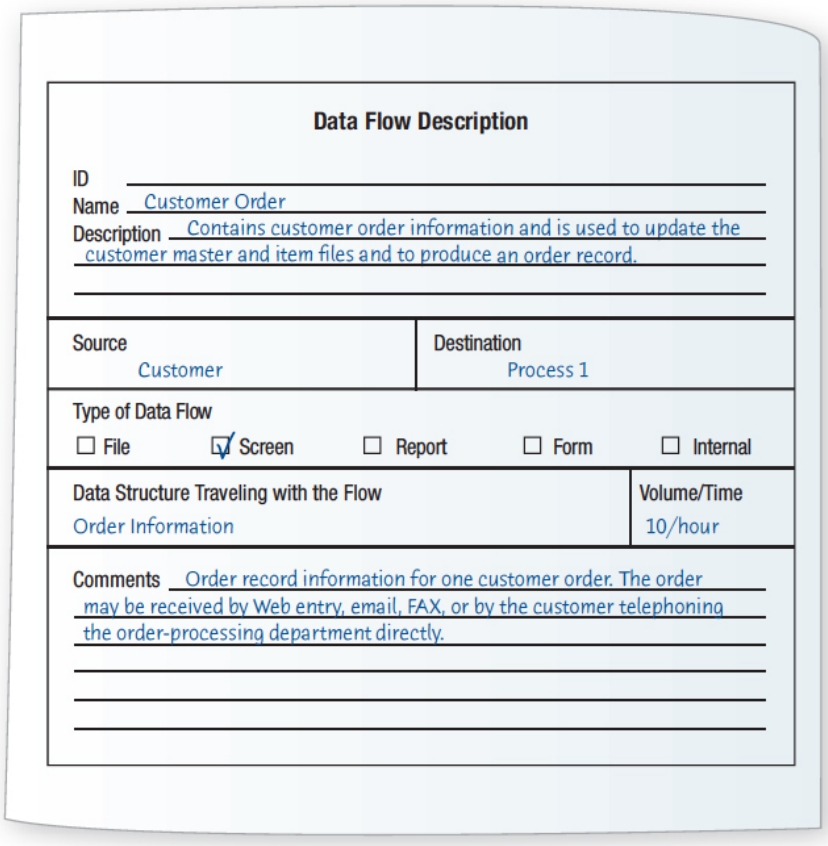
Figure 8.3 An Example of a Data Flow Description from World's Trend Catalog Division



Data Flow Description

ID

Name Customer Order

Description Contains customer order information and is used to update the customer master and item files and to produce an order record.

Source	Destination
Customer	Process 1

Type of Data Flow

☐ File ☑ Screen ☐ Report ☐ Form ☐ Internal

Data Structure Traveling with the Flow	Volume/Time
Order Information	10/hour

Comments Order record information for one customer order. The order may be received by Web entry, email, FAX, or by the customer telephoning the order-processing department directly.

Other Examples:

- Customer order data flows from the Web UI → Order Service → Payment Gateway → Inventory System."
- ETL (Extract, Transform, Load) pipeline: Raw logs → Cleansing → Data Warehouse.

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Describing Data Structures

- Defines the **schema, format**, and **organization of data** entities (tables, files, messages, APIs).
- Data structures are made up of **smaller structures** and **elements**
- An **algebraic notation** is used to **describe** data structures

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Algebraic Notation

- **Equal** sign means "**is composed of**"
- **Plus** sign means "**and**"
- **Braces** {} mean **repetitive** elements
- **Brackets** [] for an **either/or** situation
- **Parentheses** () for an **optional** element

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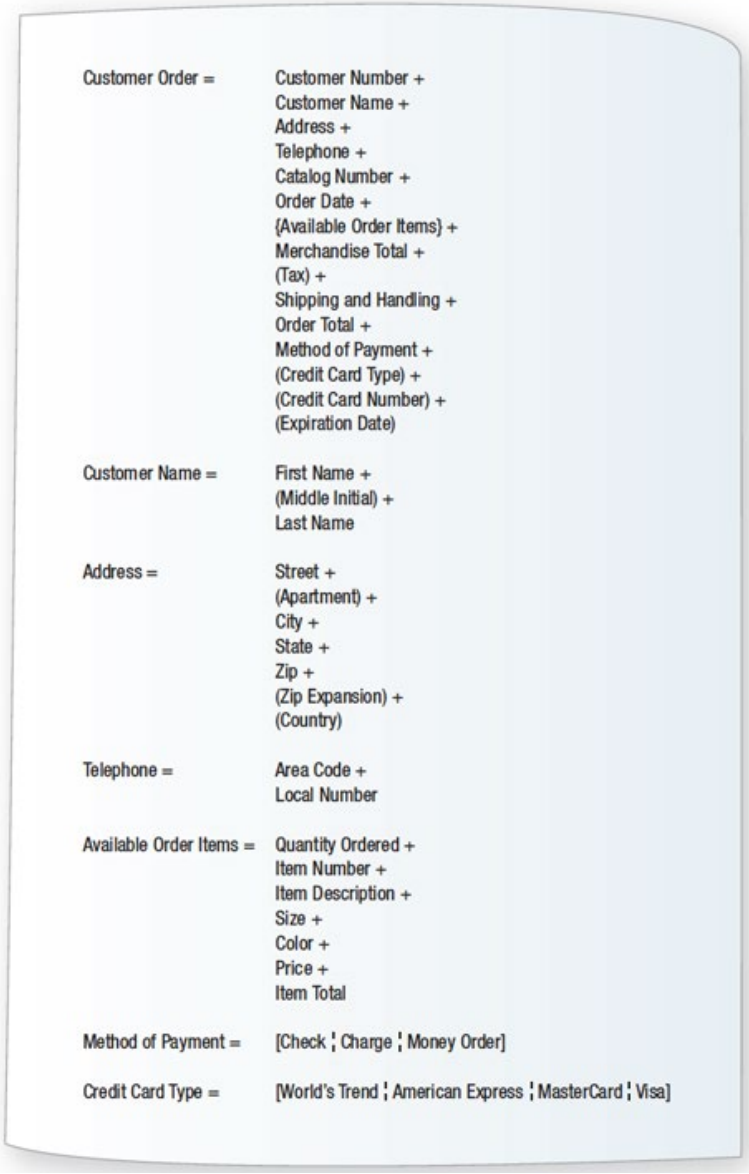
Figure 8.4 Data Structure Example for Adding a Customer Order at World's Trend Catalog Division



Customer Order =	Customer Number + Customer Name + Address + Telephone + Catalog Number + Order Date + {Available Order Items} + Merchandise Total + (Tax) + Shipping and Handling + Order Total + Method of Payment + (Credit Card Type) + (Credit Card Number) + (Expiration Date)
Customer Name =	First Name + (Middle Initial) + Last Name
Address =	Street + (Apartment) + City + State + Zip + (Zip Expansion) + (Country)
Telephone =	Area Code + Local Number
Available Order Items =	Quantity Ordered + Item Number + Item Description + Size + Color + Price + Item Total
Method of Payment =	[Check ¦ Charge ¦ Money Order]
Credit Card Type =	[World's Trend ¦ American Express ¦ MasterCard ¦ Visa]

Other Examples:

- **Database table:** Users (user_id INT, name VARCHAR(100), email VARCHAR(255))
- **JSON** message schema for an API response
- **CSV** file layout with column names, types, and constraints

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Structural Records

- A structure may consist of **elements** or **structural** records (nested groups)
- A structural record is a container that groups related data
- These are a **group of elements**, such as:
  - **Customer name**
  - **Address**
  - **Telephone**
- Each of these must be **further defined** **until** they are **broken down** into their **component elements**

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Structural Records Used in Different Systems

- Use **generic, reusable** names for common data groups so they can be shared across systems without confusion.
- Structural **records** and **elements** that are used within many different systems are given a **non-system-specific name**, such as **street**, **city**, and **zip**
- The names **do not reflect** a **functional area**
- This allows the analyst to **define** them once and **use** in **many** different **applications** (*Reduce integration errors*)

Problem	**Solution**
Marketing calls it `Client_Address`	Use neutral: `Postal_Address`
Billing calls it `Invoice_Location`	Same `Postal_Address` reused

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Logical and Physical Data Structures

- **Logical**:
  - **Show** what **data** the **business needs** for its **day-to-day operations**
  - What the business cares about

- **Physical**:
  - Include **additional elements** necessary for **implementing** the system
  - How it's actually stored in the system

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Logical Data Structures

- **Simple** & **Point** Form
- A ***business-friendly*** picture of data – no tech details.
- Focus only what the business needs to run daily.
- Audiences are managers, analysts, users – not programmers.
- Goal is easy to understand, discuss, and agree on.
- **Includes**:
  - Real-world things (Customer, Order, Product)
  - Clear relationships (One Customer → Many Orders)
- **Excludes:** IDs, keys, codes

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Physical Data Structures

- These are invisible to business users but critical for system operation (tech people).
- The real way data is stored in computers/databases.
- Key fields used to locate records (*eq: address_id*)
- Codes to identify record status (*eq: A=Active*)
- Transaction codes to identify different record types (*eq: 01 = New Order, 02 = Update*)
- Repeating group entries (*eq: 1:Many*)
- Limits on items in a repeating group (*eq: Max:10*)
- Password (*eq: Encrypted*)

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Data Element Characteristics (1 of 2)

- A data element is ONE single piece of information — like a box that holds one thing (a name, a number, a date).

Characteristic	Simple Meaning	Example
Element ID	A code number for the data (like a student ID)	***DE-001***
The name of the element	What you call it	***Student Name***
Aliases	Other nicknames people use for it	Also called: ***Full Name, Student Full Name***
A short description	One sentence: What does it mean?	"The full name of the student"
Base or Derived?	Is it entered by hand (base) or calculated (derived)?	Base: You type it Derived: ***Age = Today - Birthdate***
Element length	How long it can be	Max 50 letters
Type of data	What kind of info	Text, Number, Date, Yes/No

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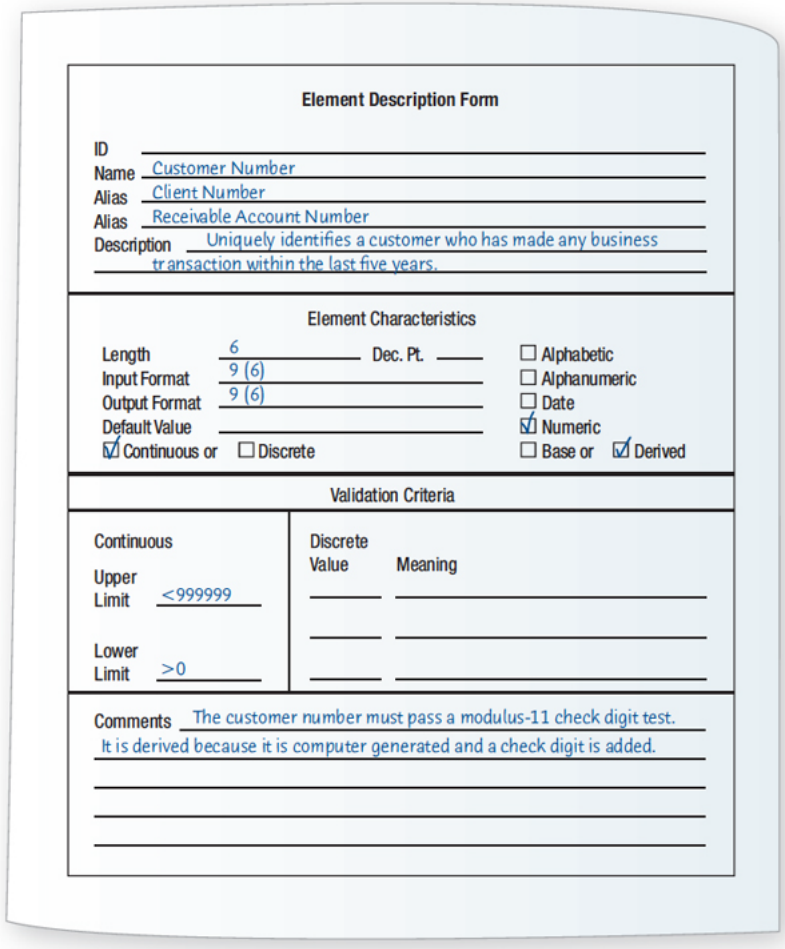
Figure 8.6 An Element Description Form Example from World's Trend Catalog Division



Element Description Form

ID
Name: Customer Number
Alias: Client Number
Alias: Receivable Account Number
Description: Uniquely identifies a customer who has made any business transaction within the last five years.

Element Characteristics

Length: 6 Dec. Pt.:
Input Format: 9 (6)
Output Format: 9 (6)
Default Value:
☑ Continuous or ☐ Discrete

☐ Alphabetic
☐ Alphanumeric
☐ Date
☑ Numeric
☐ Base or ☑ Derived

Validation Criteria

Continuous
Upper Limit: <999999
Lower Limit: >0

Discrete
Value | Meaning

Comments: The customer number must pass a modulus-11 check digit test. It is derived because it is computer generated and a check digit is added.

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Element ID

- **A secret code for the data piece**
- **Optional** entry - You don't have to use it
- Helps computers find the data fast
- Allows the analyst to **build automated** data dictionary entries
- Think of it like: A student ID number — not your name, but a code to find you!

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The Name of the Element

- What everyone calls it
- Should be:
  - **Descriptive** → *"Student Age" (not "A")*
  - **Unique** → No two fields have same name
- Based on what the element is commonly called in most programs or by the major user of the element
- Use common words most people know
- Example: "Customer Phone Number"

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Aliases

- Other names people use for the same thing
- **Synonyms** or other **names** for the element
- **Names** used by **different** **users** in **different systems**
- **A customer number** may also be called a **receivable account number or a client number**
- Different teams call it different things. **Example:**
  - Main name: ***Customer ID***
  - Aliases: ***Client Number, Account #, Receivable ID***
- **Why?** *So everyone knows: "All these mean the SAME thing!"*

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Short Description of the Element

- One sentence: What is it for?
- An example might be:
  - Uniquely **identifies** a **customer** who has made any **business transactions** within the **last five years**
- Keep it short & clear — like a tweet!

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Element is Base or Derived

- Where does the data come from?
- A **base** element is one that has been **initially keyed** into the system
- A **derived** element is one that is **created** by a **process**, usually as the **result** of a **calculation** or a **series** of **decision-making** statements
- Base = Human enters, Derived = Computer makes

Type	Meaning	Example
Base	**Typed in by a person**	Name, Phone, Address
Derived	**Calculated by computer**	Age = Today – Birthdate Total = Price × Qty

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Element Length

- How many letters/numbers can fit?
- **What** should the element **length** be?
  - Some elements have **standard lengths**, **state** abbreviations, **zip codes**, or **telephone** numbers.
  - For other elements, the **length** may **vary** and the analyst and **user community** must **decide** the **final** length.

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Element Length Considerations

- Numeric **amount** lengths
- **Name** and **address** fields
- **Other** fields – YOU decide
  - Ask: “What’s the longest name/address we’ll see?”
  - Then **add a little extra**
- Don’t make it too small → **Truncation**!
-

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Name and Address Length

Element	Length	Percent of Data That Will Fit (United States)
Last Name	11	98
First Name	18	95
Company Name	20	95
Street	18	90
City	17	99

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Type of Data

- What kind of info goes in?

Type	Examples
Text (Alphanumeric)	Names, addresses, emails
Number	Age, price, quantity
Date	Birthdate, order date
Yes/No	Is student active?

- **Alphanumeric** or **text** data
- Formats
  - ***Mainframe***: **packed, binary, display**
  - ***Microcomputer*** (PC) formats
  - ***PC formats***, such as **Currency**, **Number**, or **Scientific**, depend on how the data will be used

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Data Element Characteristics (2 of 2)

Characteristic	Simple Meaning	Example
Input format	How it must be typed in	Date must be ***DD/MM/YYYY***
Output format	How it shows on screen/report	Show age as “15 years”
Validation criteria	Rules to check if it’s correct	- Age must be 5–100 - Email must have **@**
Default value	Auto-fill if left blank	Country = “USA”
Comment / Remark	Extra note for future	“Do not share email without permission”

Why Do We Care?

- Because if we **don’t define** these clearly: *Wrong data goes in → Age = "banana“*
- A data element is **one piece of data,** and its characteristics are **the rules and labels** that keep it **correct** and **clear**.

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Figure 8.7 Some Examples of Data Formats Used in PC Systems

Data Type	Meaning
Bit	A value of 1 or 0, a true/false value
Char, varchar, text	Any alphanumeric character
Datetime, smalldatetime	Alphanumeric data, several formats
Decimal, numeric	Numeric data that are accurate to the least significant digit; can contain a whole and decimal portion
Float, real	Floating-point values that contain an approximate decimal value
Int, smallint, tinyint	Only integer (whole digit) data
Currency, money, smallmoney	Monetary numbers accurate to four decimal places
Binary, varbinary, image	Binary strings (sound, pictures, video)
Cursor, timestamp, uniqueidentifier	A value that is always unique within a database
Autonumber	A number that is always incremented by one when a record is added to a database table

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Figure 8.8 Format Character Codes

Formatting Character	Meaning
X	May enter or display/print any character
9	Enter or display only numbers
Z	Display leading zeros as spaces
,	Insert commas into a numeric display
.	Insert a period into a numeric display
/	Insert slashes into a numeric display
-	Insert a hyphen into a numeric display
V	Indicate a decimal position (when the decimal point is not included)

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Validation Criteria

- Rules to make sure data is CORRECT
- Goal: Only good data gets in!
- **Ensure** that **accurate** **data** are **captured** by the system
- Elements are either:
  - **Discrete**, meaning they have ***fixed*** values
  - **Continuous**, with a smooth ***range of values***

Type	Examples	How to Check
Discrete	Gender: *M, F, Other* State: *CA, NY, TX*	Use **drop-down list**
Continuous	Age: *5 to 100 Price*: *$0.01 to $999.99*	Use **min/max rules**

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Default Value

- Auto-fill when nothing is typed. Saves time & reduces mistakes
- Include any default value the **element** may have
- The default value is **displayed** on **entry** screens
- Reduces the **amount** of **keying**
  - Default values on **GUI** screens
    - Initially display in **drop-down** lists
    - Are selected when a group of **radio buttons** are used

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Comment or Remarks Area

- Extra notes for future people. Like sticky notes on the data
- This might be used to indicate the **format** of the **date**, **special validation** that is required, the **check-digit method** used, and so on

Use	Example
Format	“Use MM/DD/YYYY”
Special Rule	“Only teachers can edit”
Check-digit	“Last digit = sum of others”
Warning	“Do NOT share email”

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Data Truncation

- **Truncation:** Cutting off the end of data because the box is too small.
- If the **element** is **too small**, the data **will** be **truncated**
- The analyst must **decide** how this will **affect** the system outputs
- If a **last name** is **truncated**, **mail** would usually still be **delivered but a truncated email** address or **Web** address is **not usable**

truncated email	Real email = sarah.johnson@example.com. Box allows 10 letters → Saved as sarah.joh
not usable	The email is **broken** — you **can't send mail** to sarah.joh

- **How to Fix It:**
  - **Make the box bigger → length = 50 for names**
  - **Check before saving → Show error: "Name too long!"**
  - **Test with real data → Try O'Connor, Nguyen, Vanderbilt**

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Data Stores

- Data stores are created for each different data entity being stored
- When data flow base elements are **grouped** together to form a structural record, a data store is created for each **unique** structural record
- Because a given data flow may only show part of the **collective** data that a **structural record contains**, many different data flow structures may need to be **examined** to arrive at a **complete** data store description

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Describing the Data Store (1 of 2)

- The data store ID
- The data store name
- An alias for the table
- A short description of the data store
- The file type
- File format

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Describing the Data Store (2 of 2)

- The **maximum** and **average** number of records on the file as well as the growth per year
- The ***file*** or ***data set*** name specifies the ***file*** name, if known
- The **data structure** should use a name **found** in the data dictionary
- **Primary** and **secondary** keys
- **Comments**

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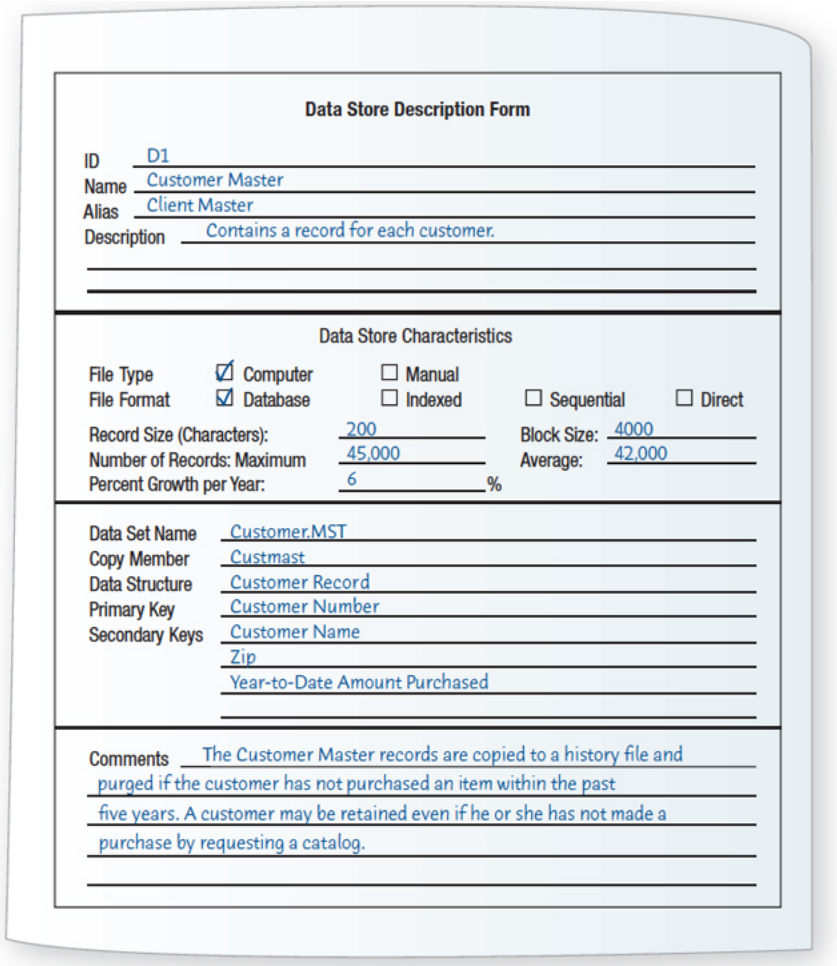
Figure 8.9 Example of a Data Store Form for World's Trend Catalog Division



Data Store Description Form

ID: D1
Name: Customer Master
Alias: Client Master
Description: Contains a record for each customer.

Data Store Characteristics

File Type: ☑ Computer ☐ Manual
File Format: ☑ Database ☐ Indexed ☐ Sequential ☐ Direct
Record Size (Characters): 200 Block Size: 4000
Number of Records: Maximum 45,000 Average: 42,000
Percent Growth per Year: 6 %

Data Set Name: Customer.MST
Copy Member: Custmast
Data Structure: Customer Record
Primary Key: Customer Number
Secondary Keys: Customer Name
Zip
Year-to-Date Amount Purchased

Comments: The Customer Master records are copied to a history file and purged if the customer has not purchased an item within the past five years. A customer may be retained even if he or she has not made a purchase by requesting a catalog.

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Creating the Data Dictionary

- Like creating a “cheat sheet” that explains every piece of data in the system. Tells what, how, why & etc.
- When we create a Data Dictionary?:

Option	Simple Meaning
After DFD	Finish the flowchart first, then list all data
During DFD	Build it **as you draw** – faster & no forgetting any details.
Top-Down	Start with **big parts**, then break into **small details**

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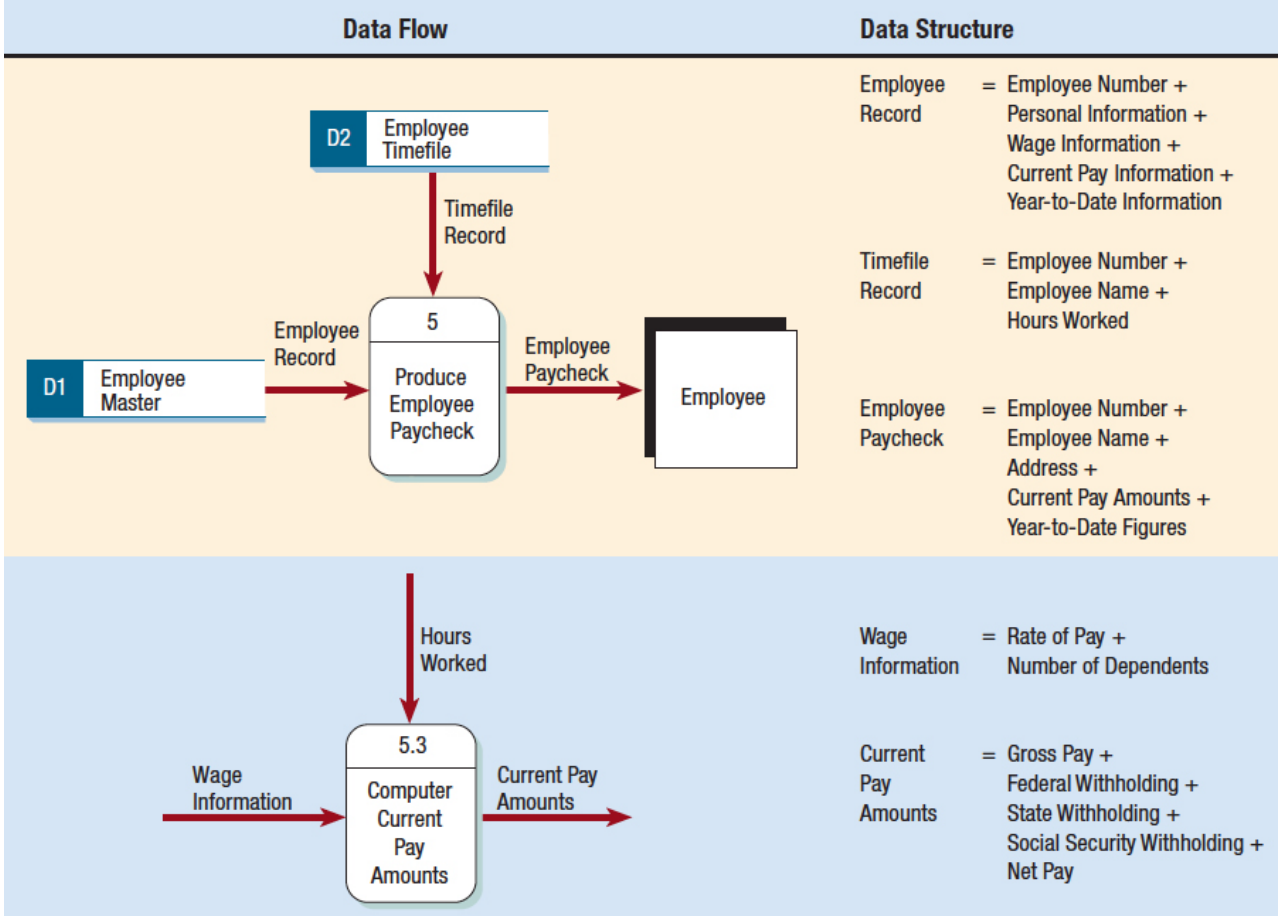
Figure 8.10 Two Data Flow Diagrams and Corresponding Data Dictionary Entries for Producing an Employee Paycheck



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Analyzing Input and Output Form

- Purpose: Checks for Mistakes
- A **descriptive name** for the input or output
- The user **contact responsible**
- **Whether** the data is **input** or **output**
- The **format** of the data flow
- Elements indicating the **sequence** of the data on a report or screen
- A **list** of elements

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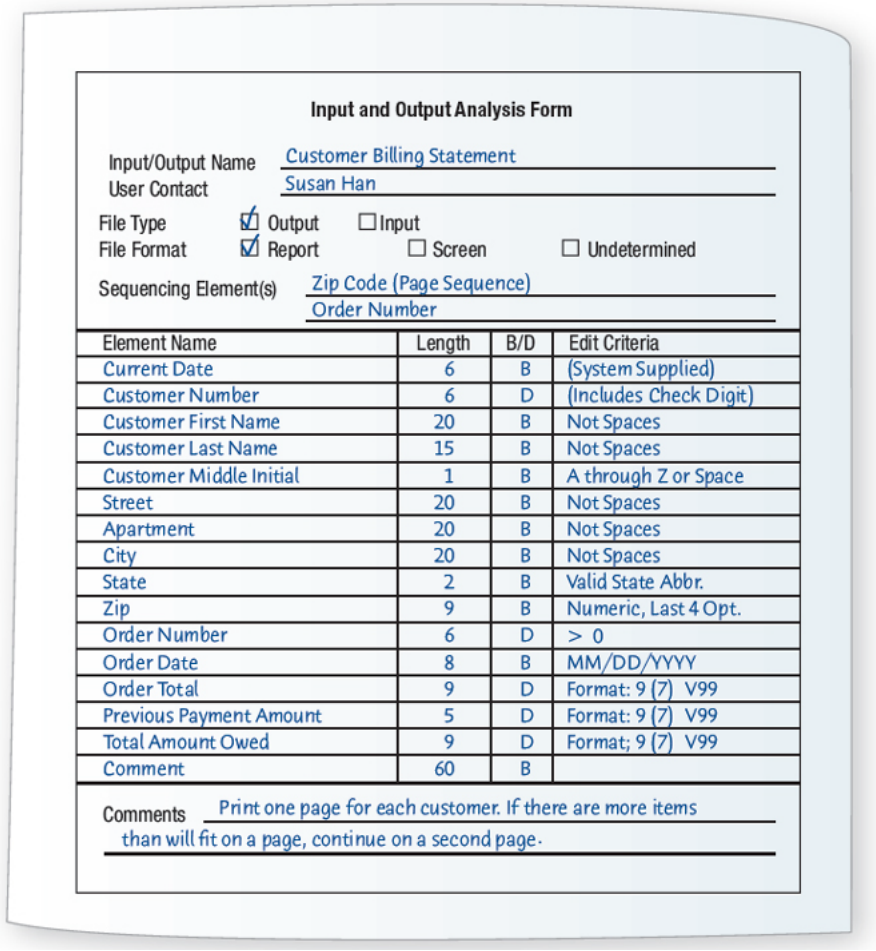
Figure 8.11 An Example of an Input/Output Analysis Form for World's Trend Catalog Division



Input and Output Analysis Form

Input/Output Name: Customer Billing Statement
User Contact: Susan Han

File Type: ☑ Output ☐ Input
File Format: ☑ Report ☐ Screen ☐ Undetermined

Sequencing Element(s): Zip Code (Page Sequence), Order Number

Element Name	Length	B/D	Edit Criteria
Current Date	6	B	(System Supplied)
Customer Number	6	D	(Includes Check Digit)
Customer First Name	20	B	Not Spaces
Customer Last Name	15	B	Not Spaces
Customer Middle Initial	1	B	A through Z or Space
Street	20	B	Not Spaces
Apartment	20	B	Not Spaces
City	20	B	Not Spaces
State	2	B	Valid State Abbr.
Zip	9	B	Numeric, Last 4 Opt.
Order Number	6	D	> 0
Order Date	8	B	MM/DD/YYYY
Order Total	9	D	Format: 9 (7) V99
Previous Payment Amount	5	D	Format: 9 (7) V99
Total Amount Owed	9	D	Format; 9 (7) V99
Comment	60	B	

Comments: Print one page for each customer. If there are more items than will fit on a page, continue on a second page.

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Developing Data Stores

- Where data “sleeps” when not being used. Represent data at **rest**
- Contain information of a **permanent** or **semipermanent** (temporary) nature
- When data stores are created for only one **report** or **screen**, we refer to them as “**user views**”

Rule	Simple Meaning
Permanent	Stays forever (e.g., Student Records)
Semi-Permanent	Kept for a while (e.g., This Year’s Grades)
Temporary	Only for one report → called **User View**

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Using the Data Dictionary

- Super useful for many jobs!
- To have **maximum power**, the **data dictionary** should be **tied** into **several systems programs**
- May be used to
  - Create **screens**, **reports**, and **forms**
  - Generate computer language **source code**
  - **Analyze** the system **design**, **detecting flaws** and **areas** that need **clarification**
- **Goal**: One dictionary → used by everyone (designers, coders, testers)

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Create Screens, Reports, and Forms

- Use the element **definition** and their **lengths**
- Arrange the elements in a **pleasing** and **functional** way using **design guidelines** and **common sense**
- Repeating **groups** become **columns**
- Structural records are **grouped together** on the **screen**, **report**, or **form**

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Analyze the System Design, Detecting Flaws and Areas That Need Clarification

- All base elements on an **output data flow** must be **present** on an **input data** flow to the **process producing** the **output**
- A **derived element** should be **created** by a **process** and should be output from **at least one process** into which it is not input
- The **elements** that are present in a **data flow** coming into or going out of a **data store** must be **contained** in the data store

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What is an XML?

- XML stands for eXtensible Markup Language
- XML is a markup language much like HTML
- XML was designed to store and transport data
- XML was designed to be self-descriptive

Problem	XML Fix
Company A uses Excel	→ Sends data in XML
Company B uses Oracle	→ Still reads it!
Different software	→ XML works everywhere

- Why use XML?

Share data	Between businesses, apps, websites
Change format	Turn into PDF, HTML, app screen
Sort / Filter	Find “all customers in CA” easily
Make anywhere	From database, form, or type by hand

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Using Data Dictionaries to Create XML (1 of 2)

- XML is used to exchange data between businesses
- XML addresses the problem of sharing data when users have different computer systems and software or different database management systems
- XML documents may be transformed into different output formats
- XML is a way to define, sort, filter, and translate data into a universal data language that can be used by anyone
- XML may be created from databases, a form, software programs, or keyed directly into a document, text editor, or XML entry program

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Using Data Dictionaries to Create XML (2 of 2)

- Data Dictionary → Recipe & XML → Cooked Meal using that recipe
- The data dictionary is an ideal starting point for developing XML content
- A standard definition of the data is created using a set of tags that are included before and after each data element or structure
- XML elements may also include attributes
- The XML document tends to mirror the data dictionary structure

Rule	Simple Example
Tags have start & end	<name>John</name>
Nested like folders	Customer → Name → First
Can add attributes	<order id="0123">
Mirrors dictionary	Same structure, just in XML form

Figure 8.15 Using a Data Dictionary Entry to Develop XML Content: The XML Document Mirrors the Data Dictionary Structure

Data Dictionary

Customer =	Name + Address + Current Balance + {Order Information} + Payment
Name =	Last Name + First Name + (Middle Initial)
Address =	Street + (Apartment) + City + State + Zip + Country
Order Information =	Order Number + Order Date + Ship Date + Total
Payment =	[Check ¦ Credit Card] + Payment Date + Payment Amt
Check =	Check Number
Credit Card =	Credit Card Number + Expiration Date

XML

```
<?xml version="1.0">
<customers>
<customer number="C15008">
  <name type="l">
     <lastname>Stadler</lastname>
     <firstname>Karen</firstname>
     <middle_initial>L</middle_initial>
  </name>
  <address>
     <street>123 Oak Street </street>
     <apartment>Suite 16</apartment>
     <city>Madison</city>
     <state>WI</state>
     <zip>43704</zip>
     <country>United States</country>
  </address>
  <current_balance>123.45</current_balance>
  <order customer_number="C15008">
     <order_number>00123</order_number>
     <order_date format="yyyymmdd">2008-06-23</order_date>
     <ship_date format="yyyymmdd">2008-06-25</ship_date>
     <total>1345.89</total>
  </order>
  <order customer_number="C15008">
     <order_number>00127</order_number>
     <order_date format="yyyymmdd">2008-09-18</order_date>
     <ship_date format="yyyymmdd">2008-09-26</ship_date>
     <total>240.00</total>
  </order>
  <payment>
     <check>
        <check_number>7234</check_number>
     </check>
     <payment_date format="yyyymmdd">2008-09-30</payment_date>
     <payment_amt>1585.89</payment_amt>
  </payment>
</customer>
</customers>
```

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XML Document Type Definitions (DTD)

- DTD → "XML Police" Checks if XML follows the exact rules
- Used to determine if the XML document content is valid
- DTDs may be created using the data dictionary
- DTD may be used to validate the XML document
- **Think:** DTD → **Answer key** → XML must match or it's **wrong**!

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Figure 8.16 A Document Type Definition for the Customer XML Document

```
<!DOCTYPE   customers    [
<!ELEMENT   customers                 (customer) + >
<!ELEMENT   customer                  (name, address, current_balance, order*) >
<!ATTLIST   customer number           ID #REQUIRED>
<!ELEMENT   name                      (lastname, firstname, middle_initial?) >
<!ATTLIST   name type                 (I|C)  #REQUIRED>
<!ELEMENT   lastname                  (#PCDATA) >
<!ELEMENT   firstname                 (#PCDATA) >
<!ELEMENT   middle_initial            (#PCDATA) >
<!ELEMENT   address                   (street, apartment?, city, state, zip, country) >
<!ELEMENT   street                    (#PCDATA) >
<!ELEMENT   apartment                 (#PCDATA) >
<!ELEMENT   city                      (#PCDATA) >
<!ELEMENT   state                     (#PCDATA) >
<!ELEMENT   zip                       (#PCDATA) >
<!ELEMENT   country                   (#PCDATA) >
<!ELEMENT   current_balance           (#PCDATA) >
<!ELEMENT   order                     (order_number, order_date, ship_date, total) >
<!ATTLIST   order customer_number     IDREF  #REQUIRED>
<!ELEMENT   order_number              (#PCDATA) >
<!ELEMENT   order_date                (#PCDATA) >
<!ATTLIST   order_date format         (mmddyyyy|yyyymmdd|ddmmyyyy)  #REQUIRED>
<!ELEMENT   payment                   (check|credit_card) >
<!ELEMENT   check                     (check_number) >
<!ELEMENT   credit_card               (credit_card_number, expiration_date) >
<!ATTLIST   credit_card type          (M|V|A|D|O)  #REQUIRED>
<!ELEMENT   credit_card_number        (#PCDATA) >
<!ELEMENT   expiration_date           (#PCDATA) >
<!ELEMENT   payment_date              (#PCDATA) >
<!ATTLIST   payment_date format       (mmddyyyy|yyyymmdd|ddmmyyyy)  #REQUIRED>
<!ELEMENT   payment_amt               (#PCDATA) >
    ] >
```

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XML Schemas

- Better than DTD → more precise!
- A more precise way to define the content of an XML document
- Includes exact number of times an element may occur
- Includes type of data within elements

Feature	Simple Meaning
How many times?	orders can appear 1–10 times
Data type	balance must be **number**, not text
Exact rules	zip = 5 digits only

```
<xs:element name="zip" type="xs:string" pattern="[0-9]{5}"/>
```

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What is a JSON?

- JSON stands for JavaScript Object Notation.
- Lightweight, human-readable way to store and send data
- JSON is often used when data is sent from a server to a web page. Like XML's simpler cousin – uses curly braces {} and brackets [ ]

XML	JSON
<name>John</name>	"name": "John"
Needs **tags**	Just **key-value pairs**
Heavy & wordy	**Light & fast**

- Think: XML → Formal letter & JSON → Quick text message

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Why use JSON?

Feature	Simple Example
Super fast	Less text → loads quicker
Easy to read	Humans and computers love it
Works everywhere	Web, apps, APIs, databases
Perfect for web	Used by **99% of websites**

- Scenario
  - Send student info from school app → parent phone → JSON does it in 1 line!

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Using Data Dictionaries to Create JSON (1 of 2)

- Data Dictionary → Blueprint & JSON → Data using that blueprint

From Your Dictionary (Fig 8.15)

```text
Customer = Name + Address + Balance + Orders
Name = First + Middle + Last
Address = Street + City + State + Zip
```

```json
{
  "customer": {
    "name": {
      "first": "Karen",
      "middle": "L.",
      "last": "Stadler"
    },
    "address": {
      "street": "123 Oak Street",
      "apartment": "Suite 16",
      "city": "Madison",
      "state": "WI",
      "zip": "43701",
      "country": "United States"
    },
    "balance": 123.45,
    "orders": [
      {
        "order_number": "0123",
        "order_date": "2008-06-23",
        "ship_date": "2008-06-25",
        "total": 1345.89
      },
      {
        "order_number": "0127",
        "order_date": "2008-09-18",
        "total": 240.00
      }
    ]
  }
}
```

- "+" → becomes nested {}
- Repeating → becomes [ ] array

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Using Data Dictionaries to Create JSON (2 of 2)

- JSON rules (Data is in name/value pairs) :

Rule	Example
Keys in quotes	"name":
Values: text, number, true/false, null	"age": 25, "active": true
Objects in {}	{ "name": "Ali" }
Array Lists in []	["Math", "English"]
Comma separates	"city": "NY", "state": "NY"

- Bad JSON (Error):

```
json

{ name: "John" }  ← No quotes on key!
```

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JSON Schemas:The Rule Checker (Like DTD)

- JSON Schema → “JSON Police” that checks if JSON follows your data dictionary rules.
- JSON Schema Example →

```json
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Customer",
  "type": "object",
  "required": ["name", "address"],
  "properties": {
    "name": {
      "type": "object",
      "required": ["first", "last"],
      "properties": {
        "first": { "type": "string", "minLength": 1 },
        "last": { "type": "string", "minLength": 1 }
      }
    },
    "address": {
      "type": "object",
      "required": ["street", "city", "zip"],
      "properties": {
        "zip": { "type": "string", "pattern": "^[0-9]{5}$" }
      }
    },
    "balance": { "type": "number" }
  }
}
```

XML vs JSON

Feature	XML	JSON
Tags	<name>John</name>	"name": "John"
Arrays	<order>...</order> <order>...</order>	[{...}, {...}]
Size	Bigger file	**Smaller & faster**
Readability	OK	**Super easy**
Validation	DTD / XSD	**JSON Schema**
Best for	Old systems, documents	**Web, apps, APIs**

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Summary (1 of 2)

- The data dictionary
  - A reference work containing data about data
  - Includes all data items from data flow diagrams
- Repository
  - A larger collection of project information
- Defining data structures
- Defining elements

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Summary (2 of 2)

- Defining data stores
- Data dictionary entries
- Using the data dictionary
- Data dictionary analysis
- Data dictionary to XML
- Data dictionary to JSON

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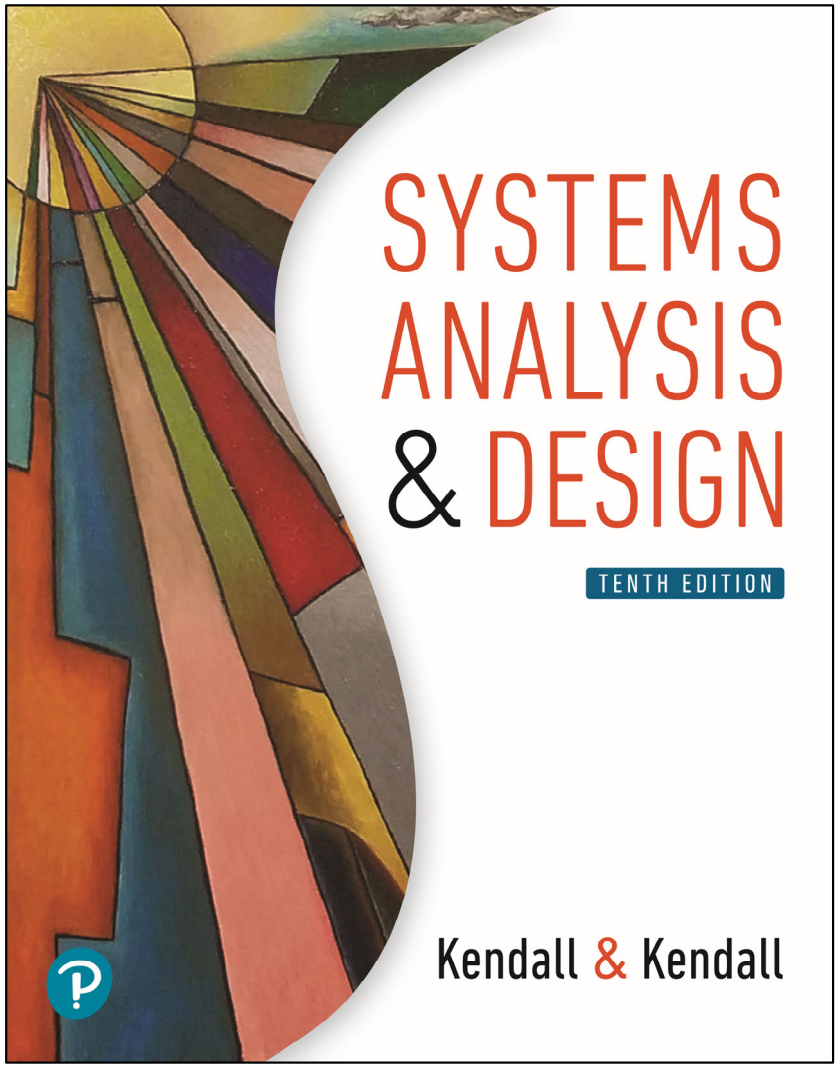
Systems Analysis & Design

Tenth Edition



Chapter 7

Process Specifications and
Structured Decisions

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Learning Objectives

7.1 Understand the purpose of process specifications.

7.2 Recognize the difference between structured and semistructured decisions.

7.3 Use structured English, decision tables, and decision trees to analyze, describe, and document structured decisions.

7.4 Choose an appropriate decision analysis method for analyzing structured decisions and creating process specifications.

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Logic of Decisions

- A set of rules that controls how a system responds to different inputs or situations
- It answers questions like:
  - ✓ If a customer is new, then create an account.
  - ✓ If the total purchase > $100, then apply a discount.
  - ✓ If the password is incorrect 3 times, then lock the account.
- In other words, decision logic is the brain of a system's **business rules.**

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Why decision logic important?

- Ensures consistent system behavior
- Automates business rules
- Reduces manual decision-making
- Helps define workflow, validation, and **process flows**
- Supports designing algorithms, program flow, and system behavior

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What is Process Logic?

- A clear, structured specification of the rules, conditions, and workflow steps that determine how a system processes inputs, makes decisions, and responds to events within a business process.
- A formal representation of:
  - ✓ what the system should do
  - ✓ how it should respond
  - ✓ which conditions trigger which actions
  - ✓ how data is validated, transformed, or routed
  - ✓ how decisions are made consistently

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Decision Logic vs Process Logic

- Decision Logic (What choice to make)
  - ✓ **Focus**: Rules, conditions, and choices the system makes.
  - ✓ It determines which action the system should take based on conditions.
  - ✓ Examples: If credit score < 600 → Reject loan
- Process Logic (How to perform tasks)
  - ❖ **Focus**: Steps, sequences, and workflow of the system.
  - ❖ It defines the order of activities the system follows to accomplish a task.
  - ❖ Examples: Validate input → Store data → Generate report

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Major Topics

- Process specifications
- Business rules
- Structured English
- Decision tables
- Decision trees

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Process Specifications

- It is the **formal document** that developers use to build the system.
- Sometimes called mini-specs that contains both process logic and decision logic
- Created for primitive processes as well as for some higher-level processes on a data flow diagram
- Created for class methods in object-oriented design and for the steps in a use case

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Goals of Producing Process Specifications

- Reduce process ambiguity
- Obtain a precise description of what is accomplished
- Validate the system design
- Developers use them as the primary source to write code or configure ERP/CRM rules.
- Testers use them to create test cases (every path in a decision table becomes a test).
- Prevent misunderstandings between analysts, users, and programmers.

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Process Specifications Are Not Created for

- Processes that represent physical input and/or output
  - E.g., “Scan barcode”, “Print invoice”, “Fill out paper form”
- Processes that represent simple data validation
  - E.g., “Check that Customer-ID is not blank”, “Verify ZIP code is 5 digits”
- Processes that use prewritten code
  - E.g., “Call Tax Calculation Web Service”, “Invoke SAP Pricing Engine”, “Run standard credit-score module”

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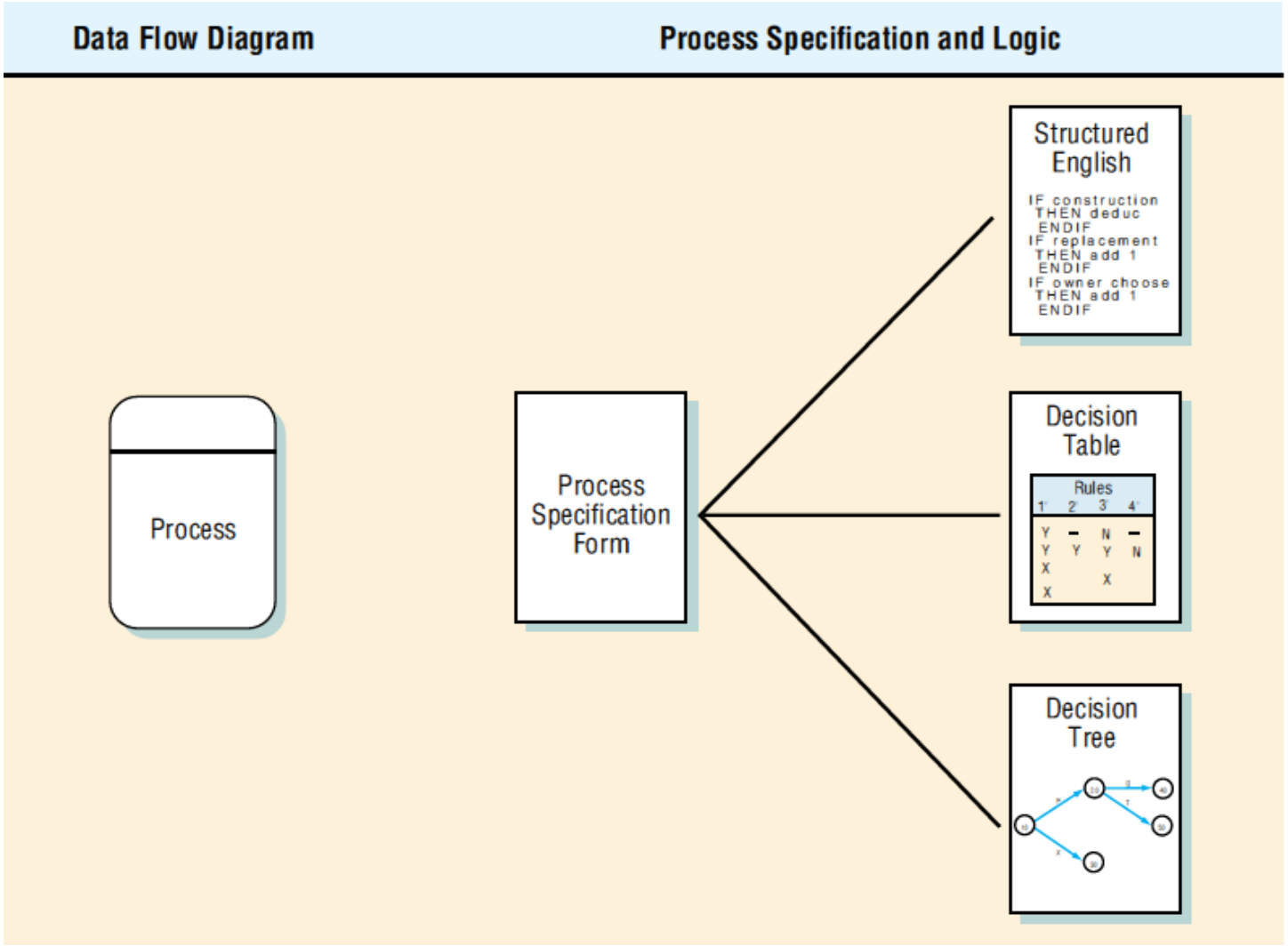
Figure 7.1 How Process Specifications Relate to the Data Flow Diagram



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Process Specification Format Information (1 of 2)

- The process number
- The process name
- Description of what the process accomplishes
- A list of input data flow
- Output data flows

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Process Specification Format Information (2 of 2)

- Type of process
- Uses prewritten code
- Process logic description
- Logic method reference
- List any unresolved issues

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The Process Number

- Must match the process ID on the data flow diagram
- Allows the analyst to work on or review any process, and to locate the data flow diagram containing the process easily
- E.g.: 3.2 Validate Customer Order

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The Process Name

- The same as displays within the process symbol on the DFD
- E.g.: Validate Customer Order

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Description of What the Process Accomplishes

- Short 1–2 sentence business objectives
- Example:
  - Determine if an item is available for sale. If it is not available, create a backordered item record. Determine the quantity available.

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List of Input Data Flow

- Data flows coming into the bubble (with source)
- Uses the names found on the data flow diagram
- Data names used in the formula or logic should match the data dictionary, for consistency and good communication

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Output Data Flows

- Data flows leaving the bubble (with destination)
- Uses data flow diagram and data dictionary names

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Type of Process

- Batch
- Online
  - Require screen design or web pages
- Manual
  - Should have well-defined procedures for employees performing the process tasks
- E.g.:Validation process

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Uses Prewritten Code

- Include the name of the subprogram or function containing the code
- E.g.: Uses existing “CheckCustomerID()” function.

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Process Logic Description

- Step-by-step explanation of what the process does.
- This should state policy and business rules, not computer language pseudo-code
- Business rules are the procedures that allow a corporation to run its business

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Logic Method Reference

- If there is not enough room for a complete structured English description include a reference to the structured English description, decision table, or tree depicting the logic
- E.g.: See attached Decision Table DT-02.

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List Any Unresolved Issues

- Incomplete portions of logic
- These issues form the basis of the questions used for follow-up interviews with users or business experts you have added to your project team
- E.g.: Discount percentages not confirmed by management.

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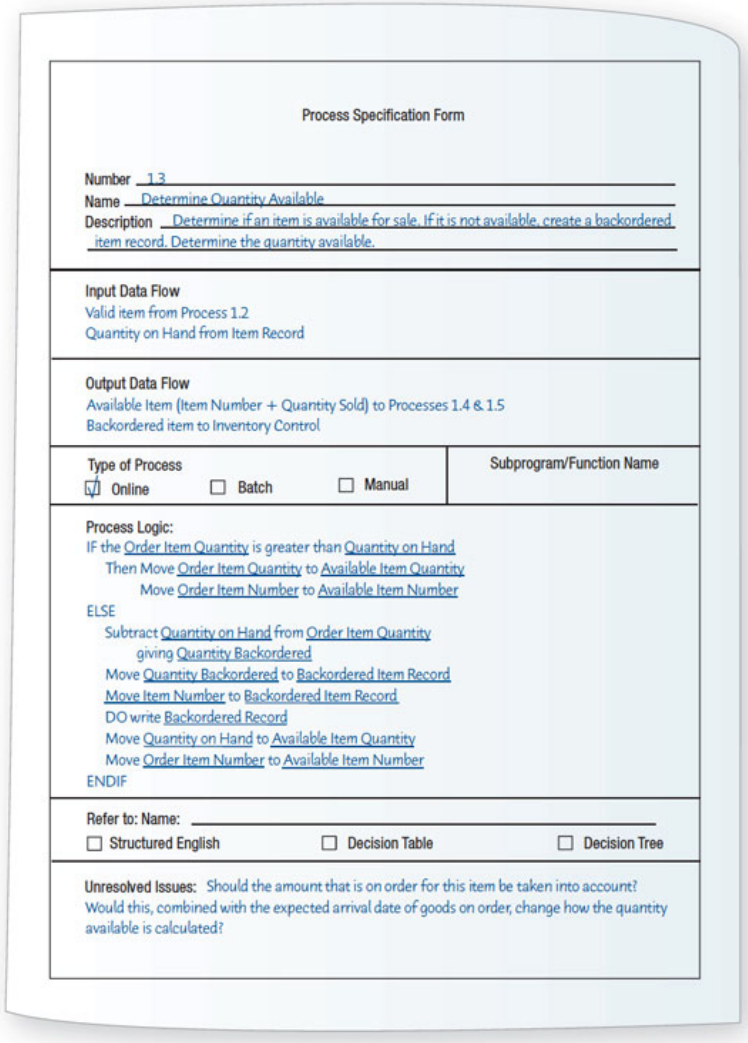
Figure 7.2 An Example of a Completed Process Specification (Mini-Specs) Form



Process Specification Form

Number 1.3
Name Determine Quantity Available
Description Determine if an item is available for sale. If it is not available, create a backordered item record. Determine the quantity available.

Input Data Flow
Valid item from Process 1.2
Quantity on Hand from Item Record

Output Data Flow
Available Item (Item Number + Quantity Sold) to Processes 1.4 & 1.5
Backordered item to Inventory Control

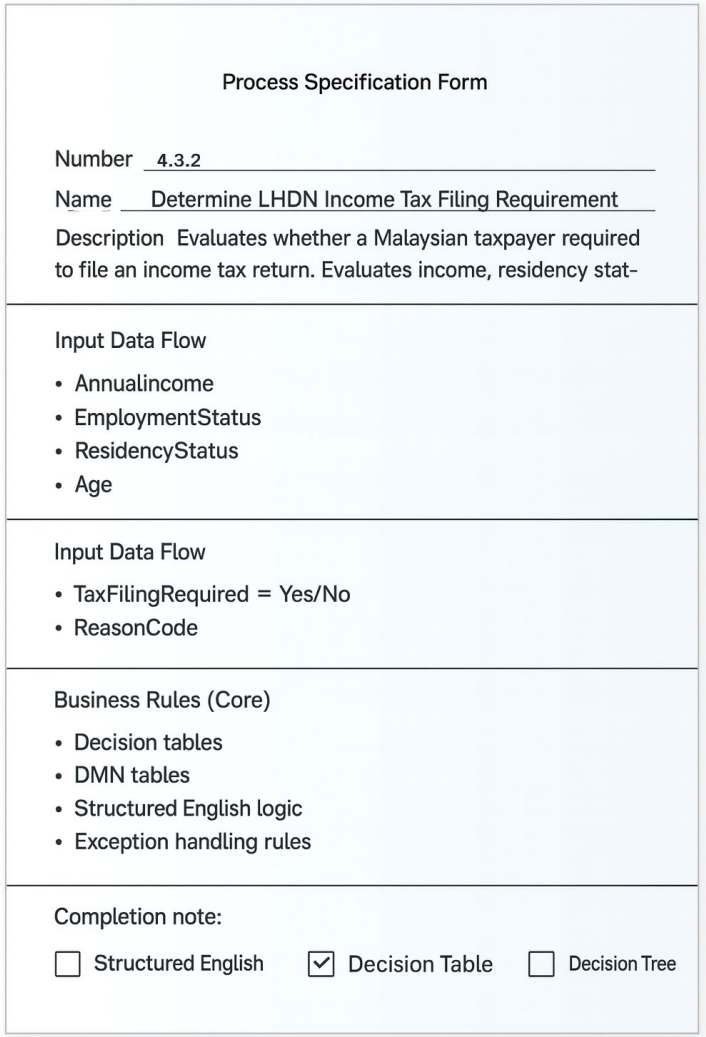
Type of Process
☑ Online ☐ Batch ☐ Manual

Subprogram/Function Name

Process Logic:
IF the Order Item Quantity is greater than Quantity on Hand
Then Move Order Item Quantity to Available Item Quantity
Move Order Item Number to Available Item Number
ELSE
Subtract Quantity on Hand from Order Item Quantity
giving Quantity Backordered
Move Quantity Backordered to Backordered Item Record
Move Item Number to Backordered Item Record
DO write Backordered Record
Move Quantity on Hand to Available Item Quantity
Move Order Item Number to Available Item Number
ENDIF

Refer to: Name:
☐ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: Should the amount that is on order for this item be taken into account? Would this, combined with the expected arrival date of goods on order, change how the quantity available is calculated?



Process Specification Form

Number 4.3.2
Name Determine LHDN Income Tax Filing Requirement
Description Evaluates whether a Malaysian taxpayer required to file an income tax return. Evaluates income, residency stat-

Input Data Flow
- Annualincome
- EmploymentStatus
- ResidencyStatus
- Age

Input Data Flow
- TaxFilingRequired = Yes/No
- ReasonCode

Business Rules (Core)
- Decision tables
- DMN tables
- Structured English logic
- Exception handling rules

Completion note:
☐ Structured English ☑ Decision Table ☐ Decision Tree

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What is Business Rules in Mini-Specs?

- They guide the **decision points** and **actions** inside the process.
- Provide the logic that drives decisions in a process
  - (e.g., approval thresholds, eligibility criteria).
- Define constraints the system must enforce
  - (e.g., “A customer must have an active account to place an order”).
- Clarify conditions and exceptions that shape process flows.
- Ensure the system behaves in a way that is consistent with organizational policies and real-world requirements.

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How Business Rules Drive Mini-Specs

- Business Rules are the **Raw Material** that process specification translate into logic
- Without business rules, there is no process specifications
- Process specs simply *package* business rules into a structured form:
  - ✓ Structured English
  - ✓ Decision tables
  - ✓ Decision trees

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Common Business Rule Formats

- Definitions of business terms
- Business conditions and actions
- Data integrity constraints
- Mathematical and functional derivations
- Logical inferences
- Processing sequences
- Relationships among facts about the business

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Methods for Documenting Process Logic

- There are three classic structured methods used which are:-
  - Structured English
  - Decision Tables
  - Decision Trees
- Each method is suited for different complexity levels.

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Structured English

- Used when the process logic involves formulas or iteration, or when structured decisions are not complex
- A simplified form of pseudocode that uses:
  - Plain English keywords
  - Structured logic constructs
  - Indentation for hierarchy
  - Capitalized control keywords

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Writing Structured English

- Express all logic in terms of sequential structures, decision structures, case structures, or iterations
- Use and capitalize accepted keywords such as IF, THEN, ELSE, DO, and PERFORM
- Indent blocks of statements to show their hierarchy (nesting) clearly
- Underline words or phrases that have been defined in a data dictionary
- Clarify the logical statements

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Rules for Structured English Syntax

- Keep sentences short
- Use business terms (from data dictionary)
- Use indentation to show blocks
- Avoid programming language symbols ({}, ;, etc.)
- State WHAT the system must do (not HOW to code it)

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Figure 7.3 Examples of Structured English Types

Structured English Type	Example
Sequential Structure A block of instructions in which no branching occurs	Action #1 Action #2 Action #3
Decision Structure Only IF a condition is true, complete the following statements; otherwise, jump to the ELSE	IF Condition A is True THEN implement Action A ELSE implement Action B ENDIF
Case Structure A special type of decision structure in which the cases are mutually exclusive (if one occurs, the others cannot)	IF Case #1 Implement Action #1 ELSE IF Case #2 Implement Action #2 ELSE IF Case #3 Implement Action #3 ELSE IF Case #4 Implement Action #4 ELSE print error ENDIF
Iteration Blocks of statements that are repeated until done	DO WHILE there are customers. Action #1 ENDDO

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Advantages of Structured English

- Clarifying the logic and relationships found in human languages
- An effective communication tool, it can be taught to and understood by users in the organization
- Easy for business users to understand
- Very flexible and fast to write
- Good for algorithms and sequences
- Acts as a “bridge” between natural language and decision tables

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Data Dictionary and Process Specification

- The data dictionary is a starting point for creating structured English:
  - Sequence—a simple sequence of statements MOVE, ADD, and SUBTRACT
  - Selection—[] entries become IF…THEN...ELSE statements
  - Iteration { } entries become DO WHILE, DO UNTIL, or PERFORM UNTIL

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Decision Tables

- A table of rows and columns, separated into four quadrants:
  - Conditions
  - Condition alternatives
  - Actions to be taken
  - Rules for executing the actions

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Figure 7.7 Standard Format Used for Presenting a Decision Table

Conditions and Actions	Rules
Conditions **(What to test)**	Condition Alternatives **(Yes/No, values, ranges)**
Actions **(What system does)**	Action Entries **(X marks which action executed)**

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Figure 7.8 Customer Checkout Decision Table

Conditions and Actions	Rules 1	2	3	4
Under $50	Y	Y	N	N
Pays by check with two forms of ID	Y	N	Y	N
Uses credit card	N	Y	N	Y
Complete the sale after verifying signature.	X			
Complete the sale. No signature needed.		X		
Call supervisor for approval.			X	
Communicate electronically with bank for credit card authorization.				X

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Figure 7.9 Constructing a Decision Table for Deciding Which Catalog to Send to Customers Who Order Only from Selected Catalogs

Conditions and Actions	Rules 1	2	3	4	5	6	7	8
Customer ordered from Fall catalog.	Y	Y	Y	Y	N	N	N	N
Customer ordered from Christmas catalog.	Y	Y	N	N	Y	Y	N	N
Customer ordered from specialty catalog.	Y	N	Y	N	Y	N	Y	N
Send out this year's Christmas catalog.		X		X		X		X
Send out specialty catalog.			X				X	
Send out both catalogs.	X				X			

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Developing Decision Tables (1 of 2)

- Determine conditions that affect the decision
- Determine possible actions that can be taken
- Determine condition alternatives for each condition
- Calculate the maximum number of columns in the decision table

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Developing Decision Tables (2 of 2)

- Fill in the condition alternatives
- Complete table by inserting an X where rules suggest actions
- Combine rules where it is apparent
- Check for impossible situations
- Rearrange to make more understandable

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Developing Decision Tables Step 1

- Determine the number of conditions that may affect the decision
- Combine rows that overlap, such as conditions that are mutually exclusive
- The number of conditions becomes the number of rows in the top half of the decision table

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Developing Decision Tables Step 2

- Determine the number of possible actions that can be taken
- That number becomes the number of rows in the lower half of the decision table

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Developing Decision Tables Step 3

- Determine the number of condition alternatives for each condition
- In the simplest form of decision table, there would be two alternatives (Y or N) for each condition
- An extended entry table may have many alternatives for each condition
- Make sure that all possible values for the condition are included

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Developing Decision Tables Step 4

- Calculate the maximum number of columns in the decision table by multiplying the number of alternatives for each condition
- If there were four conditions and two alternatives (Y or N) for each of the conditions, there would be 16 possibilities

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Developing Decision Tables Step 5

- Fill in the condition alternatives
- Start with the first condition and divide the number of columns by the number of alternatives for that condition
- If there are 16 columns and two alternatives (Y or N), then 16 divided by 2 is 8
- Choose one of the alternatives, say Y, and write it in the first eight columns
- Finish by writing N in the remaining eight columns

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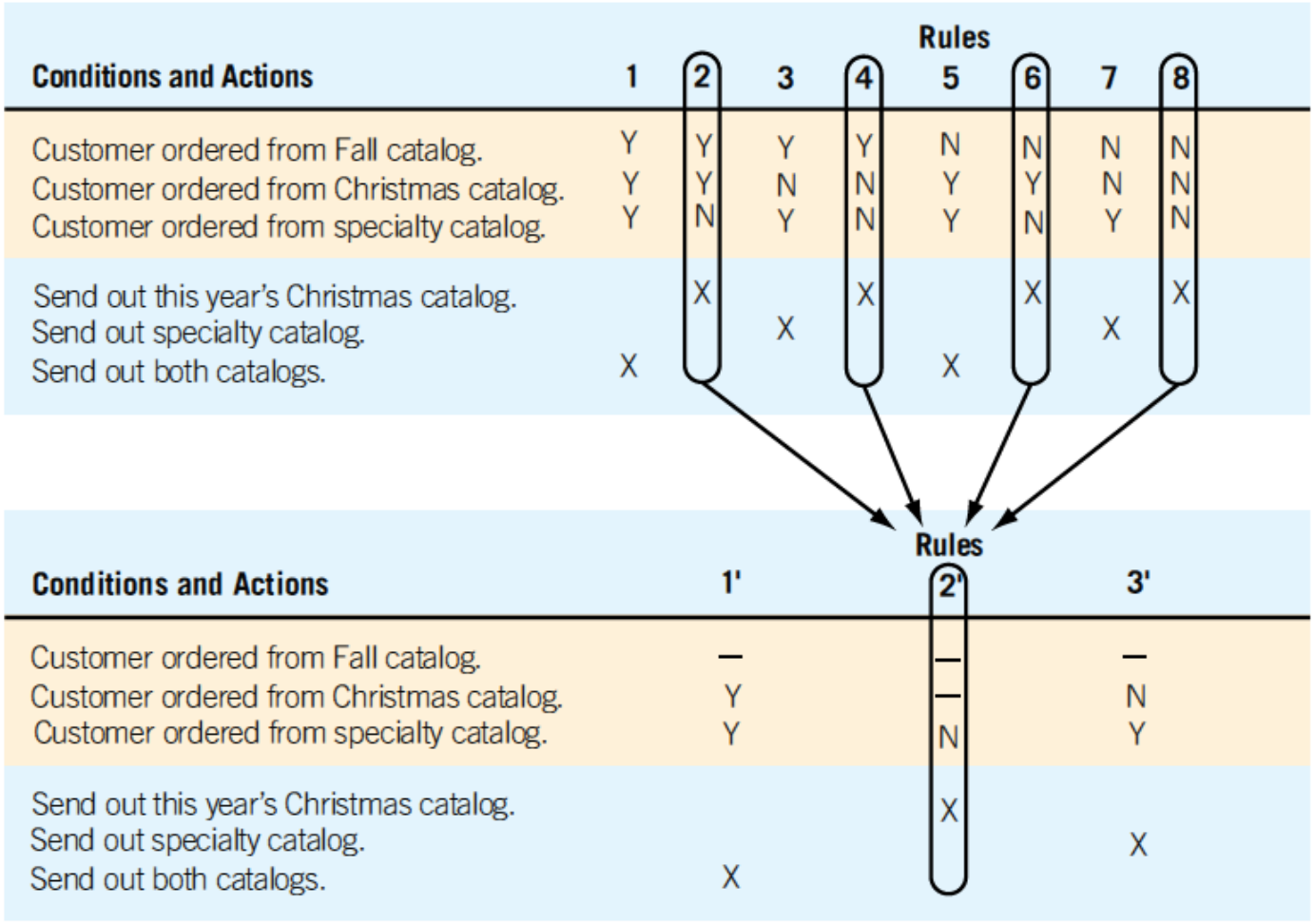
Figure 7.10 Combining Rules to Simplify the Decision Table



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Checking for Completeness and Accuracy

- Four main problems:
  - Incompleteness
  - Impossible situations
  - Contradictions
  - Redundancy

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Figure 7.12 Checking the Decision Table for Impossible Situations

Conditions and Actions	Rules 1	2	3	4
Salary > $50,000/year	Y	Y	N	N
Salary < $2,000/month	Y	N	Y	N
Action 1				
Action 2				

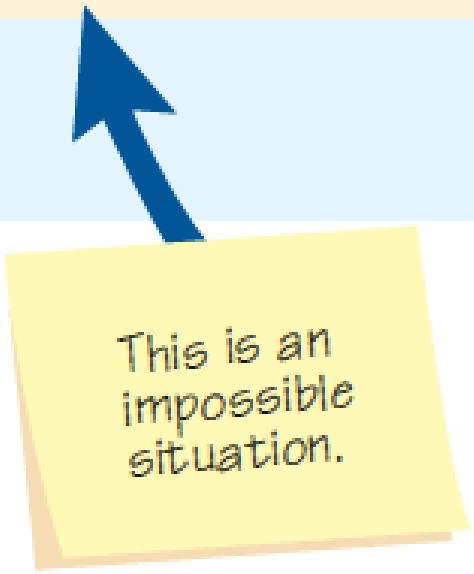

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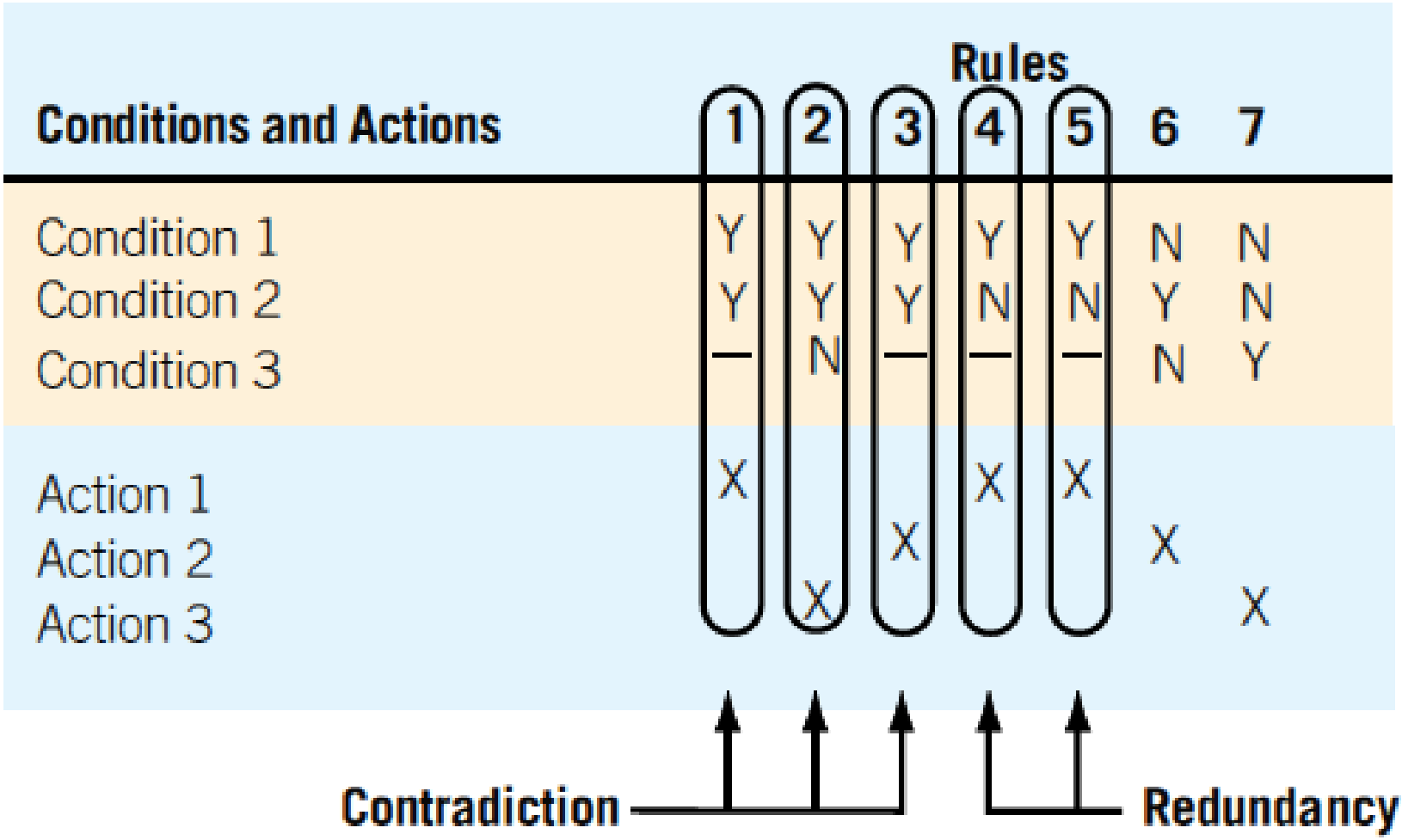
Figure 7.13 Checking for Contradictions and Redundancy



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Decision Table Advantages

- Help the analysis ensure completeness
- Easy to check for possible errors
  - Impossible situations
  - Contradictions
  - Redundancy

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Decision Trees

- Decision trees are used when complex branching occurs in a structured decision process
- Trees are also useful when it is essential to keep a string of decisions in a particular sequence

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Decision Trees Format

- **How is Decision Trees look?** It visually representing sequential decisions that shows:
  - Branching conditions
  - Paths
  - Leaves (end decisions)
- A decision tree structure normally includes the following:
  - Node = condition
  - Branch = Yes/No, True/False, or value
  - Leaf = action/output

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When to use Decision Tree?

- Order of decisions matters
- Branches differ significantly
- Logic resembles a flow
- Some conditions apply only on certain paths
- There are procedural steps mixed with decisions
- You want an intuitive visual representation

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Drawing Decision Trees

- Identify all conditions and actions and their order and timing (if they are critical)
- Begin building the tree from left to right, making sure you list all possible alternatives before moving to the right

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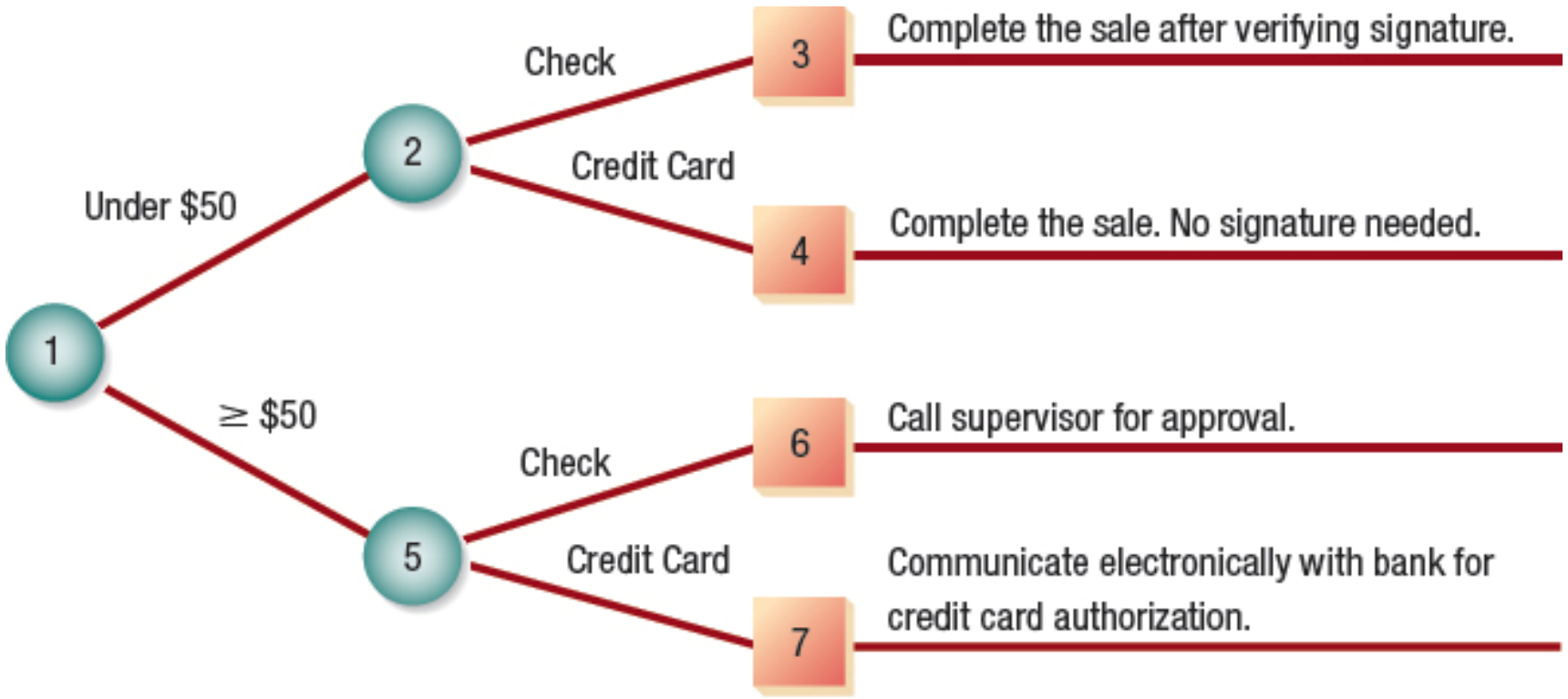
Figure 7.14 Drawing a Decision Tree to Show the Noncash Purchase Approval Actions for a Department Store



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Decision Tree Advantages

- The order of checking conditions and executing actions is immediately noticeable
- Conditions and actions of decision trees are found on some branches but not on others
- Compared to decision tables, decision trees are more readily understood by others in the organization

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Selecting a Structured Decision Analysis Technique

- Use structured English when there are many repetitious actions or when communication to end users is important
- Use decision tables when a complex combination of conditions, actions, and rules are found or you require a method that effectively avoids impossible situations, redundancies, and contradictions
- Use decision trees when the sequence of conditions and actions is critical or when not every condition is relevant to every action (the branches are different)

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When to Choose Which Method

Structured English – Use When:

- Logic is simple
- Flow is mostly squential
- Conditions a few
- Human readiability required
- Early-stage requirement drafting

Examples:

- Calculate shipping fee
- Validate form data
- Apply memberstip discount

Decision Tables (DMN) – Use When:

- Logic depends on multiple conditions
- Many combinations of inputs
- High risk of errors if unchecked
- Rules must be complete, auditable, versioned
- Stakeholders require clarity

Examples:

- LHDN lax filing rules
- Insurance premium rules
- Eligibility testing

Decision Trees Use When:

- Logic is sequential
- Branches differ significanty
- Some paths require unique conditions
- Visual clarity is needed

Examples:

- Bank loan approval
- Customer service routing
- Medical triage
- Fraud detection escalation

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Structured English vs Decision Table vs Decision Tree

Criteria	Structured English	Decision Table (DMN)	Decision Tree
Complexity Level	Low → Medium	Medium → High	Medium
Best For	Sequential logic, simple rules	Multi-condition, exhaustive rules	Branching logic, order-dependent
Visibility	Text-based	Tabular, formal	Highly visual
Completeness Check	Weak	Very strong	Medium
Detect Contradictions	Manual only	Automatic (DMN engines)	Medium
Business User Friendly?	Yes	Yes	Very
Machine-Executable?	No (but AI can convert)	Yes	No (but convertible)
Industry Usage	Requirements writing	Compliance, banking, insurance	Customer flows, triage systems
Real-World Example	Fee calculation	Tax rules, eligibility checks	Loan approval workflows

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Summary (1 of 3)

- Process specifications
- Decision analysis
  - Structured English
    - Logic is expressed in sequential structures, decision structures, case structures, or iterations

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Summary (2 of 3)

- Decision tables
  - Four quadrants are used to:
    - Describe the conditions
    - Identify possible decision alternatives
    - Indicate which actions should be performed
    - Describe the actions
- Decision trees
  - Consist of nodes and branches

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Summary (3 of 3)

- Decision analysis advantages
  - Structured English is useful when many actions are repeated and when communicating with others is important
  - Decision tables provide complete analysis of complex situations while limiting the need for change attributable to impossible situations, redundancies, or contradictions
  - Decision trees are important when proper sequencing of conditions and actions is critical and when each condition is not relevant to each action

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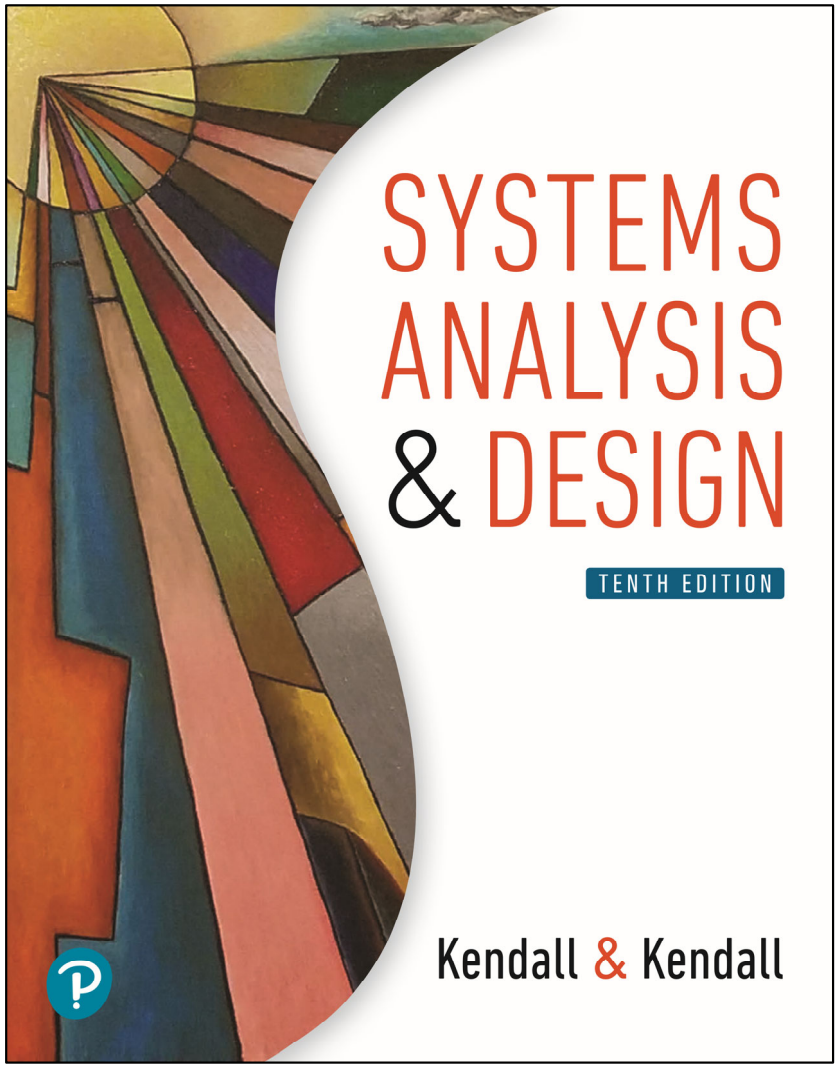
Systems Analysis & Design

Tenth Edition



Chapter 8

Object-Oriented Systems Analysis and Design Using UML

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Learning Objectives (1 of 2)

8.1 Understand what object-oriented systems analysis and design is and appreciate its usefulness

8.2 Comprehend the concepts of Unified Modeling Language (UML), the standard approach for modeling a system in the object-oriented world

8.3 Apply the steps used in UML to break down the system into a use case model and then a class model

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Learning Objectives (2 of 2)

8.4 Diagram systems with the UML toolset so they can be described and properly designed

8.5 Document and communicate the newly modeled object-oriented system to users and other analysts

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Object-Oriented Analysis and Design (OOAD)

- Everything is a thing (an object).
- Each of these objects has:
  - **attributes** (data about it)
  - **behaviors** (things it can do)
  - Just like people.

- OOAD = Object-Oriented Analysis (OOA) + Object-Oriented Design (OOD)

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Object-Oriented Analysis and Design (OOAD)

- OOA is the process of understanding a real-world problem by **identifying** the **objects involved and the relationships between them**.
- Focus on the problem, not the code yet
- We design the system using these objects, we use OOD.
- OOD is designing how these **objects** will **work together** as a software system.
- OOAD → A way of thinking about systems as a collection of interacting objects — just like how things interact in real life.

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What is Object-Oriented Analysis and Design (OOAD)?

- OOAD is a software development approach that models a system as a group of objects.
- Each object represents some real-world entity within the system being modeled, and has its own attributes and operations
- Object-oriented systems describe **entities as objects** where **objects are part of a general concept** called **classes**
- A range of models can be created using such objects to reflect the structure and behaviors of the system

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Object-Oriented Analysis and Design (1 of 2)

- Works well in situations where complicated systems are undergoing continuous maintenance, adaptation, and design
- Offer an approach that facilitates logical, rapid, and thorough methods for creating new systems responsive to a changing business landscape
- Objects, classes are reusable
- The Unified Modeling Language (UML) is an industry standard for modeling object-oriented systems.

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Why Object-Oriented Analysis and Design
(2 of 2)

- Reusability
  - Recycling of program parts should reduce the costs of development in computer-based systems
- Maintaining systems
  - Making a change in one object has a minimal impact on other objects

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Major Topics

- Object-oriented concepts
- CRC cards and object think
- Unified Modeling Language
- Use case and other UML diagrams
- Packages
- Using UML

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Object-Oriented Concepts

- Objects
- Classes
- Inheritance

Inheritance

(IS-A Relationship)

A new class (Child) reuses and extends another class (Parent). Helps code reuse + avoids duplication.

Example: DeliveryRider IS-A Staff

Relationships Between Classes

- Association — works with
- Aggregation — has-a a(weak whole/part)
- Composition — owns-a (strong whole/parart)
- Generalization — parent -+ child

These show how classes interact to complete a workflow.

Encapsulation

(**Data** Hiding. Differecnt Behaviors)

- Attributes are protected
- Access only through methods
- Prevents misuse & mainthains data integrity

"You cannnot simply open a staffs salary variable – you must use approved methods."

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Objects

- Persons, places, or things that are relevant to the system being analyzed
- May be customers, items, orders, and so on
- May be GUI displays or text areas on a display

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Classes

- Defines the set of shared attributes and behaviors found in each object in the class
- Should have a name that differentiates it from all other classes
- **Instantiate** is when an object is **created** from a class
- **An attribute** describes some property that is possessed by all objects of the class
- **A method** is an action that can be requested from any object of the class

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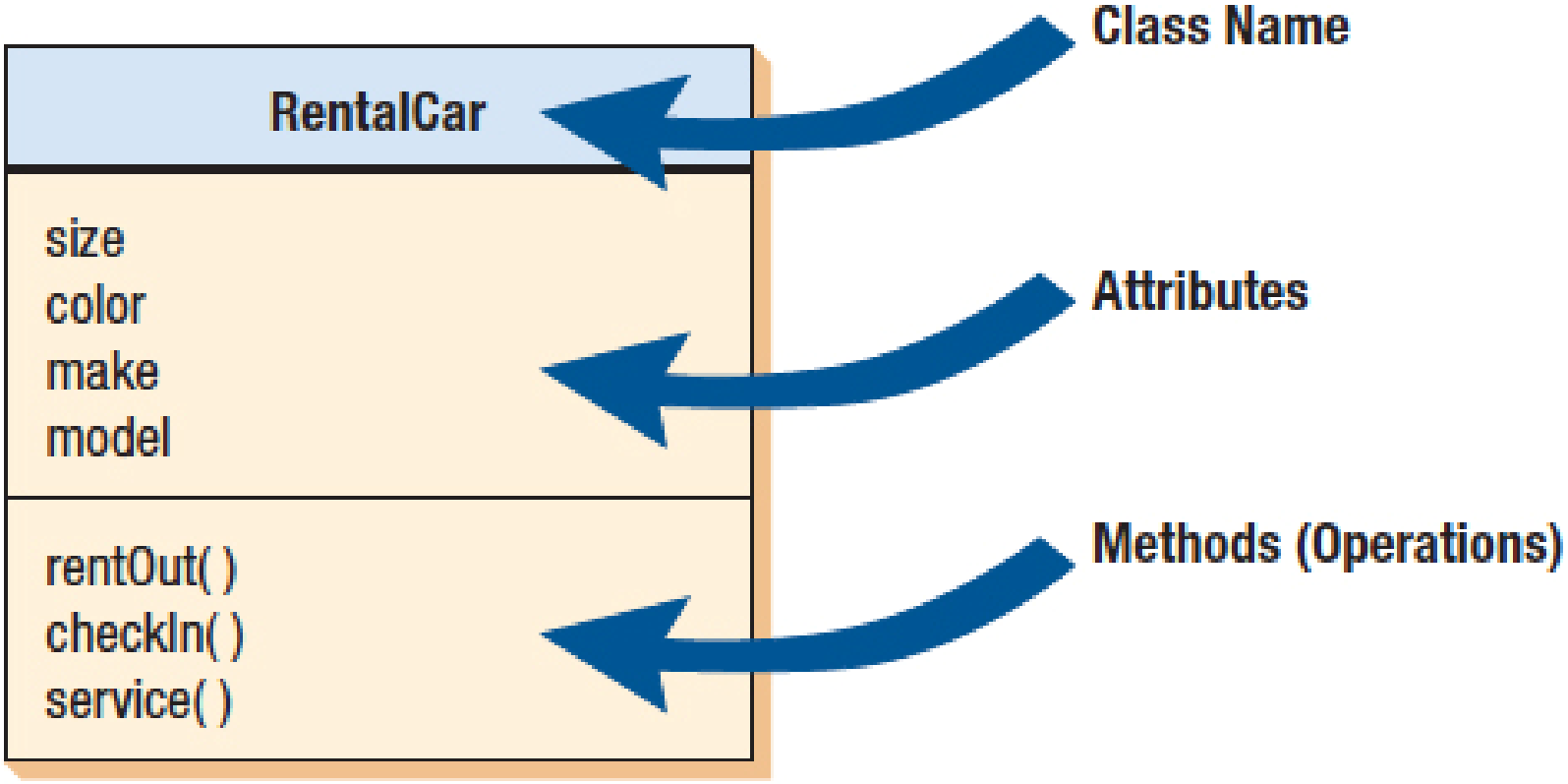
Figure 8.1 An Example of a UML Class



- Class names are usually nouns or short phrases and begin with an **uppercase letter.**

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Inheritance (1 of 2)

- When a derived class inherits all the attributes and behaviors of the base class
- Reduces programming labor by using common objects easily
- A feature only found in object-oriented systems

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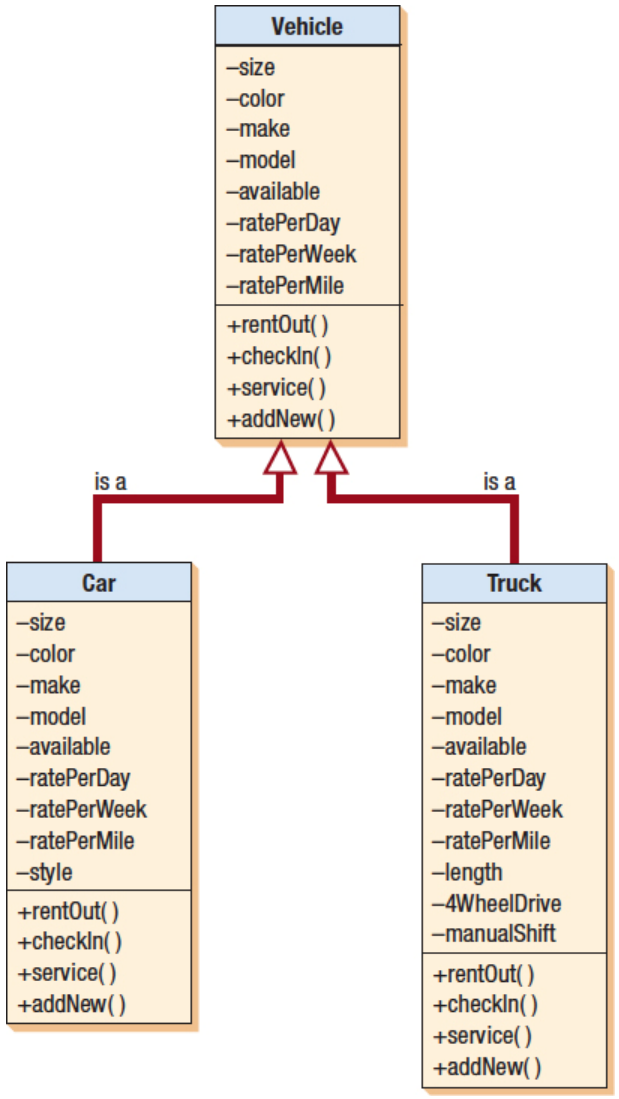
Figure 8.2 A Class Diagram Showing Inheritance

Car and truck are specific examples of vehicles and inherit the characteristics of the more general class vehicle.



- The parent class is called the base class.
- The child class is called the derived class.

- Allows the programmer to define once but use many times.

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CRC Cards and Object Think

- CRC: Class–Responsibility–Collaborator
  - Class
    - The name of the object (e.g., Order, Customer, Rider).
  - Responsibilities
    - What the class must do:
      - ✓ Actions
      - ✓ Decisions
      - ✓ Information it must know
  - Collaborators
    - Other classes this class depends on to fulfill its responsibilities.

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CRC Cards and Object Think

- CRC is a **lightweight, brainstorming technique** used to identify and refine potential classes in an object-oriented system **before drawing UML diagrams**.
- CRC cards are used to represent the responsibilities of classes and the interaction between the classes
- CRC cards are created based on scenarios that outline system requirements.

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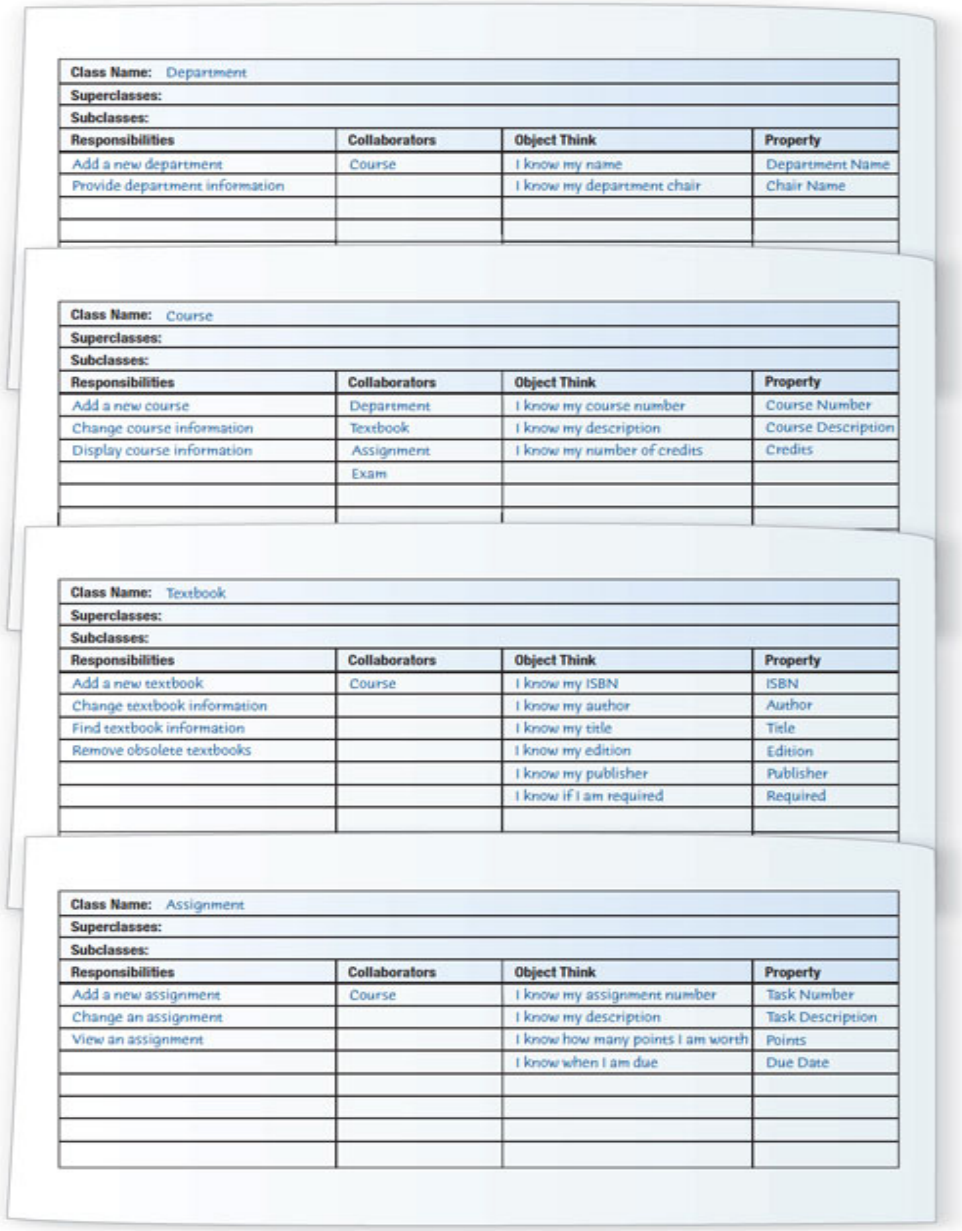
Figure 8.3 Four CRC Cards



Class Name: Department			
Superclasses:			
Subclasses:			
Responsibilities	**Collaborators**	**Object Think**	**Property**
Add a new department	Course	I know my name	Department Name
Provide department information		I know my department chair	Chair Name

Class Name: Course			
Superclasses:			
Subclasses:			
Responsibilities	**Collaborators**	**Object Think**	**Property**
Add a new course	Department	I know my course number	Course Number
Change course information	Textbook	I know my description	Course Description
Display course information	Assignment	I know my number of credits	Credits
	Exam		

Class Name: Textbook			
Superclasses:			
Subclasses:			
Responsibilities	**Collaborators**	**Object Think**	**Property**
Add a new textbook	Course	I know my ISBN	ISBN
Change textbook information		I know my author	Author
Find textbook information		I know my title	Title
Remove obsolete textbooks		I know my edition	Edition
		I know my publisher	Publisher
		I know if I am required	Required

Class Name: Assignment			
Superclasses:			
Subclasses:			
Responsibilities	**Collaborators**	**Object Think**	**Property**
Add a new assignment	Course	I know my assignment number	Task Number
Change an assignment		I know my description	Task Description
View an assignment		I know how many points I am worth	Points
		I know when I am due	Due Date

**** Each CRC card represents one class**

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Interacting during a CRC Session

- Identify all the classes you can - when all classes have been identified, the analysts can then compile them, weed out the illogical ones, and write each one on its own card.
- Create scenarios - actual walkthroughs of system functions.
- Identify and refine responsibilities - The responsibilities will eventually evolve into methods in the UML.
- The think statements can later be converted to attributes in the UML.

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The Unified Modeling Language (UML) Concepts and Diagrams

- UML provides a standardized set of tools to document the analysis and design of a software system.
- Things – the objects
- Relationships – the glue that holds things together
- Diagrams – categorized as either structural or behavioral

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Things

- Things are the primary elements of the UML (objects).
- In UML, **structural things** and **behavioral things** are the two broad categories used to describe a system.
- Structural things represent the **static**, tangible parts of the model — the nouns of the system. ***(What the system is)***
- Used in Diagrams Like
  - Class diagram
  - Object diagram
  - Component diagram
  - Deployment diagram
  - Package diagram
- They define architecture, data, and static relationships.

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Things

- Behavioral things represent the dynamic, action-oriented aspects — the verbs of the system. ***(What the system does)***
- Behavioral things describe how things work - interactions and state machines
- Used in Diagrams Like
  - Sequence diagram
  - Activity diagram
  - State machine diagram
- They describe actions, events, changes, and workflows over time.
- Group things → Used to define boundaries
- Annotational things → Can add notes to the diagrams

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Relationships (1 of 2)

- Structural relationships → describe static, long-lasting connections between model elements. *(How things are related).*
- Tie things together in structural diagrams

- Behavioral relationships → describe dynamic interactions *(How elements communicate, trigger, or depend on actions).*
- Used in behavioral diagrams

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Types of Structural Relationships

Relationship	Meaning
Association	A general connection between classes (A uses/knows B)
Aggregation	Whole–part relationship (weak ownership)
Composition	Strong whole–part (lifecycle dependency)
Generalization	Inheritance (is-a relationship)
Realization	A class implements an interface

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Type of Behavioral Relationships

Relationship	Meaning
Dependency	A temporary or dynamic reliance (A depends on B)
Usage (a subtype of dependency)	One element uses another at runtime
Message sending	A sends a message to B (sequence diagrams)
Control flow / object flow	In activity diagrams, how actions are ordered
Event → state transition	In state machines, behavior change based on events

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Diagrams

- Structural diagrams
  - Used to describe the relation between classes
  - Structure relationships describe the architecture — what is connected to what.
- Behavioral diagrams
  - Used to describe the interaction between people (actors) and a use case (how the actors use the system)
  - Behavior relationships describe how elements interact over time.

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Structural Diagrams

- Class diagrams
- Object diagrams
- Component diagrams
- Deployment diagrams

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Behavioral Diagrams

- Use case diagrams
- Sequence diagrams
- Collaboration diagrams
- Statechart diagrams
- Activity diagrams

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Figure 8.4 An Overall View of UML and Its Components: Things, Relationships, and Diagrams (1 of 2)

UML Category	UML Elements	Specific UML Details
Things	Structural Things	Classes Interfaces Collaborations Use Cases Active Classes Components Nodes
	Behavioral Things	Interactions State Machines
	Grouping Things	Packages
	Annotational Things	Notes
Relationships	Structural Relationships	Dependencies Aggregations Associations Generalizations

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Figure 8.4 An Overall View of UML and Its Components: Things, Relationships, and Diagrams (2 of 2)

UML Category	UML Elements	Specific UML Details
	Behavioral Relationships	Communicates Includes Extends Generalizes
Diagrams	Structural Diagrams	Class Diagrams Component Diagrams Deployment Diagrams
	Behavioral Diagrams	Use Case Diagrams Sequence Diagrams Communication Diagrams Statechart Diagrams Activity Diagrams

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Commonly Used UML Diagrams (1 of 2)

- Use case diagram
  - Describing how the system is used
  - The starting point for UML modeling
- Use case scenario
  - A textual description of a single use case execution.
  - Main flow + alternative flows
- Activity diagram
  - Illustrates the overall flow of activities
  - Shows workflow, process, or business logic from start to finish.

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Commonly Used UML Diagrams (2 of 2)

- Sequence diagrams
  - Show the sequence of activities and class relationships
  - Shows time-ordered interactions between objects.
- Class diagrams
  - Show classes and relationships
  - Shows the static structure of the system.
- Statechart (State Machine) diagrams
  - Show the state transitions.
  - Models the lifecycle of an object (or system) through states and transitions.

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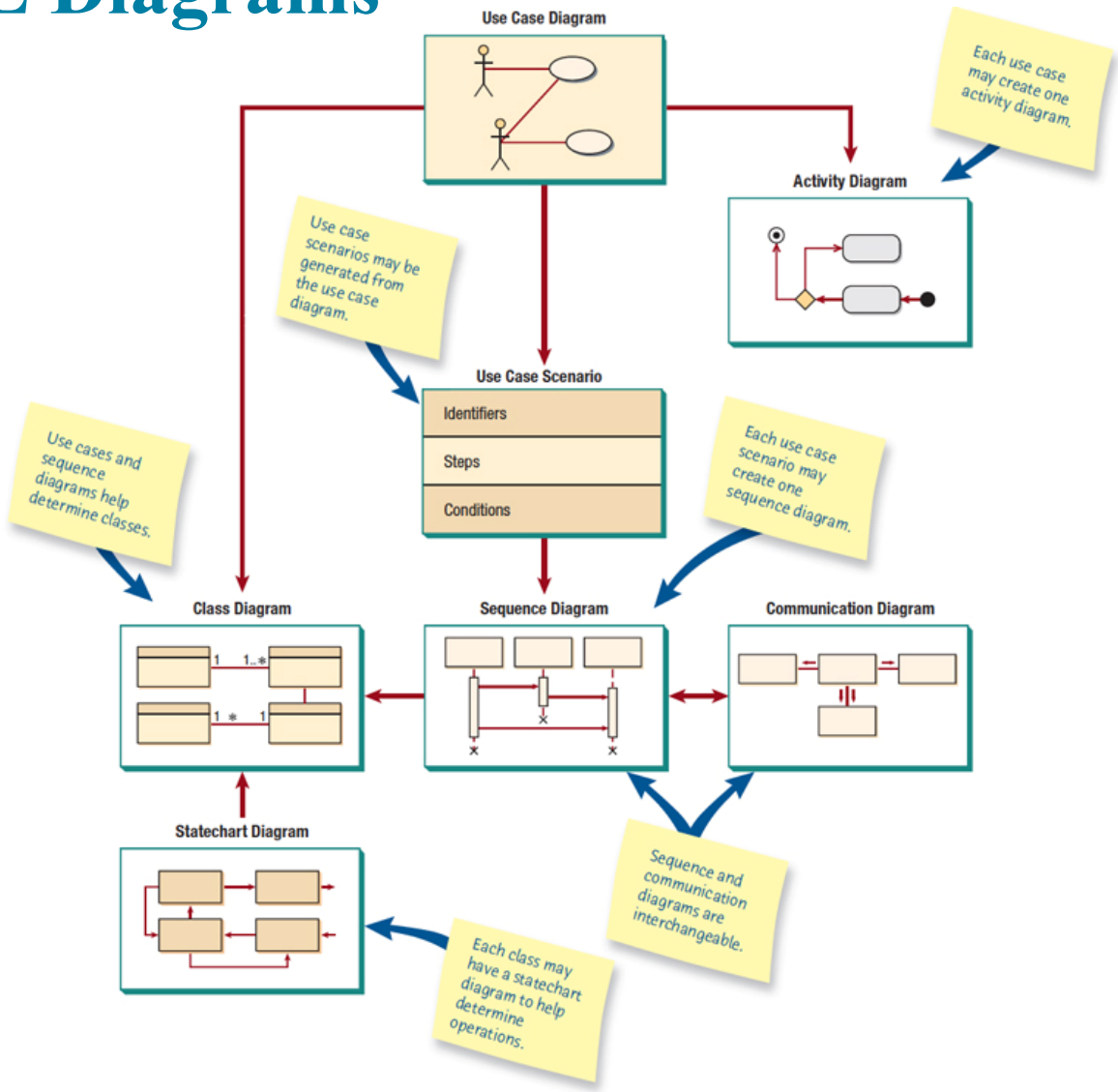
Figure 8.5 An Overview of UML Diagrams Showing How Each Diagram Leads to the Development of Other UML Diagrams



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1. Use Case Modeling

- Describes what the system does, without describing how the system does it
- Based on the interactions and relationships of individual use cases
- Use case describes
  - Actor
  - Event
  - Use case

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Figure 8.6 A Use Case Example of Student Enrollment

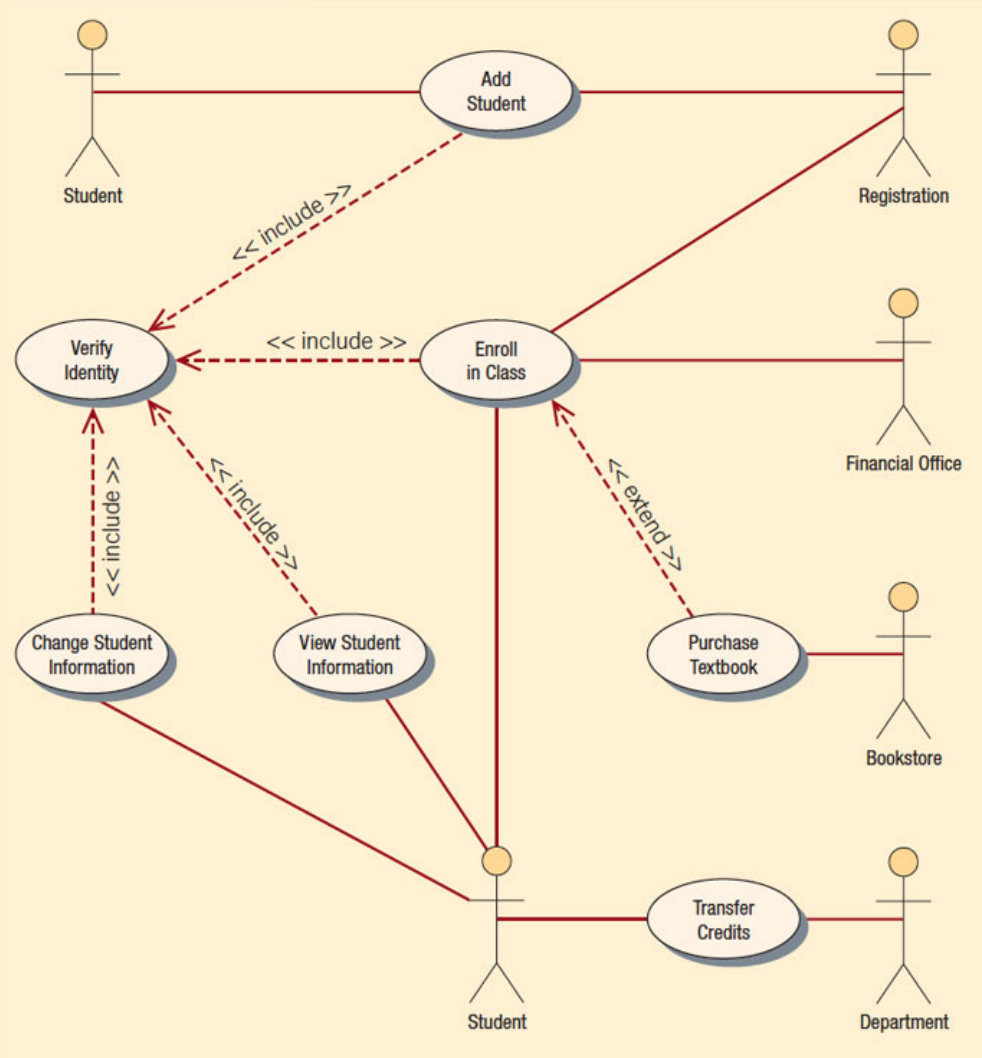

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Figure 8.7 A Use Case Scenario is Divided into Three Sections: Identification and Initiation, Steps Performed, and Conditions, Assumptions, and Questions

Use case name:	Change Student Information	**UniqueID:**	Student UC 005
Area:	Student System		
Actor(s):	Student		
Description:	Allow student to change his or her own information, such as name, home address, home telephone, campus address, campus telephone, cell phone, and other information using a secure website.		
Triggering Event:	Student uses Change Student Information website, enters student ID and password, and clicks the **Submit** button.		
Trigger type:	☒ External ☐ Temporal		
Steps Performed (Main Path)		**Information for Steps**	
1. Student logs on to the secure web server.		Student ID, Password	
2. Student record is read and password is verified.		Student Record, Student ID, Password	
3. Current student personal information is displayed on the Change Student web page.		Student Record	
4. Student enters changes on the Change Student Web form and clicks **Submit** button.		Change Student Web Form	
5. Changes are validated on the web server.		Change Student Web Form	
6. Change Student Journal record is written.		Change Student Web Form	
7. Student record is updated on the Student Master.		Change Student Web Form, Student Record	
8. Confirmation web page is sent to the student.		Confirmation Page	
Preconditions:	Student is on the Change Student Information web page.		
Postconditions:	Student has successfully changed personal information.		
Assumptions:	Student has a browser and a valid user ID and password.		
Requirements Met:	Allow students to be able to change personal information using a secure website.		
Outstanding Issues:	Should the number of times a student is allowed to logon be controlled?		
Priority:	Medium		
Risk:	Medium		

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2. Activity Diagrams

- Show the sequence of activities in a process, including sequential and parallel activities, and decisions that are made
- Symbols
  - Rectangle with rounded ends
  - Arrow
  - Diamond
  - Long, flat rectangle
  - Filled-in circle
  - Black circle surrounded by a white circle
  - Swimlanes

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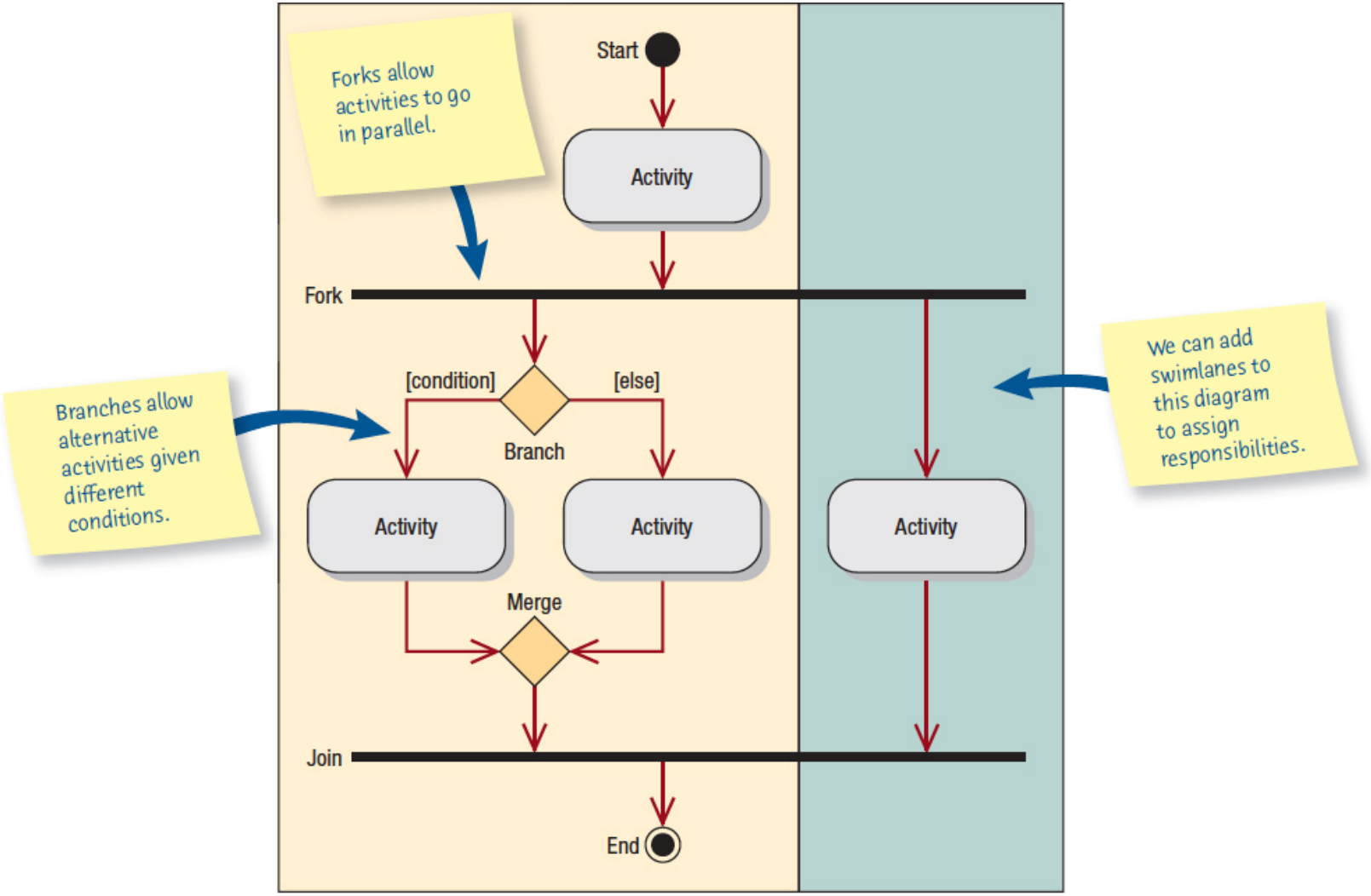
Figure 8.8 Specialized Symbols are Used to Draw an Activity Diagram



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Creating Activity Diagrams

- Created by asking what happens first, what happens second, and so on
- Must determine what activities are done in sequence or in parallel
- The sequence of activities can be determined from physical data flow diagrams
- Can be created by examining all the scenarios for a use case

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Swimlanes

- Useful to show how the data must be transmitted or converted (e.g.: Web → server → from server →mainframe).
- Help to divide up the tasks in a team
- Makes the activity diagram one that people want to use to communicate with others
- The use of unique symbols and swimlanes makes this diagram one that people want to use to communicate with others.

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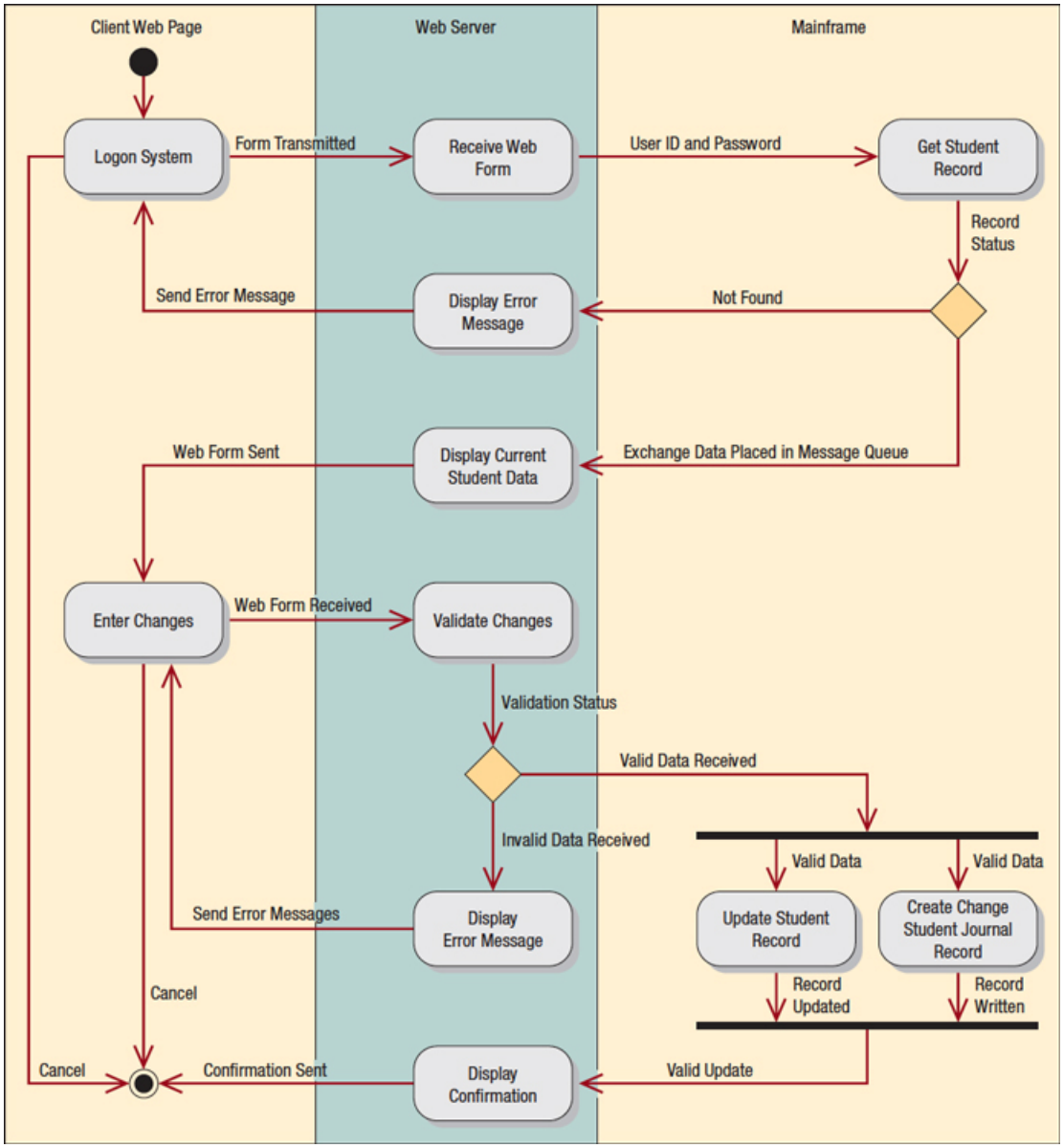
Figure 8.9 This Activity Diagram Shows Three Swimlanes: Client Web Page, Web Server, and Mainframe



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Activity Diagrams and Test Plans

- Activity diagrams may be used to construct test plans
- Each event must be tested to see if the system goes to the next state
- Each decision must be tested

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Activity Diagrams Not Created for All Use Cases

- Use an activity diagram when:
  - It helps to understand the activities of a use case
  - The flow of control is complex
  - There is a need to model workflow
  - When all scenarios for a use case need to be shown

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3. Sequence Diagrams

- Illustrate a succession of interactions between classes or object instances over time *(explaining object collaboration)*
- Often used to show the processing described in use case scenarios *(designing methods and logic)*
- Used to show the overall pattern of the activities or interactions in a use case *(explaining object collaboration)*
- Each use case scenario may create one sequence diagram, although sequence diagrams are not always created for minor scenarios.

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Figure 8.10 Specialized Symbols Used to Draw a Sequence Diagram

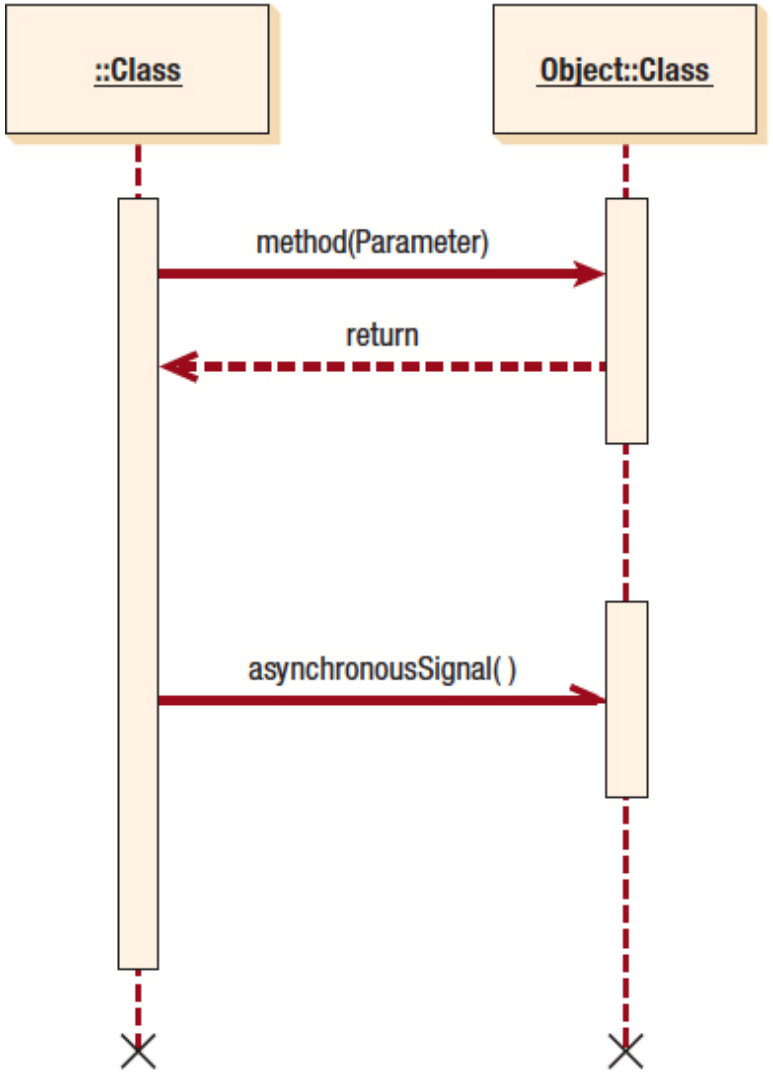

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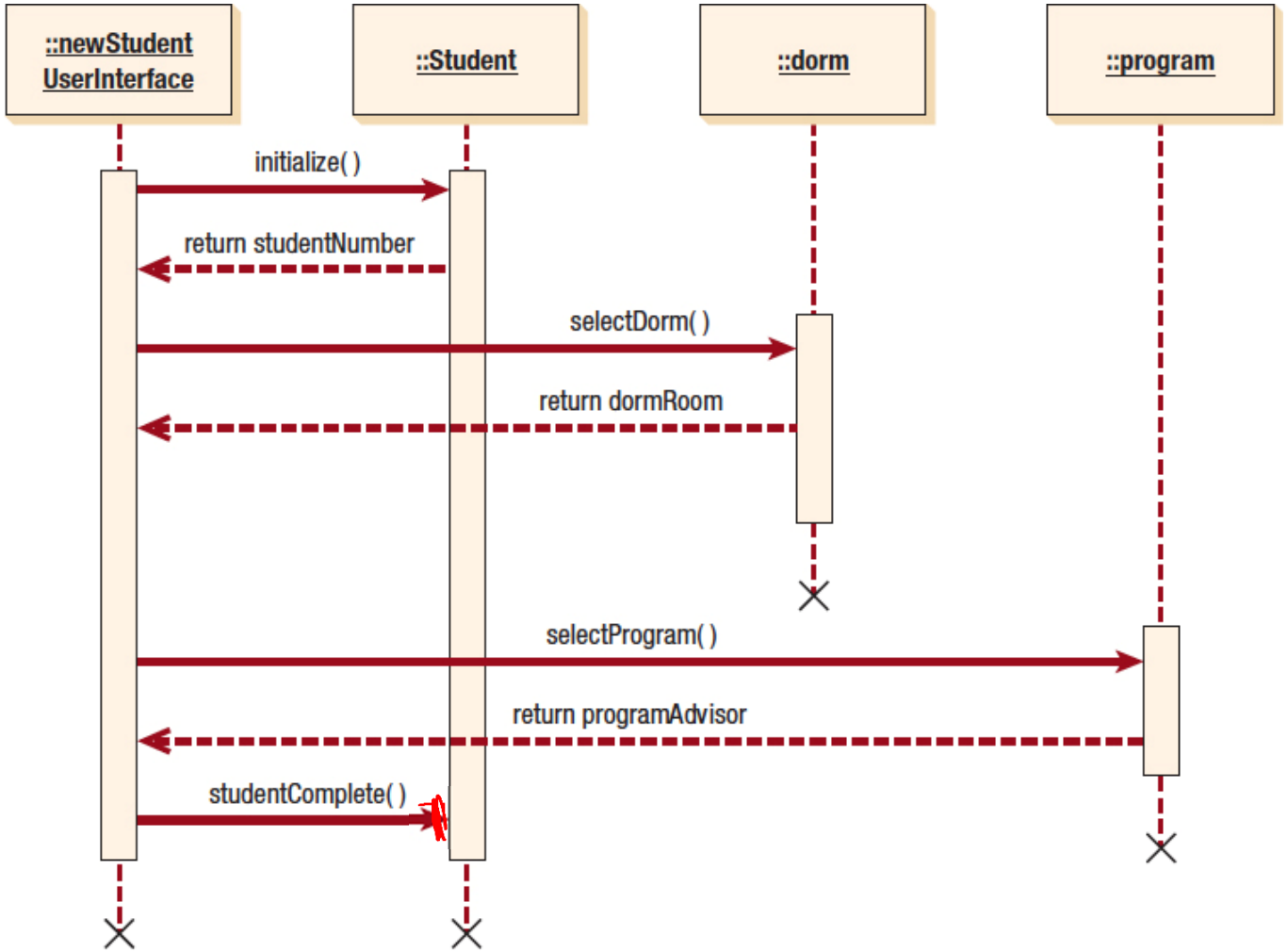
Figure 8.11 A Sequence Diagram for Student Admission: Sequence Diagrams Emphasize the Time Ordering of Messages



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4. Communication Diagrams

- Describes the interactions of two or more things in the system that perform a behavior that is more than any one of the things can do alone
- Shows the same information as a sequence diagram, but may be more difficult to read
- Emphasizes the organization of objects
- Made up of objects, communication links, and the messages that can be passed along those links

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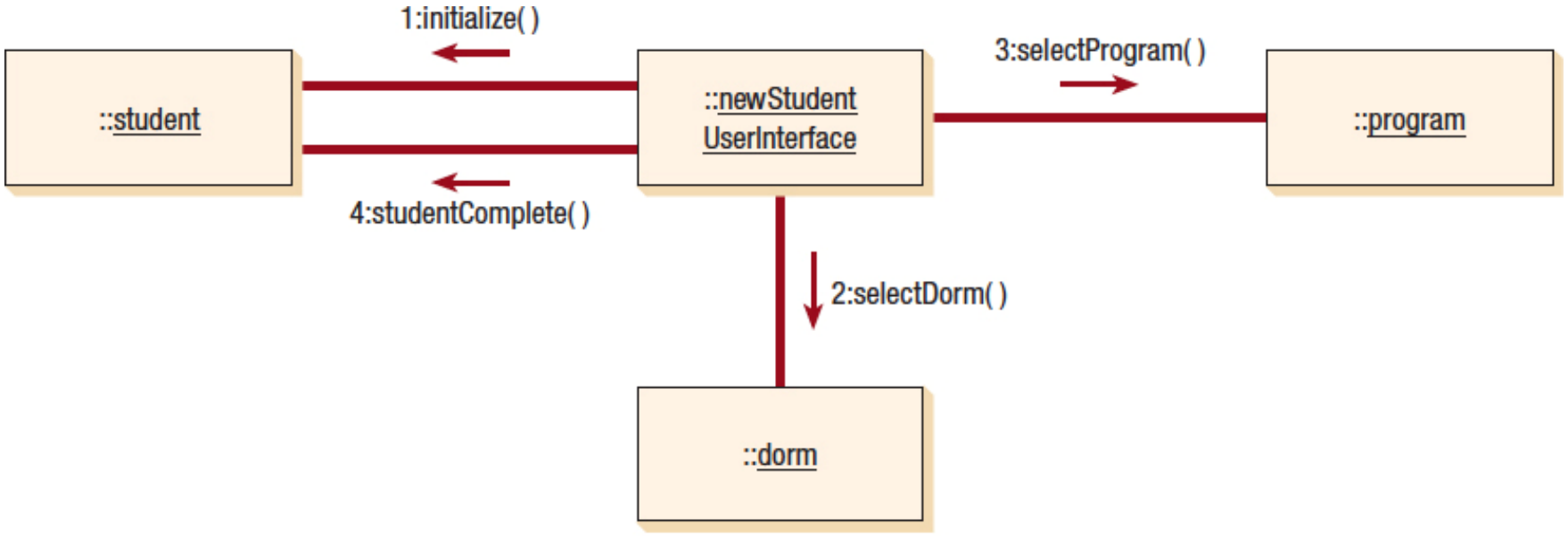
Figure 8.12 A Communication Diagram for Student Admission

Communication diagrams show the same information that is depicted in a sequence diagram but emphasize the organization of objects rather than the time ordering.



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5. Class Diagrams (1 of 2)

- Show the static features of the system and do not represent any particular processing
- Show the nature of the relationships between classes
- Show data storage requirements as well as processing requirements

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Class Diagrams (2 of 2)

- Classes —represented by a rectangle on the class diagram.
- Attributes —what the class knows about the characteristics of the objects:
  - Private → only available in the object
  - Public → visible to other objects outside its class
  - Protected →hidden from all classes except immediate subclasses
- Methods —small sections of code that work with the attributes:
  - Standard →basic things that all classes of object know how to do
  - Custom → designed for a specific class

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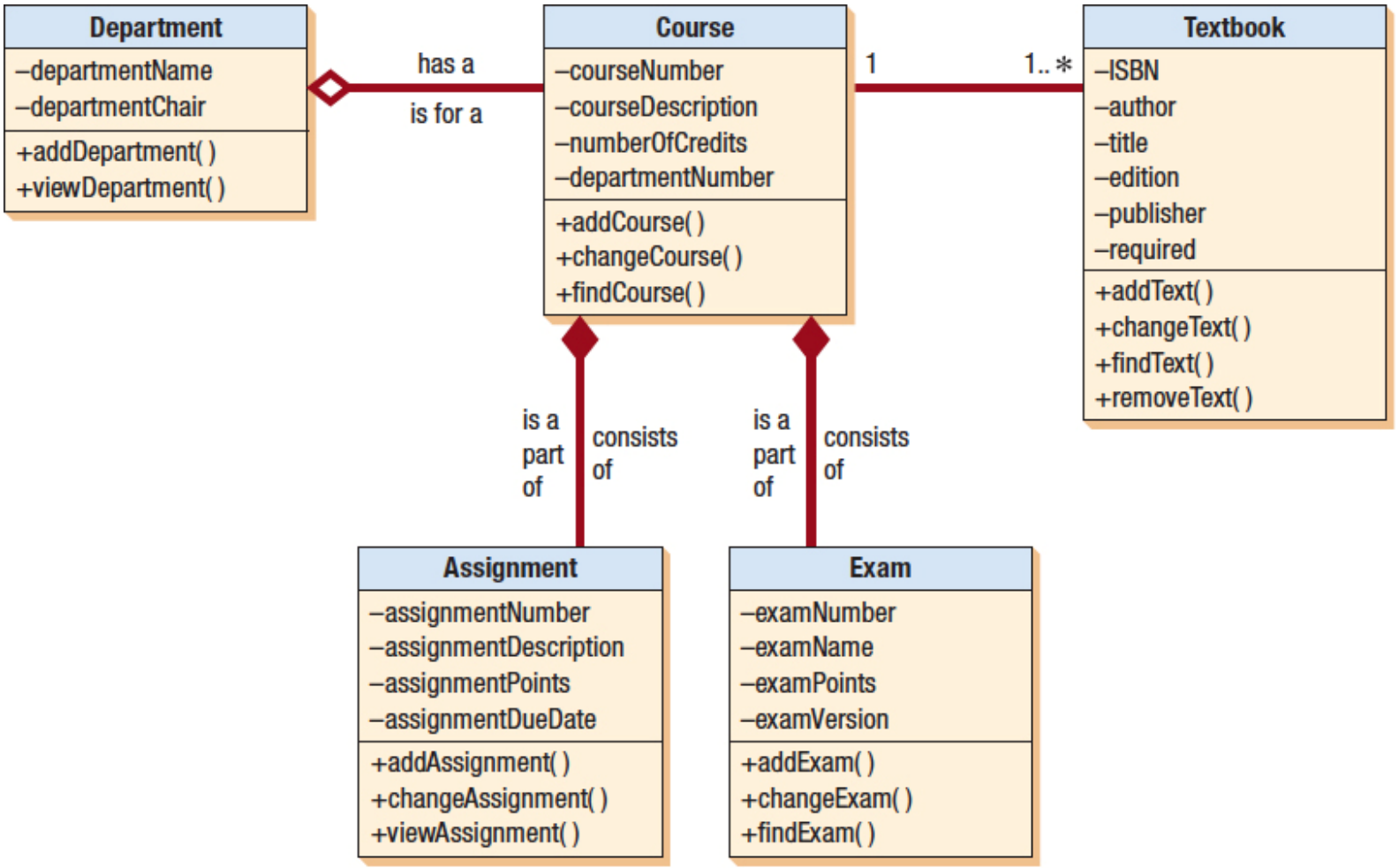
Figure 8.13 A Class Diagram for Course Offerings: the Filled-In Diamonds Show Aggregation and the Empty Diamond Shows a Whole-Part Relationship



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Types of Classes

- Entity classes
- Interface classes
- Abstract classes
- Control classes

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Entity Classes

- Represent real-world items
- The entities represented on an entity-relationship diagram

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Interface or Boundary Classes

- Provide a means for users to work with the system
- Human interfaces may be a display, window, Web form, dialogue box, touch-tone telephone, or other way for users to interact with the system
- System interfaces involve sending data to or receiving data from others

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Control Classes

- Used to control the flow of activities
- Many small control classes can be used to achieve classes that are reusable

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Method Overloading

- Including the same method (or operation) several times in a class
- The same method may be defined more than once in a given class, as long as the parameters sent as part of the message are different
- Method Overloading means having more than one method with the same name in a class, but each method has different parameters.
- Same method name + different inputs = Method Overloading

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Create Sequence Diagrams

- Include the actor from the use case diagram
- Define one or more interface classes for each actor
- Each use case should have one control class
- Examine the use case to see what entity classes are required
- The sequence diagram may be modified when doing detailed design

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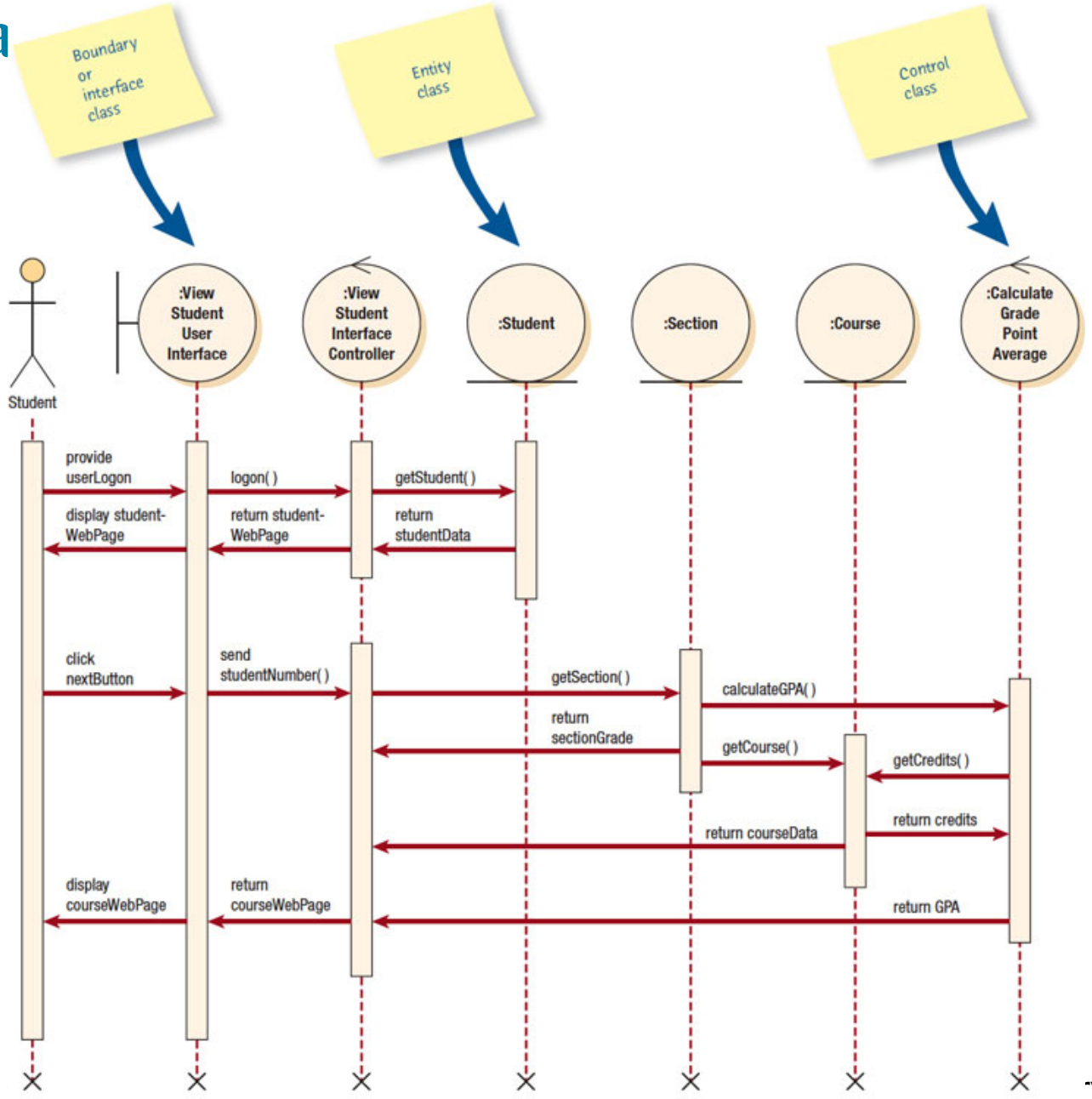
Figure 8.15 A Sequence Diagram for Using Two Web Pages: One for Student Information, One for Course Informa



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Creating a Test Plan from a Sequence Diagram

- Does each method return correct results?
- Ensure that entity classes store or obtain the correct attribute values
- Verify that all JavaScript paths work correctly
- Ensure that the server control classes work correctly
- Ask, “What may fail?”
- Determine what to do if something can fail

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Relationships (2 of 2)

- The connections between classes
  - Associations →a structural connection between classes or objects.
  - Whole/part → when one class represents the whole object and other classes represent parts.

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Associations

- The simplest type of relationship
- Association classes are those that are used to break up a many-to-many association between classes
- Association means two classes are connected, but neither owns the other.
- They are **linked**, but not dependent.
- Example: A Teacher and a Student
  - A teacher teaches students → A student learns from teachers → But if the teacher leaves the school→ the student still exists. If a student leaves school → the teacher still exists

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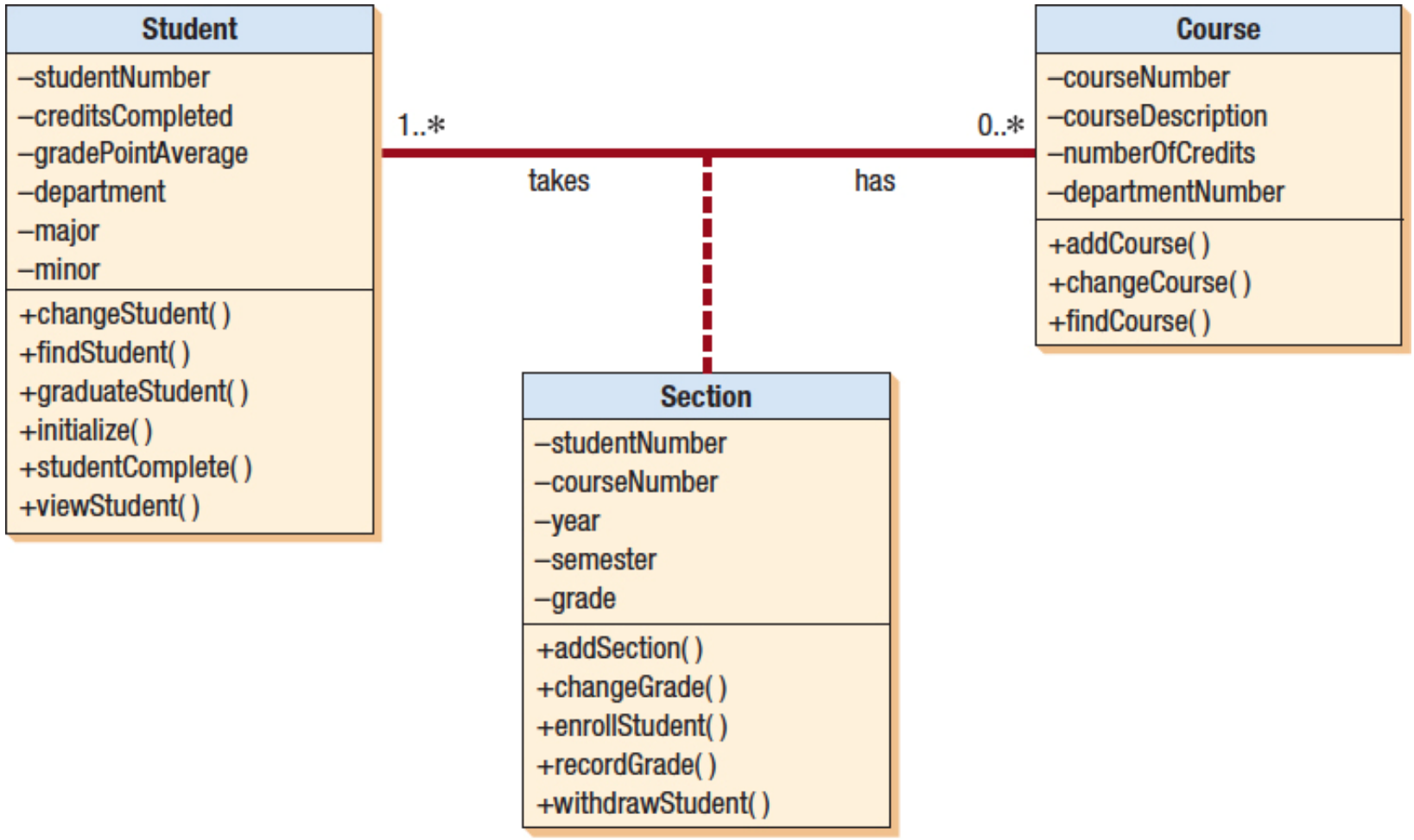
Figure 8.18 An Example of an Associative Class in Which a Particular Section Defines the Relationship between a Student and a Course



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Whole/Part Relationships

- When one class represents the whole object, and other classes represent parts
- But there are different levels of “how strongly the parts belong to the whole”.
- Categories
  - Aggregation (Weak Whole–Part)
  - Collection (Medium)
  - Composition (Strong Whole-Part)

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Aggregation

- A “has a” relationship
- Provides a means of showing that the whole object is composed of the sum of its parts
- The “part” can exist **even if the “whole” is gone**.
  - Example: A Classroom and Students. A classroom contains students → If the classroom is deleted, the students still exist. Students can move to another classroom
  - So, the “part” is not fully owned by the “whole”.

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Collection

- Consists of a whole and its members
- A “whole” holds a **group** (collection) of “parts”. The collection relationship is mainly about **grouping**, not ownership.
- Members may change, but the whole retains its identity
  - Example: A Library has a Collection of Books. Books are grouped inside the library → But books can be moved to another library. Books do not depend on the library to exist.
- Collections are similar to aggregation but focus on the idea of grouping.

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Composition

- The whole has a responsibility for the parts, and is a stronger relationship.
- The “part” **cannot exist without** the “whole”. The “part” cannot exist without the “whole”.
- If the whole is deleted, all parts are deleted
- Example: A House and Rooms. Rooms exist only inside a house → If the house is destroyed, the rooms disappear too. Rooms cannot be moved to another house

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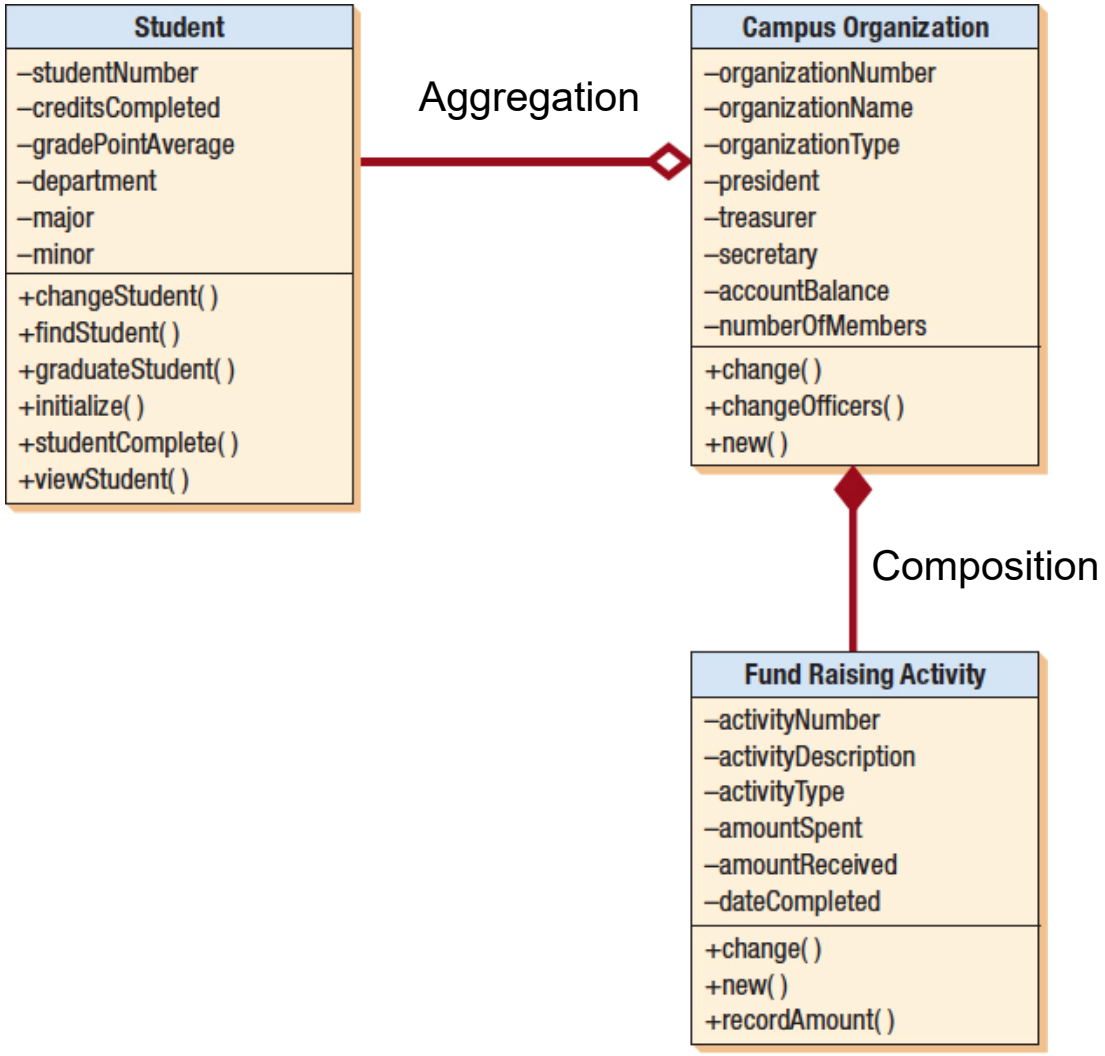
Figure 8.19 An Example of Whole-Part and Aggregation Relationships



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Summary of Relationships

Relationship	Strength	Can the part live without the whole?	Example
Association	Weak	✓ Yes	Teacher ↔ Student
Aggregation	Medium	✓ Yes	Classroom → Students
Collection	Medium	✓ Yes	Library → Book collection
Composition	Strong	✗ No	House → Rooms

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Generalization/Specialization Diagrams

- Generalization
- Inheritance
- Polymorphism
- Abstract classes
- Messages

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Generalization (1 of 5)

- Describes a relationship between a general kind of thing and a more specific kind of thing
- Described as an “is a” relationship
- Used for modeling class inheritance and specialization
- General class is a parent, base, or superclass
- Specialized class is a child, derived, or subclass
- Example: a car is a vehicle; a truck is a vehicle. *(Superclass: Vehicle → Generalization. Subclass: Car & Truck → Specialization)*

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Inheritance (2 of 5)

- **Inheritance** means a child class (specialized class) **receives** the properties and behaviors of the parent class (general class).
- The child class **inherits**:
  - attributes (data)
  - methods (functions)
- Helps to foster reuse → because the code is used many times.
- Helps to maintain existing program code → this allows the analyst to define attributes and methods once but use them many times, in each inherited class.

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Polymorphism (3 of 5)

- **Polymorphism** means “one name, many forms.”
- The capability of an object-oriented program to have several versions of the same method with the same name within a superclass/subclass relationship
- The subclass method overrides the superclass method
- The same method name can behave differently depending on the object that uses it.
  - Example: ***Animal*** has a method: Each child class overrides it:
  - **Dog**.*sound()* → “Woof”; **Cat.***sound()* → “Meow”; **Cow**.*sound()* → “Moo”
  - When you call *sound(),* the correct version runs based on the object type.
- Make program flexible & easier to extend

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Abstract Classes (4 of 5)

- Abstract classes are general classes
- An **abstract class** is a class that **cannot create objects**, but is used as a **Blueprint** for other classes.
- No direct objects or class instances, and is only used in conjunction with specialized classes
- Usually have attributes and may have a few methods

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Overview

Concept	Simple Meaning	Example
Inheritance	Child gets parent's features	Dog inherits from Animal
Polymorphism	Same method, different behavior	Dog.sound(), Cat.sound()
Abstract Class	Cannot create object, used as a blueprint	Shape class
Message	Object calling a method	car.start()

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Messages (5 of 5)

- A **message** is how objects **communicate** with each other in OOP.
- Used to send information by an object in one class to an object in another class
- Acts as a command, telling the receiving class to do something
- Consists of the name of the method in the receiving class, as well as the attributes that are passed with the method name
- May be thought of as an output or an input

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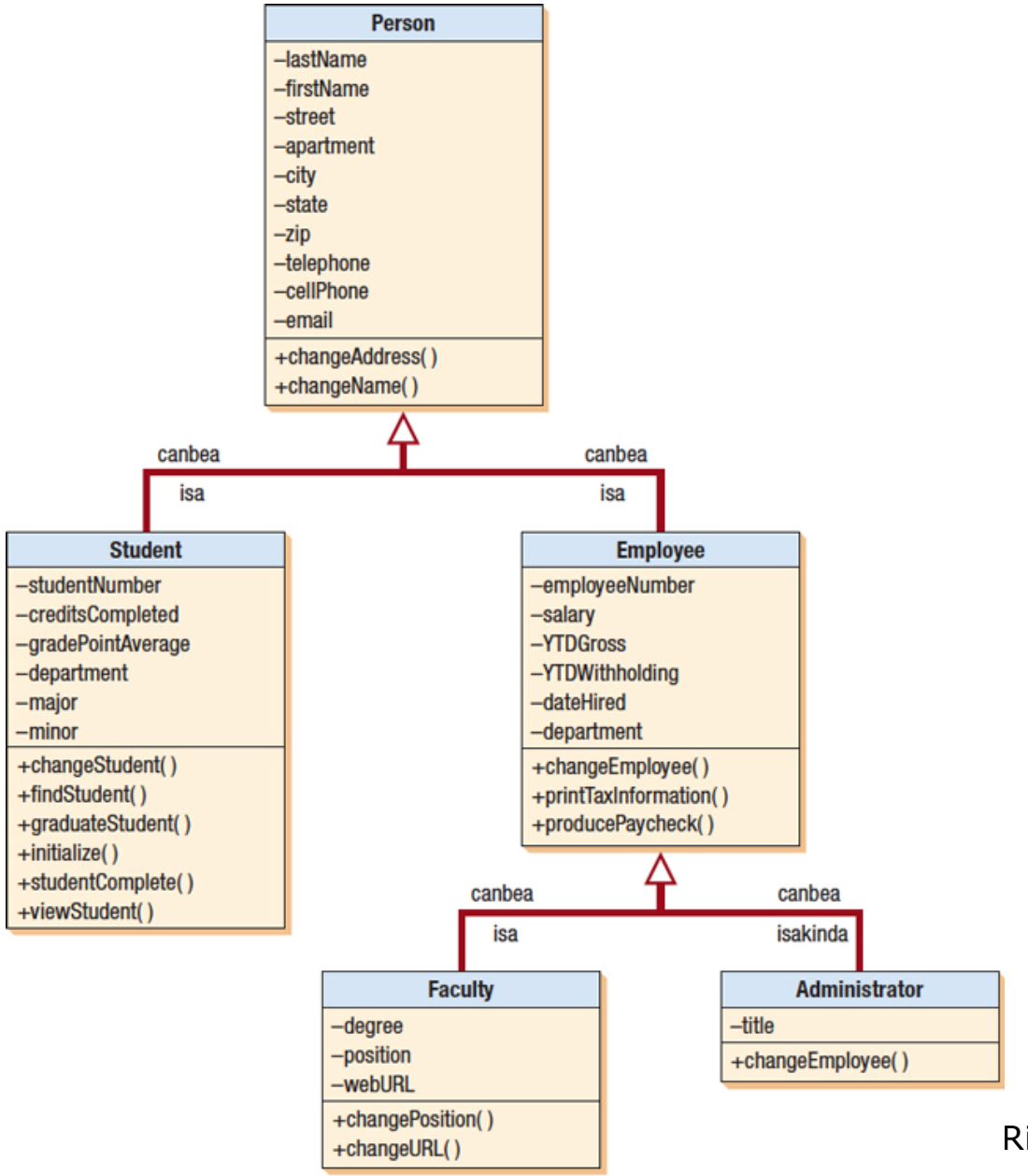
Figure 8.20 A Generalization/Specification Diagram is a Refined Form of a Class Diagram



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Finding Classes

- During interviewing or JAD sessions
- During facilitated team sessions
- During brainstorming sessions
- Analyzing documents and memos
- Examining use cases, looking for nouns

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6. Statechart Diagrams (1 of 2)

- A Statechart Diagram shows how an object **changes** its state **based on events** that happen to it.
- It answers three questions:
  - ✓ **Where can the object be?** (States)
  - ✓ **What makes it move from one state to another?** (Events/Transitions)
  - ✓ **What actions happen when it changes?** (Actions)

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6. Statechart Diagrams (1a of 2)

- A **state** is a situation or condition of an object at a particular time.
- **Examples of states:**
  - A **phone** can be: *Locked* → *Unlocked* → *In Call* → *Call Ended*
  - A **student record** can be: *Created* → *Verified* → *Approved* → *Archived*
  - A **package** can be: *Ordered* → *Packed* → *Shipped* → *Delivered*
- A state is usually drawn as a **rounded rectangle**

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6. Statechart Diagrams (1b of 2)

- A **transition** is a movement from one state to another.
- It happens when an **event** occurs.
- **Example:**
  - Pressing the **power button** moves a laptop from *Off* → *On*
  - Clicking **Submit** moves a form from *Editing* → *Submitted*
- Transitions are drawn as **arrows**.

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6. Statechart Diagrams (1c of 2)

- An **event** is something that happens and triggers a state change.
- **Example events:**
  - Button click
  - Payment completed
- Events are written **on the arrow** between states.
- Event something that happens at a specific time and place.
  - Signals or asynchronous messages—occur when the calling program does not wait for a returning message.
  - Synchronous messages—calls to functions or subroutines.
  - Temporal events—occur at a predetermined time.

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6. Statechart Diagrams (1d of 2)

- Actions are things that happen **during** a transition or inside a state.
- **Examples:**
  - Display a message
  - Save data
  - Start a timer

- **Initial State**: Where the object starts
  Drawn as a **filled black circle**
- **Final State**: Where the object ends
  Drawn as a **black circle inside a white circle**

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Statechart Diagrams (2 of 2)

- Created when:
  - A class has a complex life cycle
  - An instance of a class may update its attributes in a number of ways through the life cycle
  - A class has an operational life cycle
  - Two classes depend on each other
  - The object's current behavior depends on what happened previously

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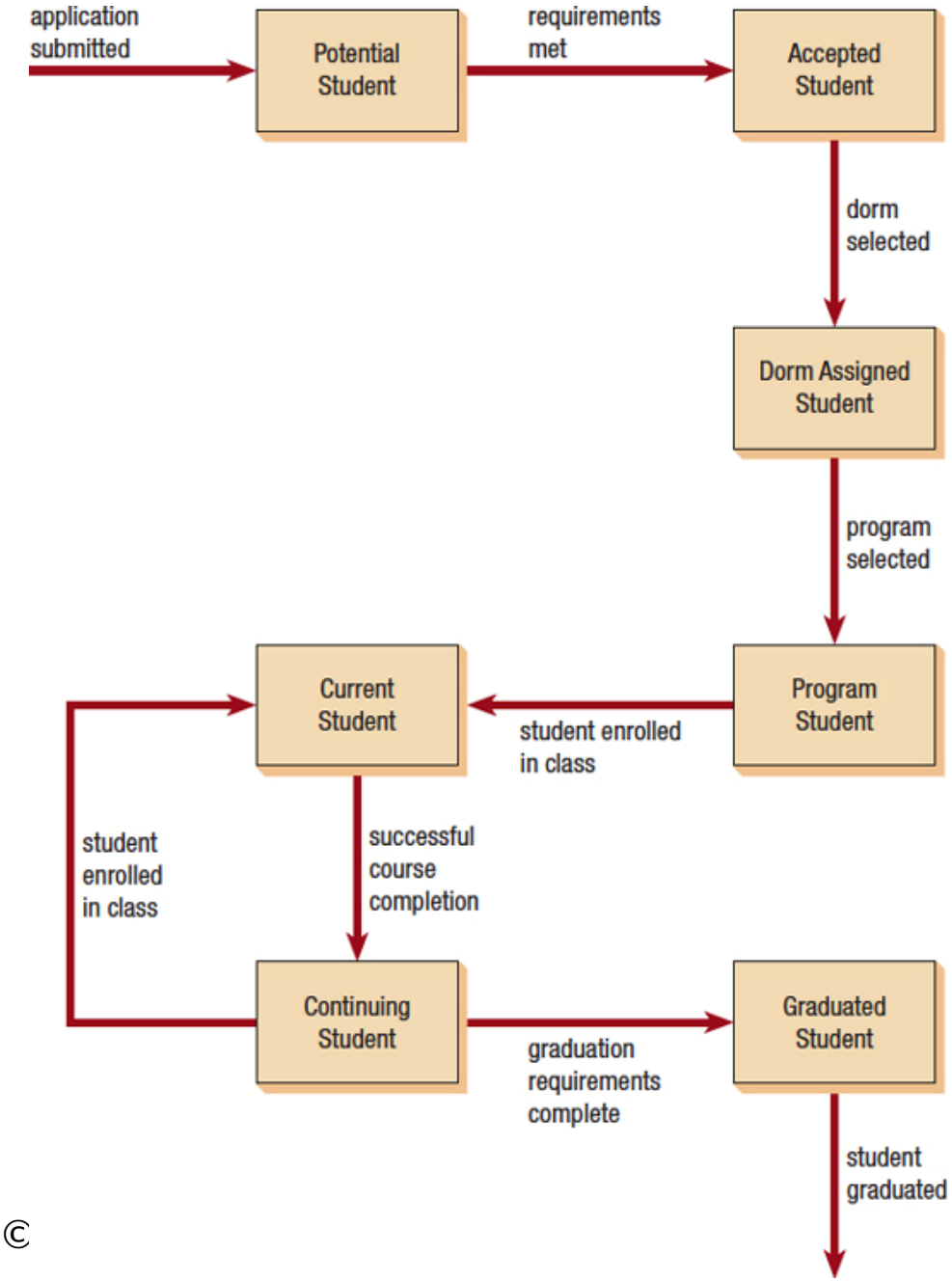
Figure 8.22 A Statechart Diagram Showing How a Student Progresses from a Potential Student to a Graduated Student



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Packages

- Containers for other UML things → such as use cases or classes.
- Show system partitioning → indicating which classes or use cases are grouped into a subsystem (logical packages).
- Can be component packages → contain physical system components, or use case packages, containing a group of use cases.
- Can be physical subsystems
- Use a folder symbol
- May have relationships

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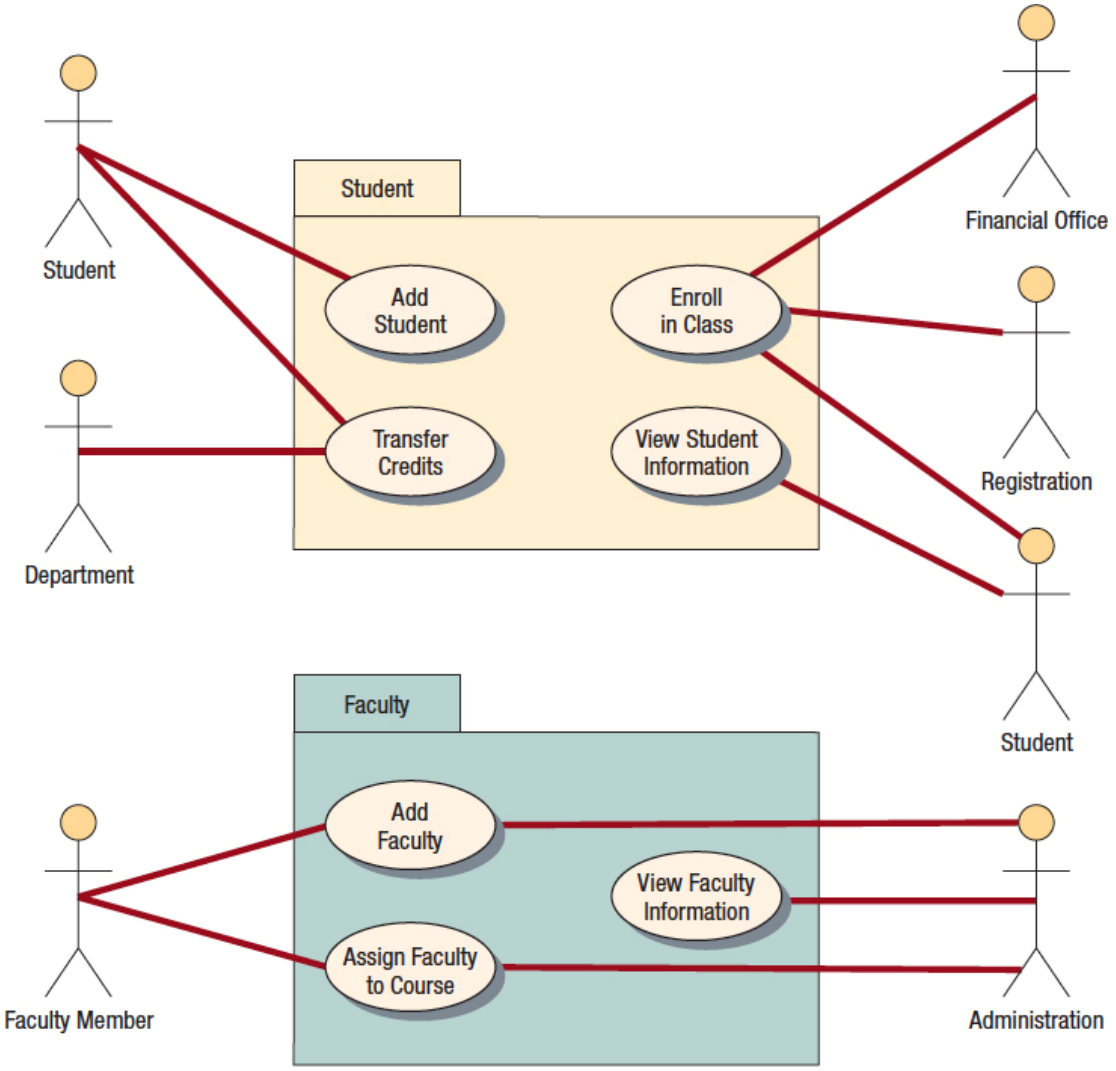
Figure 8.23 Use Cases Can Be Grouped into Packages



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Putting UML to Work

- The steps used in UML are:
  - Define the use case model
  - Continue UML diagramming to model the system during the systems analysis phase
  - Develop the class diagrams
  - Draw statechart diagrams
  - Begin systems design by refining the UML diagrams
  - Document your system design in detail

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Summary (1 of 2)

- Object-oriented systems
  - Objects
  - Classes
  - Inheritance
- CRC cards
- UML and use case modeling
- Components of UML
  - Things
  - Relationships
  - Diagrams

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Summary (2 of 2)

- UML diagrams
  1) Use case diagrams
  2) Activity diagrams
  3) Sequence diagrams
  4) Communication diagrams
  5) Class diagrams
  6) Statechart diagrams
- Using UML

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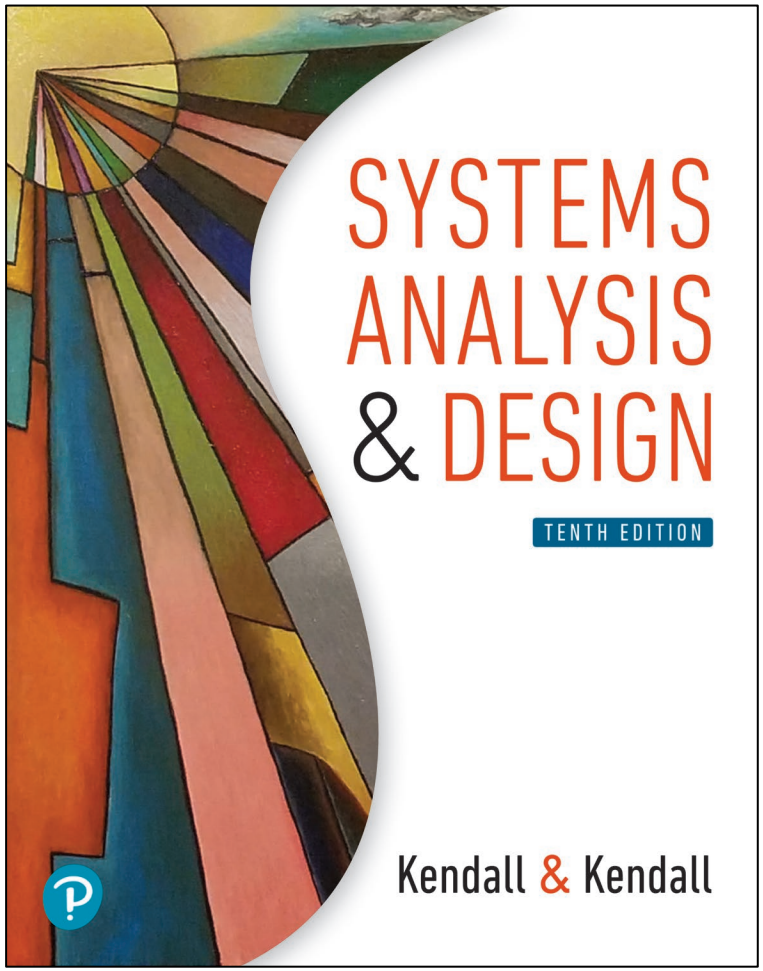
Systems Analysis & Design

Tenth Edition



Chapter 9

Designing Effective Output

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Learning Objectives

11.1 Understand the objectives for effective output design

11.2 Relate output content to output methods

11.3 Realize how output bias affects users

11.4 Design display output

11.5 Design dashboard, widgets, and gadgets

11.6 Design a websites for ecommerce and corporate uses that include Web 2.0 technologies

11.7 Design output for social media

11.8 Understand the development process for apps used on smartphones and tablets

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Output

- Information delivered to users
- Output can be delivered via:
  - intranets
  - extranets
  - the Web
- Output forms
  - Hard-copy—printed reports
  - Soft-copy—computer screens, microforms, and audio
- To create output, the analyst works interactively with the user until the output is satisfactory

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Major Topics

- Designing output
- Output technologies
- Factors in choosing an output technology
- Report design
- Screen design
- Web site design
- Smartphone and tablet design
- Responsive Web design

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Output Design Objectives

- Output is essential to ensuring the use and acceptance of the information system.
- Serve a specific user or organizational purpose → if the output is not functional, it should not be created, there are costs of time and materials with all output from the system.
- Meaningful to the user → on the basis of interviews, observations, cost considerations, and prototypes, it is possible to design output that addresses what users need and prefer.

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Output Design Objectives

- Deliver the appropriate quantity of output → the system must provide what each person needs to complete his or her work.
- Make sure the output is where it is needed → to be used and useful, output must be presented to the right user.
- Provide output on time → accurate timing of output can be critical to business operations.
- Choosing the most effective output method → there are tradeoffs in output methods; costs differ, accessibility, flexibility, durability, distribution, storage, retrieval, transportability, overall impact.

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Output Design Objectives

- What the user wants to see?
- What information the user needs?
- What format the user can use?
- What the user will do after seeing the output?

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Relating Output Content to Method

- Content of output must be considered as interrelated to the output method
  - External—going outside the business
  - Internal—staying within the business

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External Output

- Output that is sent outside the organization to customers, suppliers, partners, government bodies, or the public.
- **Purpose:**
  - Communicate information to **people outside** the business
  - Support **transactions**, **marketing**, **branding**, **compliance**, and **customer service**
- Differs from internal output in:
  - Distribution
  - Design
  - Appearance

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External Output

- **Characteristics of External Output:**

Feature	Explanation
Highly polished	Professional design, branding, readable formats
Customer-friendly	Simple, short, easy to understand
Legally compliant	Must follow regulations (tax invoice, SOPs)
Error-sensitive	Mistakes damage reputation
Delivered quickly	Customers expect real-time feedback

- **Common Types of External Output:**
  - Customer receipts & invoices
  - Delivery notifications
  - Reports to government agencies (e.g., tax, compliance)

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Internal Output

- Output intended for **employees, managers, and internal stakeholders** inside the organization.
- **Purpose:**
  - Support **operations**, **decision-making**, **performance monitoring**, and **problem solving**
  - Provide detailed information not meant for customers
- Might consist of material available on an intranet

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Internal Output

- **Characteristics of Internal Output:**

Feature	Explanation
More detailed	Contains raw data for analysis
Operational or strategic	Helps staff run processes efficiently
Used frequently by teams	Daily, hourly, real-time
Confidential	Not for public view; may contain sensitive data
Functionally designed	Focus on clarity, not aesthetics

- **Common Types of Internal Output**
  - Sales reports (daily/weekly)
  - Staff performance reports
  - Shift schedules

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Figure 9.1 A Comparison of Output Methods (1 of 2)

Output Method	Advantages	Disadvantages
Printer	Affordable for most organizations Flexible in types of output, location, and capabilities Handles large volumes of output Highly reliable with little down time	Still requires some operator intervention Compatibility problems with computer software May require special, expensive supplies Depending on model, may be slow Environmentally unfriendly
Display screen	Interactive Online, real-time transmission Quiet Takes advantage of computer capabilities for movement within databases and files Good for frequently accessed, ephemeral messages	May require cabling and setup space Requires system for taking "snapshots" of screen and storing them for future use

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Figure 11.1 A Comparison of Output Methods (2 of 2)

Output Method	Advantages	Disadvantages
Audio output and podcasts	Good for individual user Good for transient messages Good where worker needs hands free Good if output needs to be widely distributed	Needs earbuds where output will interfere with other tasks Has limited application
Mobile devices	Highly portable Very interactive using gestures Zoom and magnification are possible	Screen may be too small for text Icons and buttons may be confusing May be lost more easily
Electronic output (email, websites, blogs, and RSS feeds)	Reduces paper Can be updated very easily Can be "broadcast" Can be made interactive	Is not conducive to formatting (email) Is difficult to convey context of messages (email) Websites need diligent maintenance

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Factors to Consider When Choosing Output Technology (1 of 2)

- Who will use the output?
- How many people need the output?
- Where is the output needed?
- What is the purpose?
- What is the speed with which output is needed?
- How frequently will the output be accessed?

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Factors to Consider When Choosing Output Technology (2 of 2)

- How long will the output be stored?
- Regulations depicting output produced, stored, and distributed
- Initial and ongoing costs of maintenance and supplies
- Human and environmental requirements

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Green IT Initiatives

- May limit the quantity of paper reports that are printed
- Electronic transactions
- Encourage customers to think about environmental savings

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UN 17 Goals

SUSTAINABLE DEVELOPMENT GOALS
17 GOALS TO TRANSFORM OUR WORLD
1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE, JUSTICE AND STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS
SUSTAINABLE DEVELOPMENT GOALS

Output Bias

- Bias is present in everything that humans create.
- Analysts must avoid unnecessarily biasing output and make users aware of the possible biases in output
- Bias is introduced in three main ways:
  - How information is sorted
  - Setting of acceptable limits
  - Choice of graphics

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Avoiding Bias in the Design Output

- Be aware of the sources of bias
- Design of output that includes users
- Work with users so that they are informed of the output's biases
- Creating output that is flexible and allows users to modify limits and ranges
- Train users to rely on multiple output for conducting "reality tests" on system output

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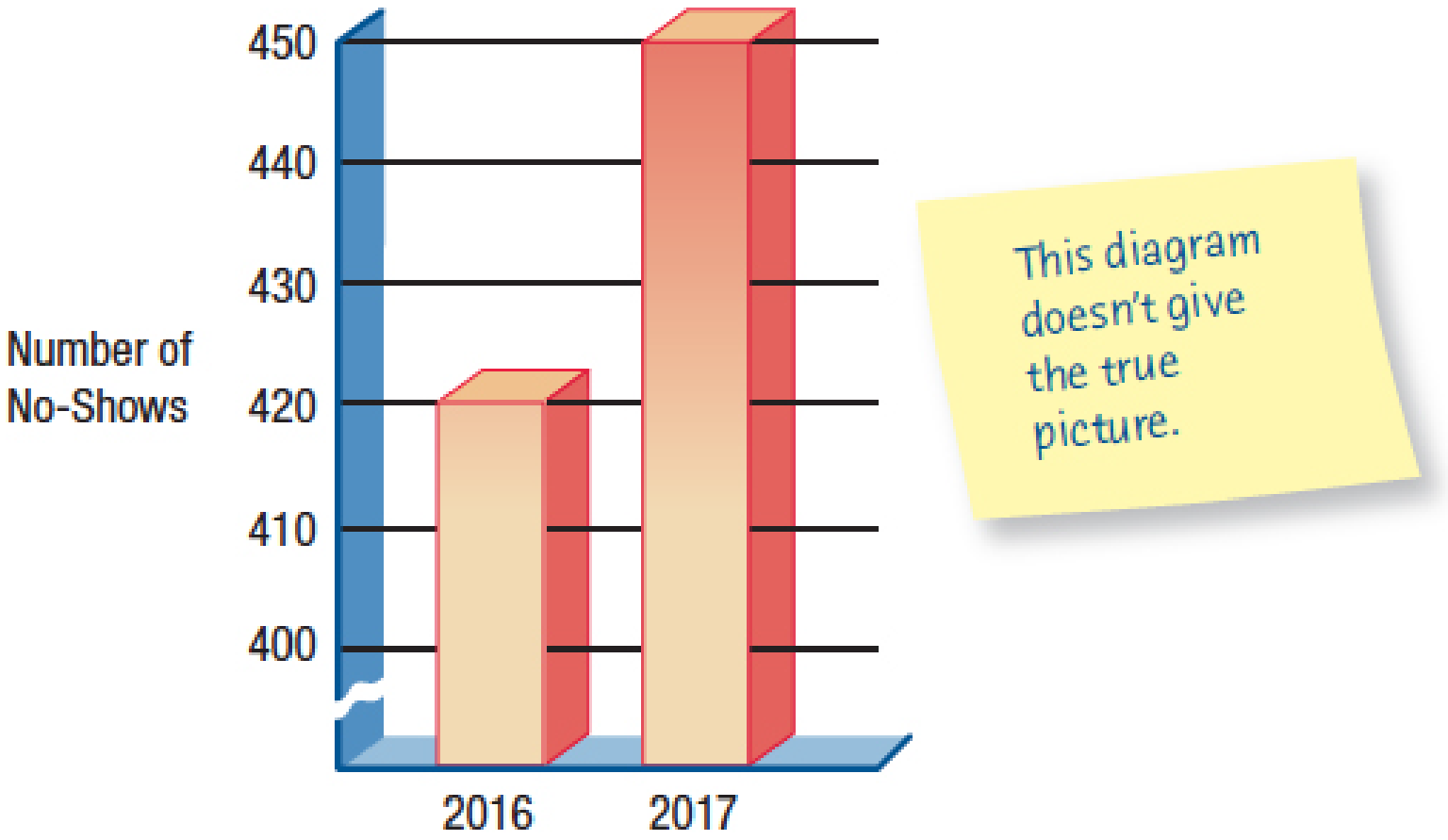
Figure 11.4 A Misleading Graph Will Most Likely Bias the User



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Designing Printed Output

- Detailed reports
  - Print a report line for every record on the master file
- Exception reports
  - Print a line for all records that match a certain condition
- Summary reports
  - Print one line for a group of records that are used to make decisions

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Designing Output for Displays

- Keep the display simple
- Keep the presentation consistent
- Facilitate user movement among displayed output
- Create an attractive and pleasing display

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Designing Output for Displays

- Keep the display simple
- Keep the presentation consistent
- Facilitate user movement among displayed output
- Create an attractive and pleasing display

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Graphical Output in Screen Design

- The purpose of the graph
- The kind of data to be displayed
- The audience
- The effects on the audience of different kinds of graphical output

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Printed Output vs Display Output

Aspect	Printed Output	Display Output
Medium	Paper (hard copy)	Electronic screen
Update Speed	Static, one-time	Dynamic, real-time
Interactivity	None	High (scroll, click, filter)
Durability	Permanent	Temporary (unless saved)
Environment Needs	No electricity	Requires device + power
Design Focus	Simplicity & readability	Usability, responsiveness, visual design
Cost	Ink, paper, storage	Device cost; low per-output cost
Use Cases	Legal docs, receipts, kitchen chits	Dashboards, mobile apps, notifications
Best For	Busy physical environments	Analysis, monitoring, quick updates

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Dashboards (1 of 3)

- Make sure the data has context
- Display the proper amount of summarization and precision
- Choose appropriate performance measures for display
- Present data fairly

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Dashboards (2 of 3)

- Choose the correct style of graph or chart for display
- Use well-designed display media
- Limit the variety of item types
- Highlight important data

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Dashboards (3 of 3)

- Highlight important data
- Arrange the data in meaningful groups
- Keep the screen uncluttered
- Keep the entire dashboard on a single screen
- Allow flexibility

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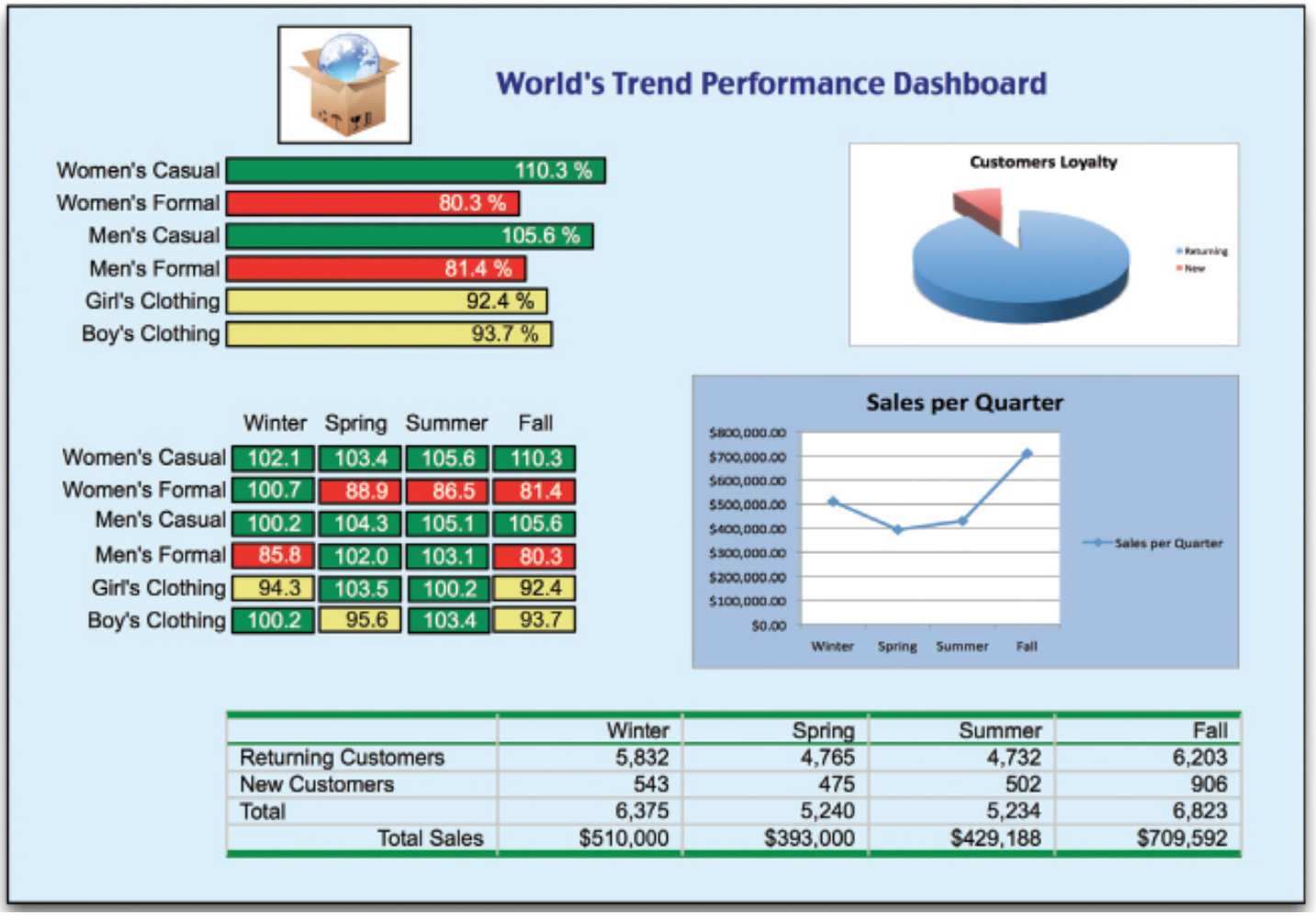
Figure 9.8 Dashboard Example



** https://trends.google.com/trends/

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Infographics

- Combine numbers, charts, and different types of graphs together in one view at the same time in an infographic
- Infographic can mean any graph, chart, or picture that communicates the data better than words or tables
- Infographics should be newsworthy

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Creating Infographics (1 of 2)

- Start the infographic with a headline, something brief but catchy
- Then artfully design a small page of information
- Mix in visuals that together tell a story

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Creating Infographics (2 of 2)

- Design elements include charts and graphs, images (mostly icons, cartoons)
- Photos should be rarely used
- Summary information in brief text
- Contact information
- Some action the reader might want to take

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Figure 9.9 An Infographic Using a Variety of Design Elements

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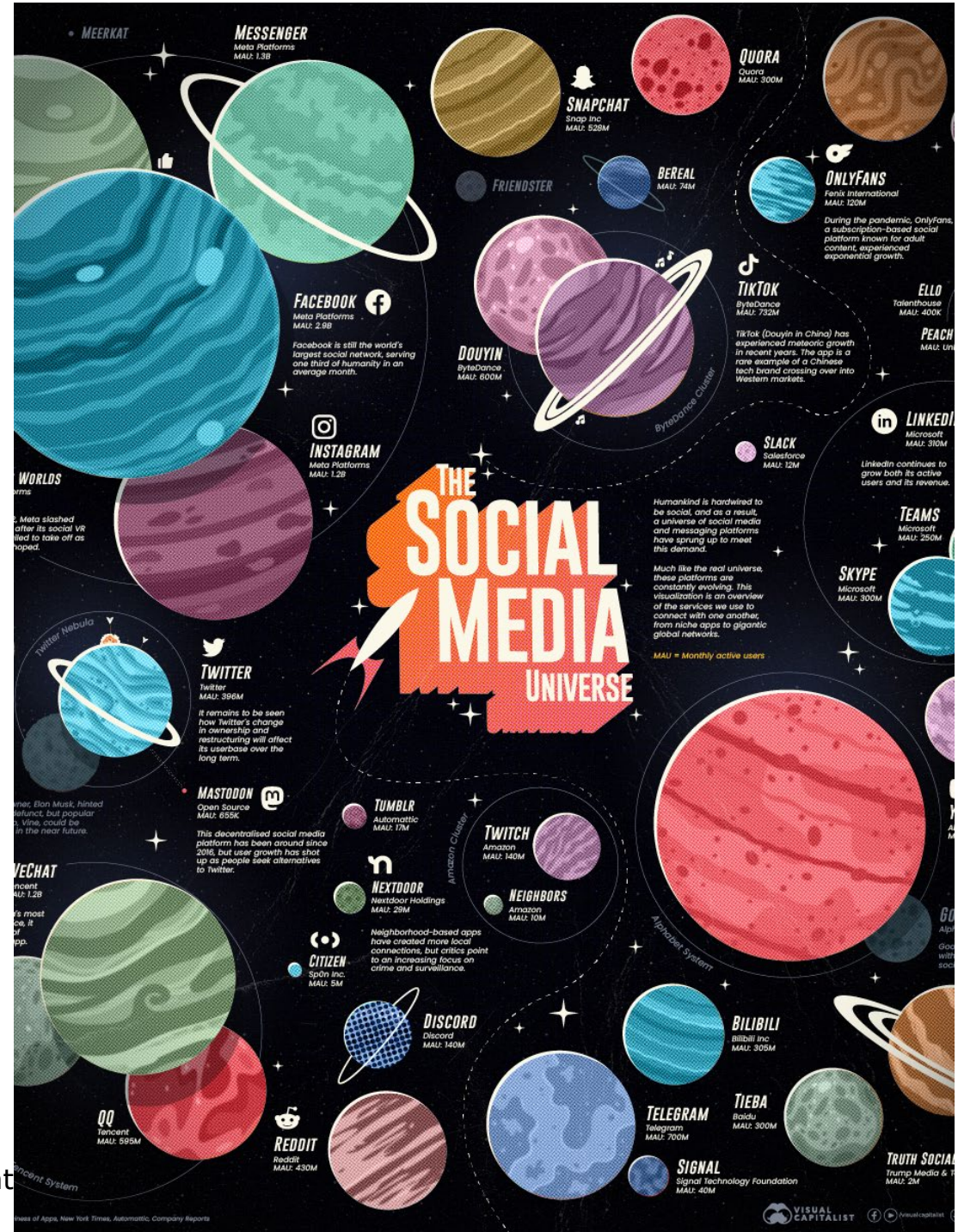
Figure 9.9a An Infographic Using a Variety of Design Elements



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Responsive Web Design (1 of 4)

- Responsive Web Design (RWD) is a **design approach that ensures a website automatically adapts to any device**, whether desktop, laptop, tablet, or mobile phone.
- There is no “standard” screen resolution anymore
- Individuals may set a browser to use different fonts and may disable the use of JavaScript, cookies, and other Web programming elements

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Responsive Web Design (2 of 4)

- Responsive Web Design ensures that **one single website**:
  - Looks good
  - Reads comfortably
  - Functions smoothly
  - Adjusts its layout
  - …no matter what device size the user has.
- This means seeing all of the content, experiencing the same design concepts, and being able to perform all of the tasks on any device

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Responsive Web Design (3 of 4)

- Websites are designed in percentages rather than a fixed number of pixels
- **The website *responds* to:**
  - Screen width and height
  - Orientation (portrait/landscape)
  - Pixel density
  - Browser window resizing
  - User settings (font size, dark mode)
- The Web designer might need to develop different landing pages for each type of device
- Ajax can recognize a device and display different sized advertising components on a page

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Responsive Web Design (4 of 4)

- **Core Techniques Behind Responsive Web Design**
  - **Fluid Grids:** Instead of fixed pixel layouts, designers use **percentages**, **flexbox**, and **CSS grid** so elements resize naturally.
  - **Flexible Images & Media:** Images scale based on device width using CSS
  - **Media Queries (CSS Breakpoints):** Media queries change the layout based on screen size:
  - **Responsive Typography:** Text automatically scales based on screen width using units like:
  - **Touch-Friendly Design:** Google recommends **minimum 48px touch targets**.

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Flat Web Design (1 of 2)

- Web design trends moved away from skeuomorphic design to flat Web design
- Skeuomorphic design means that objects were meant to look realistic and three-dimensional
- Flat Web design is a minimalistic approach to design

Flat Web Design (2 of 2)

- Web design is clean, two-dimensional, and simple in many ways
- Using a bright color palette that is adhered to in all design
- Concentrate on only what is necessary

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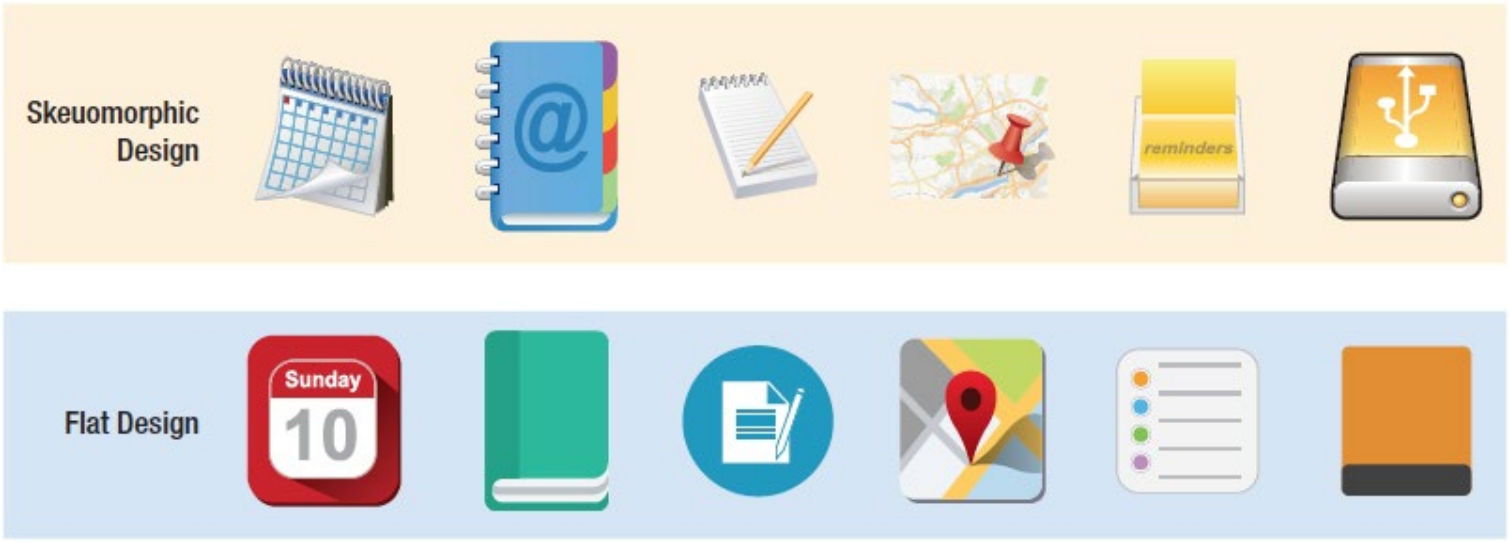
Figure 9.10 Comparing Skeuomorphic Icons with Flat Web Design



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Designing a Website (1 of 3)

- Use professional tools
- Study other sites
- Examine the sites of professional website designers
- Use the tools you've learned
- Use storyboarding, wireframing, and mockups

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Designing a Website (2 of 3)

- Creating Web templates
  - Style sheets allow you to format all Web pages in a site consistently
- Using plug-ins, audio, and video sparingly → remember that everyone does not have new plug-ins.

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Storyboarding

- A storyboard is a **visual sequence of screens** that shows *how the user moves through a system*.
- In developing a website or app, a storyboard could be used to show the differences between screens
- It can show how a visitor to the site would navigate the website
- Perfect for early discussions because clients understand it immediately.

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Wireframing (1 of 2)

- Page design can be accomplished using a process called wireframing.
- It focuses on **structure**, **placement**, and **function**, not colours or graphics. It shows only the basics
- There is no color, no type style; graphics are shown as a simple box with an X drawn in
- In this way, each of the items acts as a placeholder

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Wireframing (2 of 2)

- Wireframing allows the designer to plan:
  - The overall design, showing what element appears at each position on the page
  - The navigational design, showing how to move from one page to the next using buttons, tabs, links, and pull-down menus
  - The interface design, showing how to interact with the Web site by inputting data or responding to questions

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Mockups (1 of 2)

- A mockup is a **high-fidelity visual design** of the final interface.
- It shows **exact colours, fonts, images, spacing, and branding**.
- Mockups show what the output and input will look like
- Abundant software is available to help a systems designer develop a mockup

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Mockups (2 of 2)

- The software has objects that can be dragged and dropped onto the screen
- Templates are available for any type of display including:
  - Desktops
  - Notebooks
  - Smartphones
  - Tablets
- When designing for smartphones and tablets, both screen orientations are included

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Storyboarding vs WireFrame vs Mockups

Feature	Storyboarding	Wireframing	Mockups
Focus	User journey flow	Screen structure/layout	Visual design
Detail Level	Low	Medium	High
Shows Navigation?	✓ Yes	✓ Yes	✓ Yes
Shows Exact Colors?	✗ No	✗ No	✓ Yes
Shows Real Images?	✗ No	✗ No	✓ Yes
Primary Audience	Users, stakeholders	Analysts, designers	Clients, developers
Best Used When	Planning flow	Planning layout	Finalizing UI

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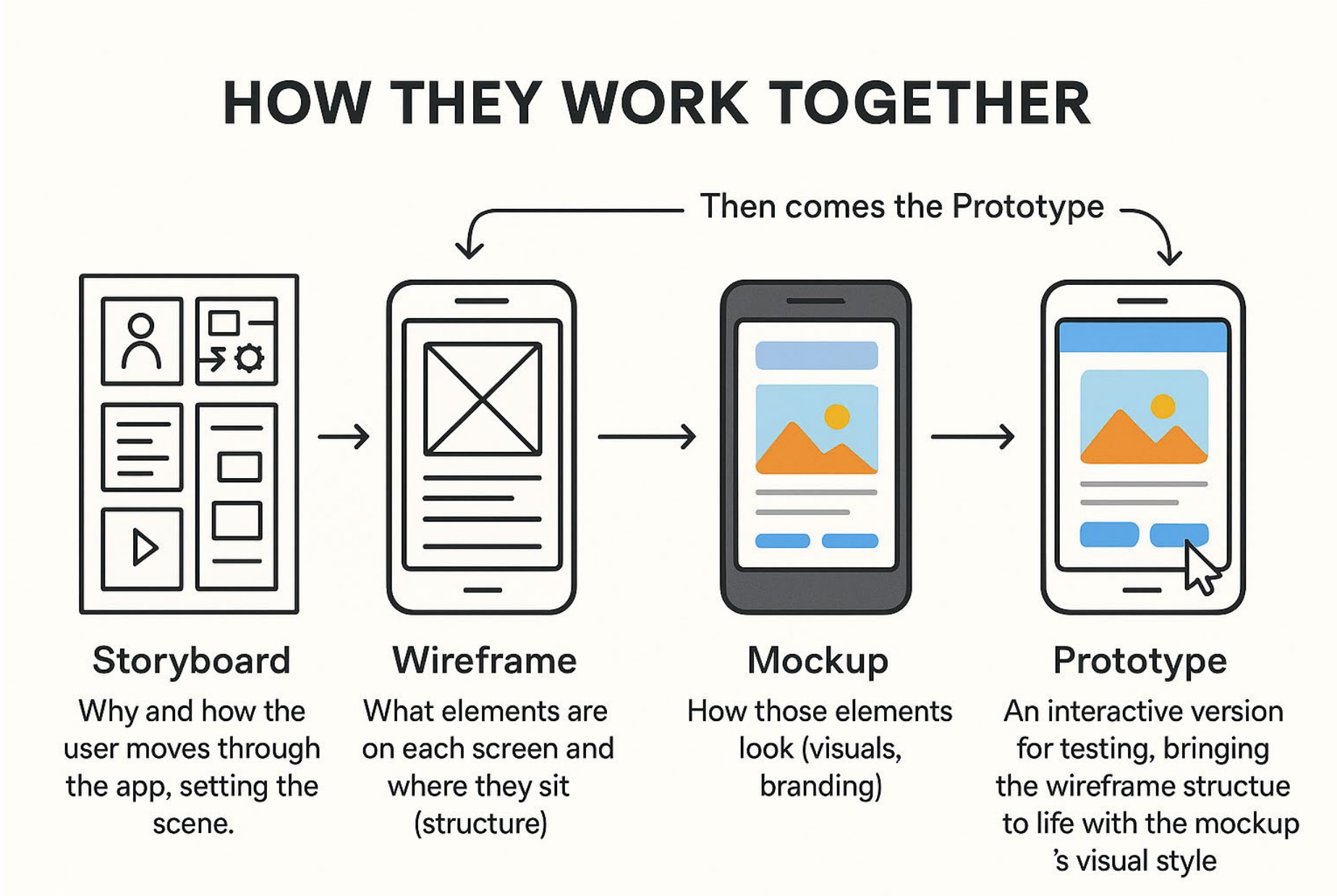
How’s they work together?



Designing a Website (3 of 3)

- Plan ahead, pay attention to:
  - ✓ Structure
  - ✓ Content
  - ✓ Text
  - ✓ Graphics
  - ✓ Presentations style
  - ✓ Navigation
  - ✓ Promotion

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Structure

- One of the most important steps in developing a professional Web site
- Think about your goals and objectives.
- Each page in the Web structure should have a distinct message
- Can benefit from using Web site diagramming and mapping tools
- Look for applications that do Web functionality testing or error checking

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Content

- Appropriate content is needed to keep the user interested
- If a user stays at your site for a long period of time, your site has a high degree of stickiness.
- **Good Content Should Be:**
  - ✓ Relevant
  - ✓ Up-to-date
  - ✓ Engaging
  - ✓ Clear and concise
  - ✓ Optimised for SEO (keywords, headings)
- Should include a FAQ page
- May take advantage of prewritten software

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Text

- Text must be easy to read, well formatted, and written in a user-friendly tone.
- Each Web page should have a title
- Place meaningful words in the first sentence appearing on your Web page
- Clear writing is important
- Text directly affects **usability, accessibility**, and **information retrieval**.

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Content Management Systems

- Content on ecommerce sites needs to be constantly updated
- Content management systems (CMSs) are software tools that help to develop and maintain websites and online applications

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Graphics

- Use either JPEG, GIF, or PNG formats
- Keep the background simple and readable
- Create a few professional-looking graphics for use on your pages
- Keep images small and reuse bullets or navigational buttons
- Include tooltip help for images and image hot spots
- Examine your Web site on a variety of displays and screen resolutions and on mobile devices

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Presentation Style (1 of 2)

- Provide a home page
- Keep the number of graphics to a reasonable minimum
- Use large and colorful fonts for headings
- Use interesting images and buttons for links
- Use CSS to control the formatting and layout of the Web page
- Use divisions and cascading styles or tables to enhance a layout
- Use the same graphics image on several Web pages
- Use Javascript to enhance Web page layout

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Presentation Style (2 of 2)

- Avoid overusing animation, sound, and other elements
- **Best Practices:**
  - ✓ Maintain brand identity (same colours everywhere)
  - ✓ Keep design clean
  - ✓ Ensure spacing so the layout doesn't feel cramped
  - ✓ Use animations sparingly
  - ✓ Maintain consistency across desktop and mobile

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Navigation

- Navigation helps users **move around the website efficiently**.
- The three-clicks rule → a user should be able to move from the page they are currently on to the page containing the information they want in three clicks of the mouse button.
- Include a navigation bar and links to the home page on every page on the Web site

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Promotion

- Promotion refers to how the website is marketed and how it attracts traffic.
- Promote your site
- Submit often to search engines
- Include key words in metatags
- Encourage your readers to bookmark your Web site

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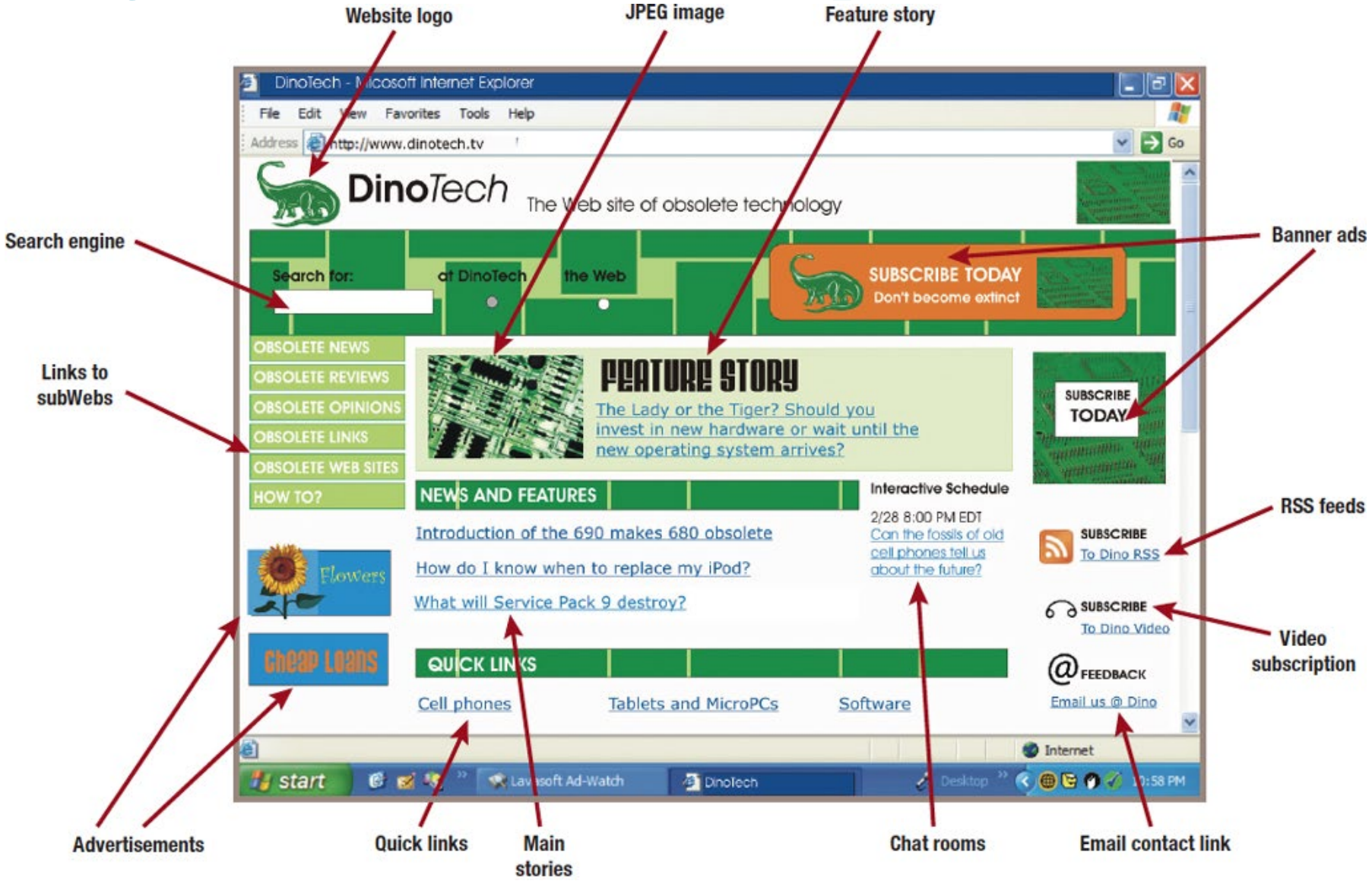
Summary of Designing a Website

Component	Goal	Example (MaxPizza)
Structure	Organize content logically	Menu → Pizza → Add-ons
Content	Provide useful information	Updated promotions, pizza descriptions
Text	Ensure readability	Clear product names & short descriptions
Graphics	Improve visual appeal	High-quality pizza photos
Presentation Style	Maintain branding consistency	Red/yellow theme, rounded buttons
Navigation	Help users find things easily	Search bar, fixed “Order Now” button
Promotion	Attract visitors & boost sales	SEO, discounts, social media integration

Figure 9.12 Website Example



Web 2.0 Technologies and Social Media Design

- It is important to include Web 2.0 technologies that focus on enabling and facilitating user-generated content and collaboration
- Types of technologies you should think about including:
  - Blogs
  - Wikis
  - Links to social networks on which the company has a presence
  - Tagging

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Tagging

- Tagging or social bookmarking provides useful pointers to online resources such as:
  - Web sites
  - Content on corporate intranets
  - Corporate documents, or photos that are relevant to the organization and to users

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Reasons for Using Collaborative Tools

- Companies use collaborative tools to:
  - Communicate an integrated branding and messaging strategy across multiple platforms
  - To gauge consumer opinion
  - To gather feedback
  - To create a community of users

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Internal Use of Social Media

- Inward-facing Web technologies can be useful in:
  - Building employee relationships
  - Maintaining trust
  - Sharing knowledge
  - Innovating among employees and groups of employees
  - Locating corporate resources more readily
  - Nurturing corporate culture and subcultures inside the organization

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Figure 9.13 Many Different Web 2.0 Sets of Icons

Name of Icon Set	Designer or Supplier	Facebook	LinkedIn	RSS Feed	Twitter
Social Networks Pro Icons	Artbees				
Circular Social Media	BlogPerfume.com				
Free Social Media Icon Set	Elegant Themes				
Amazing 3-D Social Icons	pinkmoustache.net				
Vintage Icons for Bloggers	Nikola Lazarevic				

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Social Media Design (1 of 3)

- Businesses use social media for a number of reasons:
  - Growing their audiences
  - Increasing traffic to already well-established Web sites
  - Reinforcing brand awareness
  - Developing an online persona that can be trusted

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Social Media Design (2 of 3)

- Numerous options, including:
  - Facebook
  - Instagram
  - Twitter
  - Snapchat
  - TikTok
  - YouTube
  - Others

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Social Media Design (3 of 3)

- Designing content for social media is highly visual
- Visual content is 40 times more likely to be shared on social media
- Sharing is what allows a designer of social media to leverage social media for its best networking aspects

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Social Media Design Guidelines

- Emphasize the goal of the design
- Develop a consistent look
- Create appealing design flow
- Simplify to use space in a positive way

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Designing for Smartphones and Tablets (1 of 12)

1. Set up a developer account

- To publish apps or test advanced device features, developers need platform accounts.
- **Apple Developer Program**
  - Required for iOS/iPadOS app distribution
  - Cost: USD $99 per year
  - Includes TestFlight for beta testing
    https://developer.apple.com/programs/
- **Google Play Developer Account**
  - Required for publishing Android apps
  - One-time fee: USD $25
  - https://play.google.com/console/about/

- **Why This Matters:** Access SDKs, emulators, testing tools. Publish apps for smartphones, tablets, foldables. Receive analytics & crash reports

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Designing for Smartphones and Tablets (2 of 12)

2. Choose a Development Process

- Modern mobile development favors **agile** and **rapid prototyping**, allowing early feedback and continuous improvements.
- **Recommended Processes:**

- **Prototyping-first approach**
  Use tools like Figma, Adobe XD, Sketch → get stakeholder approval early.
- **Agile / Scrum**
  Small iterative updates work best for mobile apps where user feedback is critical.
- **Flutter (Google)** – fast UI, native compilation: https://flutter.dev
- **React Native (Meta)** – JavaScript-based, widely used: https://reactnative.dev
- **SwiftUI / Jetpack Compose** – next-gen native UI frameworks

- **Why This Matters:** Users expect apps to update frequently with new features, bug fixes, and improved UI/UX.

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Designing for Smartphones and Tablets (3 of 12)

3. Be an original

- A mobile app must stand out to avoid becoming "just another app" in the store.
- **How to Ensure Originality**
  - Provide a unique value proposition
  - Solve a real pain point with fewer steps
  - Use branded UI patterns (colors, icons, tone)
  - Avoid copying competitor interfaces directly
  - Prioritize performance → smooth animations, fast loading
- **Example (MaxPizza)**
  - Instead of copying FoodPanda's layout:
  - Add **1-tap reordering** based on user history
  - Smart suggestions like "Your favorite drink?"
  - Visual cooking progress animations

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Designing for Smartphones and Tablets (4 of 12)

4. Determine How You Will Price the App

- Mobile Pricing Models:
  - Free with ads
  - Freemium (basic features free, premium subscription unlocks extra features)
  - One-time paid app
  - Subscription model (most sustainable)
  - In-app purchases (IAP)
  - Free but requires account login (data-driven model)
- **Reference:**
  - Google Play Monetization Guidelines: https://support.google.com/googleplay/android-developer/answer/9858738
  - Apple App Store Business Guidelines 🔗 https://developer.apple.com/app-store/business-models/

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Designing for Smartphones and Tablets (5a of 12)

5. Determine Modern Output Design Rules

→Mobile design today follows established UX frameworks:

A. **Mobile-First Design:** Always design for the smallest screen first.

- **Best Practices:**
  - ✓Start with 360px width (common mobile width)
  - ✓Prioritize content vertically
  - ✓Minimize text input
  - ✓Use collapsible menus
- Google Mobile-Friendly Test: https://search.google.com/test/mobile-friendly

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Designing for Smartphones and Tablets (5b of 12)

5. Determine Modern Output Design Rules

→ Modern guidelines require:

B. Touch-Friendly Interaction:

- **Touch Target Size:**
  - ✓Minimum 48 x 48 dp (Android)
  - ✓Minimum 44 x 44 pt (Apple)
- Source:
  - ✓Android Material Design → https://m3.material.io/
  - ✓Apple Human Interface Guidelines → https://developer.apple.com/design/human-interface-guidelines/

- **Add Gesture Support**: Swipe, Tap & long-press, Drag & Pinch to zoom

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Designing for Smartphones and Tablets (5c of 12)

5. Determine Modern Output Design Rules

→ Modern UI frameworks use responsive auto-layout systems:

C. Layout & Readability:

- **Use:**
  - ✓Auto-Layout (iOS)
  - ✓ConstraitLayout (Android)
  - ✓Grid/Flex layout for web apps
  - ✓Adaptive UI for foldable devices
- Latest Trends:
  - ✓More Whitespaces
  - ✓Rounded Card Components
  - ✓High-Contrast Clean UI
  - ✓Simplified Icons

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Designing for Smartphones and Tablets (5d of 12)

5. Determine Modern Output Design Rules

→ Mobile users often experience weak or unstable connections:

D. Optimizing for Network Conditions:

- **Modern Standard:**
  - ✓Offline Cache (LocalStorage/IndexedDB/SQLite)
  - ✓Lightweight images (WebP/AVIF)
  - ✓Lazy Loading
  - ✓Background Sync

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Designing for Smartphones and Tablets (5e of 12)

5. Determine Modern Output Design Rules

→ Mobile users often experience weak or unstable connections:

E. Performance Optimization:

- **Focus Areas:**
  - ✓60 FPS smooth animations
  - ✓Load Core content under 2 seconds
  - ✓Minimize app size (<50MB recommended)
  - ✓Use prefetching

- **Reference:** Google's Core Web Vitals → https://web.dev/vitals/

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Designing for Smartphones and Tablets (6 of 12)

6. Design your icon (Logo brandings)
7. Choose a name for the app (brandings)
8. Design for a variety of devices
   - ➢ **Adaptations to Consider:**
   - ✓ Horizontal vs vertical layouts
   - ✓ Multi-window mode
   - ✓ Split screen on tablets
   - ✓ Aspect ratio variations

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Designing for Smartphones and Tablets (9 of 12)

9. Design the output for the app

Output Type	Example
Visual UI	menus, images, product cards
Notifications	"Your order is arriving soon"
Audio output	alert tones, spoken updates
Vibration	haptic feedback
Maps	delivery tracking
Micro-interactions	button animations

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Designing for Smartphones and Tablets (10 of 12)

10. Design for Gestures

- Mobile apps rely heavily on intuitive gesture-based navigation.
- **Essential Gestures:**
  - Swipe → navigation or delete
  - Pinch → zoom content
  - Pull-to-refresh
  - Tap & hold for menu options
- **Reference:**
  Google Material Gestures:
  https://m3.material.io/foundations/gestures/overview

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Designing for Smartphones and Tablets (11 of 12)

11. Protect your property

- Modern app security requirements include:
  - SSL/TLS mandatory
  - Biometric authentication (FaceID, TouchID)
  - App sandboxing
  - Secure storage for tokens
  - Zero-trust access
  - Anti-tampering / anti-debugging tools
  - OWASP Mobile Security Guide:
    https://owasp.org/www-project-mobile-top-10/

12. Market your app

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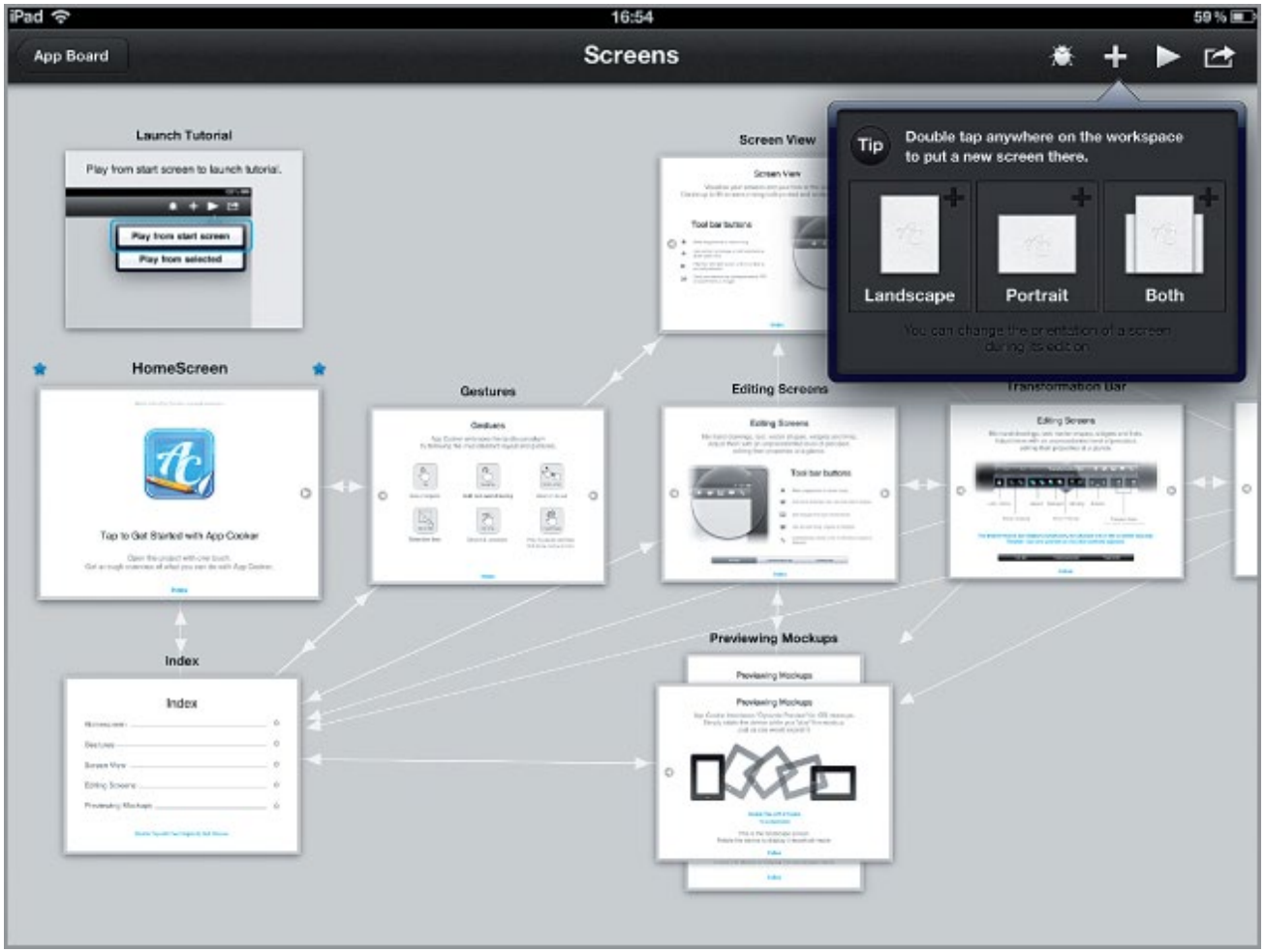
Figure 9.17 App Cooker Plans Movement



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Choose a Development Process (summary)

- Prototyping is most likely the best way to develop your app
- Quick releases are important
- Quality should not be sacrificed, but you can introduce an app and then add features later
- Advantages of introducing an app first:
  - It allows you to gain an advantage
  - Revise the app adding new features
  - Increases visibility because the app appears on a list of apps that have been updated

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Determine Pricing the App (summary)

- There are six basic options for pricing:
  1. Choose a low-cost strategy
  2. Introduce an app as a “premium” app
  3. Adopt a “freemium” model
  4. Offer an app for free
  5. Promote an app by reducing its price
  6. Accept advertising

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Figure 9.15 Design the App Icon (summary)

Icon Required	Size for Older iPhones in (pixels)	Size for High-Resolution iPhone (pixels)	Size for Older iPad (pixels)	Size for High-Resolution iPad (pixels)
Display size	320 × 480	640 × 1136	1024 × 768	2048 × 1536
Application icon (Home screen icon)	57 × 57	114 × 114	72 × 72	144 × 144
App icon for the app store	512 × 512	1024 × 1024 recommended	512 × 512	1024 × 1024 recommended
Launch image	320 × 480	640 × 960	768 × 1004 and 1024 × 748	1536 × 2008 and 2048 × 1496
Spotlight search icon	29 × 29	58 × 58	50 × 50	100 × 100
Settings icon	29 × 29	50 × 50	29 × 29	50 × 50
Documents icon	22 × 29	44 × 58	64 × 64 and 320 × 320	128 × 128 and 640 × 640

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Design the App's Logic (summary)

- Tablets and smartphones fit in well with the prototyping method of development
- Sometimes the best way is to sketch out the logic using structured decision-making techniques

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Design Movement (summary)

- Moving from screen to screen may make a big difference in a user's experience
- Plan movement carefully
- Use storyboards or an app

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Create the User Interface Using Gestures (summary)

- Smartphones and tablets have innovative user interfaces
- Technically called touchscreen capacitive sensing
- Design apps assuming that users will demand touch-sensitive interfaces
- Use gestures such as:
  - Swipes
  - Pinches
  - Tugs
  - Shakes

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Market Your App (summary)

- Need to convince a person to pay for and download your app
- To market your app, you will need:
  - A large icon
  - A description
  - A section explaining what is new in the current version
  - A sample set of screenshots

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Ajax

- Uses both JavaScript and XML to obtain small amounts of data from a server without leaving the Web page
- It refers to a technique that allows a webpage to **send or receive data from a server in the background *without reloading the entire page*.**
- AJAX allows web pages to **update just part of the screen** instead of refreshing the whole page.

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Summary (1 of 2)

- Output
- Output design objectives
- Output content
- Output technologies
- Presentation of output
- Responsive Web design

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Summary (2 of 2)

- Flat Web design
- Display output
- Web design
- Social media
- Smart phone and tablet design
- AJAX

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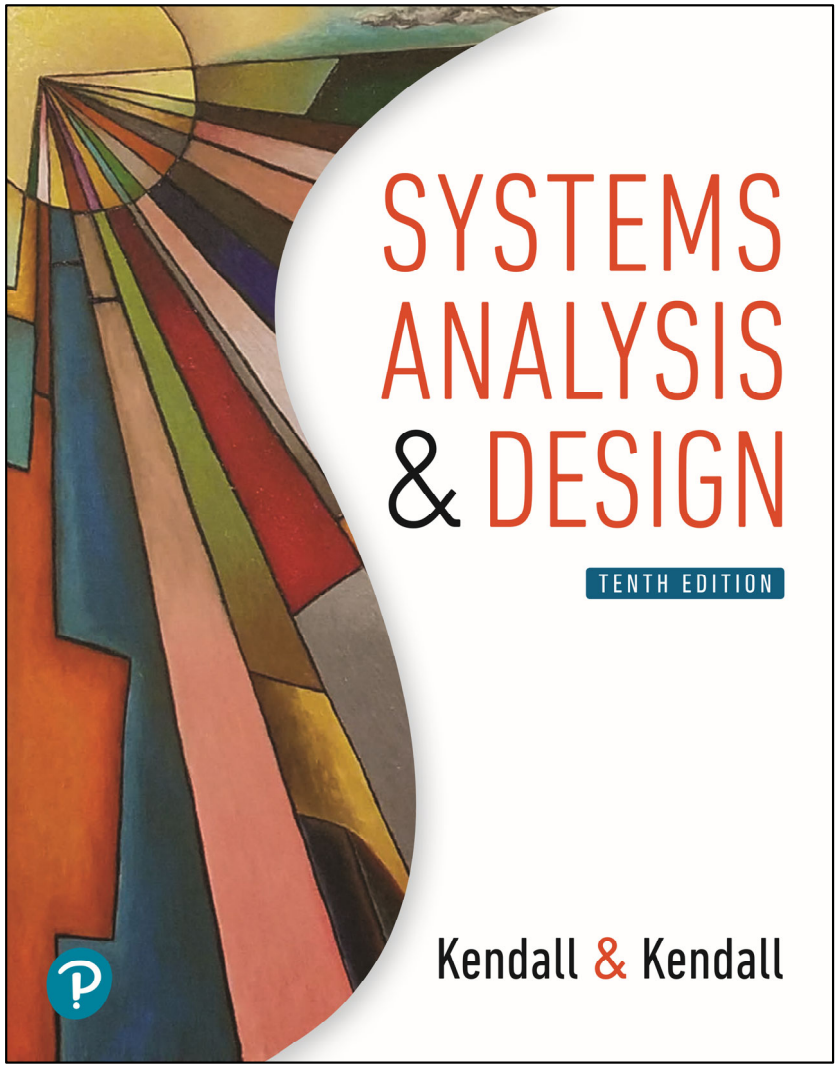
Systems Analysis & Design

Tenth Edition



Chapter 10

Designing Effective Input

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Learning Objectives (1 of 2)

10.1 Design functional input forms for users of business systems

10.2 Design engaging input displays for users of information systems

10.3 Use pervasive navigational elements and icons to give Web designs a professional look

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Learning Objectives (2 of 2)

10.4 Design useful input forms for people interacting on the Web

10.5 Design useful input pages for users of intranets, the Web, smartphones, and tablets

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Input Design Objectives (1 of 2)

- The quality of system input determines the quality of system output
- Input design objectives:
  - Effectiveness
  - Accuracy
  - Ease of use
  - Consistency
  - Simplicity
  - Attractiveness

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Input Design Objectives (2 of 2)

- Use standard navigational elements and icons to give Web designs a professional look
- Context-sensitive help
- Input validation
- A breadcrumb trail
- Fat footers

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Major Topics (1 of 2)

- Input design
- Form design
- Display design
- GUI screen design
- GUI controls
- Web design guidelines

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Good Form Design

- Make forms easy to fill in
- Ensure that forms meet the purpose for which they are designed
- Design forms to assure accurate completion
- Keep forms attractive

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1. Make Forms Easy to Fill in

- Form flow
- Seven sections of a form
  - Heading (name, address..)
  - Identification and access (codes to files report)
  - Instructions
  - Body
  - Signature and verification
  - Totals
  - Comments
- Captioning

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Form Flow

- Can minimize the time and effort expended in form completion
- Should flow from left to right and top to bottom

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Seven Sections of a Form

- Heading
- Identification and access
- Instructions
- Body
- Signature and verification
- Totals
- Comments

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Caption Types

- Line caption
  - Putting the caption on the same line or below the line
- Boxed caption
  - Providing a box for data instead of a line
- Check off caption
  - Lining up choices or alternatives vertically
- Horizontal check off caption
  - Lining up choices or alternatives horizontally
- Table caption
  - Work well in the body of a form
- Combination

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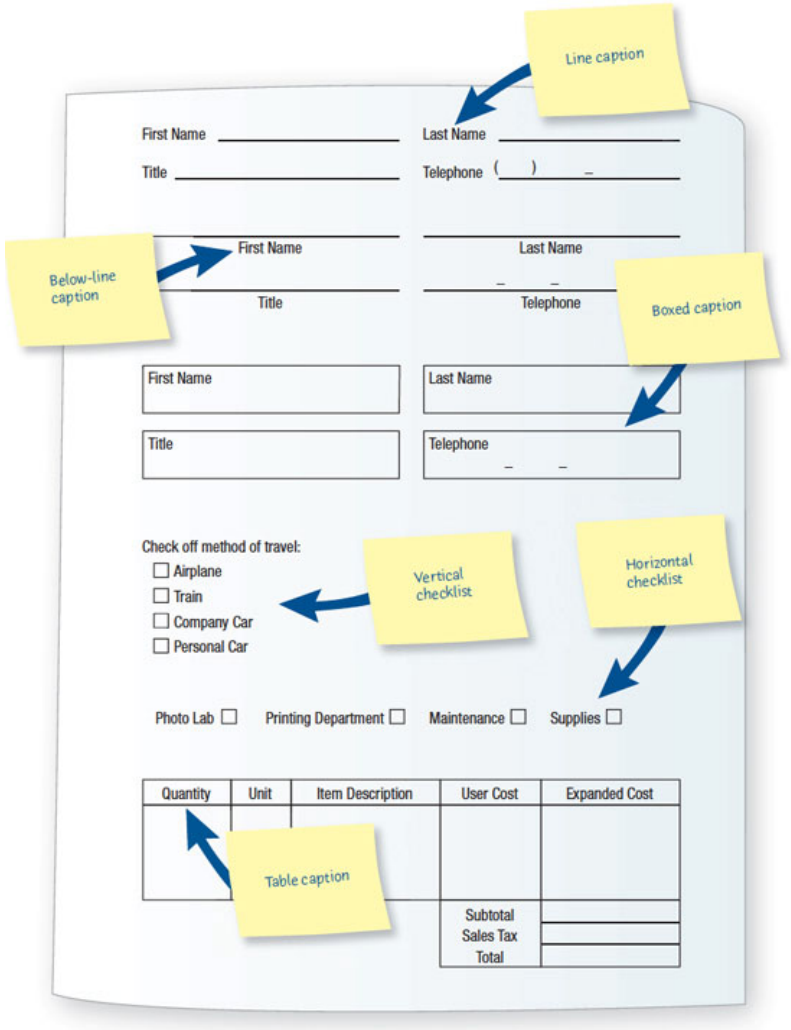
Figure 10.2 Major Captioning Alternatives



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2. Meeting the Intended Purpose

- Systems analysts may use different types of specialty forms for different purposes
- Specialty forms
  - Multiple-part
  - Continuous-feed
  - Perforated

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3. Ensuring Accurate Completion

- To reduce error rates associated with data collection, forms should be designed to assure accurate completion
- Design forms to make people do the right thing with the form

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4. Keeping Forms Attractive

- Aesthetic forms draw people into them and encourage completion
- Forms should look uncluttered, and elicit information in the expected order
- Using different fonts and line weights within the same form can help make it more attractive for users

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Computer-Assisted Form Design

- Numerous form design packages are available for desktop systems—there are tools to set up:
  - Fields
  - Check boxes
  - Lines
  - Boxes

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Controlling Business Forms

- Make sure that each form in use fulfills its specific purpose
- Make sure that the specified purpose is integral to organizational functioning
- Prevent duplication of information collected and of the forms that collect it
- Design effective forms
- Decide on how to reproduce forms in the most economical way
- Establish procedures that make forms available, at the lowest possible cost

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Good Display and Web Forms Design

- Keep the display simple
- Keep the display presentation consistent
- Facilitate user movement among display screens and pages
- Create an attractive and pleasing display

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1.Keeping the Display Simple

- Heading
- Body
- Comments and instructions

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2. Keeping the Display Consistent

- Locate information in the same area each time a new display is accessed
- Information that logically belongs together should be consistently grouped together
- Information should not overlap from one group to another

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3. Facilitating Movement

- The three-clicks rule says that users should be able to get to the screens they need within three mouse or keyboard clicks
- Movement among screens:
  - Scrolling by using arrows or PgDn keys
  - Context-sensitive pop-up windows
  - Onscreen dialogue

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4. Designing an Attractive and Pleasing Display

- Should draw users into them and hold their attention
- Use logical flows in the plan to your display pages
- Color or shaded boxes and creating three-dimensional boxes and arrows

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Using Icons in Screen Design

- Icons are pictorial, onscreen representations symbolizing computer actions that users may select using a mouse, keyboard, lightpen, touch screen, or joystick
- Shapes should be readily recognizable
- Icons for a particular application should be limited to 20 recognizable shapes
- Use icons consistently throughout

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Graphical User Interface (GUI) Controls

- GUI Controls:
  - Text boxes
  - Check boxes
  - Option or radio buttons
  - List and drop-down list boxes
  - Sliders and spin buttons
  - Image maps
  - Text area
  - Message boxes

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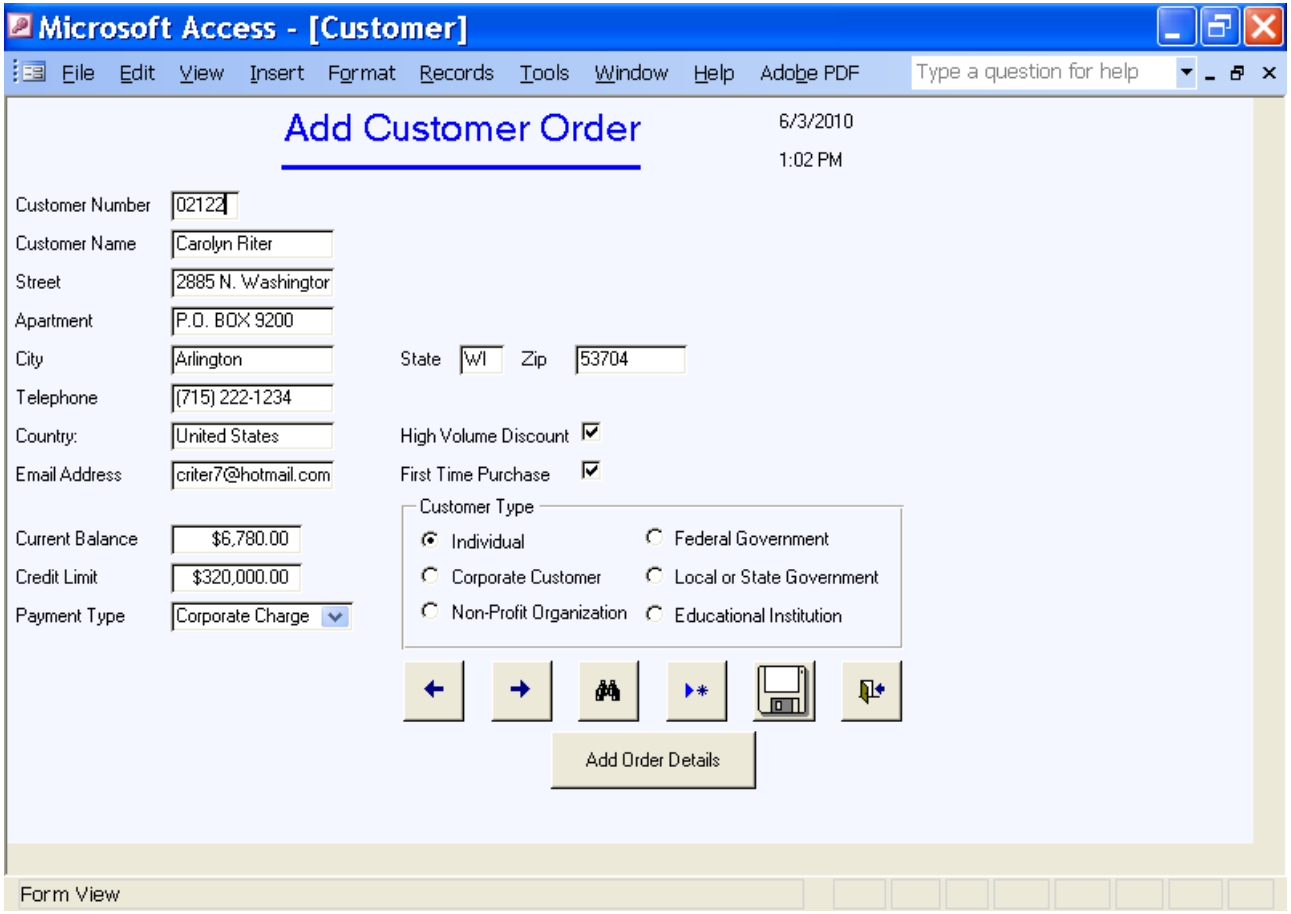
Figure 10.3 The Designer Has Many GUI Components That Allow Flexibility in Designing Input Screens for the Web or Other Software Packages

This example is from Microsoft Access



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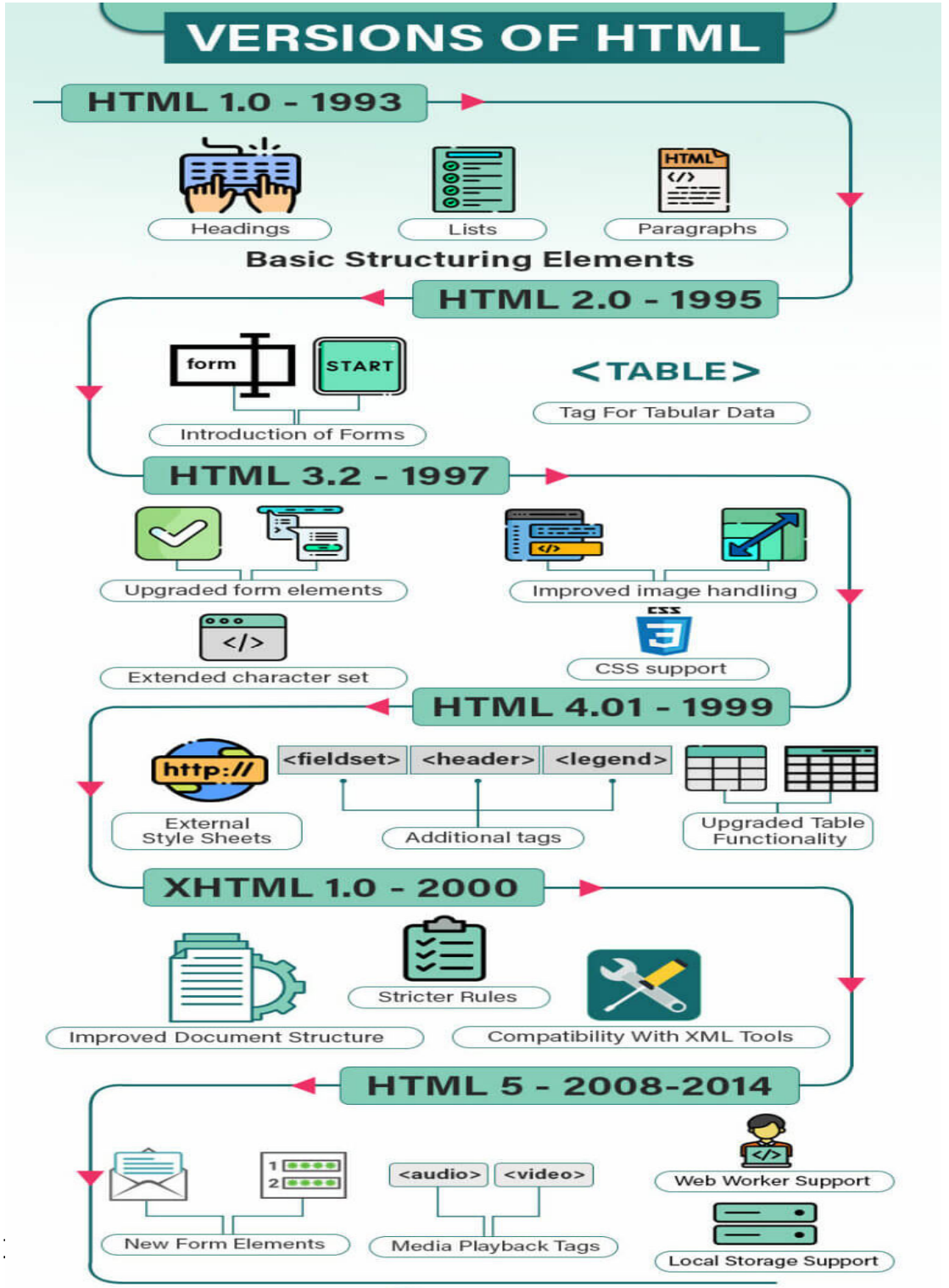
What is HTML5?

- **HTML5** is the **latest major version of HTML (HyperText Markup Language)**, the standard language used to **structure and present content on the web.**



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What makes HTML5 important?

- HTML5 expanded HTML beyond just text and images, making modern web apps possible **without extra plugins** (like Flash).

Browser	Icon	HTML5 Support	Notes
Google Chrome	🌐	✅ Full	Fast, frequent updates, excellent dev tools
Mozilla Firefox	🦊	✅ Full	Open-source, strong standards & privacy
Apple Safari	🧭	✅ Full	Optimized for macOS & iOS
Microsoft Edge	🔵	✅ Full	Chromium-based, replaces IE
Chrome (Android)	📱🌐	✅ Full	Best HTML5 support on Android
Safari (iOS)	📱🧭	✅ Full	Default browser on iPhone & iPad
Samsung Internet	📱📘	✅ Full	Popular on Samsung devices
Internet Explorer	❌	❌ Limited	Deprecated, not recommended

HTML5's Key Features

- ✓ New semantic elements
  - Make web pages more meaningful and easier to read for browsers, search engines, and screen readers
- ✓ Native audio and video
  - HTML5 allows media playback without plugins
- ✓ Better support for graphics
- ✓ Improved Forms
- ✓ APIs for web applications that works with JavaScript to support:
  - Local storage (Offline Data)
  - Geolocation
  - Drag and drop
  - Web workers (background processing)

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HTML5 Browsers Features (1 of 4)

- Placeholder puts a small amount of hint text into a text input box, displaying in a lighter color
- When the cursor is placed in the field the hint text vanishes. Reappears if field is empty.
- Input Types Upgrades: introduced **semantic input types** that browsers understand.
- Common Input Types:

Input Type	Purpose	Example
email	Email format validation	user@email.com
tel	Phone number input	012-3456789
url	Website address	https://example.com
number	Numeric input	Quantity
date	Date selection	Booking date

HTML5 Browsers Features (2 of 4)

- These appear as normal text boxes on a computer
- When used on a tablet or smartphone, they can be used to customize the pop-up keyboard
  - Telephone number, the keyboard layout changes to a number pad
  - A URL, the keyboard includes a .com button
  - An email address, the keyboard includes an @ symbol
- Helps the user enter data quickly and accurately

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HTML5 Browsers Features (3 of 4)

- A datalist displays a drop-down list of predefined suggestions to make entry easier for the user
- As the user begins to type the first few letters, the datalist is displayed
- The user can choose one of the items in the list to make a selection
- This is used in the autocomplete function
- Real-life example
  - City names
  - Product search
  - Country selection

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HTML5 Browsers Features (4 of 4)

- HTML5 provides **built-in date pickers**. A calendar control to select a date, a date and time, or a local date and time
- Selecting dates from a pop-up calendar is easier and less error-prone than entering text
- **Variations:** `date, datetime-local, month & time`
- Built-In Validation (Client-Side): supports **automatic validation** before form submission.
- **Browser checks for:** Empty required fields, Invalid email format & Numbers out of range
- Example:

```
<input type="email" required>
<input type="number" min="1" max="10">
```

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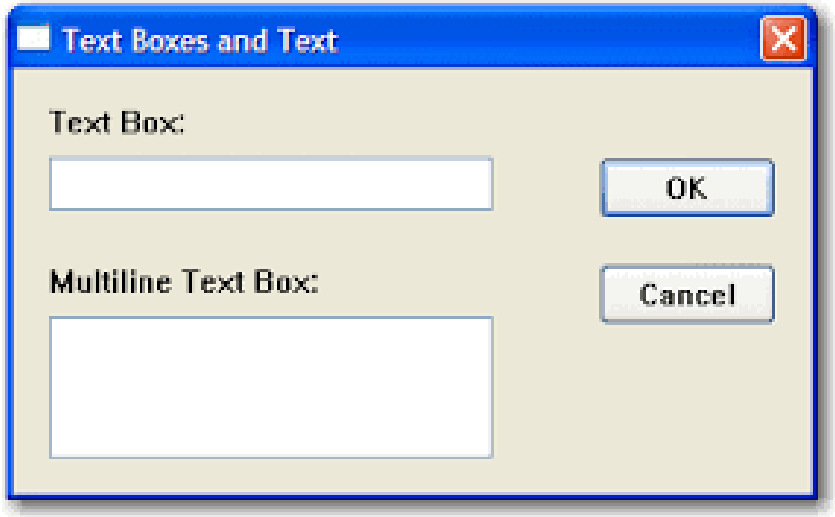
Text Boxes

- Text boxes should be large enough to accommodate all the characters
- Captions should be to the left of the text box
- Character data is left-aligned within the box
- Numeric data is right-aligned



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Check Boxes

- Check boxes are used for nonexclusive choices
- Check box text or label is placed to the right of the check box
- If there are more than ten check boxes, group together in a bordered box

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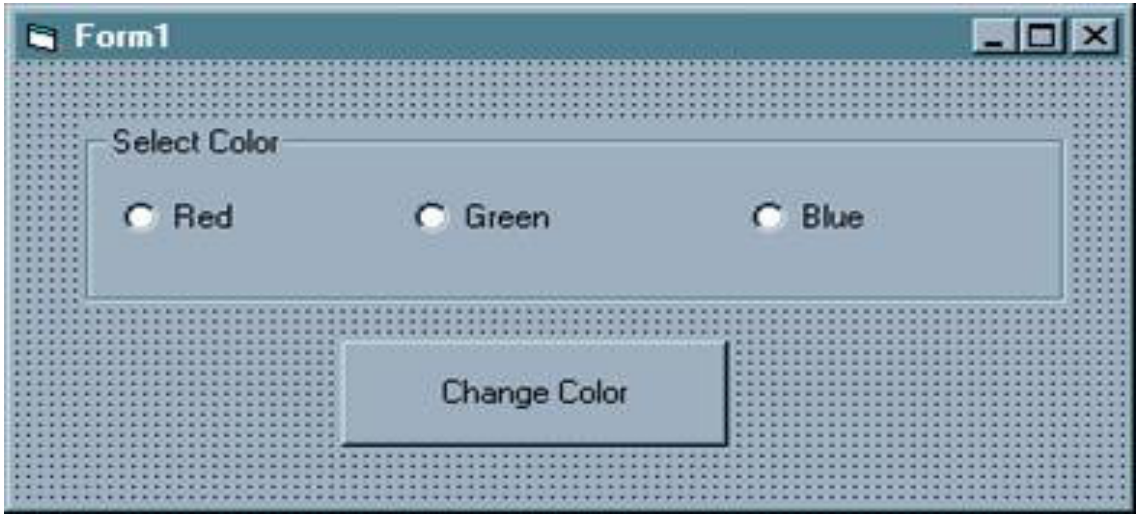
Option Buttons

- Option or radio buttons are used for exclusive choices
- Choices are listed to the right of the button, in some sequence
- Often they are placed in a rectangle called an option group
- If more than six option buttons are used, a list box or drop-down list box should be implemented



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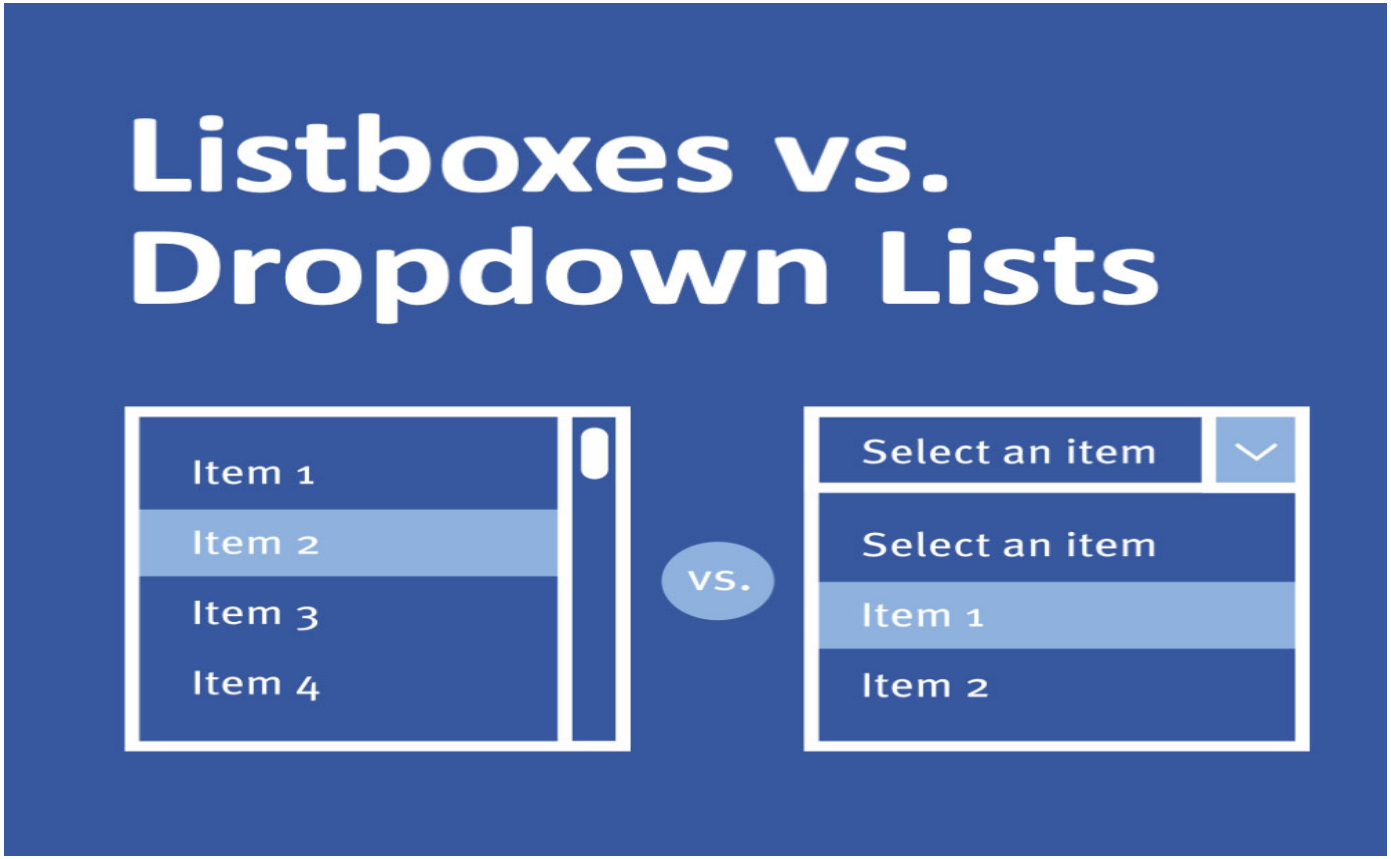
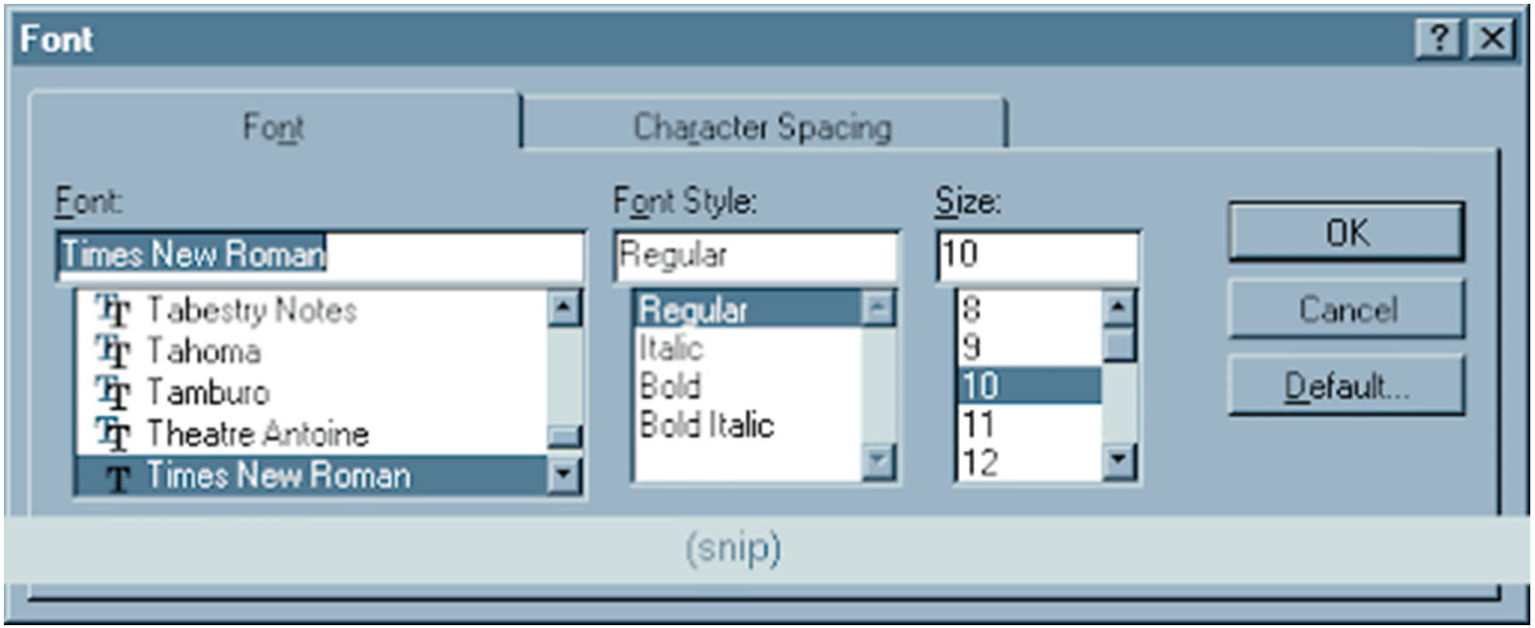
List and Drop-Down List Boxes

- Used when there is little room available on the page
- If there is a commonly selected choice, it is usually displayed in the drop-down list by default



Tab Control Dialogue Boxes

- Create a separate tab for each unique feature
- Place the most commonly used tabs in front and display them first
- Consider including three basic buttons in your design:
  - OK
  - Cancel
  - Help



Sliders, Spin Buttons, and Image Maps

- Sliders and spin buttons are used to change data that have a continuous range of values
- Image map fields are used to select values within an image
- Example:
  - ✓ Sliders: https://cssslider.com/image-slider-jquery-30.html
  - ✓ Spin Buttons: https://jqueryui.com/spinner/
  - ✓ Image map: https://www.w3schools.com/html/html_images_imagemap.asp

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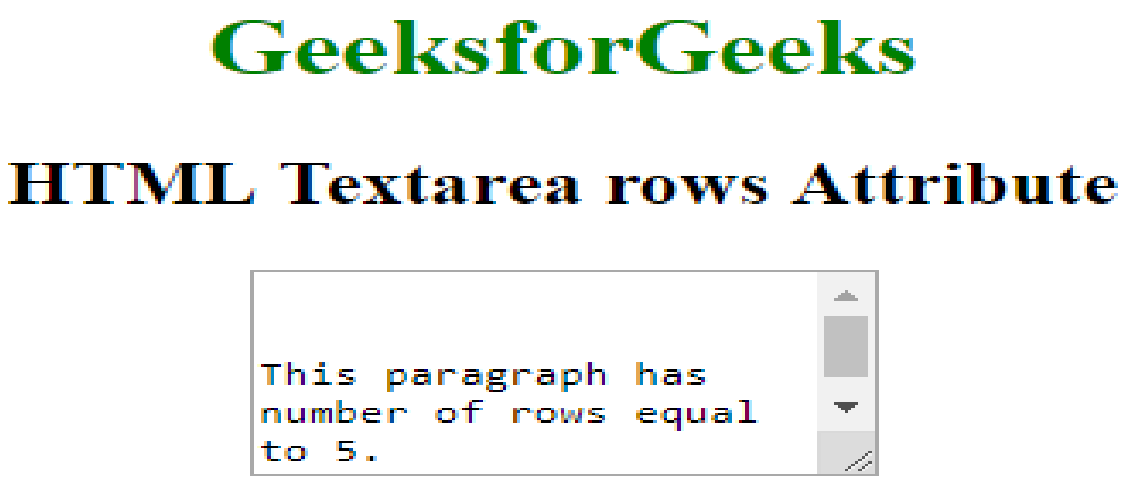
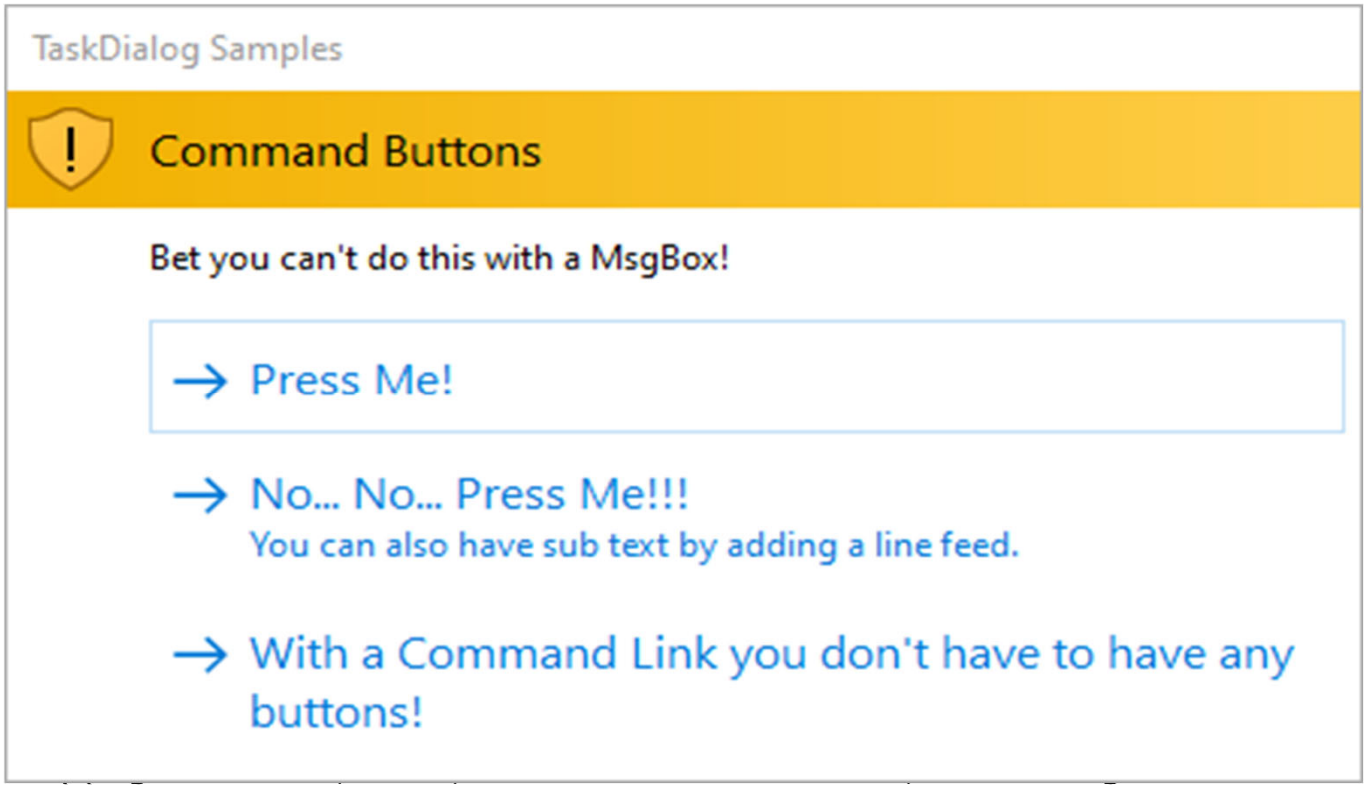
Text Area

- A text area is used for entering a larger amount of text
- Can view data larger then the box area
- Handling text:
  - Hard return is used to force new lines
  - Use word wrap within the text area



Message Boxes and Command Buttons

- Message boxes are used to warn users and provide feedback messages in a dialog box
- Command buttons perform an action when the user selects it



Form Controls and Values

- Each control in a GUI interface stores data associated with the control
- Web pages use a name and value pair that are transmitted to the server or in an email sent along with the form

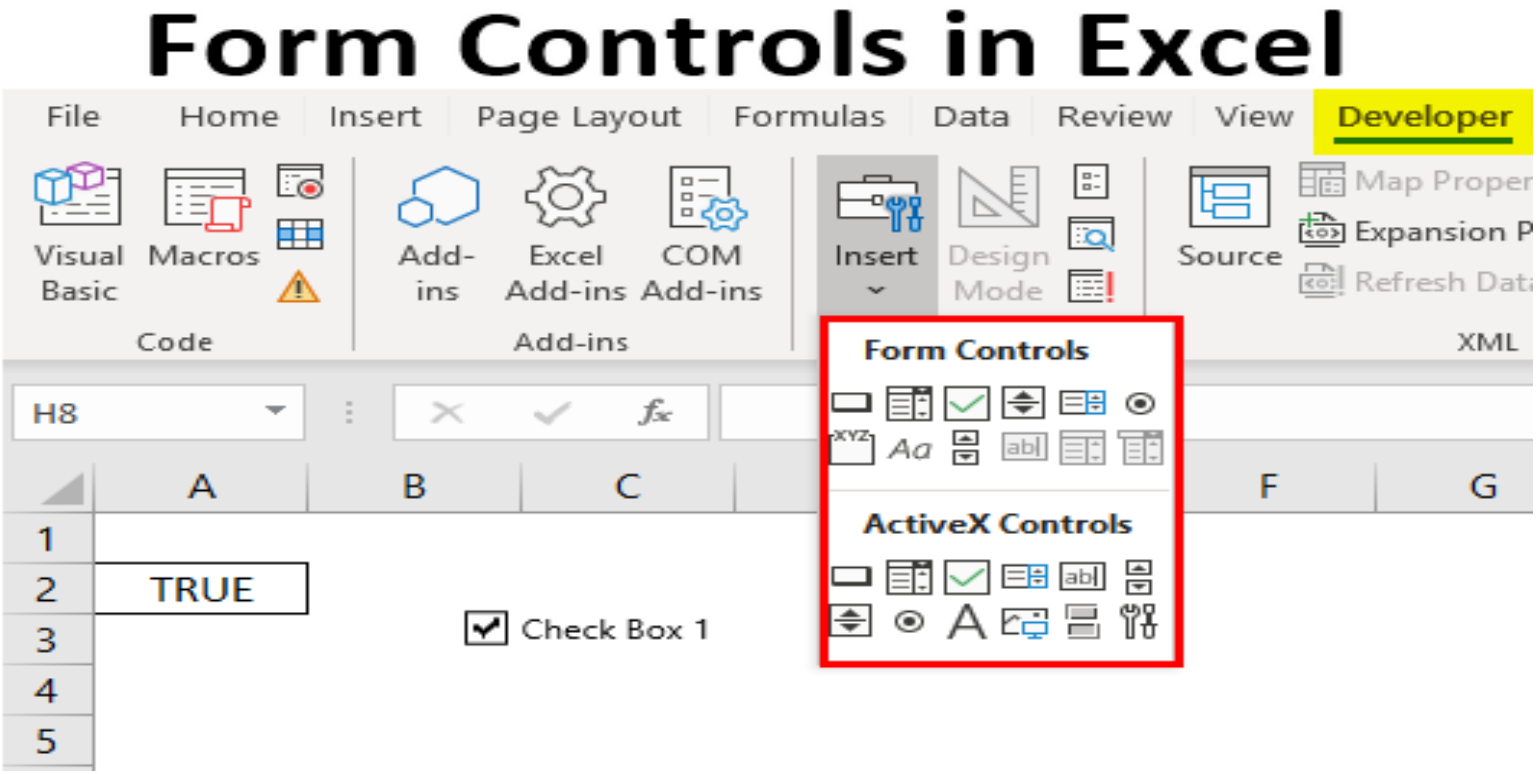


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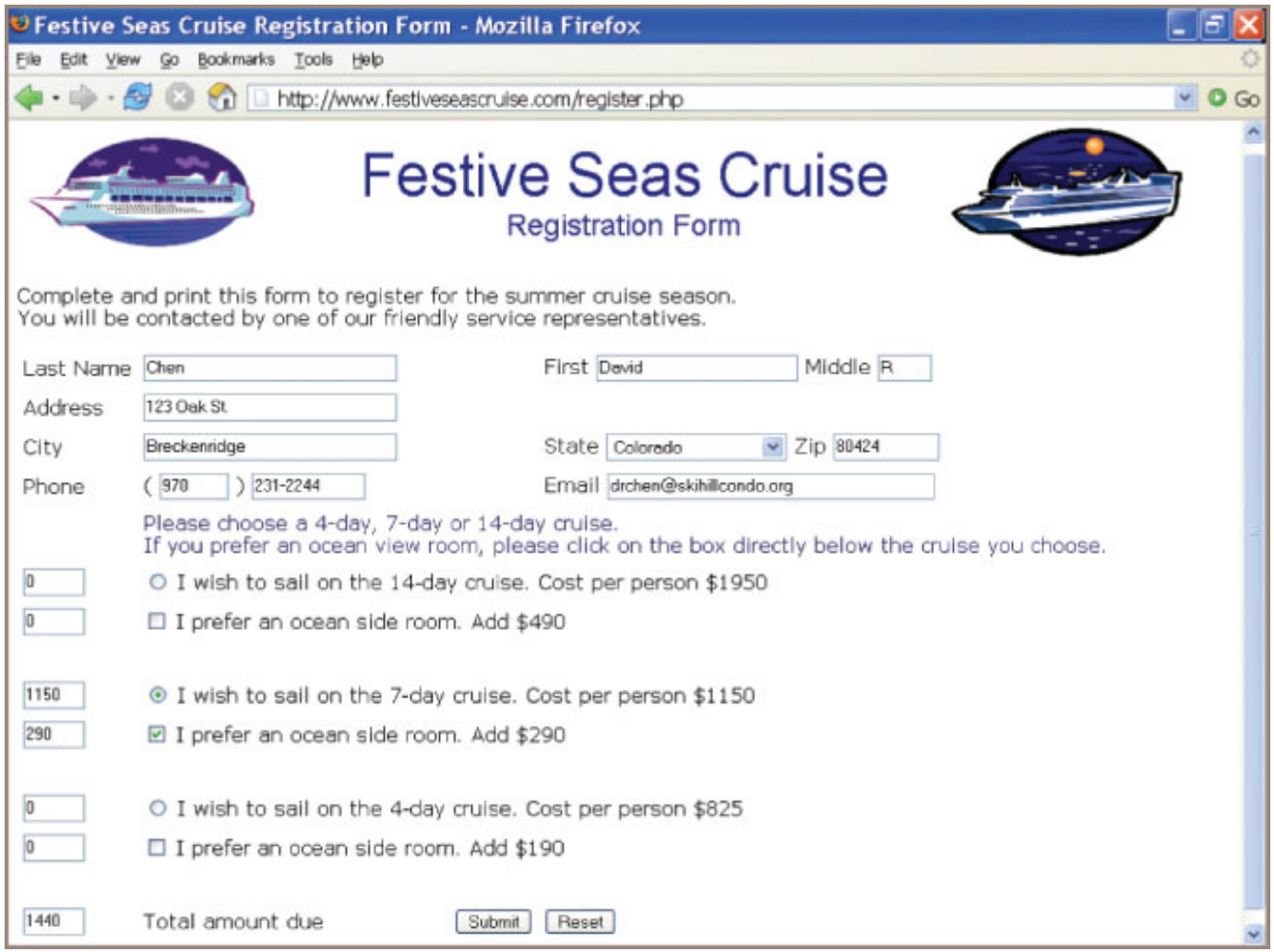
Figure 10.4 Web Based Input Form



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Hidden Fields

- Not visible to the viewer
- Do not take up any space on the Web page
- Can only contain a name and value
- Used to store values sent from one Web form to the server
- E.g.: https://www.wufoo.com/guides/using-hidden-fields-forms/

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Event-Response Charts

- Used to:
  - List the variety of events that can occur
  - Show what should happen
  - Build a Web form that requires minimal action from the user
  - Explore improvements to the web page
- Events may be used to:
  - Control navigation between web pages
  - Change the contents of drop-down lists

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Event-Response Charts

Event	Source	Trigger	Activity	Response	Destination
Customer logs on	Customer	Customer number and password	Find customer record and verify password. Send Welcome Web page.	Welcome Web page	Customer
Customer browses items at Web storefront	Customer	Item information	Find item price and quantity available. Send Item Response Web page.	Item Response Web page	Customer
Customer places item into shopping basket at Web storefront	Customer	Item purchase (item number and quantity)	Store data on Order Detail Record. Calculate shipping cost using shipping tables. Update customer total. Update item quantity on hand.	Items Purchased Web page	Customer
Customer checks out	Customer	Clicks "Check Out" button on Web page	Display Customer Order Web page.	Verification Web page	
Obtain customer payment	Customer	Credit card information	Verify credit card amount with credit card company. Send.	Credit card data Customer feedback	Credit card company Customer
Send customer email		Temporal, hourly	Send customer an email confirming shipment.		Customer

Figure 10.5 Website to Estimate Resort Cost

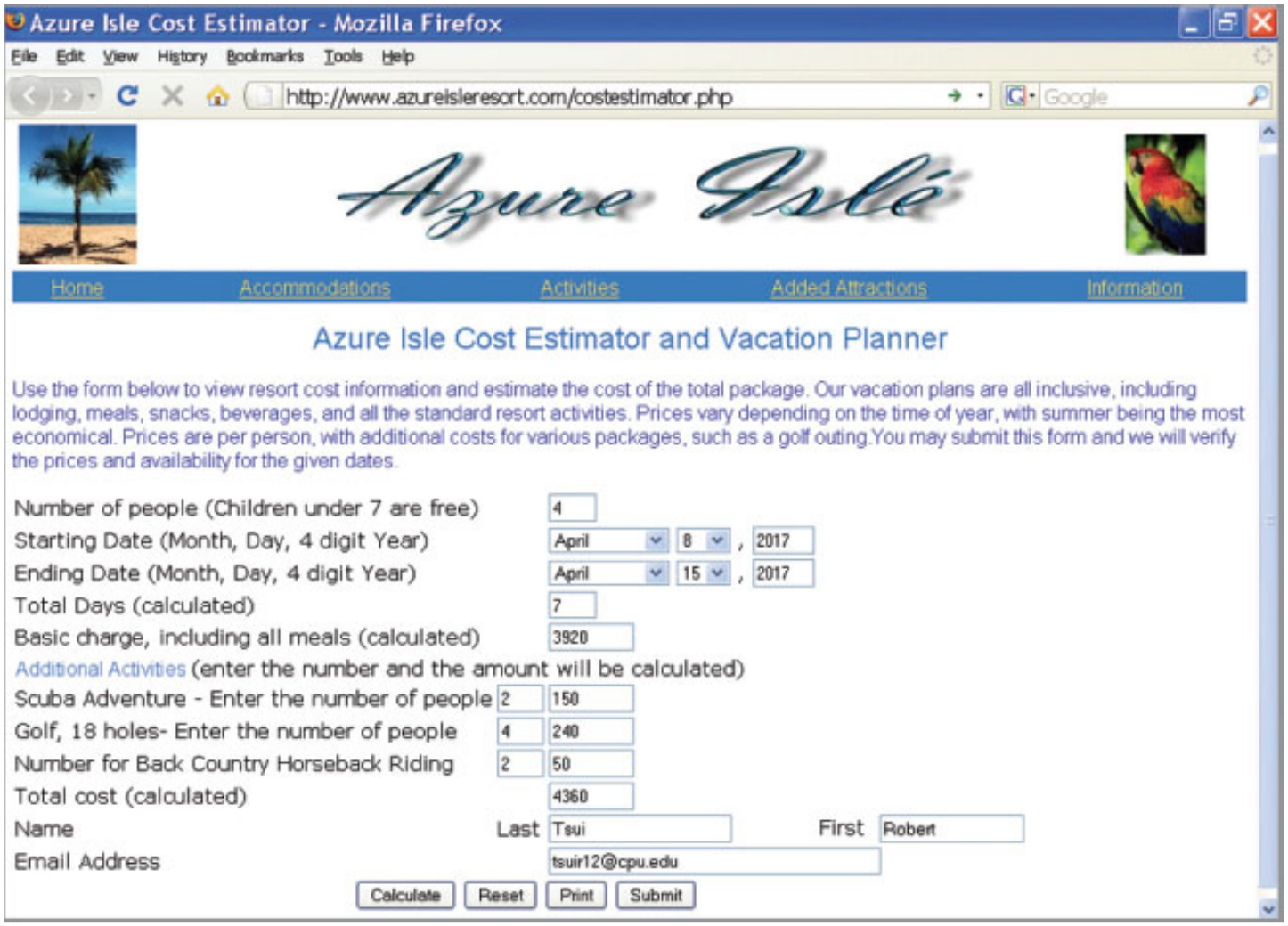

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Figure 10.6 Event-Response Chart

Form Control	Event	Response
Web Page	Page loads	Place the current year in the **Starting Year** and **Ending Year** fields. Place the cursor in the **Number of People** field.
Number of People	Value changes	Verify that **Number of People** contains a number greater than zero.
Starting Month	Selection changes	Set the **Ending Month** in the drop-down list to the **Starting Month**. If the month is less than the current month, change the **Starting Year** and **Ending Year** values to the next year.
Starting Day	Selection changes	Set the **Ending Day** in the drop-down list to the **Starting Day**. Use the **Starting Month** value to set the **Starting Year** and **Ending Year** values.
Starting Year	Receives focus	Use the **Starting Month** value to calculate the **Starting Year** and **Ending Year** values.
Number of Days	Receives focus; value changes	Calculate the **Number of Days** that the customer is staying. If the number is less than or equal to zero, display an error message.
Basic Charge	Receives focus	Calculate the **Basic Charge** and put the cursor in the **Scuba Adventure Number of People** field.
Scuba Adventure Number of People	Receives focus	Select the current amount displayed (zero) so the customer may replace it.
Scuba Adventure Number of People	Value changes	Calculate the **Scuba Cost** based on the value of **Scuba Adventure Number of People** and put the cursor in the **Golf Number of People** field.
Scuba Cost	Value changes	Recalculate the **Scuba Cost** and put the cursor in the **Golf Number of People** field.
Golf Number of People	Receives focus	Select the current amount (zero) so the customer may replace it.
Golf Number of People	Value changes	Calculate the **Golf Cost** and put the cursor in the **Horseback Riding Number of People** field.
Golf Cost	Value changes	Recalculate the **Golf Cost** and put the cursor in the **Horseback Riding Number of People** field.
Horseback Riding Number of People	Receives focus	Select the current amount displayed (zero) so the customer may replace it.
Horseback Riding Number of People	Value changes	Calculate the **Horseback Riding Cost** based on the value of **Horseback Riding Number of People** and put the cursor in the **Last Name** field.
Horseback Riding Cost	Value changes	Recalculate the **Horseback Riding Cost** and put the cursor in the **Last Name** field.
Total Cost	Value changes	Calculate the total cost and put the cursor in the **Last Name** field.
Calculate Button	Button clicked	Validate the form data and display an error message if any errors occur. Calculate the **Total Cost** if there are no errors.
Reset Button	Button clicked	Clear the form and place the current year in the **Starting Year** and **Ending Year** fields. Place the cursor in the **Number of People** field.
Print Button	Button clicked	Validate the form data and display an error message if any errors occur. Calculate **Total Cost** if there are no errors. Use a Web cookie to pass the data to a confirmation page that does not allow the users to change any data.
Submit Button	Button clicked	Validate the form data and display an error message if any errors occur. Calculate **Total Cost** if there are no errors. Send the form to the server and send confirmation to the user.

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Dynamic Web Pages

- Web pages that change themselves as the result of some user action
- Advantage
  - Modify themselves quickly
- Disadvantage
  - Will not work if JavaScript is turned off
  - Dynamic Web pages may not be compliant with the American Disabilities Act

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What are Three Dimensional Web Pages?

- Do **not** mean true 3D graphics like games.
- Instead, they refer to **layered, depth-based interfaces** that create a *sense of depth* using:
  - Overlapping layers
  - Pop-ups, modals, cards
  - Sliding panels
  - Drop-downs, drawers, overlays
- Think in terms of **"UI layers stacked on top of each other"**, not 3D models.

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How does Three Dimensional Web Pages?

- 3D web pages are created using **front-end technologies**, mainly:
- **Core Technologies**
  - **HTML** → Structure of elements
  - **CSS** → Layering, shadows, depth
  - **JavaScript** → Triggers & interactions (*controls when layers appear or disappear).*
- Layered input ***reduces cognitive (memory) load*** and improves ***input accuracy***.

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Three Dimensional Web Pages (1 of 2)

- Use stacked layers
- Code is moved in front of the Web page
- Code may be created using JavaScript
- Analyst must determine when to use layers

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Three Dimensional Web Pages (2 of 2)

- Analyst must determine:
  - How is the layer built?
  - What events cause the layer to be created?
  - What events remove the layer?
  - Where should the block be placed?
  - The size of the block?
  - How to frame the region?
  - What happens when an option is selected?

- Example: https://wpdean.com/css-modals/

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Ajax (Asynchronous JavaScript and XML)

- Uses JavaScript and XML
- Allows Web developers to build a Web page that works like a traditional desktop program
- AJAX allows a web page to **send or receive data** from a server in the background, **without reloading the whole page**
- The data may be either a small text file or an XML document containing many customers or repeating data
- Has the advantage of making the Web work faster and of providing a smoother viewing experience for users
- The disadvantages are that JavaScript must be enabled

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AJAX's Working Flows & Technologies

Step	User / System Action	What Happens	Technology Involved	Purpose
1	User action	User types, clicks, selects, or scrolls	HTML, CSS	Capture user interaction
2	Event triggered	Browser detects event (e.g., click, keyup)	JavaScript	Monitor and respond to actions
3	AJAX request sent	Data sent to server in background	JavaScript, AJAX	Avoid full page reload
4	Server processing	Server validates or retrieves data	Server, Database	Apply business logic
5	Data returned	Server sends small data response	JSON	Fast, lightweight data transfer
6	Partial page update	Only part of page updates	JavaScript, HTML, CSS	Improve speed & user experience

Example of AJAX websites

Website / System	User Action	What User Sees	AJAX Role	Event → Response (Event-Response Chart)	Why It Matters
Google Search	User types keywords	Search suggestions appear instantly	Sends typed characters and retrieves suggested terms	**Event:** User types letters → **Response:** Display suggestion list	Faster searching, no page reload
Gmail	User opens inbox / marks email as read	Inbox updates immediately	Retrieves and updates email data in background	**Event:** User clicks email → **Response:** Load email content instantly	Desktop-like experience, efficient email management
Google Maps	User drags or zooms map	New map areas load smoothly	Requests new map tiles without reload	**Event:** User drags map → **Response:** Load new map data	Smooth navigation, real-time interaction
Online Shopping Cart (Shopee / Lazada)	User adds item to cart	Cart count updates, side panel slides out	Sends product ID and updates cart data	**Event:** Click “Add to Cart” → **Response:** Update cart & show confirmation	Faster checkout, improved user experience

Figure 10.7 Dynamic Web Page

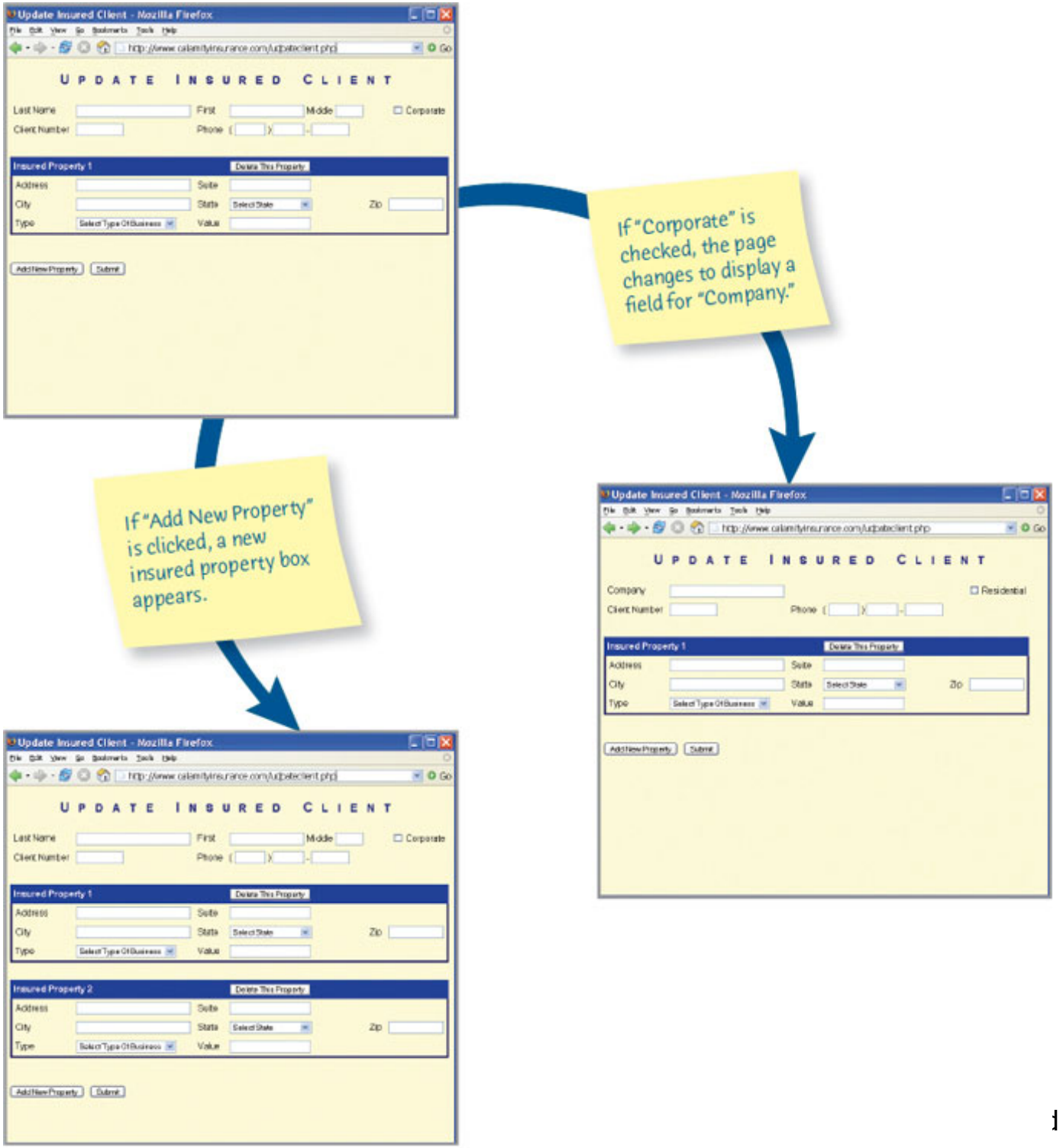
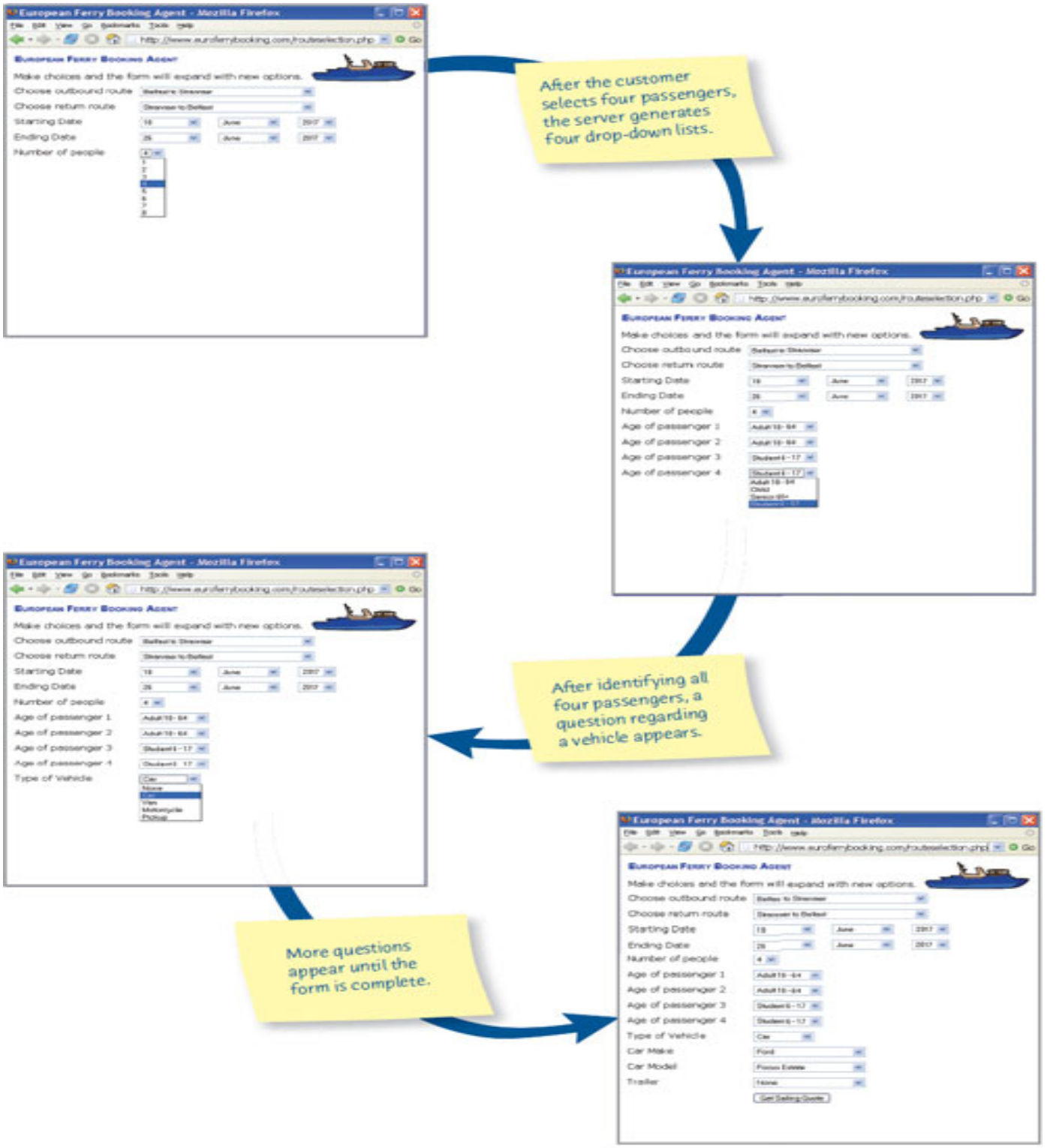


Figure 10.8 Ajax Techniques and Dynamic Web Page



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Color

- Communicating Meaning (Semantic Colors)

Color	Common Meaning	Input Design Usage
Red	Error / danger	Invalid input, missing required field
Green	Success / valid	Form submitted successfully
Blue	Action / trust	Submit button, links
Yellow / Orange	Warning	Confirmation needed
Grey	Disabled / inactive	Unavailable fields

- The five most legible foreground/background color (**high-contrast foreground/background pairs**):

Foreground	Background	Why It Works
Black	Yellow	Very high contrast
Green	White	Easy on eyes
Blue	White	Clean & professional
White	Blue	Strong emphasis
Yellow	Black	Attention-grabbing

Website Design

- Effective Web design using:
  - Hamburger icon (☰)
  - Context-sensitive help ⓘ
  - Input validation
  - Using a breadcrumb trail
  - Fat footer

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Hamburger Icon (☰)

- The hamburger icon / hamburger menu is depicted with three stacked bars
  - The three bars are stacked together in a parallel position within a square
  - Place this icon at the top of a graphical display
  - User can click on the hamburger icon to display a menu of the Web site

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Context Sensitive Help (ⓘ or ❓)

- Help users as they progress through an application or a Web site
  - Cursor changes to a question mark as the user hovers over a word that is part of the context-sensitive help feature
  - Clicking the question mark will bring up a specific comment designed to help the user with the feature they are currently trying to use
- Why it matters in input design: Reduces user errors, Eliminates need for long instructions & Supports first-time user

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Input Validation

- A mechanism that **checks data correctness** before allowing submission.
- Input validation is a way to prevent the user from entering the wrong type of data or incorrectly formatted data into the file or database
- The design is such that the user is not permitted to advance to the next input field until the problem is corrected

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Breadcrumb Trail

- A breadcrumb trail is a navigational element to let the user know where they are on a large Web site, in a program, or in a complex document
- Show the path the user has used to arrive at that page, such as:
  - Home>Help>Contact Us>Email>Live Chat
- Where it is most useful
  - ✓ E-commerce websites
  - ✓ Large information systems
  - ✓ Multi-step processes

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Fat Footer

- Bottom of a Web page is called the footer
- Fat footers are an entire section of the Web site and can feature:
  - Social media links
  - Recommend what visitors should check out next on the Web site
  - Add search engine keywords and tags
- Typical contents: Contact information, Help & FAQ, Policies (Privacy, Terms) or Sitemap shortcuts

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How these elements work together?

Element	Main Purpose	Contribution to Input Design
Hamburger icon	Navigation control	Reduces distraction
Context-sensitive help	Guidance	Reduces input errors
Input validation	Accuracy	Ensures correct data
Breadcrumb trail	Orientation	Prevents user confusion
Fat footer	Secondary navigation	Improves accessibility

- Effective web design is not about adding features, but about guiding users smoothly through input and navigation tasks.

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Website Page Design (1 of 2)

- Provide clear instructions
- Demonstrate a logical entry sequence for fill-in forms
- Use a variety of text boxes, push buttons, radio buttons, drop-down lists, and other GUI features
- Provide a scrolling text box if you are uncertain how much text will be entered

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Website Page Design (2 of 2)

- Include two basic buttons: Submit and Clear
- If the form is lengthy, divide it into several simpler forms on separate pages
- Create a feedback screen that lists error messages if a form has not correctly been filled out
- Website page design focuses on guiding users through clear, logical, and user-friendly input processes to ensure accuracy and usability.

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Website Page Design – Summary table

Design Principle	Purpose	Benefit
Clear instructions	Guide users	Fewer errors
Logical sequence	Natural flow	Faster completion
Proper controls	Match data type	Higher accuracy
Scrolling text box	Handle long input	Clean layout
Action buttons	User control	Clear submission
Multi-page forms	Reduce overload	Better completion
Feedback messages	Error correction	Better data quality

- Good: https://www.justinmind.com/blog/form-examples-web-mobile/
- Bad: https://www.zuko.io/blog/7-examples-of-bad-web-form-designs

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Ecommerce Applications

- Ecommerce applications involve more than just good designs of Web sites.
- Customers need to feel confident in the site
  - Shopping cart
  - Customer can edit the quantity of the item ordered or can remove the item entirely

Aspect	GOOD E-commerce Design	BAD E-commerce Design
Navigation	Simple & intuitive	Confusing & cluttered
Product Images	High-quality & zoomable	Low-quality or missing
Mobile Experience	Fully responsive	Broken or unreadable
Checkout	Short & simple	Long & complex
Trust Signals	Reviews, SSL, clear pricing	No reviews, hidden fees
Visual Design	Clean & consistent	Outdated or messy

Summary (1 of 4)

- Guidelines for well-designed input forms:
  - Forms must be easy to fill out
  - Forms must meet the purpose for which they are designed
  - Forms must be designed to ensure accurate completion
  - Forms must be pleasing and attractive

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Summary (2 of 4)

- Guidelines for well-designed displays:
  - Displays must be kept simple
  - Displays must be consistent in presentation
  - Design must facilitate movement between pages
  - Displays must be attractive

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Summary (3 of 4)

- Guidelines for Web fill-in forms:
  - Provide clear instructions
  - Demonstrate a logical entry sequence for fill-in forms
  - Use a variety of text boxes, push buttons, drop-down menus, check boxes, and radio buttons
  - Provide a scrolling text box if you are uncertain about how much space users will need to respond to a question

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Summary (4 of 4)

- Guidelines for Web fill-in forms: [continued]
  - Prepare two basic buttons on every Web fill-in form: Submit and Clear Form
  - If the form is lengthy and the users must scroll extensively, divide the form into several simpler forms on separate pages
  - Create a feedback screen that highlights errors in an appropriate color and refuses submission of the form until mandatory fields are correctly filled in

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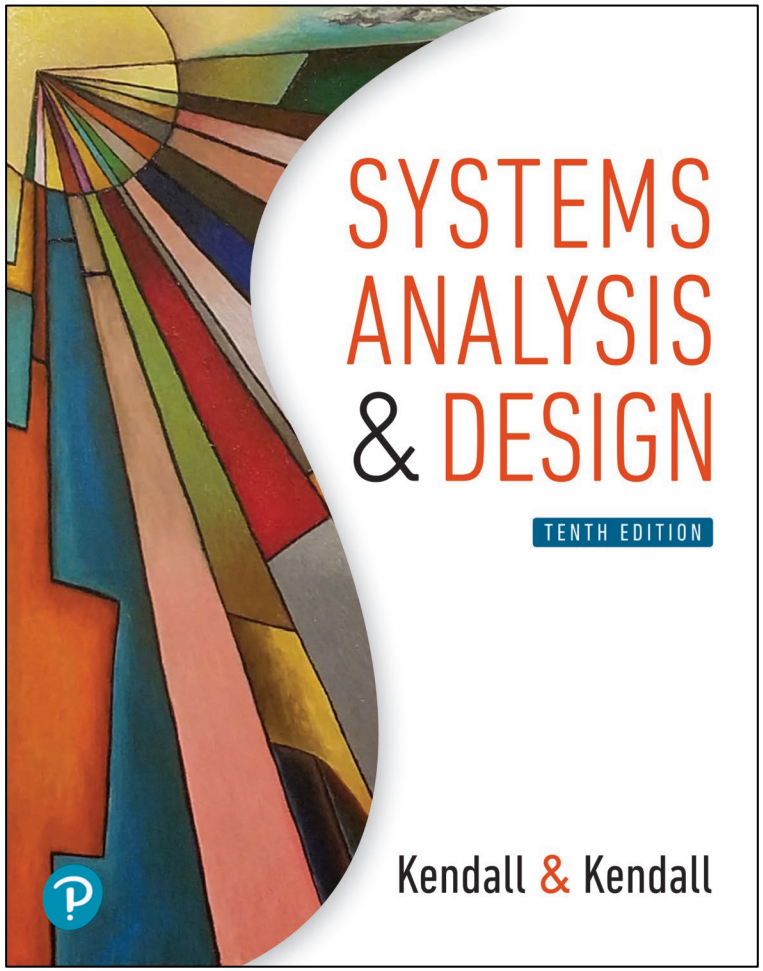
Systems Analysis & Design

Tenth Edition



Chapter 11

Designing Accurate Data Entry Procedures

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Learning Objectives

11.1 Understand the uses of effective coding to support users in accomplishing their tasks.

11.2 Design effective and efficient data capture approaches for people and systems.

11.3 Recognize how to ensure data quality through validation.

11.4 Articulate accuracy advantages of user input on ecommerce Web sites .

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Accurate Data-Entry Objectives

- Effective coding → creating meaningful coding for data.
- Efficient data capture → designing efficient data capture approaches
- Effective data capture → assuring complete and effective data capture
- Assuring data quality through validation

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Major Topics

- Effective coding
- Types of codes
- Guidelines for coding
- Validation methods
- Check digits
- Ecommerce accuracy

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Effective Coding

- Coding—the process of putting ambiguous or cumbersome data into short, easily entered digits or letters.
- Data that are coded require less time to enter
- Coding helps to reduce the number of items entered
- Coding can help in sorting data during the data transformation process
- Coded data can save valuable memory and storage space

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Human Purposes for Coding

- Specific types of codes allow us to treat data in a particular manner.
- Keeps track of something
- Classifies information
- Conceals information
- Reveals information
- Requests appropriate action

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Keeping Track of Something

- Sometimes we want to identify a person, place, or thing just to keep track of it.
- Simple sequence code
- Alphabetic derivation codes

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Simple Sequence Codes

- A **Single (Simple) Sequence Code** is a **system-generated running number** assigned to each record **in the order it is created**, with **no embedded meaning**.
- A number that is assigned to something if it needs to be numbered
- No relation to the data itself

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Figure 11.1 Simple Sequence Code Example

Order #	Product	Customer
5676	Rocking Chair/with Leather	Arthur Hook Jr.
5677	Dining Room Chair/Upholstered	Millie Monice
5678	Love Seat/Upholstered	J. & D. Pare
5679	Child's Rocking Chair/Decals	Lucinda Morely

- Key Characteristic:
  - ✓ Numeric only (e.g., 0001, 0002, 0003)
  - ✓ Auto-incremented by the system
  - ✓ Unique and stable
  - ✓ No human-readable meaning

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Simple Sequence Codes (Advantages)

- Eliminates the possibility of assigning the same number
- It gives users an approximation of when the order was received
- Fast & error-free input (system generated)
- Excellent as primary keys in databases
- Easy to sort, index, and retrieve
- Ideal for high-volume transactions

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Simple Sequence Codes (Disadvantages)

- When you do not wish to have someone read the code to figure out how many numbers have been assigned
- Can reveal business volume (security concern)
- When a more complex code is desirable to avoid a costly mistake
- Not meaningful to humans
- Cannot classify data without joining other tables

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Alphabetic Derivation Codes

- An **Alphabetic Derivation Code** is created by **extracting letters from a meaningful attribute**, usually a **name**, **category**, or **description**.
- A commonly used approach in identifying an account number
- **Key Characteristics**
  - Alphabetic or alphanumeric
  - Derived from human-readable data
  - Often mnemonic (easy to remember)
  - Meaningful to users

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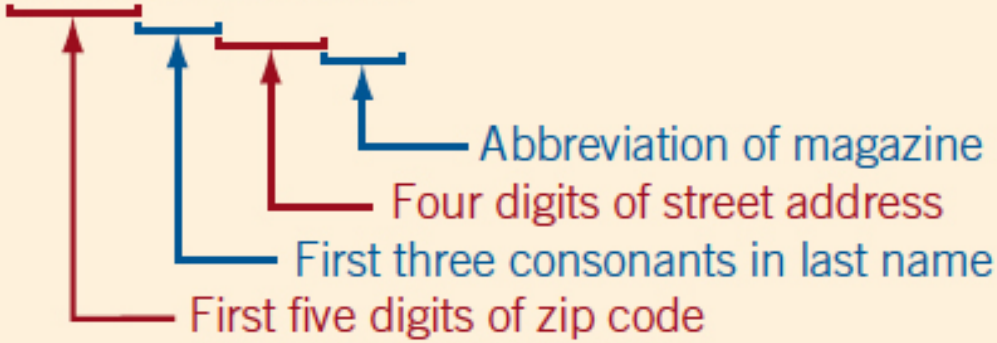
Figure 11.2 Alphabetic Derivation Code Example

Code	Explanation of Code
68506KND7533TVG	99999XXX9999XXX



- First five digits come from the first five digits of the subscribers zip code.
- Next three are the first three consonants in the subscribers name.
- Next four numbers are the street address.
- Last three make up the code for the magazine.

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Alphabetic Derivation Codes (Advantages)

- Easy for humans to remember
- Helpful for **manual entry & lookup**
- Meaning visible without opening full record
- Reduces cognitive load for users

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Alphabetic Derivation Codes (Disadvantages)

- Breaks when names change
- When the alphabetic portion is small or when the name contains fewer consonants than the code requires
  - When the alphabetic portion is small: the name Po.
- Some of the data may change
- **Not guaranteed unique**
- Limited combinations (collisions)
- Risk of bias or cultural mismatch

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Why Modern Systems Use BOTH?

- Modern systems **separate technical identity from human readability**.
- Example: E-commerce Order System

Field	Purpose
order_id = 105923	**Single Sequence Code** (Primary Key)
order_ref = ORD-DAN-2025	**Alphabetic / Mnemonic Code** (User-Facing)

- This gives:
  - ✓ System efficiency **AND**
  - ✓ User friendliness **WITHOUT risk**

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Classification Information

- **Classification information** uses **codes to group data into meaningful categories** based on shared characteristics.
- Affords the ability to distinguish between classes of items
- Must be mutually exclusive → data are unclear and not as readily interpretable when coding classes are not mutually exclusive.
- Classification codes
- Block sequence codes

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Classification Codes

- Used to distinguish one group of data with special characteristics from another
- Can consist of either a single letter or a number
- A shorthand way of describing a person, place, thing, or event
- Listed in manuals or posted so that users can locate them easily

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Figure 11.3 Classification Code Example

Code	Tax-Deductible Item
I	**I**nterest Payments
M	**M**edical Payments
T	**T**axes
C	**C**ontributions
D	**D**ues
S	**S**upplies

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Block Sequence Codes

- An extension of the sequence code
- Data are grouped according to common characteristics
- Simplicity of assigning the next available number (within the block) to the next item needing identification

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Figure 11.5 Using a Block Sequence Code to Group Similar Software Packages

Code	Name of Software Package	Type
100	Apple Safari	Browser
101	Mozilla Firefox	
102	Microsoft Internet Explorer	
103	Google Chrome	
.		
.		
.		
200	Microsoft Access	Database
201	MySQL	
202	Oracle	
.		
.		
.		
300	Adobe Dreamweaver	Web design
301	Freeway Pro	
302	BlueGriffon	

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Concealing Information

- Codes may be used to conceal or disguise information
- Cipher Codes → the direct substitution of one letter for another, one number for another, or one letter for a number.

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Cipher Codes

- A **Cipher Code** disguises or conceals information by **substituting symbols, letters, or numbers** so the original meaning is **not directly visible**.
- The direct substitution of one letter for another, one number for another, or one letter for a number
- It is used when:
  - Information should not be obvious
  - Users are **not authorized** to interpret raw data
- Cipher codes are **NOT encryption** in the modern security sense.

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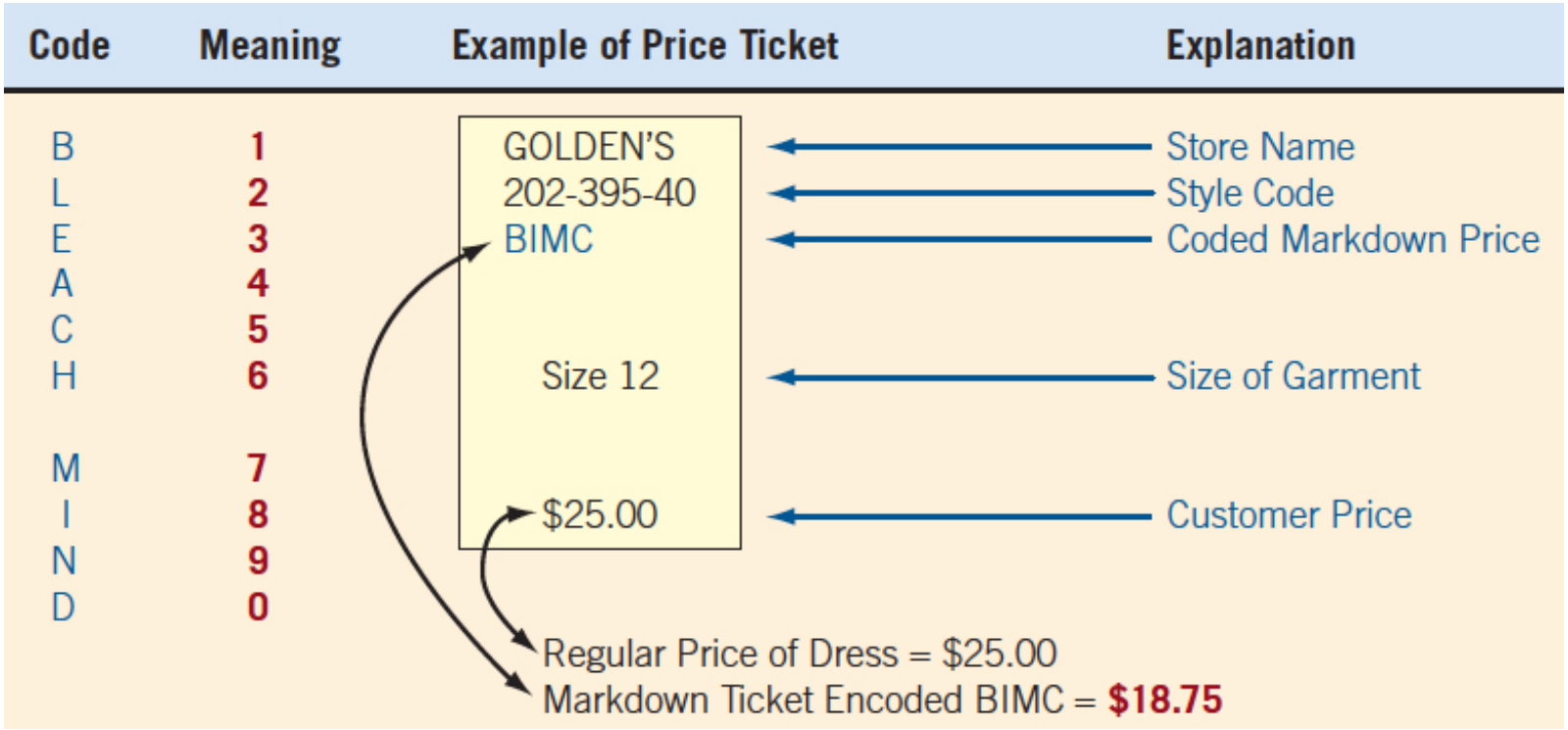
Figure 11.6 Encoding Markdown Prices with a Cipher Code is a Way of Concealing Price Information from Customers



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Revealing Information

- Sometimes it is desirable to reveal information to specific users through a code
- Makes the data entry more meaningful for humans
- Significant-digit subset codes
- Mnemonic codes
- Unicode

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Significant-Digit Subset Codes

- Used to help describe a product by virtue of its membership in many subgroups
- Possible to locate items that belong to a certain group or class
- Inquiries may be performed on a portion of the code
- Useful for a marketing product
- A numeric item description may consist of digits representing the:
  - Department
  - Product
  - Color
  - Size

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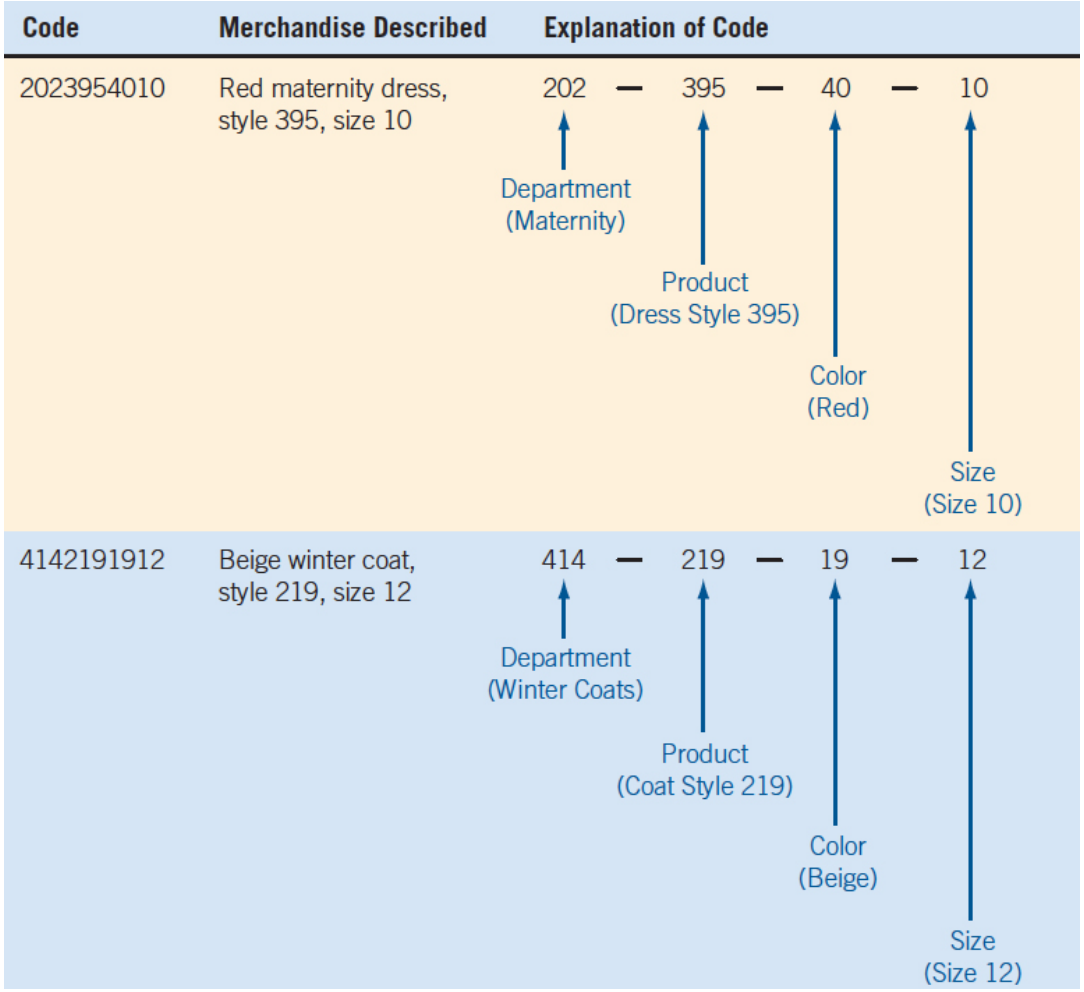
Figure 11.7 Using a Significant-Digit Subset Code Helps Employees Locate Items Belonging to a Particular Department



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Mnemonic Codes

- A mnemonic (pronounced nî-môn'-ïk) is a human memory aid
- A **Mnemonic Code** uses **letters, abbreviations, or familiar words** to help users **remember and recognize information easily**
- Any code that helps the data-entry person remember how to enter the data or the user remember how to use the information is a mnemonic
- Using a combination of letters and symbols affords a clear way to code a product so that the code is easily seen and understood

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Figure 11.8 Mnemonic Codes Function as Memory Aids by Using a Meaningful Combination of Letters and Numbers

Code	City Hospitals
BGH	**B**uffalo **G**eneral **H**ospital
ROS	**Ros**well Park Memorial Institute
KEN	**Ken**more Mercy
DEA	**Dea**coness Hospital
SIS	**Sis**ters of Charity
STF	**S**ain**t F**rancis Hospital
STJ	**S**ain**t J**oseph's Hospital
OLV	**O**ur **L**ady of **V**ictory Hospital

Code	Meaning
KUL	Kuala Lumpur
PEN	Penang
ADM	Administrator
INV	Invoice

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Unicode

- It is a **universal character encoding standard** that allows systems to **store, process, and display text from all languages consistently**.
- Unicode answers *"How do we represent characters?"*
- Includes all standard language symbols
- Has room for 1.1 millions characters
- The full set of Unicode characters are grouped by language and may be found at www.unicode.org

Character	Unicode
A	U+0041
中	U+4E2D
😊	U+1F60A

Requesting Appropriate Action

- Instruct either the computer or the decision maker about what action to take
- Function codes

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Function Codes

- Short numeric or alphanumeric codes used to spell out precisely what activities are to be accomplished
- Unlike IDs (who/what) or classification codes (type/group), **function codes trigger behavior**.
- Function codes are used to:
  - ✓ Control **processing logic**
  - ✓ Drive **business rules**
  - ✓ Trigger **system workflows**
  - ✓ Reduce input complexity
  - ✓ Improve processing speed

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Figure 11.9 Function Codes Compactly Capture Functions That the Computer Must Perform

Code	Function
1	Delivered
2	Sold
3	Spoiled
4	Lost or Stolen
5	Returned
6	Transferred Out
7	Transferred In
8	Journal Entry (Add)
9	Journal Entry (Subtract)

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General Guidelines for Coding

- Be concise
- Keep the codes stable
- Ensure that codes are unique
- Allow codes to be sortable
- Avoid confusing codes
- Keep the codes uniform
- Allow for modification of codes
- Make codes meaningful

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Be Concise

- Codes should be **as short as possible while still fulfilling their purpose**.
- Overly long codes mean more keystrokes and consequently more errors
- Long codes also mean that storing the information in a database will require more memory
- Short codes are easier for people to remember and easier to enter
- If codes must be long, they should be broken up into subcodes

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Keep the Codes Stable

- A code assigned to an entity **must remain the same throughout its lifetime**.
- Stability means that the identification code for a customer should not change each time new data are received
- Don’t change the codes in a mnemonic system

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Ensure That Codes Are Unique

- For codes to work, they must be unique
- Each code must **identify one and only one entity**.
- Do not assign the same code number or name to the same items

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Allow Codes to Be Sortable

- Codes should be designed so they can be **easily sorted in a meaningful order**.
- Make sure that users can do what you intend to do with the codes you create
- Numerical codes are much easier to sort than alphanumeric data

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Avoid Confusing Codes

- Avoid confusing codes to reduce human error.
- Avoid using coding characters that look or sound alike
- **Common Confusing Characters**
  - 0 (zero) vs O (letter O)
  - 1 vs l vs I
  - 5 vs S

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Figure 11.12 Combining Look-Alike Characters in Codes Can Result in Errors

Code Format for Canadian Postal Code X9X 9X9

Handwritten Code	Actual Code	City, Province	Problem
L8S 4M4	L8S 4M4	Hamilton, Ontario	S looks like a 5
T3A ZE5	T3A 2E5	Calgary, Alberta	2 looks like a Z 5 looks like an S
LOS 1JO	L0S 1J0	Niagara-on-the-Lake, Ontario	Zero and Oh look alike S looks like a 5 1 looks like an l

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Keep the Codes Uniform

- All codes of the same type should **follow the same structure and length**.
- Need to follow readily perceived forms
- Keep codes uniform among as well as within programs
- Example of not uniform:
  - Avoid using the codes MMDDYY in one application
  - YYDDMM in a second
  - MMDDCCYY in a third; CC—century

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Allow for Modification of Codes

- The coding scheme should **anticipate change and expansion**.
- The system will evolve over time
- The coding system should be able to encompass change
- Poor vs Better Design:

Poor	Better
2-digit product code	4-digit expandable code
Fixed alphabetic only	Alphanumeric mix

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Make Codes Meaningful

- Codes should **convey useful information** to users **when appropriate**.
- Effective codes contain information
- Should make sense to people using them
- Easier to understand, work with, and recall
- The job of data entry becomes more interesting when working with meaningful codes instead of just entering a series of meaningless numbers.

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Codes Guidelines Simplified

Guideline	Primary Benefit
Be concise	Speed & accuracy
Keep stable	Data integrity
Ensure unique	Reliable identification
Allow sorting	Better reporting
Avoid confusion	Fewer human errors
Keep uniform	Consistency
Allow modification	Scalability
Make meaningful	Usability

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Using Codes

- Validation programs → to ensure that only valid codes have been entered.
- Report and inquiry programs → to display code meanings. Reports and displays should not print the actual codes.
- GUI programs → to create drop-down lists.

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Effective and Efficient Data Capture

- To ensure the quality of data users enter into the system, it is important to capture data effectively.
- Deciding what to capture
- Letting the computer do the rest
- Avoiding bottlenecks and extra steps
- Starting with a good form
- Choosing a data-entry method

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Deciding What to Capture

- Data that change or vary with every transaction
- Data that concisely differentiate the particular item being processed from all other items
- The decision about what to capture precedes user interaction with the system. “Garbage in, garbage out”

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Letting the Computer Do the Rest

- By employing the best features of computers, efficient data capture design avoids needless data entry, which in turn alleviates much human error and boredom and permits people to focus on higher-level or creative tasks.
- Recording the time of the transaction
- Calculating new values from input
- Storing and retrieving data on demand

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Avoiding Bottlenecks and Extra Steps

- Data are poured rapidly into the wide mouth of the system only to be slowed in its “neck” because of an artificially created instance of insufficient processing for the volume or detail of the data being entered
- The fewer steps involved in inputting data, the fewer chances there are for the introduction of errors
- One way a bottleneck can be avoided is by ensuring that there is enough capacity to handle the data that are being entered.

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Starting with a Good Form

- With effective forms, it is not necessary to reenter information that the computer has already stored, or data such as time or date of entry that the computer can determine automatically

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Why Choosing a Data-Entry Method?

- need for speed
- accuracy
- user training
- cost
- methods currently in use
- System scalability

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Choosing a Data-Entry Method

- Keyboards
- Bar codes
- QR (Quick Response) codes
- RFID
- NFC (Near Field Communication)
- Optical character recognition
- Magnetic ink character recognition (MICR)
- Mark-sense forms

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Keyboards

- Keyboards allow users to **manually type data** into a system.
- Special function keys to open programs
- Keys used to scroll and explore the Web
- Keys that can be programmed with macros to reduce the number of keystrokes required
- Ergonomic keyboards and infrared or Bluetooth-enabled keyboards

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Bar Codes

- Bar codes represent data using **parallel lines** that are scanned by optical readers.
- Affords a high degree of accuracy for data entry
- Saves labor costs
- Allows the automatic capturing of data → can be used for reordering, more accurate inventory tracking, forecasting.
- Tracking of credit card purchases

2D Bar Codes

- Are two-dimensional matrix bar code squares
- Are optical, machine-readable
- Take up much less space than older linear bar codes
- Are cheaper than RFID tags
- Can appear in print

QR (Quick Response) Codes

- A QR (Quick Response) code is easily identifiable
  - Uses a position marker
  - Three nested squares that appear in three of its corners
- QR codes are completely free
- 2D code readers are likely to be able to read a QR code

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Figure 11.14 One and Two Dimensional Bar Codes

One-Dimensional UPC Code	
Two-Dimensional QR Code	

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Figure 11.15 QR Codes

QR Codes Can

- Take you to a specific Web page
- Find the item you want for purchase at a lower price
- Initiate an email, SMS text message, or make a phone call
- Find and capture a digital coupon
- Use a digital coupon
- Tell you more about the art gallery painting you are gazing at
- Get you information about the time and place of an event
- Show you a promotional video for a movie, play, or concert
- Allow you to view or extend the amount of money on a transit card

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RFID

- Radio Frequency Identification (RFID) uses **radio waves** to automatically identify tagged objects **without line-of-sight**.
- Allows the automatic collection of data using RFID tags that contain a chip and an antenna
- Passive RFID tags
- Active RFID tags
- Privacy is a concern

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NFC (Near Field Communication)

- NFC enables **short-range, secure, two-way communication** by tapping devices.
- Allows two-way communication
- Is used in contactless payment systems at checkout counters
- Customers use a smartphone to touch or come close to the point-of-sale device

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NFC

- Other uses:
  - Provide payment on transportation systems
  - Exchange information
  - Receive custom coupons
  - Exchange business cards
- Considered secure because of short range of communication
- **Weaknesses are** Short range only & requires compatible devices

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Optical Character Recognition

- OCR converts **printed or handwritten text** into **machine-readable data**.
- Increased speed → can speed data input from 60 to 90 percent over some keying methods.
- Eliminates many of the time-consuming and error-fraught steps of other input devices
- Decentralizes responsibility for quality data directly to the unit that is generating it
- The transformation of faxes into documents that can be edited

Magnetic Ink Character Recognition

- MICR reads **special magnetic ink characters**, mainly on **bank cheques**.
- A reliable and high-speed method that is not susceptible to accepting stray marks
- If it is required on all withdrawal checks, it serves as a security measure against bad checks
- Data entry personnel can see the numbers making up the code if it is necessary to verify it

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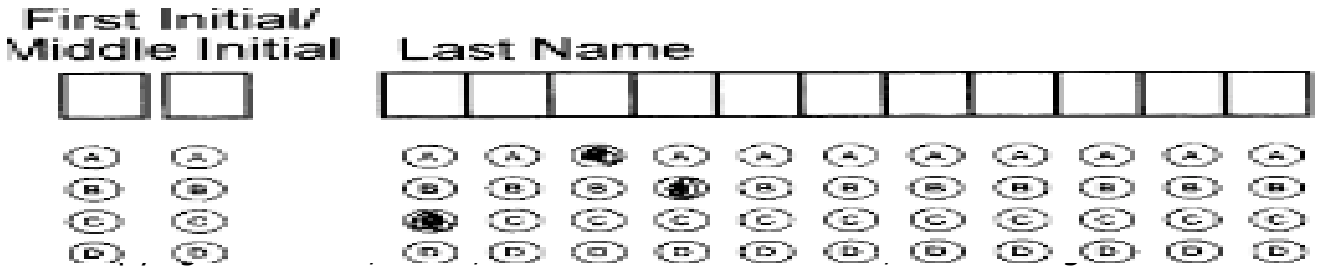
Mark-Sense Forms (OMR-Style Input)

- Mark-sense systems detect **filled marks** (✓, ●) on predefined forms.
- Little training of entry personnel is necessary
- A high volume of forms can be processed quickly
- Stray marks on form can be entered as incorrect data
- Choices are limited to the answers provided
- Difficulty in capturing alphanumeric data
- Easy to get confused and put a mark in an incorrect position



Comparison Summary

Method	Speed	Accuracy	Cost	Modern Use
Keyboard	Medium	Low–Medium	Low	Still common
Barcode	High	Very High	Low	Retail
QR Code	High	High	Very Low	Mobile systems
RFID	Very High	Very High	High	Logistics
NFC	Very High	Very High	Medium	Payments
OCR	Medium–High	Medium	Medium	Document processing
MICR	Very High	Very High	High	Banking
Mark-Sense	High	Medium	Low	Exams

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Ensuring Data Quality through Input Validation

- The critical importance of catching errors during input, prior to processing and storage cannot be overemphasized
- Potential problems
  - Validating input transactions
  - Validating input data

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Figure 11.17 Validating Input is Important to Ensure That Most Potential Problems with Data Are Eliminated Early

This Type of Validation	Can Prevent These Problems
Validating Input Transactions	Submitting the wrong data Data submitted by an unauthorized person Asking the system to perform an unacceptable function
Validating Input Data	Missing data Incorrect field length Data have unacceptable composition Data are out of range Data are invalid Data do not match with stored data

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Validating Input Transactions

- Submitting the wrong data
- Submission of data by an unauthorized person
- Asking the system to perform an unacceptable function

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Validating Input Data

- Test for missing data
- Test for correct field length
- Test for class or composition
- Test for range or reasonableness
- Test for invalid values
- Cross-reference checks
- Test for comparison with stored data
- Set up self-validating codes (check digit)

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The Process of Validation (1 of 2)

- Check first for missing data
- Check the syntax
- Check the semantics
- GUI screens can help to reduce the number of human input errors when they incorporate radio buttons, check boxes, and drop-down lists

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The Process of Validation (2 of 2)

- Regular expressions
- Validating XML documents
  - DTD
  - Schema

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Figure 11.18 These Characters Are Used in Regular Expression (Pattern) Validation

Character Code	Meaning Used in Regular Expression Validation
\d	Any digit 0–9
\D	Any nondigit character
\w	Any letter, number, or underscore
\W	Any character other than a letter, number, or underscore
.	Matches any character
[characters]	Matches the characters in the brackets
[char-char]	Matches the range of characters
[a–z][A–Z][0–9]	Will accept any letter or digit
[^characters]	Match anything other than the characters
[^char-char]	Match anything outside the range of characters
[^a–z]	Will accept anything except lowercase letters
{n}	Match exactly n occurrences of the preceding character
{n,}	Match at least n occurrences of the character
\s	Any white space formatting character (tab, new line, return, etc.)
\S	Any nonwhite space character

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Accuracy Advantages in Ecommerce Environments

- Customers generally key or enter data themselves → customers know their own information better than anyone else
- Data entered by customers are stored for later use → autocomplete and cookies
- Data entered at the point of sale are reused throughout the entire order fulfillment process → allows better supply chain management, including checking product and resource availability electronically, and automating planning, scheduling, and forecasting.
- Information is used as feedback to customers

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Cookies (1 of 2)

- A **cookie** is a **small text file** stored by a web browser on a user's device when they visit a website.
  It allows the website to **remember information about the user or their session**.
- Store information to enable faster and more accurate transactions
- Use a small file (the cookie) on the user's computer
- Once information is stored in a cookie, a Web site can suggest matches for credit card and password information

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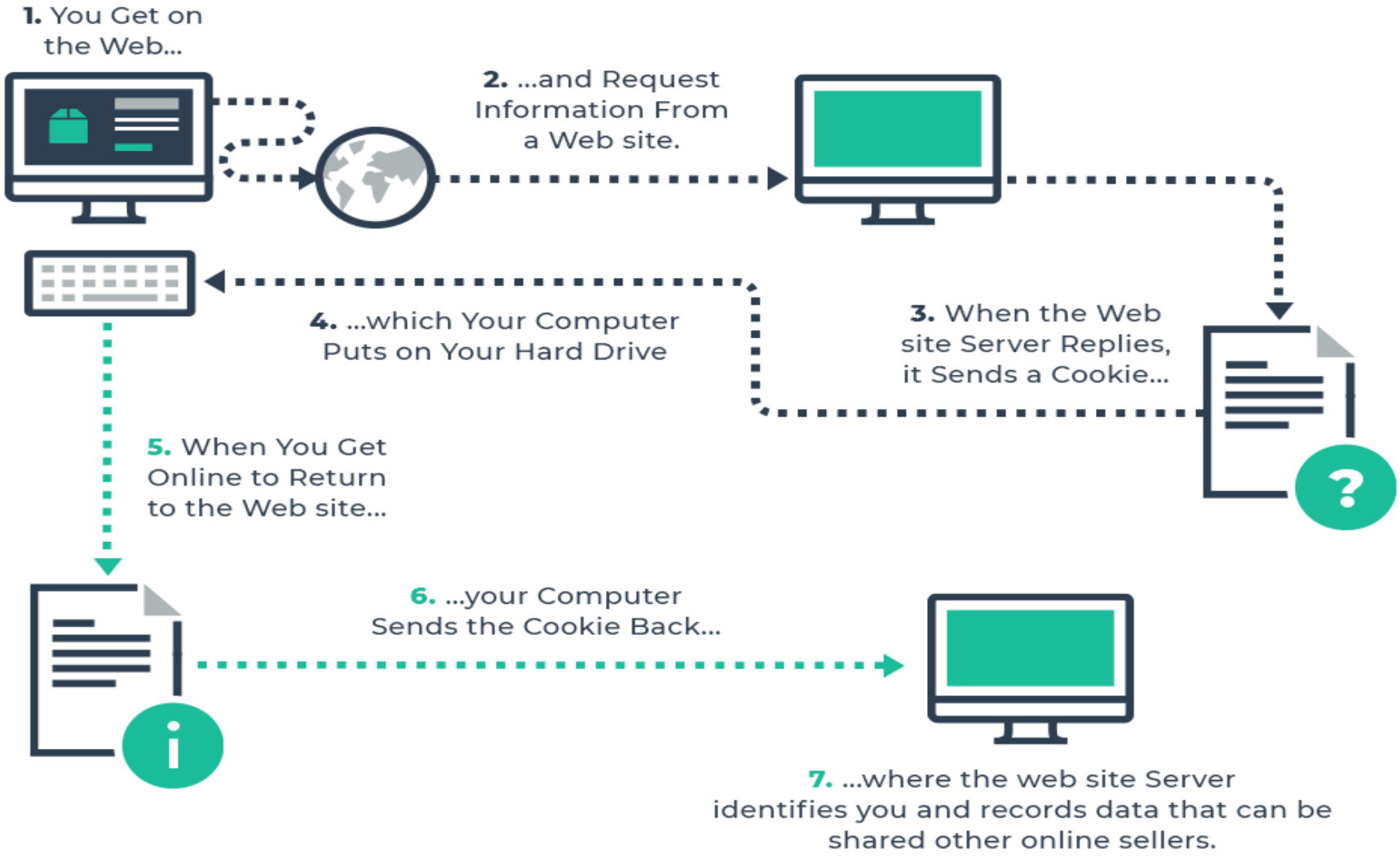
Cookies (2 of 2)

- Cookies are automatically sent back to the website **with each subsequent request**, enabling continuity.
- The information is encrypted so that other Web sites cannot read the cookie
- Personal information can only be accessed by the company that placed the cookie on the user's computer
- **First-Party Cookies** → Created by the website you visit & generally safer and necessary
  - **Example:** Shopping cart cookie from an online store
- **Third-Party Cookies** → Created by external domains, often used for tracking & ads. Increasingly restricted by browsers
  - **Example:** Advertising or analytics cookies

How Cookies works?



Summary (1 of 4)

- Quality of data
  - Effective coding
  - Effective and efficient data capture
  - Validation of data

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Summary (2 of 4)

- Coding
  - Sequence codes
  - Alphabetic derivation codes
  - Classification codes
  - Block sequence codes
  - Cipher codes
  - Significant-digit subset codes
  - Mnemonic codes
  - Uniform character set
  - Function codes

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Summary (3 of 4)

- Input devices
  - Keyboards
  - Bar codes
  - QR (Quick Response) codes
  - RFID
  - NFC (Near Field Communication)
  - Optical character recognition
  - Magnetic ink character recognition (MICR)
  - Mark-sense forms

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Summary (4 of 4)

- Input validation
  - Input transaction
  - Input data
- Ecommerce

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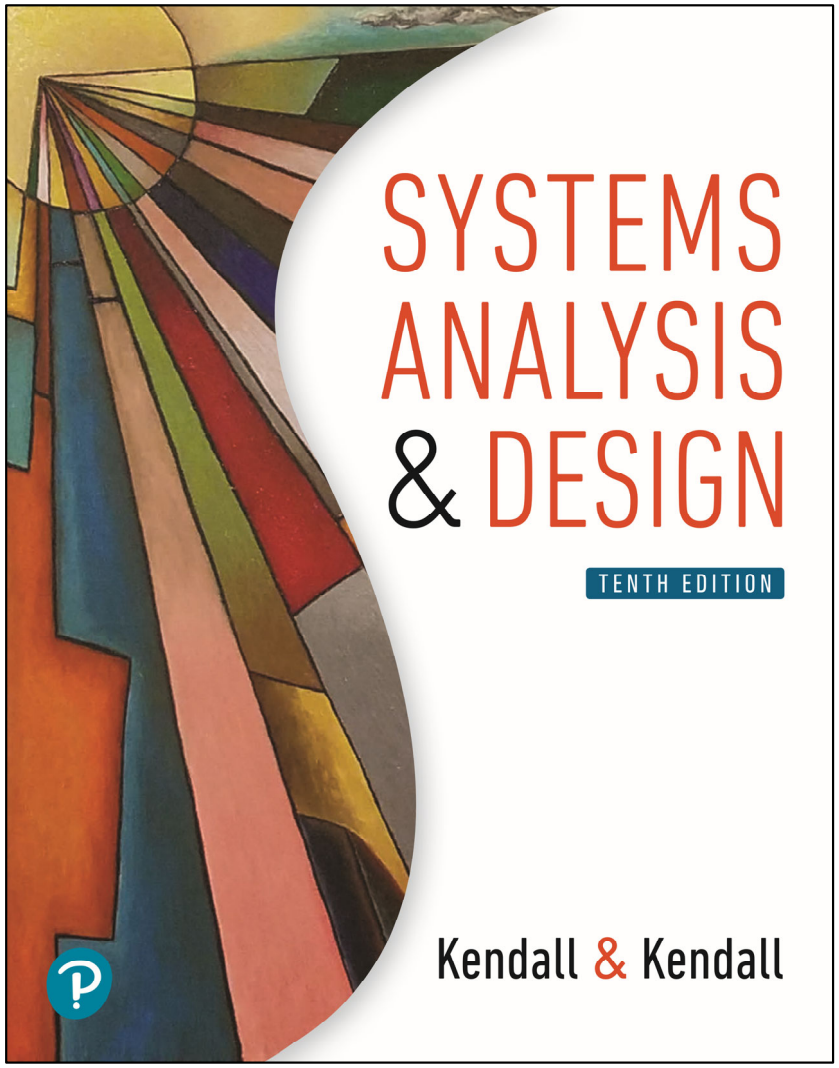
Systems Analysis & Design

Tenth Edition



Chapter 12

Quality Assurance and Implementation

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Learning Objectives (1 of 2)

12.1 Recognize the importance of users and analysts taking a total quality approach to improve the quality of software design and maintenance.

12.2 Realize the importance of documentation, testing, maintenance, and auditing.

12.3 Understand how service-oriented architecture and cloud computing is changing the nature of information system design.

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Learning Objectives (2 of 2)

12.4 Design appropriate training programs for users of the new system.

12.5 Recognize the differences among physical conversion strategies and be able to recommend an appropriate one to a client.

12.6 Address security, disaster preparedness, and disaster recovery concerns for traditional and Web-based systems.

12.7 Understand the importance of evaluating the new system, and be able to recommend a suitable evaluation technique to a client.

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Major Topics (1 of 2)

- Six Sigma
- Quality assurance
- Walkthroughs
- Structure charts
- Modules
- Documentation
- Testing

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Major Topics (2 of 2)

- Client-server computing
- Network types
- Groupware
- Training
- Security
- Organizational metaphors
- Evaluation

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Six Sigma

- Six sigma is a methodology, a philosophy and a culture.
- The goal of six sigma is to eliminate all defects.
- A culture built on quality
- Uses a top-down approach
- Project leader is called a Black Belt
- Project members are called Green Belts
- Master Black Belts have worked on many projects and are available as a resource to project teams

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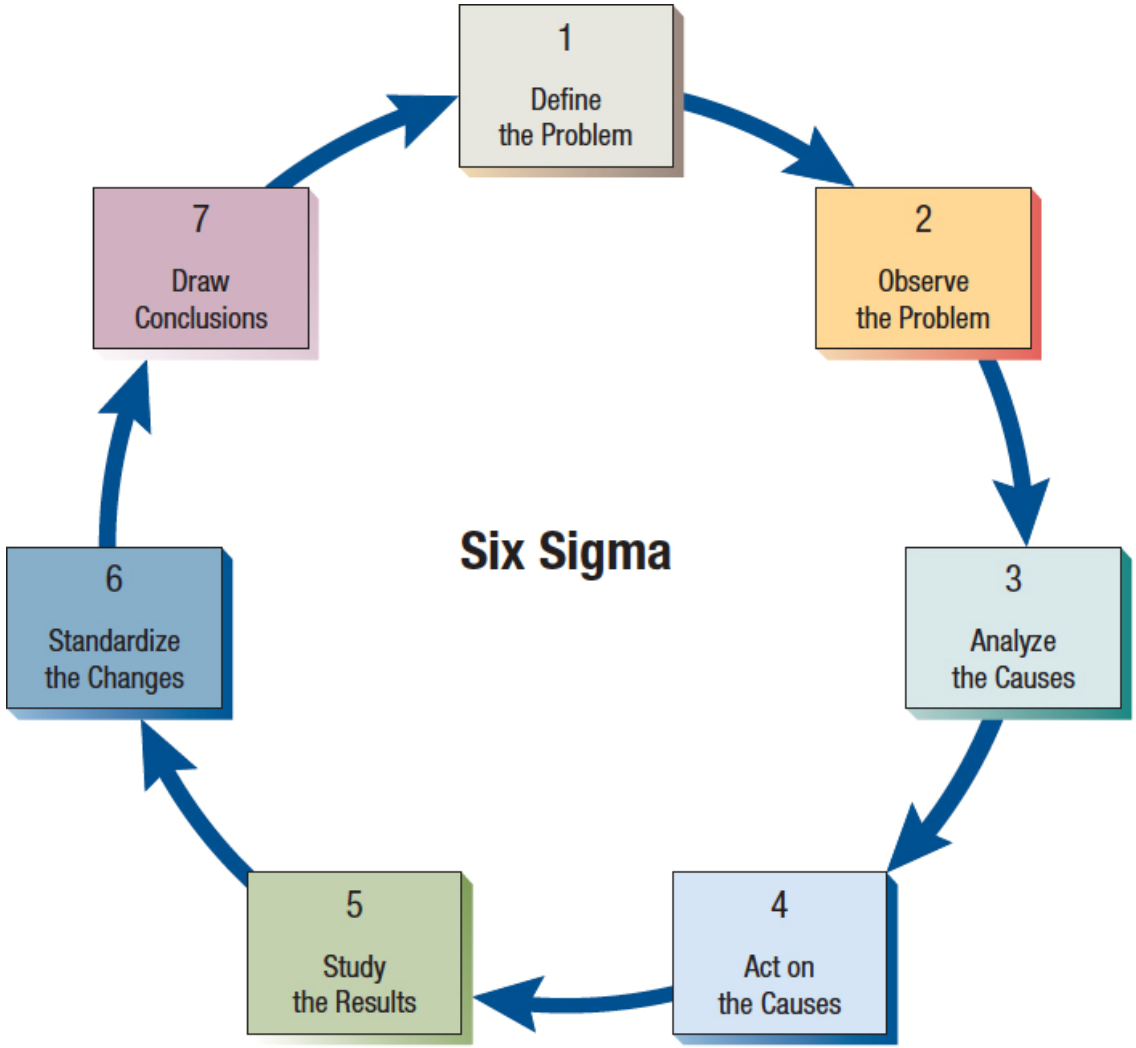
Figure 12.1 Every Systems Analyst Should Understand the Methodology and Philosophy of Six Sigma



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Responsibility for Total Quality Management

- Full organizational support of management must exist → establishing a context for management people to consider seriously how the quality of information systems and information itself affects their work
- Early commitment to quality from the analyst and business users

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Agile QA (Quality Assurance in Agile)

- **Agile QA** is a **continuous quality approach** where testing is **embedded throughout the development lifecycle**, not treated as a final phase.
- **Quality is everyone's responsibility**, not only testers.
- In Agile projects:
  - Testing starts **from Day 1**
  - QA works **alongside developers**
  - Feedback is **fast and frequent**

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Agile QA Principles

I. Shift-Left Testing

→ Test early to reduce cost of defects.

II. Test Automation First

→ Automate repetitive regression tests.

III. Customer-Centric Quality

→ Quality is defined by user satisfaction, not just “bug-free”.

IV. Fast Feedback Loops

→ Bugs are fixed within the same sprint.

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Structured Walkthroughs

- One of the strongest quality management actions is to do structured walkthroughs routinely
- Help to evaluate the product systematically on an ongoing basis.
- Use peer reviewers to monitor the system's programming and overall development
- Point out problems
- Allow the programmer or analyst to make suitable changes

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Involved in Structured Walkthroughs

- The person responsible for the part of the system being reviewed
- A walkthrough coordinator → ensures that the others adhere to any roles assigned to them and to ensure that any activities scheduled are accomplished.
- A programmer or analyst peer → there to listen.
- A peer who takes notes about suggestions → present to point out errors or potential problems.
- Note taker → records what is said so that the others present can interact without encumbrance.

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The Top-Down Approach

- The top-down approach means looking at the large picture of the system and then exploding it into smaller parts or subsystems.
- Top-down design allows the systems analyst to ascertain overall organizational objectives and how they are best met in an overall system
- The system is divided into subsystems and their requirements

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Advantages of the Top-Down Approach

- Avoiding the chaos of attempting to design a system all at once
- Enables separate systems analysis teams to work in parallel on different but necessary subsystems → can save time
- Prevents losing sight of what the system is supposed to do

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Modular Development

- Breaking the programming into logical, manageable portions or modules
- Works well with top-down design
- Each individual module should be functionally cohesive, accomplishing only one function

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Advantages of Modular Programming

- Modules are easier to write and debug
- Modules are easier to maintain
- Modules are easier to grasp because they are self-contained subsystems
- Modules are easier to write and debug:
  - Tracing an error in a module is less complicated.
  - A problem in one module should not cause problems in others.

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Guidelines for Modular Programming

- Keep each module to a manageable size
- Pay particular attention to the critical interfaces
- Minimize the number of modules the user must modify when making changes
- Maintain the hierarchical relationships set up in the top-down phases

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Using Structure Charts to Design Systems

- The recommended tool for designing a modular, top-down system is a structure chart
- A structure chart is simply a diagram consisting of rectangular boxes, representing the modules, and connecting lines
- Hollow circle arrows represent data being passed up and down the structure
- Filled-in circle arrow represents a control switch or flag

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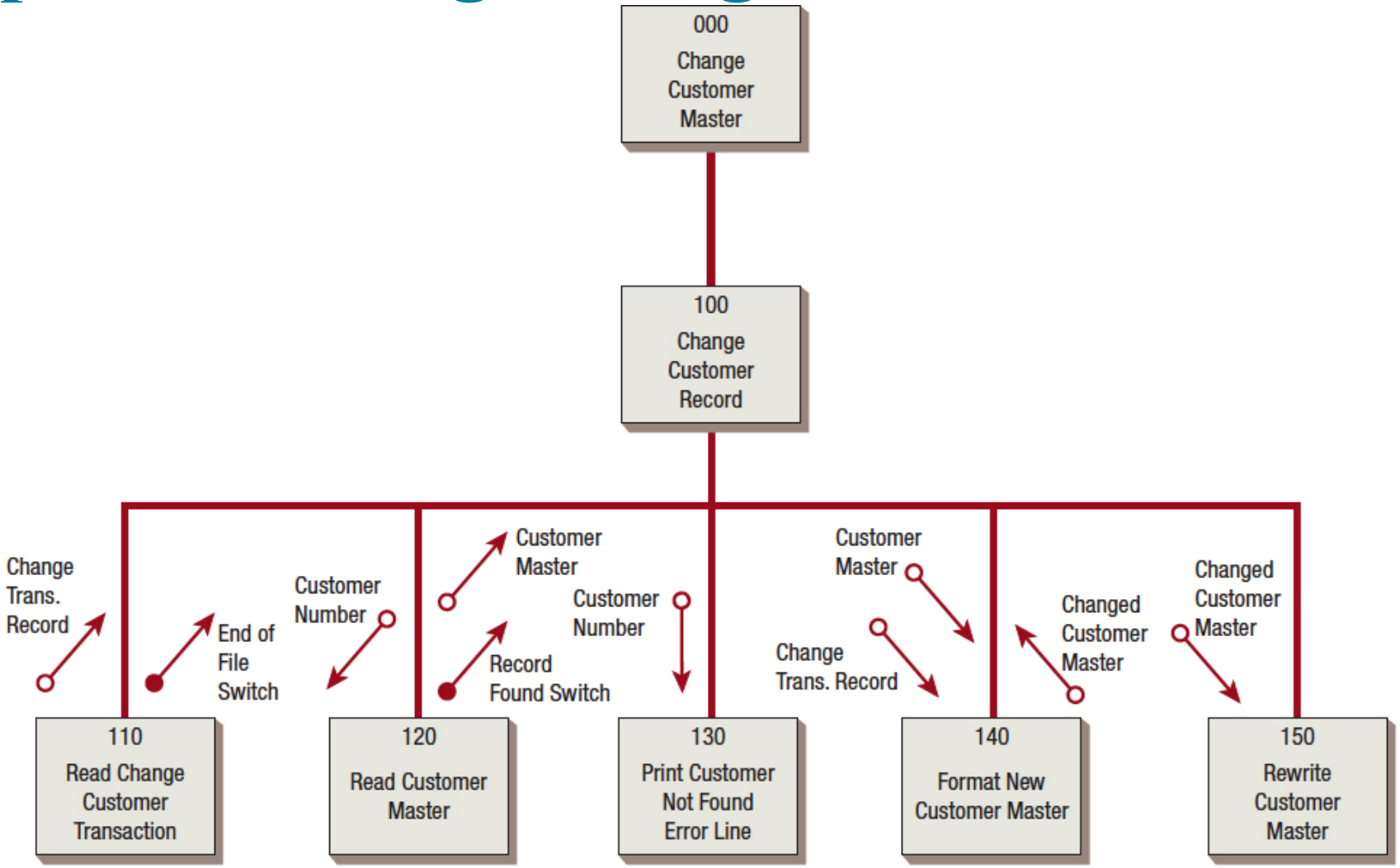
Figure 12.3 A Structure Chart Encourages Top-Down Design Using Modules



*** Modules 000 and 100 are control modules, representing the entire program.
*** Module 190 is a transformational module that formats the RESERVATION RECORD and performs module 200 to write the RESERVATION RECORD.
*** Modules 130, 140, 150, 170, 180, 200 are functional modules, performing only one task.

Service Oriented Architecture (SOA) (1 of 3)

- SOA is a design style or architectural pattern used in software engineering to build systems where software components, or **services**, are independent and interact over a network, often via web services or other communication protocols
- Each service executes one action
- Each service can be used in other applications within the organization or even in other organizations

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Service Oriented Architecture (SOA) (2 of 3)

- A service can use certain defined protocols so that it can communicate with other services
- Because services communicate using open standard protocols (e.g., SOAP, REST, HTTP, XML, JSON), SOA enables different systems, potentially built on different technologies, to interact seamlessly.
- This makes it easier to integrate various applications, even across platforms.

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Service Oriented Architecture (SOA) (3 of 3)

- In order to set up a SOA, services must be:
  - Modular
  - Reusable
  - Work together with other modules (interoperability)
  - Able to be categorized and identified
  - Able to be monitored
  - Comply with industry-specific standards

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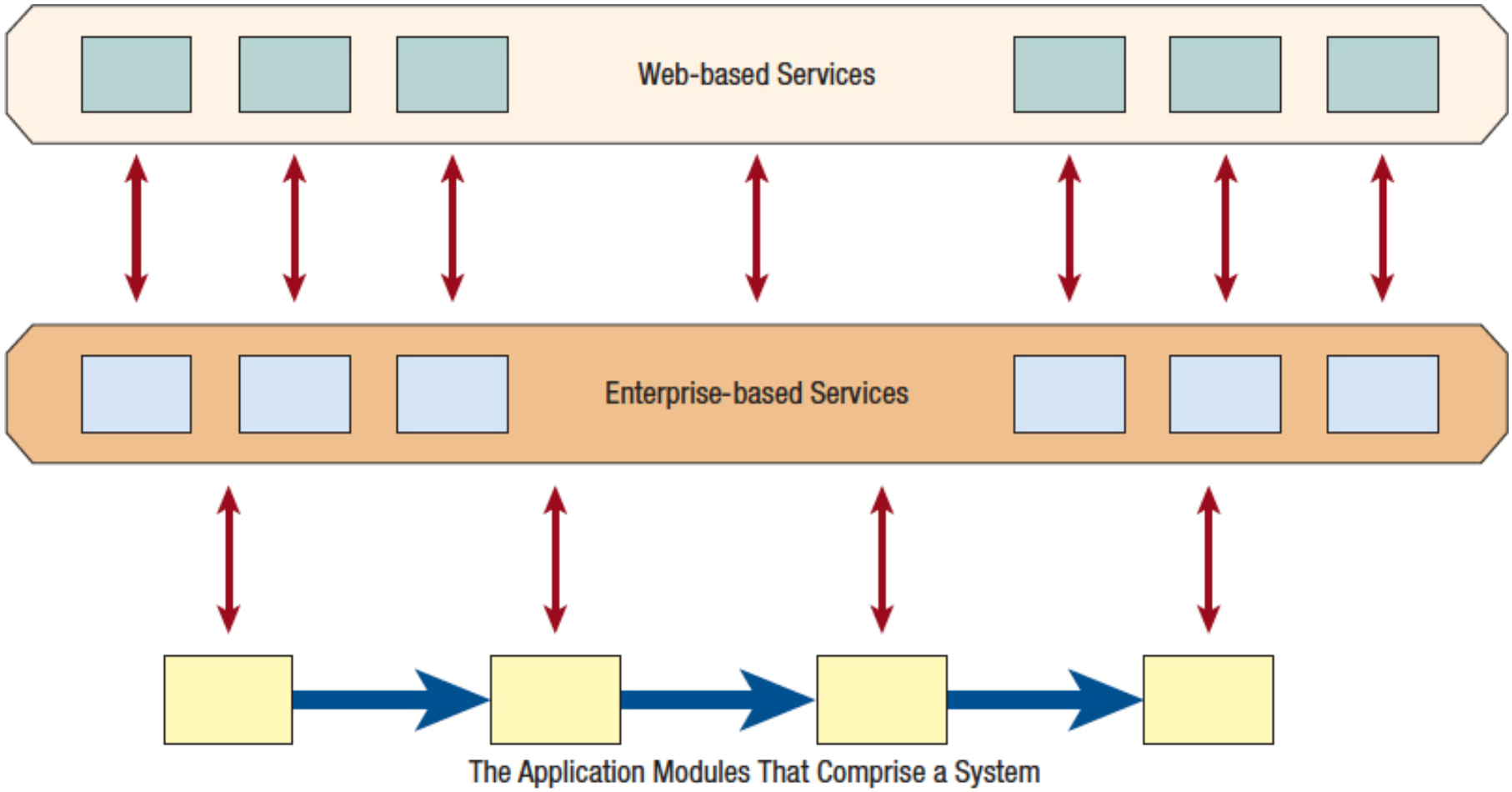
Figure 12.4 Modules in Service-Oriented Architecture Are Independent and Can Be Ubiquitous



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System Documentation

- There is no single standard design and documentation technique in use today. Each technique has its own advantages and disadvantages, because each one has unique properties.

- Procedure manuals
- The **Folklore** method

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Procedure Manuals (1 of 2)

- The English-language component of documentation
- May also contain program codes, flowcharts, and so on; intended to communicate to those who use them.
- Key sections:
  - Introduction
  - How to use the software
  - What to do if things go wrong
  - A technical reference section
  - An index
  - Information on how to contact the manufacturer

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Procedure Manuals (2 of 2)

- Procedure manual complaints:
  - They are poorly organized
  - It is hard to find needed information
  - The specific case in question does not appear in the manual
  - The manual is not written in plain English

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The FOLKLORE Method

- Created to supplement other document techniques.
- gathers information that is often shared among users but is seldom written down.
- Collects information in categories:
  - Customs → tries to capture in writing what users are currently doing to get all programs to run without problems.
  - Tales → stories that users tell regarding how the system worked.
  - Sayings → brief statements representing generalizations or advice.
  - Art forms → flowcharts, diagrams, and tables that users draw sometimes may be more useful than flowcharts drawn by the original system author.

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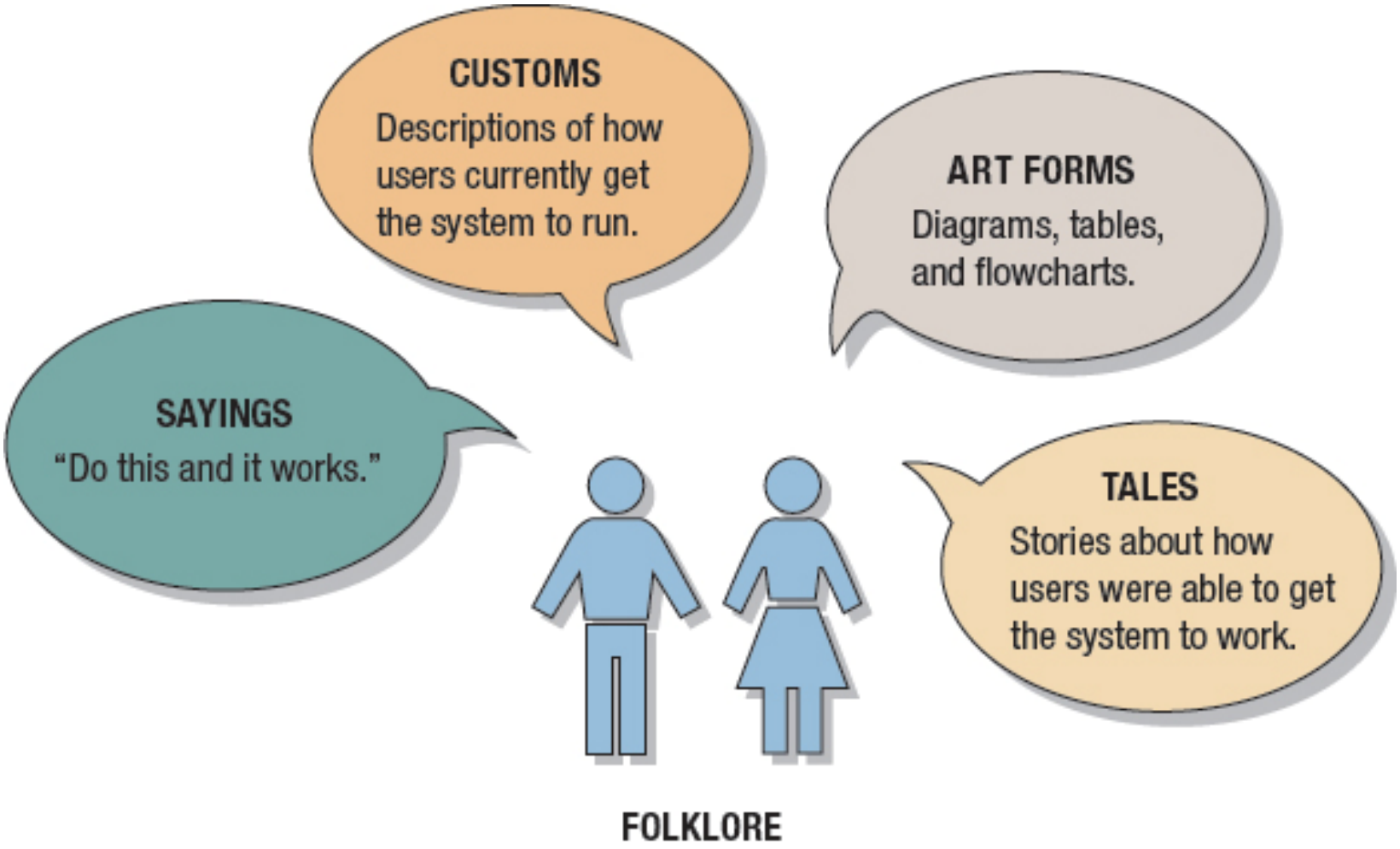
Figure 12.5 Customs, Tales, Sayings, and Art Forms Used in the FOLKLORE Method of Documentation Apply to Information Systems



** The danger of relying on FOLKLORE is that the information gathered from users may be correct, partially correct, or incorrect.

Choosing a Design and Documentation Technique (1 of 2)

- Is it compatible with existing documentation?
- Is it understood by others in the organization?
- Does it allow you to return to working on the system after you have been away from it for a period of time?

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Choosing a Design and Documentation Technique (2 of 2)

- Is it suitable for the size of the system you are working on?
- Does it allow for a structured design approach if that is considered to be more important than other factors?
- Does it allow for easy modification?

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Testing, Maintenance, and Auditing

- The testing process
- Maintenance practices
- Auditing

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The Testing Process

- Program testing with test data
- Link testing with test data
- Full system testing with test data
- Full system testing with live data

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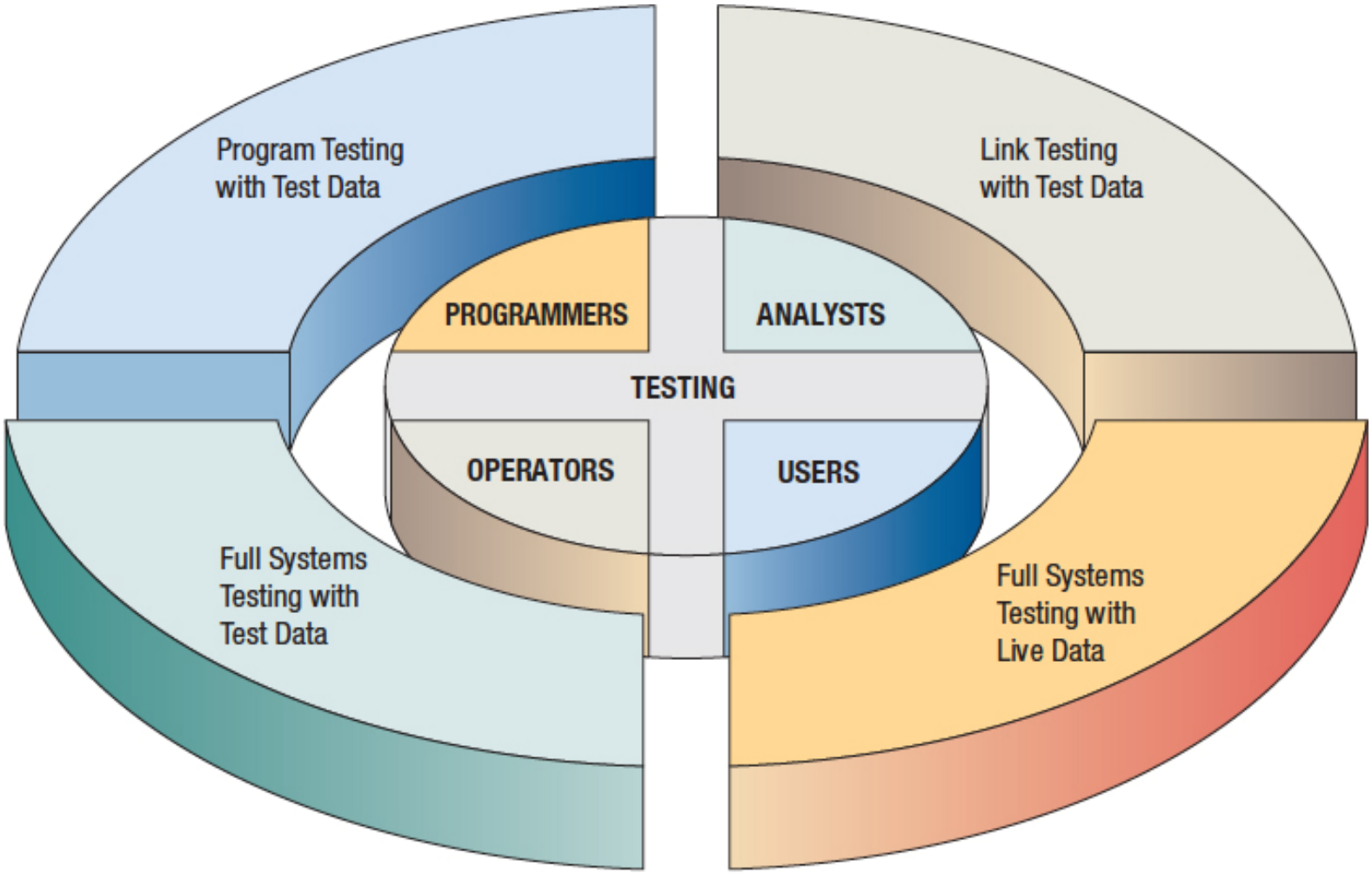
Figure 12.6 Programmers, Analysts, Operators, and Users All Play Different Roles in Testing Software and Systems



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Program Testing with Test Data

- Desk check programs → the programmer follows each step in the program on paper to check whether the routine works as it is written.
- Test with both valid and invalid data → these data are run to see if base routines work and also to catch errors.
- Check output for errors and make any needed corrections → the analyst might point out to the programmer omissions of data types to be added in later tests.

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Link Testing with Test Data

- Also referred to as **string testing**
- Checks to see if programs that are interdependent actually work together as planned
- Test for normal transactions → those that make up the bulk of the load.
- Test with invalid data → if the system works with normal transactions, variations are added, including invalid data used to ensure that the system can properly detect errors.

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Full System Testing with Test Data

- When link tests are concluded, the system as a complete entity must be tested.
- Operators and end users become actively involved in testing.
- Adequate documentation in procedure manuals
- Are procedure manuals clear enough?
- Do work flows actually “flow”?
- Is output correct and do users understand this output?

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Full System Testing with Live Data

- Live data → data that have been successfully processed through the existing system.
- Comparison of the new system's output with what you know to be correctly processed output
- Only small amounts of live data are used

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Information System Utility Approach (3 of 3)

- Form utility answers the question of what kind of output is distributed to the decision maker
- Time utility answers the question of when information is delivered
- Actualization utility involves how the information is introduced and used by the decision maker

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Website Evaluation (1 of 2)

- Know how often the Web site is visited
- Learn details about specific pages on the site
- Find out more about the Web site's visitors
- Modern evaluation uses:
  - Google Analytics
  - Heatmaps
  - Conversion tracking

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Website Evaluation (2 of 2)

- Discover if visitors can properly fill out the forms you designed
- Find out who is referring Web site visitors to the client's site
- Determine what browsers visitors are using
- Find out if the client's Web site visitors are interested in advertising

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Summary (2 of 5)

- Six Sigma
  - Define the problem
  - Observe the problem
  - Analyze the causes
  - Act on the causes
  - Study the results
  - Standardize the changes
  - Draw conclusions

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Summary (3 of 5)

- Structure charts
- Procedure manuals
- Folklore
- Testing

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